YASKAWA

NX100 OPTIONS
INSTRUCTIONS

FOR TROUBLESHOOTING

Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

MOTOMAN INSTRUCTIONS

NX100 INSTRUCTIONS
NX100 OPERATOR'S MANUAL
NX100 MAINTENANCE MANUAL

The NX100 operator's manual above corresponds to specific usage.
Be sure to use the appropriate manual.

Part Number: 162074-1CD
Revision: 1

YASKAWA
MOTOMAN ROBOTICS

MANUAL NO.
HW0483057 ⑥

MANDATORY

- This manual explains the troubleshooting for the NX100 system. Read this manual carefully and be sure to understand its contents before handling the NX100.
- General items related to safety are listed in Section 1: Safety of the NX100 Instructions. To ensure correct and safe operation, carefully read the NX100 Instructions before reading this manual.

CAUTION

- Some drawings in this manual are shown with the protective covers or shields removed for clarity. Be sure all covers and shields are replaced before operating this product.
- The drawings and photos in this manual are representative examples and differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice when necessary due to product improvements, modifications, or changes in specifications. If such modification is made, the manual number will also be revised.
- If your copy of the manual is damaged or lost, contact a YASKAWA representative to order a new copy. The representatives are listed on the back cover. Be sure to tell the representative the manual number listed on the front cover.
- YASKAWA is not responsible for incidents arising from unauthorized modification of its products. Unauthorized modification voids your product's warranty.

Notes for Safe Operation

Read this manual carefully before installation, operation, maintenance, or inspection of the NX100.

In this manual, the Notes for Safe Operation are classified as “WARNING”, “CAUTION”, “MANDATORY”, or ”PROHIBITED”.

 WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury to personnel.
 CAUTION	Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury to personnel and damage to equipment. It may also be used to alert against unsafe practices.
 MANDATORY	Always be sure to follow explicitly the items listed under this heading.
 PROHIBITED	Must never be performed.

Even items described as “CAUTION” may result in a serious accident in some situations. At any rate, be sure to follow these important items.



To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “CAUTION” and “WARNING”.

WARNING

- Before operating the manipulator, check that servo power is turned OFF when the emergency stop buttons on the front door of the NX100 and programming pendant are pressed.
  When the servo power is turned OFF, the SERVO ON LED on the programming pendant is turned OFF.

  Injury or damage to machinery may result if the emergency stop circuit cannot stop the manipulator during an emergency. The manipulator should not be used if the emergency stop buttons do not function.

Emergency Stop Button 

- Once the emergency stop button is released, clear the cell of all items which could interfere with the operation of the manipulator. Then turn the servo power ON.

  Injury may result from unintentional or unexpected manipulator motion.

Release of Emergency Stop 

- Observe the following precautions when performing teaching operations within the P-point maximum envelope of the manipulator:
  - View the manipulator from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Ensure that you have a safe place to retreat in case of emergency.

  Improper or unintended manipulator operation may result in injury.

- Confirm that no persons are present in the P-point maximum envelope of the manipulator and that you are in a safe location before:
  - Turning ON the NX100 power.
  - Moving the manipulator with the programming pendant.
  - Running the system in the check mode.
  - Performing automatic operations.

  Injury may result if anyone enters the P-point maximum envelope of the manipulator during operation. Always press an emergency stop button immediately if there is a problems. The emergency stop buttons are located on the right of the front door of the NX100 and the programming pendant.

CAUTION

- Perform the following inspection procedures prior to conducting manipulator teaching. If problems are found, repair them immediately, and be sure that all other necessary processing has been performed.

  -Check for problems in manipulator movement.
  -Check for damage to insulation and sheathing of external wires.

- Always return the programming pendant to the hook on the NX100 cabinet after use.

  The programming pendant can be damaged if it is left in the P-point maximum envelope of the manipulator, on the floor, or near fixtures.

- Read and understand the Explanation of Warning Labels in the NX100 Instructions before operating the manipulator.

Definition of Terms Used Often in This Manual

The MOTOMAN manipulator is the YASKAWA industrial robot product.
The manipulator usually consists of the controller, the programming pendant, and supply cables.
In this manual, the equipment is designated as follows.

Equipment	Manual Designation
NX100 Controller	NX100
NX100 Programming pendant	Programming pendant

Descriptions of the programming pendant keys, buttons, and displays are shown as follows:

Equipment		Manual Designation
Programming Pendant	Character Keys	The keys which have characters printed on them are denoted with [ ]. ex. [ENTER]
	Symbol Keys	The keys which have a symbol printed on them are not denoted with [ ] but depicted with a small picture. ex. page key  The cursor key is an exception, and a picture is not shown.
	Axis Keys Number Keys	“Axis Keys” and “Number Keys” are generic names for the keys for axis operation and number input.
	Keys pressed simultaneously	When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, ex. [SHIFT]+[COORD]
	Displays	The menu displayed in the programming pendant is denoted with { }. ex. {JOB}

Description of the Operation Procedure

In the explanation of the operation procedure, the expression "Select • • • " means that the cursor is moved to the object item and the SELECT key is pressed, or that the item is directly selected by touching the screen.

Explanation of Warning Labels

The following warning labels are attached to the manipulator and NX100.
Always follow the warnings on the labels.

WARNING

•The following warning labels are attached to NX100.
Always follow the warnings on the labels.

Neglect of this warning may cause a serious accident.

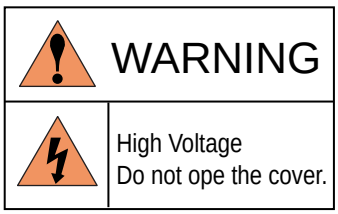
To find these labels, please refer the Manipulator Instruction book.

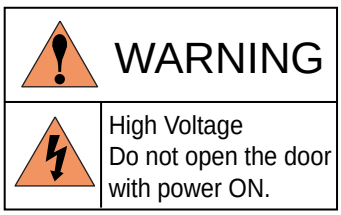


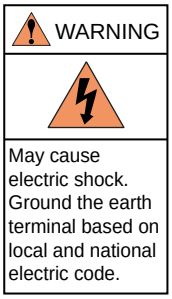


• The following warning labels are attached to the NX100.
Always follow the warnings on the labels.

Neglect of this warning may cause a serious accident.







To find these labels, please refer the Manipulator Instruction book

- Troubleshooting at the time of System Start 1
- Restoration method when PP disconnected occuuring 8
- Restoration method when TP ALARM occurring 10
- AL-0020 13
- AL-0021 14
- AL-0030 15
- AL-0060 16
- AL-0100 / 0101 / 0102 / 0103 17
- AL-0200 18
- AL-0210 19
- AL-0220 20
- AL-0230 21
- AL-0240 22
- AL-0270 23
- AL-0300 24
- AL-0310 25
- AL-0320 26
- AL-0330 27
- AL-0350 28
- AL-0390 29
- AL-0400 30
- AL-0410 31
- AL-0420 32
- AL-0500 33
- AL-0520 34
- AL-0710 35
- AL-0720 36
- AL-0800 37
- AL-0801 38
- AL-0802 39
- AL-0803 40
- AL-0900 41
- AL-0910 42
- AL-0920 43
- AL-0930 44
- AL-0950 / 0951 / 0952 / 0953 45
- AL-1001 46
- AL-1030 47
- AL-1031 48
- AL-1050 50

- AL-1051 51
- AL-1100 52
- AL-1101 52
- AL-1102 52
- AL-1103 53
- AL-1104 54
- AL-1105 55
- AL-1200 56
- AL-1204 57
- AL-1207 58
- AL-1208 58
- AL-1209 58
- AL-1210 58
- AL-1211 60
- AL-1212 61
- AL-1213 62
- AL-1214 62
- AL-1215 62
- AL-1216 62
- AL-1217 63
- AL-1218 64
- AL-1219 64
- AL-1220 65
- AL-1221 66
- AL-1223 67
- AL-1224 68
- AL-1225 69
- AL-1300 70
- AL-1301 72
- AL-1302 73
- AL-1303 74
- AL-1304 75
- AL-1306 76
- AL-1307 77
- AL-1308 78
- AL-1309 79
- AL-1310 79
- AL-1311 79
- AL-1312 79
- AL-1321 80

- AL-1322 81
- AL-1325 82
- AL-1325 83
- AL-1326 84
- AL-1327 85
- AL-1328 86
- AL-1329 87
- AL-1330 88
- AL-1331 89
- AL-1335 90
- AL-1335 91
- AL-1335 92
- AL-1335 93
- AL-1343 94
- AL-1345 95
- AL-1349 97
- AL-1352 98
- AL-1355 99
- AL-1500 100
- AL-1501 100
- AL-1502 100
- AL-1503 101
- AL-1504 102
- AL-1505 103
- AL-1506 103
- AL-1507 103
- AL-1508 103
- AL-1509 103
- AL-1510 104
- AL-1511 105
- AL-1512 106
- AL-1513 107
- AL-1514 108
- AL-1515 109
- AL-1516 109
- AL-1547 110
- AL-1592 111
- AL-1593 111
- AL-1594 111
- AL-4000 112

- AL-4001 113
- AL-4002 114
- AL-4003 115
- AL-4004 116
- AL-4005 117
- AL-4006 118
- AL-4007 119
- AL-4008 120
- AL-4009 121
- AL-4010 122
- AL-4012 123
- AL-4013 124
- AL-4014 125
- AL-4017 126
- AL-4018 127
- AL-4019 128
- AL-4020 129
- AL-4021 130
- AL-4024 131
- AL-4025 132
- AL-4028 133
- AL-4031 134
- AL-4032 135
- AL-4033 136
- AL-4034 137
- AL-4035 138
- AL-4036 139
- AL-4037 140
- AL-4038 141
- AL-4039 142
- AL-4040 143
- AL-4041 144
- AL-4042 145
- AL-4043 146
- AL-4044 147
- AL-4045 148
- AL-4046 149
- AL-4047 150
- AL-4048 151
- AL-4049 152

- AL-4050 153
- AL-4051 154
- AL-4052 155
- AL-4053 156
- AL-4054 157
- AL-4055 158
- AL-4056 159
- AL-4057 160
- AL-4058 161
- AL-4059 162
- AL-4060 163
- AL-4061 164
- AL-4062 165
- AL-4063 166
- AL-4064 167
- AL-4065 168
- AL-4069 169
- AL-4100 170
- AL-4101 171
- AL-4102 172
- AL-4103 172
- AL-4104 172
- AL-4105 172
- AL-4106 172
- AL-4107 173
- AL-4109 173
- AL-4110 174
- AL-4111 175
- AL-4112 176
- AL-4113 177
- AL-4114 178
- AL-4115 179
- AL-4116 180
- AL-4117 181
- AL-4118 182
- AL-4119 183
- AL-4121 184
- AL-4122 185
- AL-4123 186
- AL-4126 187

- AL-4130 188
- AL-4131 189
- AL-4132 190
- AL-4137 191
- AL-4200 192
- AL-4201 192
- AL-4202 192
- AL-4203 193
- AL-4204 193
- AL-4206 194
- AL-4207 195
- AL-4208 195
- AL-4209 195
- AL-4210 195
- AL-4220/4221 196
- AL-4222 197
- AL-4225 198
- AL-4228 199
- AL-4229 199
- AL-4230 199
- AL-4301 200
- AL-4302 201
- AL-4303 202
- AL-4304 203
- AL-4305 204
- AL-4306 205
- AL-4307 206
- AL-4308 207
- AL-4309 208
- AL-4310 209
- AL-4311 210
- AL-4312 211
- AL-4313 212
- AL-4314 213
- AL-4314 214
- AL-4314 215
- AL-4314 216
- AL-4315 217
- AL-4315 218
- AL-4315 219

- AL-4318 220
- AL-4320 221
- AL-4321 222
- AL-4322 222
- AL-4323 222
- AL-4324 223
- AL-4326 224
- AL-4327 225
- AL-4328 226
- AL-4328 227
- AL-1332 228
- AL-4331 229
- AL-1333 230
- AL-4334 231
- AL-4334 232
- AL-4335 233
- AL-4335 234
- AL-4335 235
- AL-4336 236
- AL-4336 237
- AL-4337 238
- AL-4337 239
- AL-4337 240
- AL-4338 241
- AL-4339 242
- AL-4340 243
- AL-4344 244
- AL-4345 244
- AL-4346 244
- AL-4347 244
- AL-4348 244
- AL-4349 244
- AL-4350 244
- AL-4352 245
- AL-4353 246
- AL-4355 247
- AL-4359 248
- AL-4364 249
- AL-4365 250
- AL-4371 251

- AL-4372 252
- AL-4400 253
- AL-4401 253
- AL-4402 253
- AL-4404 253
- AL-4406 253
- AL-4407 254
- AL-4408 254
- AL-4409 254
- AL-4410 254
- AL-4411 254
- AL-4412 254
- AL-4413 255
- AL-4414 255
- AL-4415 255
- AL-4416/AL-4417 256
- AL-4418/AL-4419 256
- AL-4420/AL-4421 256
- AL-4422/AL-4423 256
- AL-4424/AL-4425 256
- AL-4426/AL-4427 256
- AL-4428 257
- AL-4429 257
- AL-4430 257
- AL-4431 257
- AL-4432 257
- AL-4433 257
- AL-4435 258
- AL-4436 258
- AL-4437 258
- AL-4438 258
- AL-4439 259
- AL-4440 259
- AL-4441 259
- AL-4444 260
- AL-4445 261
- AL-4446 262
- AL-4447 262
- AL-4448 262
- AL-4449 264

- AL-4450 264
- AL-4451 264
- AL-4452 264
- AL-4453 264
- AL-4454 265
- AL-4455 265
- AL-4456 265
- AL-4457 265
- AL-4459 265
- AL-4460 265
- AL-4461 266
- AL-4462 266
- AL-4463 266
- AL-4464 266
- AL-4465 266
- AL-4466 267
- AL-4467 267
- AL-4468 267
- AL-4469 268
- AL-4470 268
- AL-4471 268
- AL-4473 269
- AL-4474 269
- AL-4475 269
- AL-4476 270
- AL-4477 270
- AL-4480 270
- AL-4487 271
- AL-4490 271
- AL-4491 272
- AL-4492 272
- AL-4493 272
- AL-4494 273
- AL-4495 273
- AL-4496 273
- AL-4497 274
- AL-4498 275
- AL-4499 275
- AL-4500 275
- AL-4501 275

- AL-4505 276
- AL-4506 276
- AL-4507 276
- AL-4508 277
- AL-4509 277
- AL-4510 277
- AL-4511 278
- AL-4512 279
- AL-4513 280
- AL-4514 280
- AL-4515 281
- AL-4516 281
- AL-4517 282
- AL-4518 282
- AL-4521 282
- AL-4524 282
- AL-4527 283
- AL-4528 283
- AL-4529 284
- AL-4534 285
- AL-4542 286
- AL-4543 286
- AL-4546 286
- AL-4550 287
- AL-4551 287
- AL-4566 287
- AL-4571 287
- AL-4572 288
- AL-4573 288
- AL-4574 288
- AL-4575 288
- AL-4576 288
- AL-4577 288
- AL-4578 289
- AL-4579 289
- AL-4581 289
- AL-4587 290
- AL-4591 291
- AL-4592 291
- AL-4597 291

- AL-4599 291
- AL-4601 291
- AL-4603 292
- AL-4606 292
- AL-4607 292
- AL-4608 292
- AL-4612 293
- AL-4621 293
- AL-4622 293
- AL-4652 293
- AL-4653 293
- AL-4800 294
- AL-4901 295
- AL-4332 296
- Appendix 298

■ Troubleshooting at the time of System Start

If NX100 does not start up and the alarm is not displayed, the NX100 turn off and then turn on. If this does not change the situation, take an action following the instructions below.

Table. 1

Screen status of programming pendant	7SEG LED for the robot I/F unit (NIF01)	Cause	Remedy
 *Not displayed.	Lights off	The CPU unit is not powered.	Check the state of the primary power supply.
			Check that the controller power has not been turned off from outside.
			Replace the controller power Supply (CPS-420F) (*1).
	Other than above.	Although the CPU unit is powered, the programming pendant is not powered.	Check the cable connection status between the robot I/F unit and the programming pendant.
			Replace the cable between the robot I/F unit and the programming pendant with a new one.
		The programming pendant is out of order or malfunction.	Replace the programming pendant.
		The robot I/F unit is out of order or malfunction.	Remove the robot I/F unit from the CPU unit and reinstall it.
			Replace the robot I/F unit (*2).

Table. 2

Screen status of programming pendant	7SEG LED for the robot I/F unit	NCP01 LED (two)	Cause	Remedy
 *Illustration of robot	Lights off		The robot I/F unit is out of order or malfunction.	Remove the robot I/F from the CPU unit and reinstall it.
				Replace the robot I/F unit (*2).
	Lights up all [8.]	One or more lights are off.	NCP01 is out of order or malfunction.	Remove the NCP01 unit from the CPU unit and reinstall it.
				Replace the NCP01 (*3).
			The optional board inserted into the option slot in the CPU rack is out of order or malfunction.	Remove the boards from the option slots #1 and #2 and reinstall them.
				Replace the boards in the option slots #1 and #2 with new ones.
			The backboard is out of order.	Replace the whole CPU unit (*6).
		Both lights are on.	The system program cannot be properly read from the compact flash in NCP01.	Remove the compact flash from NCP01 and reinstall it.
				Replace the compact flash of NCP01 (*4).

Screen status of programming pendant	7SEG LED for the robot I/F unit	NCP01 LED	Cause	Remedy
 *Illustration of robot	[0] or [1]		NCP01 is out of order or malfunction.	Remove the robot I/F unit from the CPU unit and reinstall it.
				Replace the NCP01 (*3).
			The system program cannot be properly read from the compact flash in NCP01.	Remove the compact flash from NCP01 and reinstall it.
				Replace the compact flash of NCP01 (*4).
			Specified data cannot be properly read from the compact flash in NCP01 or an invalid value is specified.	Remove the compact flash from NCP01 and reinstall it.
				Execute "CMOS LOAD" or "SYSTEM RESTORE" in the maintenance mode (*5).
				Replace the compact flash of NCP01 (*4).
			The robot I/F unit is out of order or malfunction.	Remove the robot I/F unit from the CPU unit and reinstall it.
				Replace the robot I/F unit (*2).
			The optional board inserted into the option slot in the CPU rack is out of order or malfunction.	Remove the boards from the option slots #1 and #2 and reinstall them.
				Replace the boards in the option slots #1 and #2.
			The backboard is out of order.	Replace the whole CPU unit (*6).
	Other than above.		A communication error has occurred between NCP01 and the programming pendant.	Check the cable connection status between the NCP01 and the programming pendant.
				Replace the cable between the NCP01 and the programming pendant.
			The programming pendant is out of order or malfunction.	Replace the programming pendant.

Table. 3

Screen status of programming pendant	7SEG LED for the robot I/F unit (NIF01)	Cause	Remedy
 *Blue gradation	[2] to [9] or [b] to [c]	NCP01 is out of order or malfunction.	Remove the NCP01 from the CPU unit and reinstall it.
			Replace the NCP01 (*3).
		The system program cannot be properly read from the compact flash in NCP01.	Remove the compact flash from NCP01 and reinstall it.
			Replace the compact flash of NCP01 (*4).

Screen status of programming pendant	7SEG LED for the robot I/F unit (NIF01)	Cause	Remedy
 *Blue gradation	[2] to [9] or [b] to [c]	Specified data cannot be properly read from the compact flash in NCP01 or an invalid value is specified.	Remove the compact flash from NCP01 and reinstall it.
			Execute "CMOS LOAD" or "SYSTEM RESTORE" in the maintenance mode (*5).
			Replace the compact flash of NCP01 (*4).
		The robot I/F unit is out of order or malfunction.	Remove the robot I/F unit from the CPU unit and reinstall it.
			Replace the robot I/F unit (*2).
		The programming pendant is out of order or malfunction.	Replace the programming pendant.
		The optional board inserted into the option slot in the CPU rack is out of order or malfunction.	Remove the boards from the option slots #1 and #2 and reinstall them.
			Replace the boards in the option slots #1 and #2.
		The backboard is out of order.	Replace the whole CPU unit (*6).
	[E]	A communication error has occurred between NCP01 and the programming pendant.	Check the cable connection status between the NCP01 and the programming pendant.
			Replace the cable between the NCP01 and the programming pendant.
		Specified data cannot be properly read from the compact flash in NCP01 or an invalid value is specified.	Remove the compact flash from NCP01 and reinstall it.
			Execute "CMOS LOAD" or "SYSTEM RESTORE" in the maintenance mode (*5).
			Replace the compact flash of NCP01 (*4).
		The programming pendant is out of order or malfunction.	Replace the programming pendant.
	[d] or [F]	A communication error has occurred between NCP01 and the programming pendant.	Check the cable connection status between the NCP01 and the programming pendant.
			Replace the cable between the NCP01 and the programming pendant.
		The version of NCP01 and programming pendant are inconsistent.	Replace it with the programming pendant with the consistent version.
		The programming pendant is out of order or malfunction.	Replace the programming pendant.

Table. 4

Screen status of programming pendant	7SEG LED for the robot I/F unit (NIF01)	Cause	Remedy
 *The maintenance mode is activated at power-on even though it is not selected.	[F]	The data in the robot I/F unit is an invalid value.	Check the battery voltage. If the battery voltage has dropped, replace the battery with a new one.
			Execute "CMOS LOAD" or "SYSTEM RESTORE" in the maintenance mode (*5).
		The robot I/F unit is out of order or malfunction.	Remove the NCP01 from the CPU unit and reinstall it.
			Replace the robot I/F unit (*2).

Table. 5

Error status determined by the function other than the programming pendant	Cause	Remedy
At least one of (two) NCP01 LEDs is off.	NCP01 is out of order or malfunction.	Remove the NCP01 from the CPU unit and reinstall it.
		Remove the NCP01 (*3).
	The backboard is out of order.	Replace the whole CPU unit (*6).
The dot of 7SEG LED in the robot I/F unit does not blink.	NCP01 is out of order or malfunction.	Remove the NCP01 from the CPU unit and reinstall it.
		Replace the NCP01 (*3).
	Specified data cannot be properly read from the compact flash in NCP01 or an invalid value is specified.	Remove the compact flash from NCP01 and reinstall it.
		Execute "CMOS LOAD" or "SYSTEM RESTORE" in the maintenance mode (*5).
		Replace the compact flash of NCP01 (*4).
	The backboard is out of order.	Replace the whole CPU unit (*6).
The 7SEG LED in the robot I/F displays "P."	After normal startup, a communication error has occurred between NCP01 and the programming pendant.	Check the cable connection status between the NCP01 and the programming pendant.
		Replace the cable between the NCP01 and the programming pendant with a new one.
	The programming pendant is out of order or malfunction.	Replace the programming pendant.
AXA01/AXA02 7SEG LEDs display "F003."	NCP01 is out of order or malfunction.	Remove the NCP01 from the CPU unit and reinstall it.
		Replace the NCP01 (*3).
	Specified data cannot be properly read from the compact flash in NCP01 or an invalid value is specified.	Remove the compact flash from NCP01 and reinstall it.
		Execute "CMOS LOAD" or "SYSTEM RESTORE" in the maintenance mode (*5).
		Replace the compact flash of NCP01 (*4).

AXA01/AXA02 7SEG LEDs display "F003."	A communication error has occurred between NCP01 and AXA01/AXA02.	Check the cable connection status between the NCP01 and the AXA01/AXA02.
		Replace the cable between the NCP01 and the AXA01/AXA02 with a new one.
	The backboard is out of order.	Replace the whole CPU unit. (*6).

*1: Replace the control power supply (CPS-420F).
- For the replacement procedures, refer to "5.1.1 Replacing Parts of the CPU Unit" in "NX100 MAINTENANCE MANUAL."

*2: Replace the robot I/F unit.
- For the replacement procedures, refer to "5.1.1 Replacing Parts of the CPU Unit" in "NX100 MAINTENANCE MANUAL."

*3: Replace NCP01.
- For the replacement procedures, refer to "5.1.1 Replacing Parts of the CPU Unit" in "NX100 MAINTENANCE MANUAL."

*4: Replace the NCP01 compact flash.
Perform replacement with the following steps.

1. Save the current data in NX100.
   - Insert the compact flash used to save data into the programming pendant, start it in the maintenance mode, and select {Tool} ⇒ {Compact Flash} ⇒ "CMOS SAVE".
2. Prepare a compact flash with the same version.
   - Prepare the compact flash containing the NX software with the same version as the currently used compact flash.
3. Remove the current compact flash from NCP01.
4. Install the new compact flash into NCP01.
5. Start it in the maintenance mode and load backup data.

<When backup data is CMOS.BIN>
- Copy the CMOS.BIN file saved in advance to the compact flash used to save data and insert the compact flash into the programming pendant.
- Change the security mode to the management mode and select {Tool} ⇒ {Compact Flash} ⇒ "CMOS LOAD".

<When backup data is CMOSBK.BIN, CMOSBK1.BIN through CMOSBK99.BIN>
- Copy the backup data saved by the automatic backup function (CMOSBK.BIN, CMOSBK1.BIN through CMOSBK99.BIN) to the compact flash used to save data and insert the compact flash into the programming pendant.
- Change the security mode to the management mode and select {Tool} ⇒ {Compact Flash} ⇒ "SYSTEM RESTORE."
- When the list of backup files is displayed, select the files you want to load to NX100 and load them.
  ∗ If the version is earlier than the one shown below, initialize in the manufacturer mode before executing "CMOS LOAD" or "SYSTEM RESTORE."
  NS3.10-00, NS2.30-02, NT2.40-02, NS2.0P-00, NS2.0P-12, NJ3.20-00

∗ If the version is earlier than NS1.8HA-00, start the compact flash in the online mode after executing "CMOS LOAD" and load all individual data except for jobs and parameters.

*5: Execute "CMOS LOAD" or "SYSTEM RESTORE" in the maintenance mode.
Perform replacement with the following steps.

1. Save the current data in NX100.
   • Insert the compact flash used to save data into the programming pendant, start it in the maintenance mode, and select {Tool} ⇒ {Compact Flash} ⇒ "CMOS SAVE".
2. Start the compact flash in the maintenance mode and load backup data.

<When backup data is CMOS.BIN>
• Copy the CMOS.BIN file saved in advance to the compact flash used to save data and insert the compact flash into the programming pendant.
• Change the security mode to the management mode and select {Tool} ⇒ {Compact Flash} ⇒ "CMOS LOAD".

<When backup data is CMOSBK.BIN, CMOSBK1.BIN through CMOSBK99.BIN>
• Copy the backup data saved by the automatic backup function (CMOSBK.BIN, CMOSBK1.BIN through CMOSBK99.BIN) to the compact flash used to save data and insert the compact flash into the programming pendant.
• Change the security mode to the management mode and select {Tool} ⇒ {Compact Flash} ⇒ "SYSTEM RESTORE."
• When the list of backup files is displayed, select the files you want to load to NX100 and load them.
∗ If the version is earlier than the one shown below, initialize in the manufacturer mode before executing "CMOS LOAD" or "SYSTEM RESTORE."
NS3.10-00, NS2.30-02, NT2.40-02, NS2.0P-00, NS2.0P-12, NJ3.20-00
∗ If the version is earlier than NS1.8HA-00, start the compact flash in the online mode after executing "CMOS LOAD" and load all individual data except for jobs and parameters.

*6:Replace the whole CPU unit.
Perform replacement with the following steps.

1. Save the current data in NX100.
   • Insert the compact flash used to save data into the programming pendant, start it in the maintenance mode, and select {Tool} ⇒ {Compact Flash} ⇒ "CMOS SAVE".
2. Shut off the power.
3. Remove all wires from the CPU unit.
4. Replace the CPU unit with a new one.
5. Install all the removed wires.
6. Set the same value as the old board to the rotary switch on each board within the CPU unit.
7. Start the compact flash in the maintenance mode and load backup data.

<When backup data is CMOS.BIN>
• Copy the CMOS.BIN file saved in advance to the compact flash used to save data and insert the compact flash into the programming pendant.
• Change the security mode to the management mode and select {Tool} ⇒ {Compact Flash} ⇒ "CMOS LOAD".

<When backup data is CMOSBK.BIN, CMOSBK1.BIN through CMOSBK99.BIN>

- Copy the backup data saved by the automatic backup function (CMOSBK.BIN, CMOSBK1.BIN through CMOSBK99.BIN) to the compact flash used to save data and insert the compact flash into the programming pendant.
- Change the security mode to the management mode and select {Tool} ⇒ {Compact Flash} ⇒ "SYSTEM RESTORE."
- When the list of backup files is displayed, select the files you want to load to NX100 and load them.
  - ∗ If the version is earlier than the one shown below, initialize in the manufacturer mode before executing "CMOS LOAD." or "SYSTEM RESTORE."
    NS3.10-00, NS2.30-02, NT2.40-02, NS2.0P-00, NS2.0P-12, NJ3.20-00
  - ∗ If the version is earlier than NS1.8HA-00, start the compact flash in the online mode after executing "CMOS LOAD" and load all individual data except for jobs and parameters.



Be sure to back up data in case that the board fails when changing the settings.

To back up data, perform the following steps.

<Manual backup>

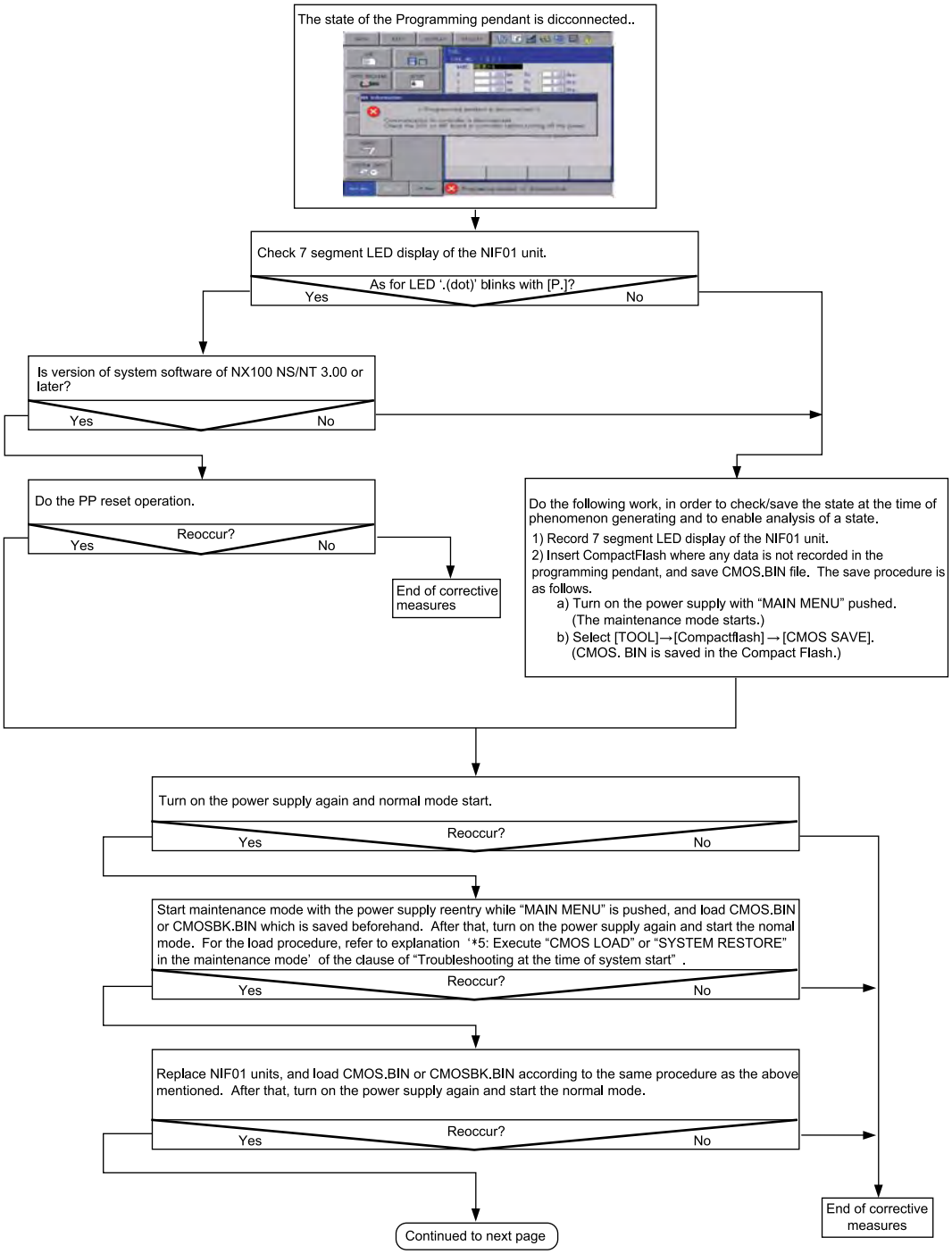
- Insert the compact flash used to save data into the programming pendant, start it in the maintenance mode, and select {Tool} ⇒ {Compact Flash} ⇒ "CMOS SAVE".

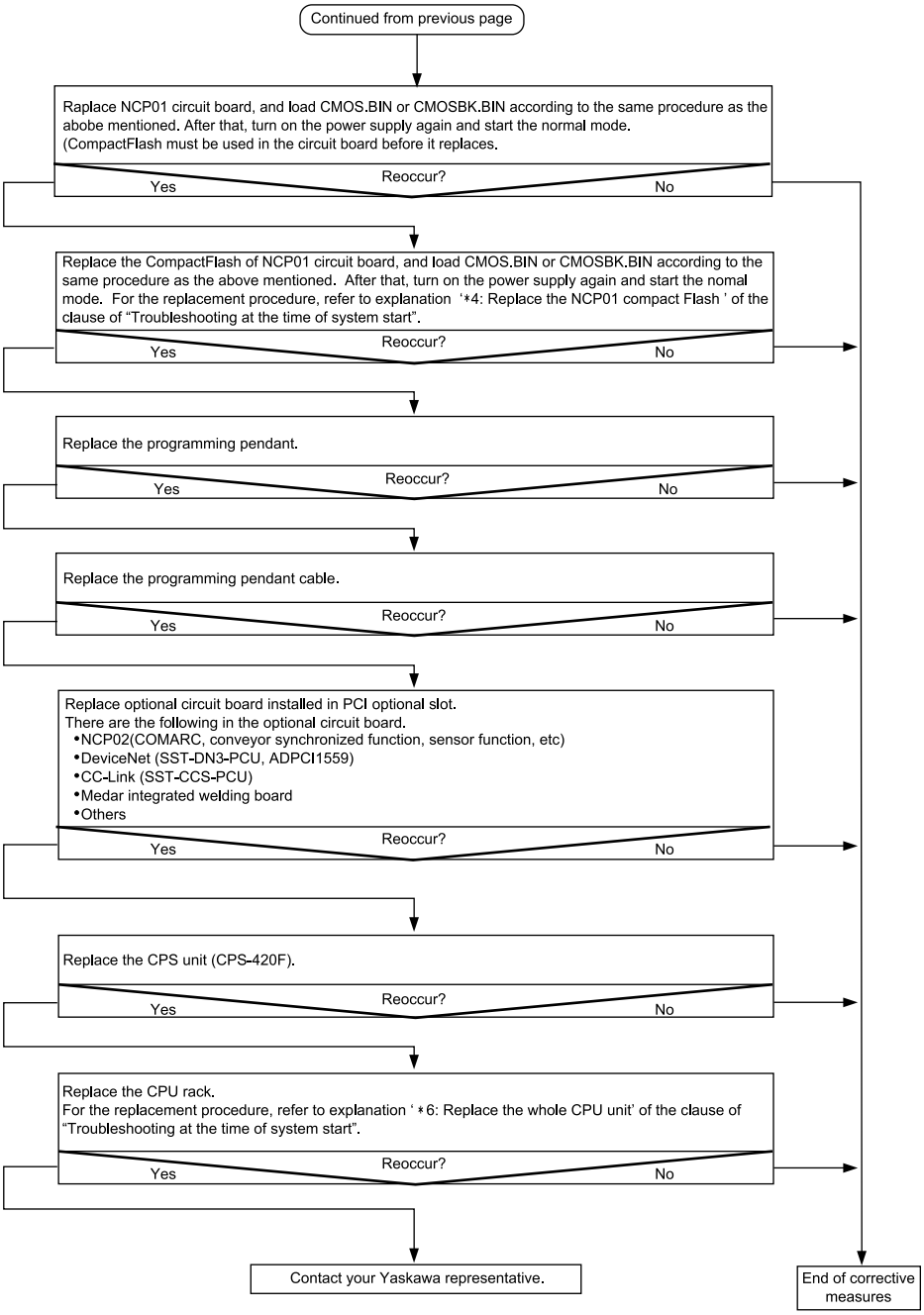
<Using the automatic backup function>

- For details on the automatic backup function, refer to "9.3 Automatic Backup Function" in "NX100 INSTRUCTIONS."
  - ∗ The automatic backup function is available in the version after NS3.00.
  - ∗ Back up individual data online if the version is earlier than NS1.8HA-00.
    Although backing up individual data is not required in other versions, this operation is recommended just to be safe.

■ Restoration method when PP disconnected occuuring

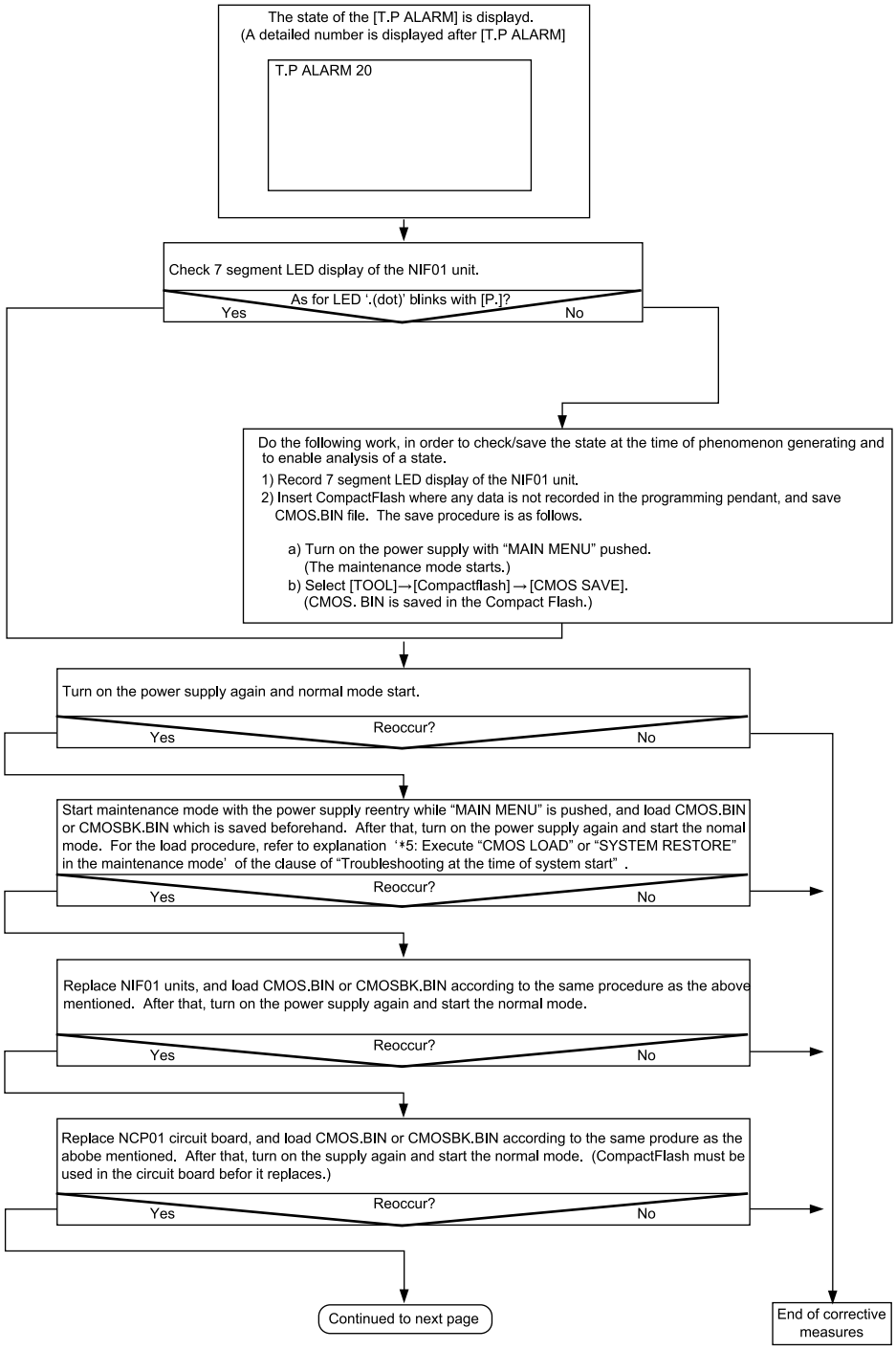
When the dialogue of 'Programming pendant is disconnected.' is displayed in the programming pendant. (standard pendant: NPP01-1), please restore in the following procedures.

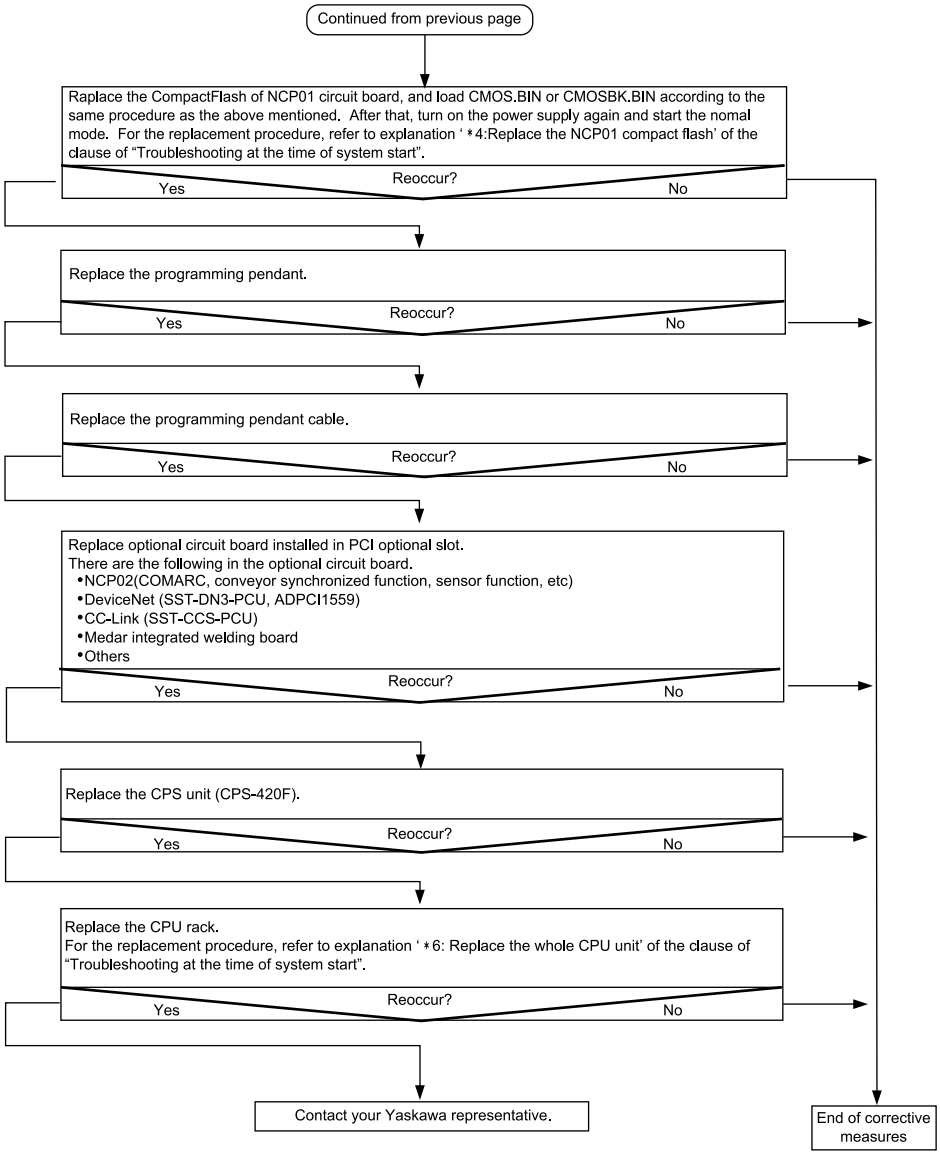



Continued from previous page
Raplace NCP01 circuit board, and load CMOS.BIN or CMOSBK.BIN according to the same procedure as the abobe mentioned. After that, turn on the power supply again and start the normal mode.
(CompactFlash must be used in the circuit board before it replaces.
Reoccur?
Yes
No
Replace the CompactFlash of NCP01 circuit board, and load CMOS.BIN or CMOSBK.BIN according to the same procedure as the above mentioned. After that, turn on the power supply again and start the nomal mode. For the replacement procedure, refer to explanation '*4: Replace the NCP01 compact Flash ' of the clause of "Troubleshooting at the time of system start".
Reoccur?
Yes
No
Replace the programming pendant.
Reoccur?
Yes
No
Replace the programming pendant cable.
Reoccur?
Yes
No
Replace optional circuit board installed in PCI optional slot.
There are the following in the optional circuit board.
•NCP02(COMARC, conveyor synchronized function, sensor function, etc)
•DeviceNet (SST-DN3-PCU, ADPCI1559)
•CC-Link (SST-CCS-PCU)
•Medar integrated welding board
•Others
Reoccur?
Yes
No
Replace the CPS unit (CPS-420F).
Reoccur?
Yes
No
Replace the CPU rack.
For the replacement procedure, refer to explanation '*6: Replace the whole CPU unit' of the clause of "Troubleshooting at the time of system start".
Reoccur?
Yes
No
Contact your Yaskawa representative.
End of corrective measures

■ Restoration method when TP ALARM occurring

When the dialogue of 'T.P ALARM' is displayed in the programming pendant (explosion-proof pendant: NPP03), please restore in the following procedures.




Continued from previous page
Raplace the CompactFlash of NCP01 circuit board, and load CMOS.BIN or CMOSBK.BIN according to the same procedure as the above mentioned. After that, turn on the power supply again and start the nomal mode. For the replacement procedure, refer to explanation ' ∗4:Replace the NCP01 compact flash' of the clause of "Troubleshooting at the time of system start".
Reoccur?
Yes
No
Replace the programming pendant.
Reoccur?
Yes
No
Replace the programming pendant cable.
Reoccur?
Yes
No
Replace optional circuit board installed in PCI optional slot.
There are the following in the optional circuit board.
•NCP02(COMARC, conveyor synchronized function, sensor function, etc)
•DeviceNet (SST-DN3-PCU, ADPCI1559)
•CC-Link (SST-CCS-PCU)
•Medar integrated welding board
•Others
Reoccur?
Yes
No
Replace the CPS unit (CPS-420F).
Reoccur?
Yes
No
Replace the CPU rack.
For the replacement procedure, refer to explanation ' ∗6: Replace the whole CPU unit' of the clause of "Troubleshooting at the time of system start".
Reoccur?
Yes
No
Contact your Yaskawa representative.
End of corrective measures

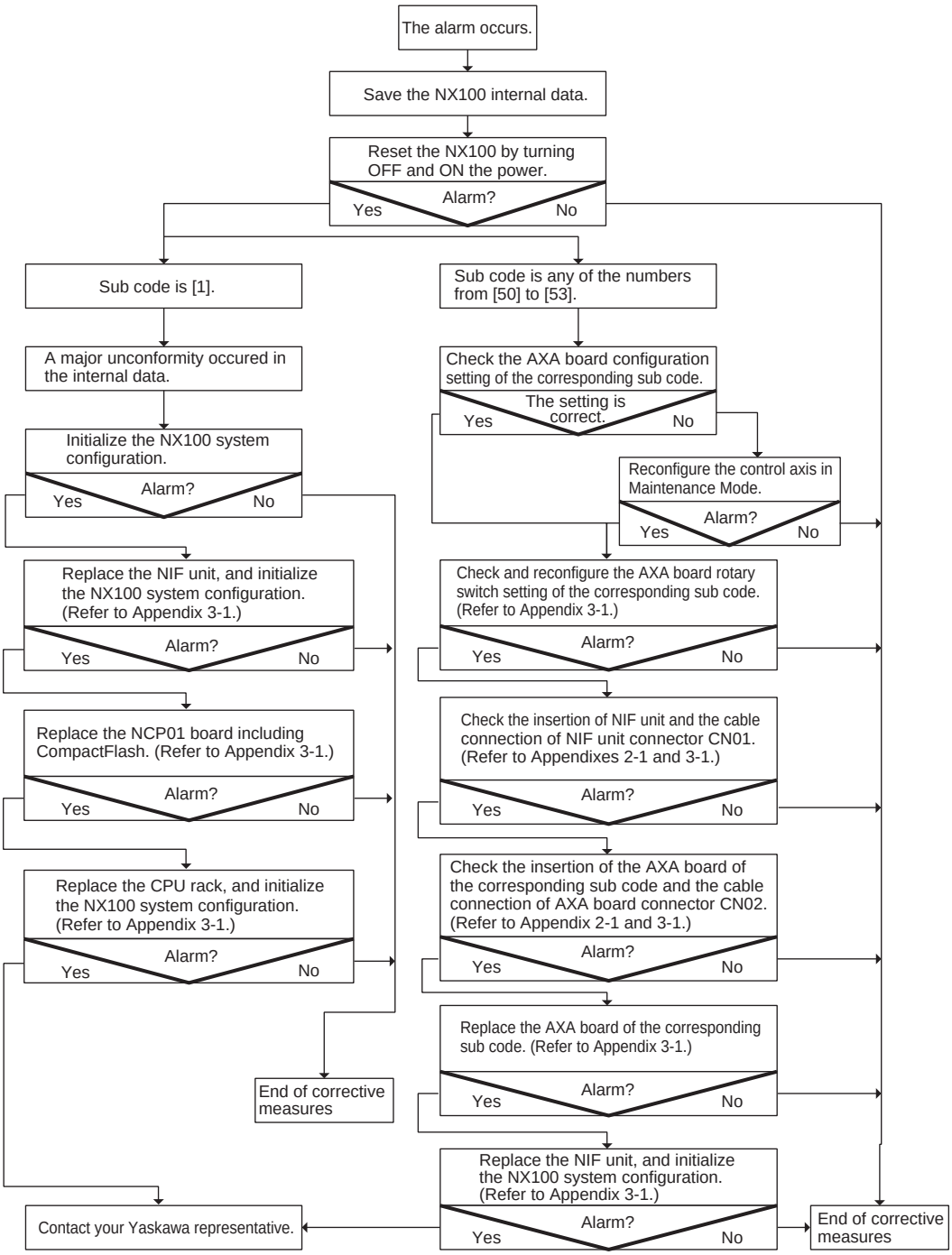
■ AL-0020

CPU COMMUNICATION ERROR: [CPU Board]

The NX100 previously stores the connected CPU boards, and check if each board properly responds on a startup. This alarm occurs if there is any CPU board which does not properly respond to the NX100.

Sub Code: CPU Board No.

01: NCP01 (Main CPU board)
10: Optional board #1
20: Optional board #2
50: AXA01/AXA02 (1st AXA board)
51: AXA01/AXA02 (2nd AXA board)
52: AXA01/AXA02 (3rd AXA board)
53: AXA01/AXA02 (4th AXA board)



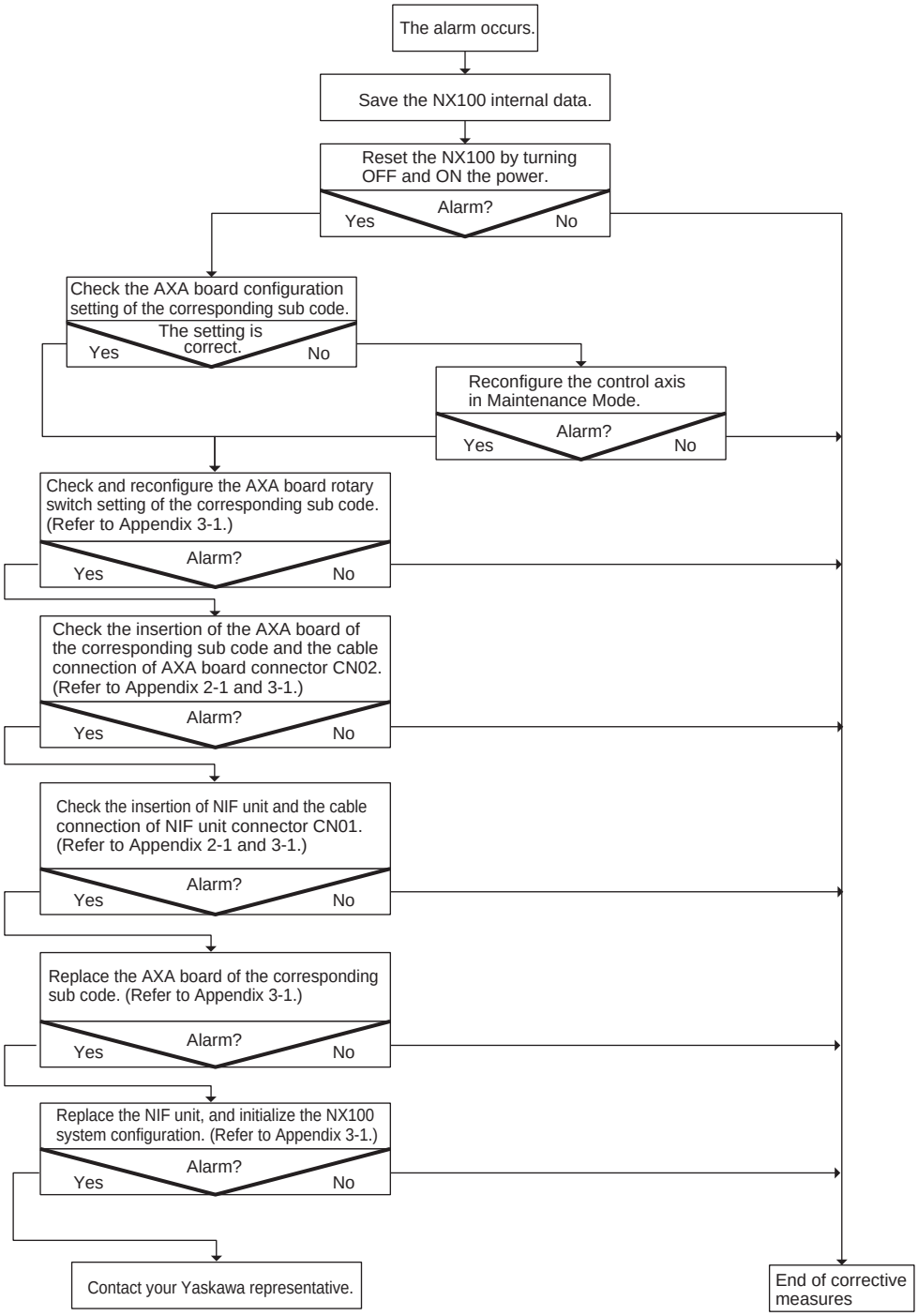
■ AL-0021

COMMUNICATION ERROR (SERVO): [CPU Board]

The NX100 transfers special commands to operate SERVO units on its startup as well as its regular operation process. This alarm occurs if there is any communication failure in transferring the special commands.

Sub Code: CPU Board No.

50: AXA01/AXA02 (1st AXA board)
51: AXA01/AXA02 (2nd AXA board)
52: AXA01/AXA02 (3rd AXA board)
53: AXA01/AXA02 (4th AXA board)



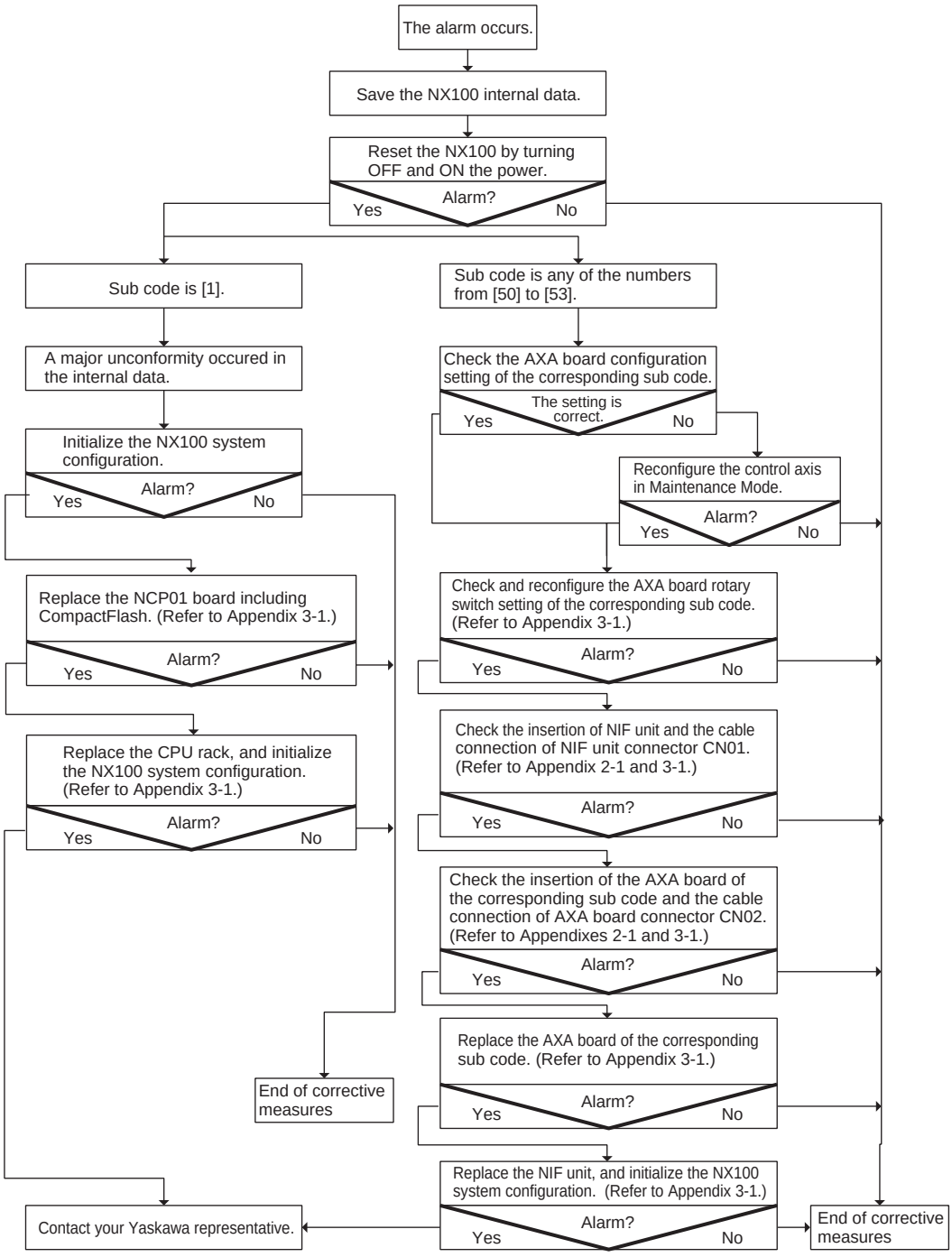
■ AL-0030

ROM ERROR: [CPU Board]

The NX100 system program (ROM) runs after RAM expansion executed on a startup. This alarm occurs if there is any failure in the RAM expansion.

Sub Code: CPU Board No.

01: NCP01 (Main CPU board)
10: Optional board #1
20: Optional board #2
50: AXA01/AXA02 (1st AXA board)
51: AXA01/AXA02 (2nd AXA board)
52: AXA01/AXA02 (3rd AXA board)
53: AXA01/AXA02 (4th AXA board)



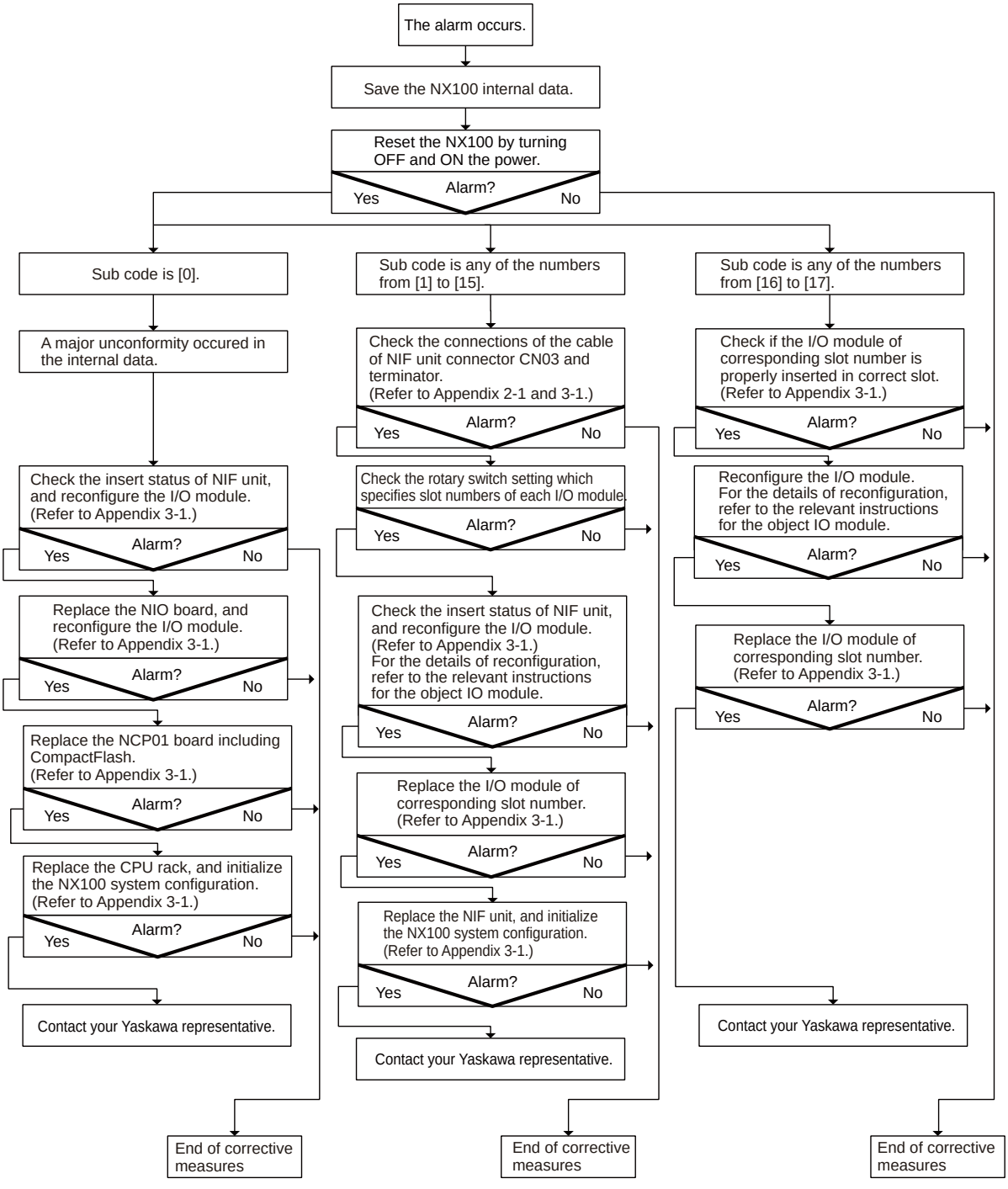
■ AL-0060

COMMUNICATION ERROR (I/O MODULE): [I/O Module Slot No.]

The NX100 previously stores the connected I/O modules, and check the presence of each module on a startup. This alarm occurs if there is any module of which presence cannot be identified.

Sub Code: Slot No. of I/O Module

- 0: NIO01/NIO02 (I/O board)
- 01 to 15: Serial-bus-connected I/O module
- 16 to 17: PCI-connected I/O module

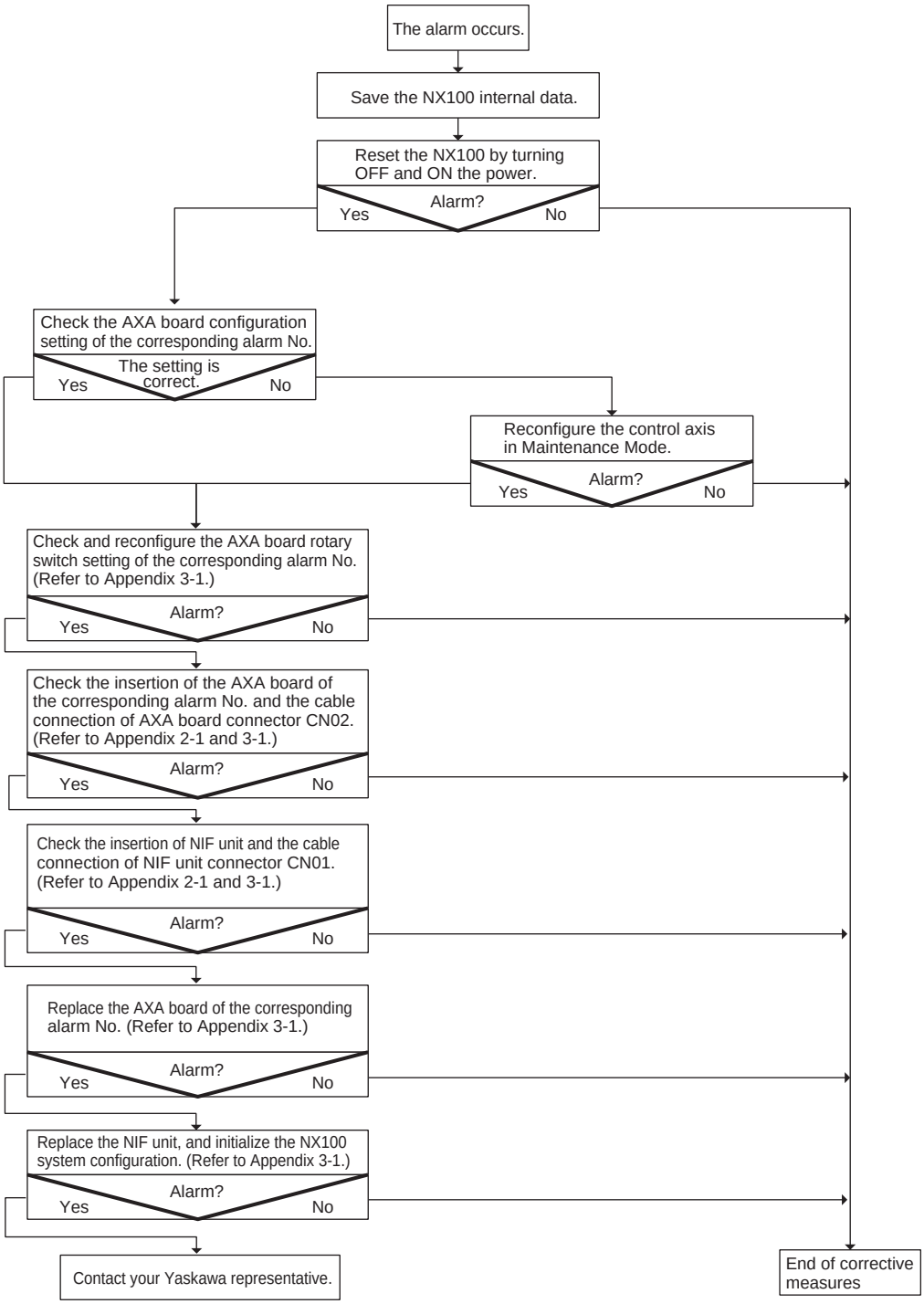


■ AL-0100 / 0101 / 0102 / 0103

COMMUNICATION ERROR (AXA#1 / #2 / #3 / #4): [Cause No.]

The serial communications between NCP01 (the main CPU board) and AXA01/AXA02 (AXA board) in the NX100 is softwarily monitored by each board.
This alarm occurs if NCP01 detects an error. The least significant digit of the alarm signifies the AXA board No. of serial communications; therefore, the station numbers of alarms 0100, 0101, 0102, and 0103 are respectively the 1st AXA board, 2nd AXA board, 3rd AXA board, and 4th AXA board.
Sub Code: Error Cause Data No.

01: Counter value received from AXA01/AXA02 (AXA board) is invalid.
02: Counter value received from AXA01/AXA02 (AXIA board) is off by one cycle.



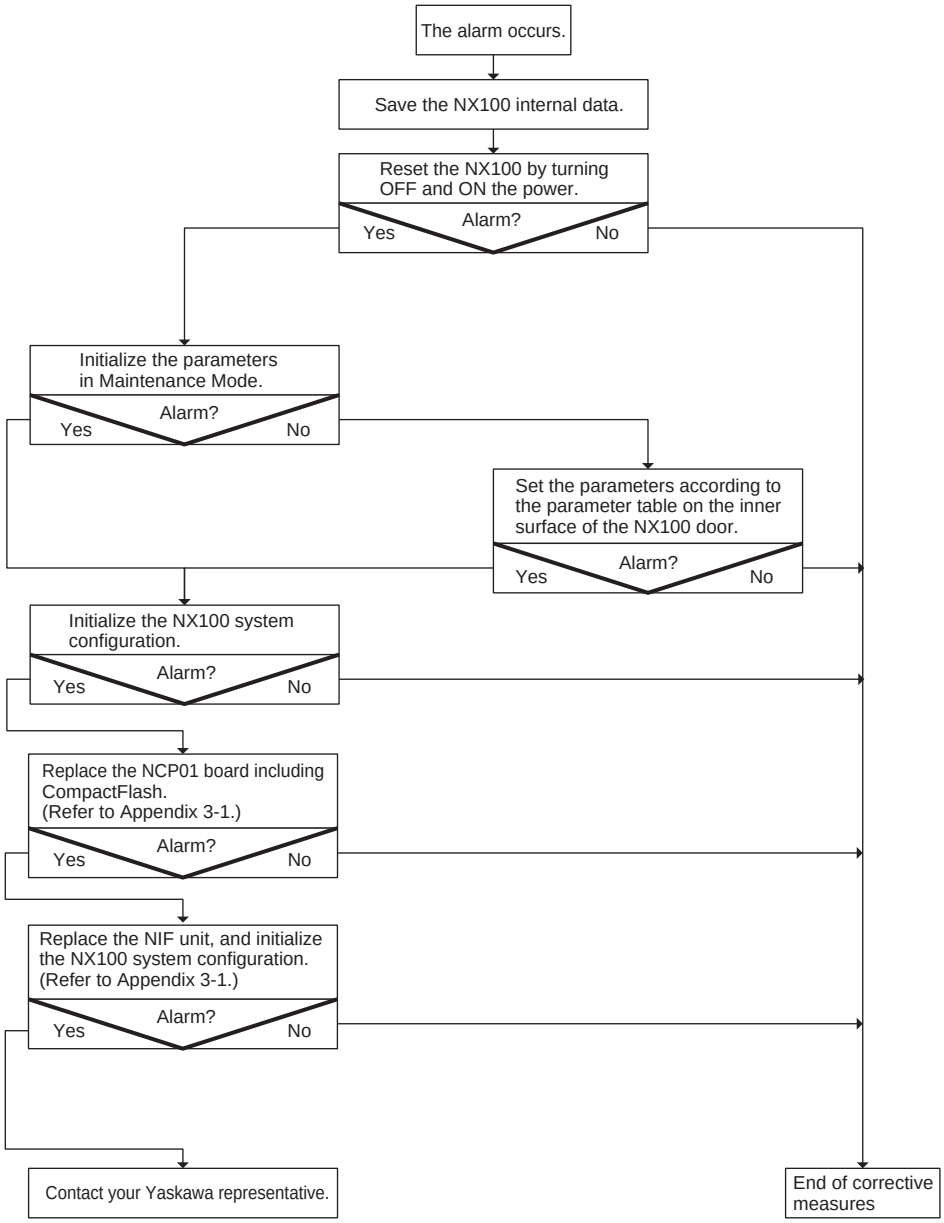
■ AL-0200

MEMORY ERROR (PARAMETER FILE): [Internal File No.]

The NX100 requires various types of parameters to operate, and performs a check sum during the startup process to ensure that the parameter files are properly retained. This alarm occurs if the NX100 detects an error in the check sum.

Sub Code: File No. Managed in NX100

00:	RC parameter	08:	AP parameter
01:	RO parameter	09:	RS parameter
02:	SV parameter	10:	SE parameter
03:	SVM parameter	11:	SVC parameter
04:	SC parameter	12:	AMC parameter
05:	SD parameter	13:	SVP parameter
06:	CIO parameter	14:	MF parameter
07:	FD parameter	15:	SVS parameter

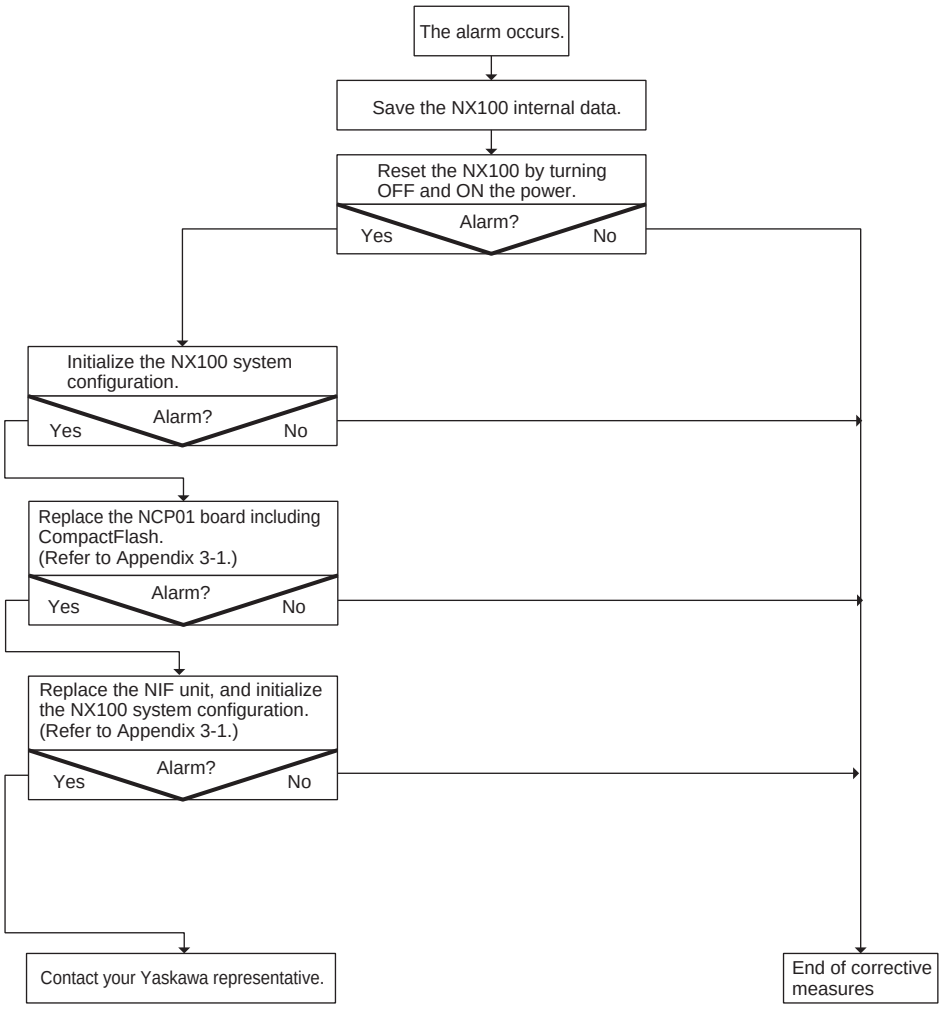


■ AL-0210

MEMORY ERROR (SYSTEM CONFIG-DATA): [0]

The NX100 carries the information to start as a system in the SYSTEM CONFIG-DATA file, and performs a check sum during the startup process to ensure that these files are properly retained. This alarm occurs if the NX100 detects an error in the check sum.

Sub Code: 0



■ AL-0220

MEMORY ERROR (JOB MNG DATA): [Cause No.]

The NX100 carries the user program as data files called JOB, and performs a check sum during the startup process to ensure that these files are properly retained. This alarm occurs if the NX100 detects an error in the check sum.

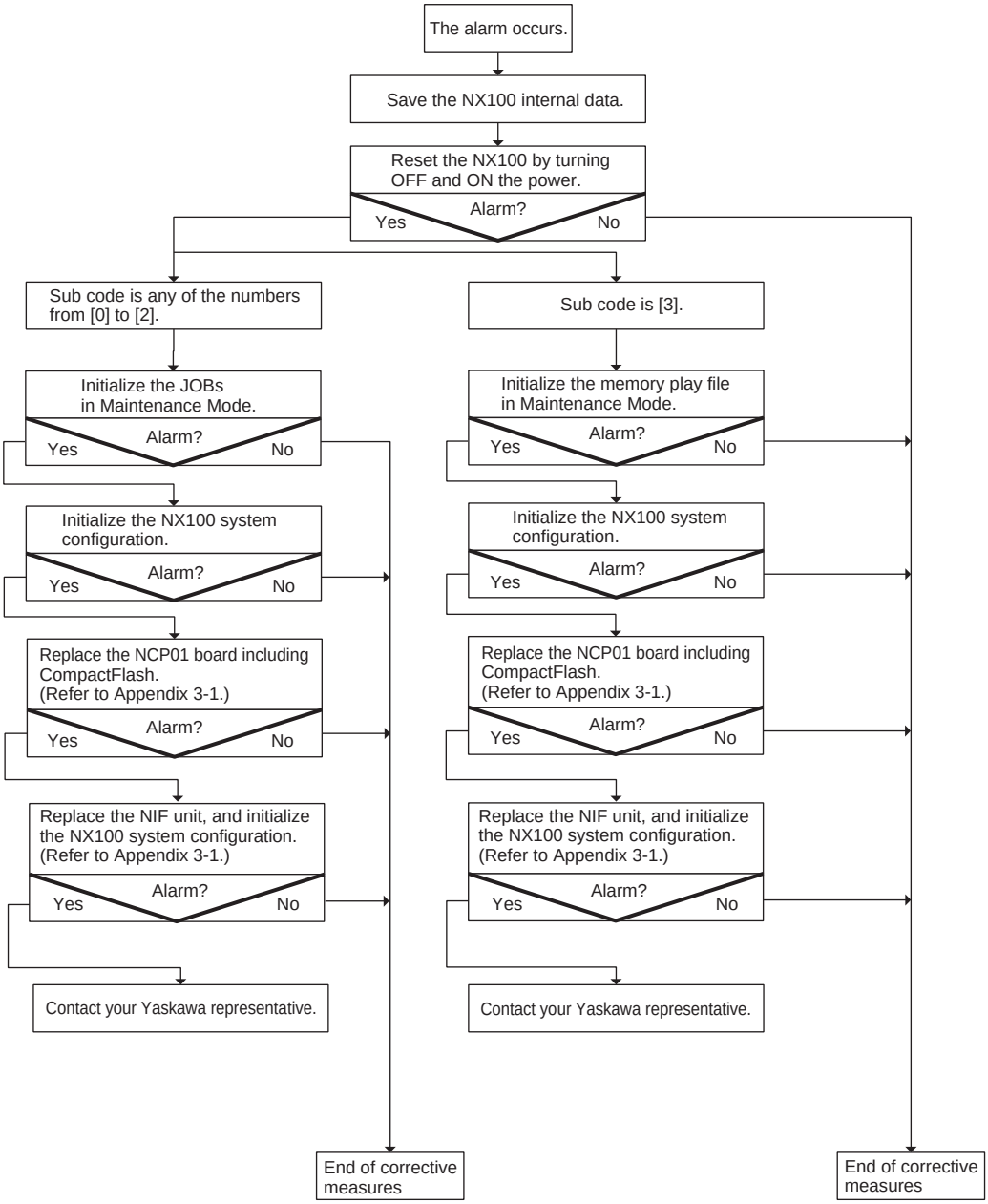
Sub Code: Error Cause Data No.

00: Error occurred during the sum check of the JOB management area

01: Error occurred during the sum check of the JOB data.

02: Error occurred during the sum check of the position data (including position variables).

03: Error occurred during the sum check of the memory play file.

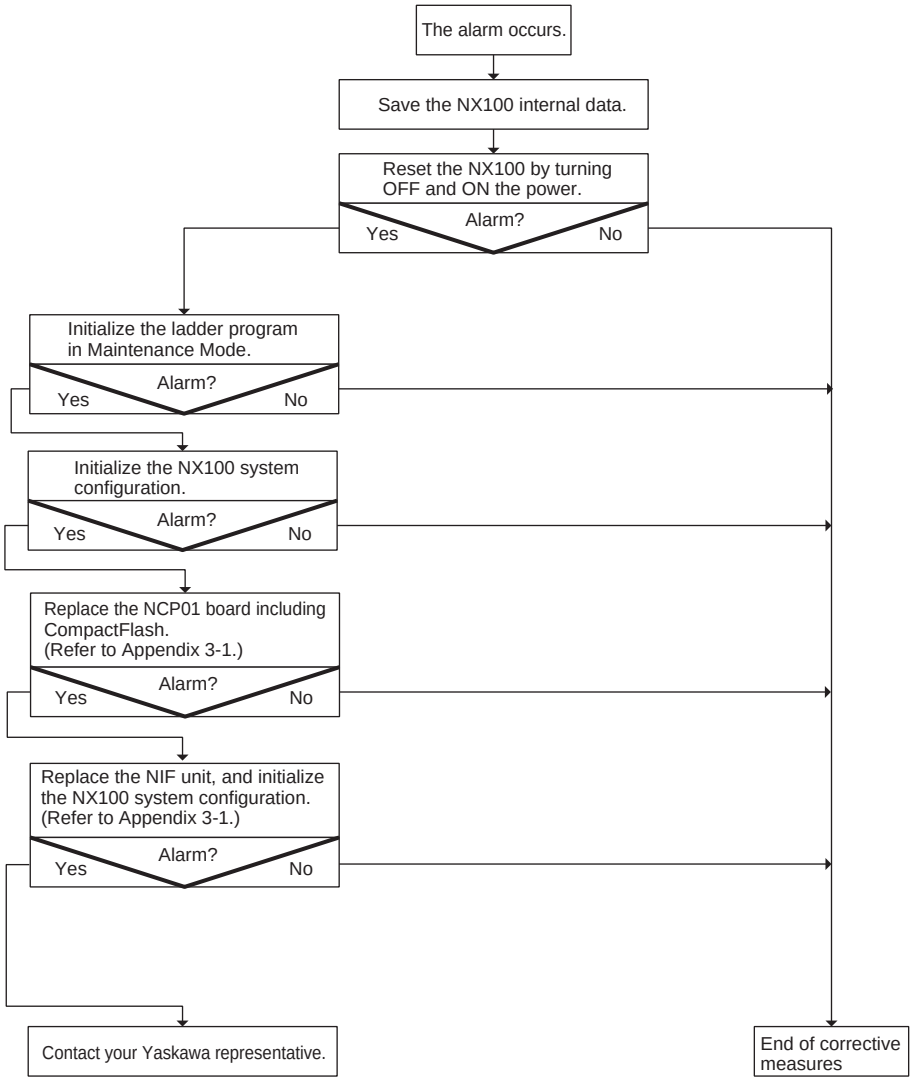


■ AL-0230

MEMORY ERROR (LADDER PRG FILE): [0]

A program software PLC which runs in the NX100 is stored in the ladder program file. The NX100 performs a check sum during the startup process to ensure that these files are properly retained. This alarm occurs if the NX100 detects an error in the check sum.

Sub Code: 0



■ AL-0240

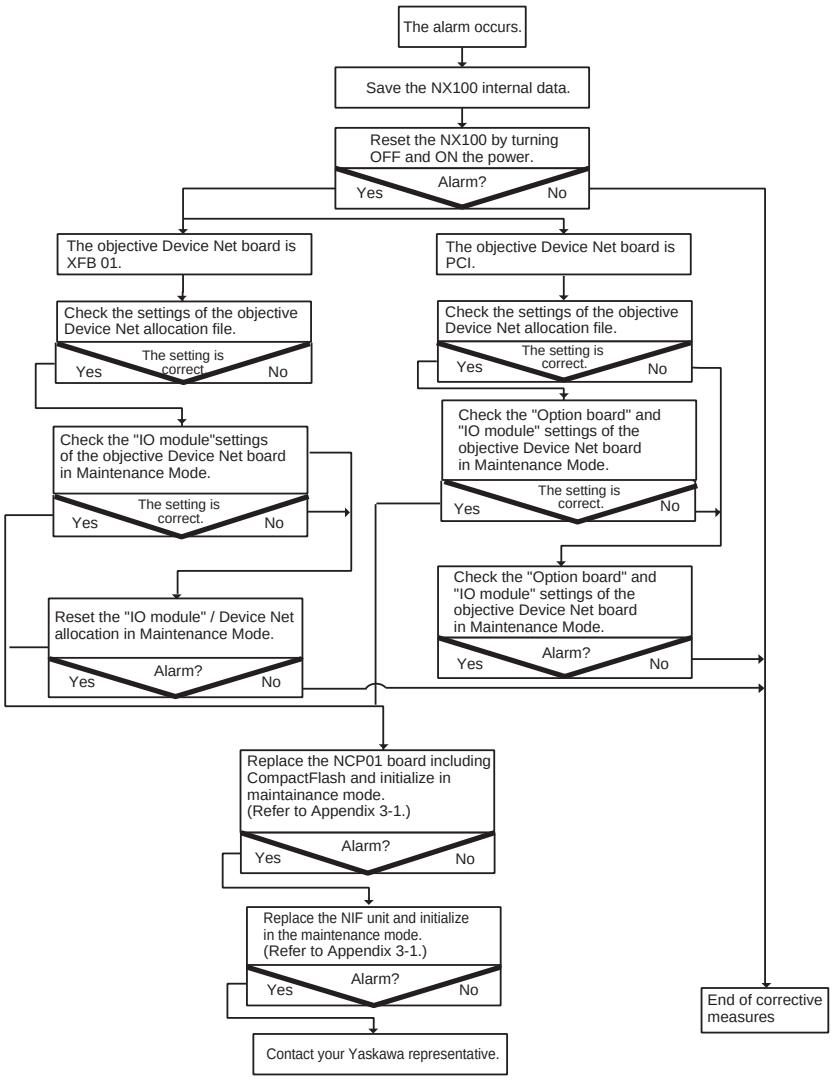
MEMORY ERROR (Device Net allollcation FILE): [File No.]

The communicatin setting of the Device Net needs to be recorded when Device Net board is used as a communication master.The file with these records are called the Device Net allocation file.It is possible to check if this Device Net allocation file is correct by the check sum when turning on the power supply. This alarm occurs if this file is found to be uncorrect.

Sub Code: File No. of DeviceNet allollcation

0: Device Net allocation file 1

1: Device Net allocation file 2



■ AL-0270

MEMORY ERROR (CF backup FILE): [File No.]

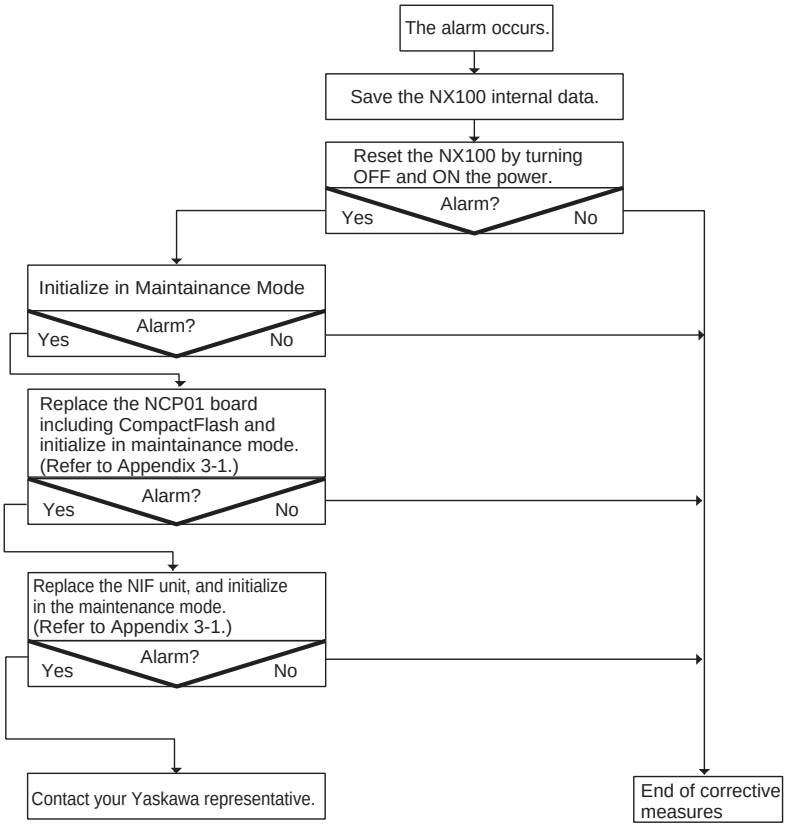
To operate the NX properly, the settings of each file (backup file) that is remained even after turning OFF the NX power supply is needed.

Part of the backup files are stored in the Compact Flash in NCP 01 board.

By retrieving these backup files stored in the Compact Flash correctly when turning on the power supply, you can check the correctness of the backup file in the compact flash.

This alarm occurs if this file is found to be uncorrect.

Sub Code: File No. of back up file in NX



■ AL-0300

VERIFY ERROR (SYSTEM CONFIG-DATA): [Cause No.]

The NX100 carries the information to start as a system in data files called "System Configuration Data", and performs a validity check on the data files if they are properly configured. This alarm occurs if the NX100 detects an error in the validity check.

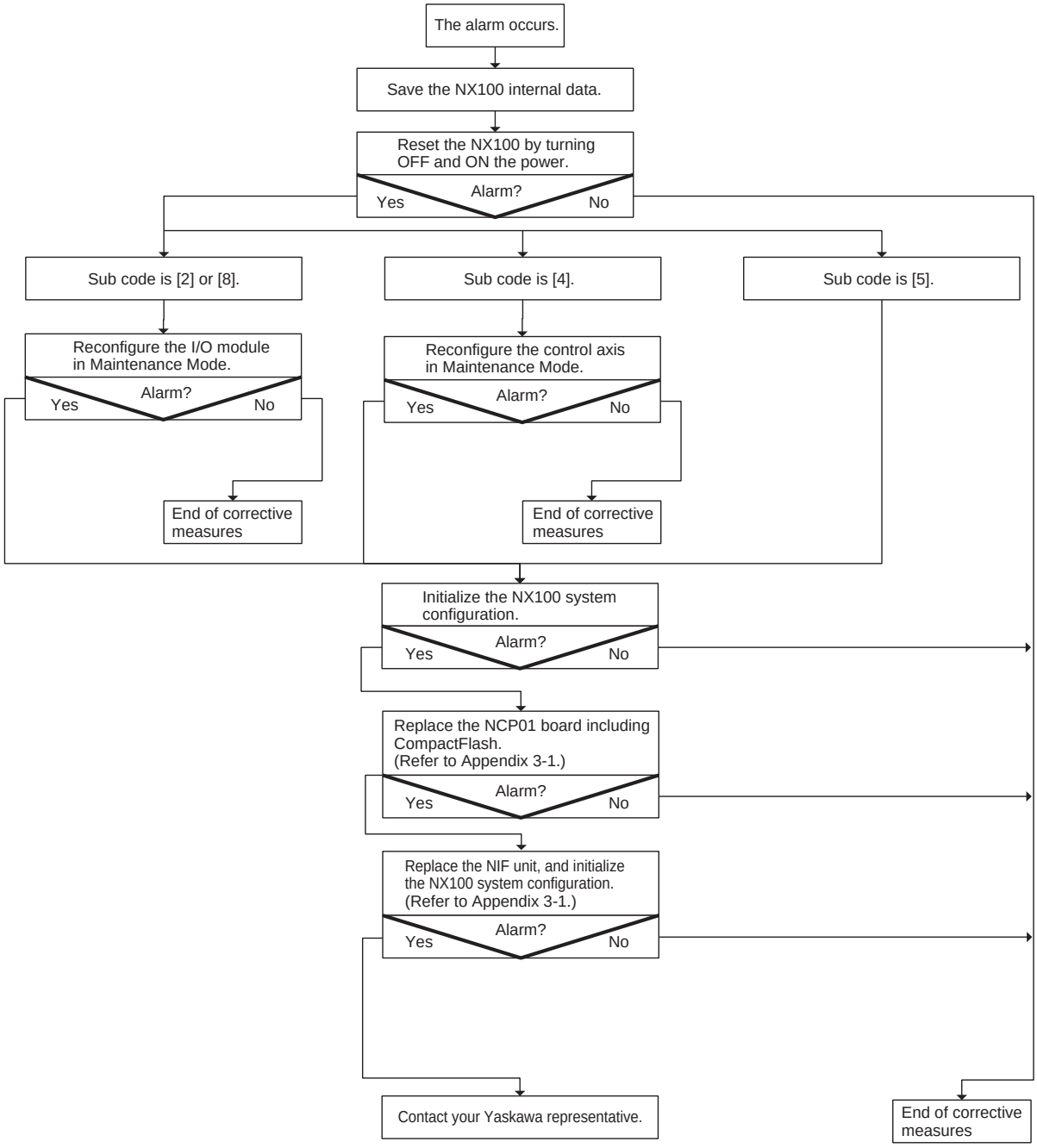
Sub Code: Error Cause Data No.

02: CIO parameter error.

04: Axis-related parameter error.

05: Sensor-use parameter error.

08: IO type error (combination impossible to coexist).

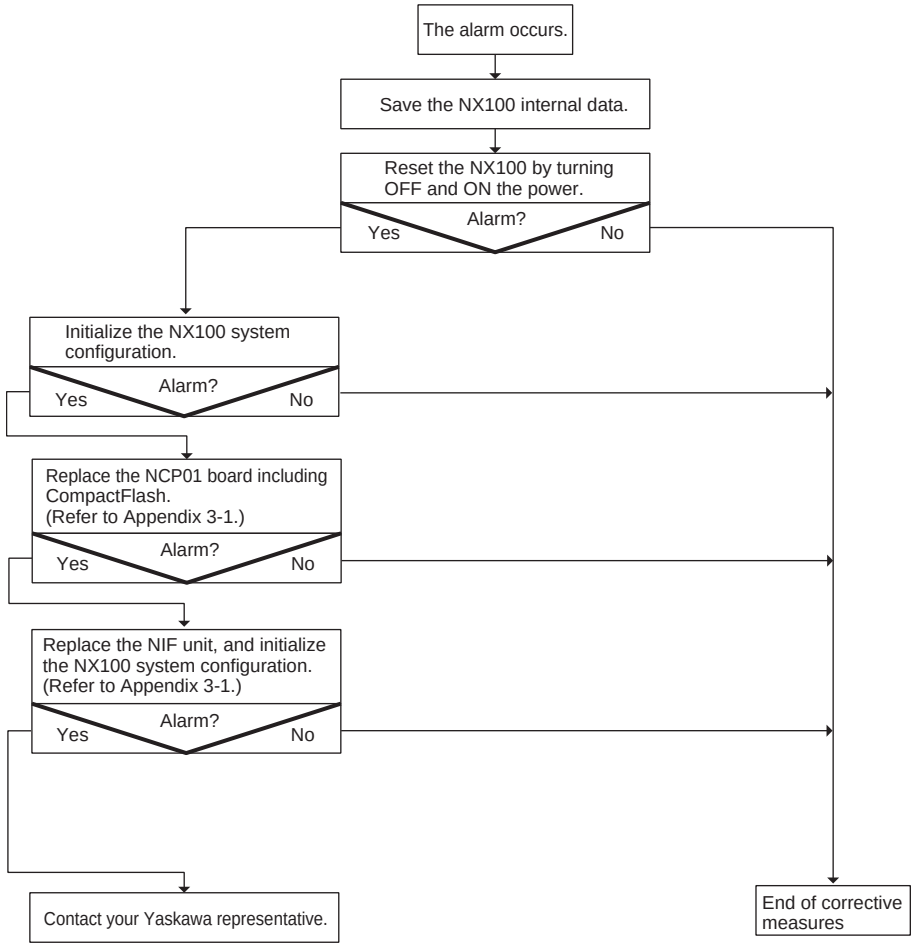


■ AL-0310

VERIFY ERROR (CMOS MEMORY SIZE): [0]

The NX100 verifies that the NIF board type (CMOS memory size) which is detected during the startup is the same as the type set at the time of system configuration. This alarm occurs if the NX100 detects an error in the verification process.

Sub Code: 0



■ AL-0320

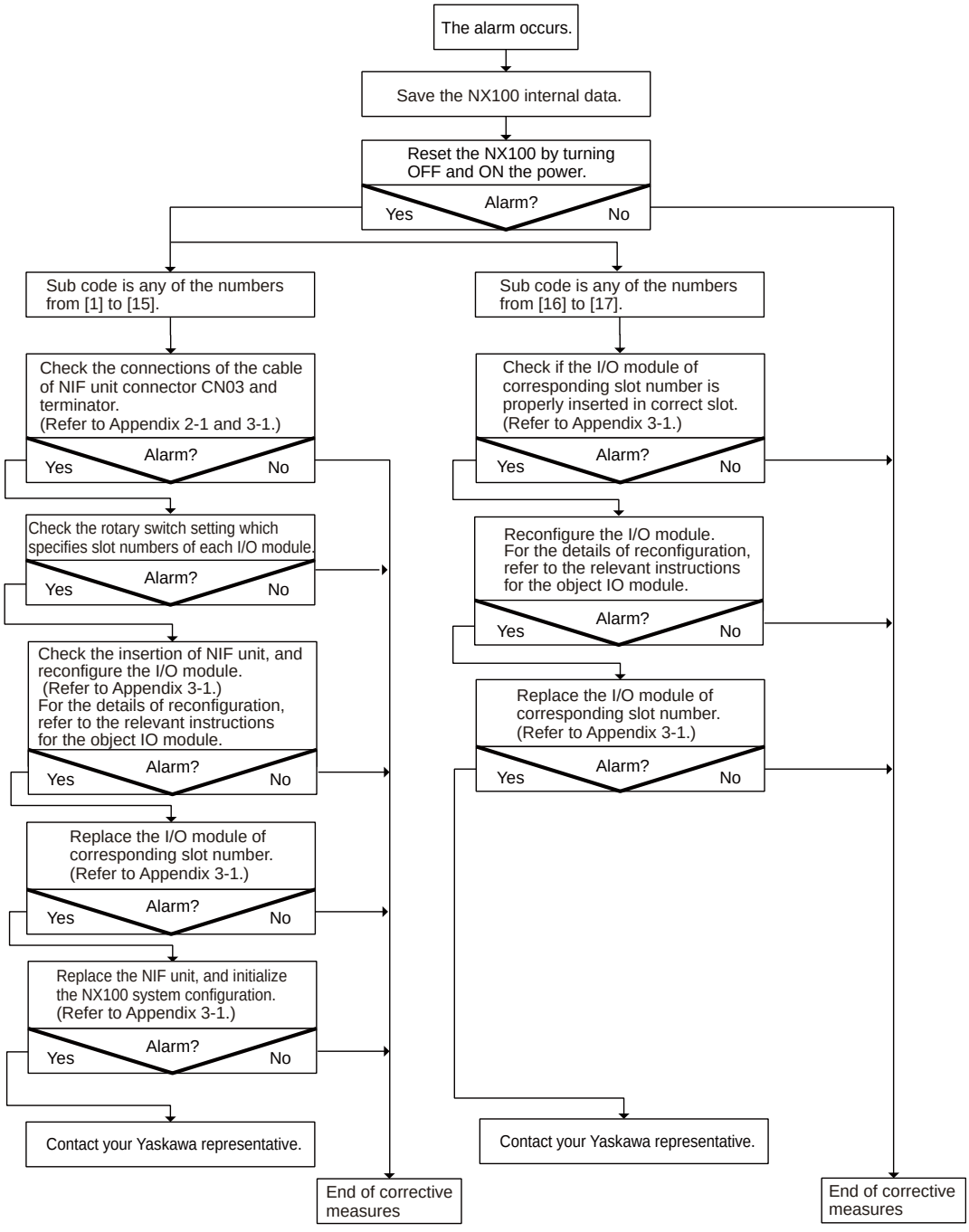
VERIFY ERROR (I/O MODULE): [I/O board Slot No.]

The NX100 verifies that the I/O module which is detected during the startup is the same as the module set at the time of system configuration. This alarm occurs if the NX100 detects an error in the verification process.

Sub Code: Slot No. of I/O Module

01 to 15: I/O module connected to serial bus.

16 to 17: I/O module connected to the PCI bus.

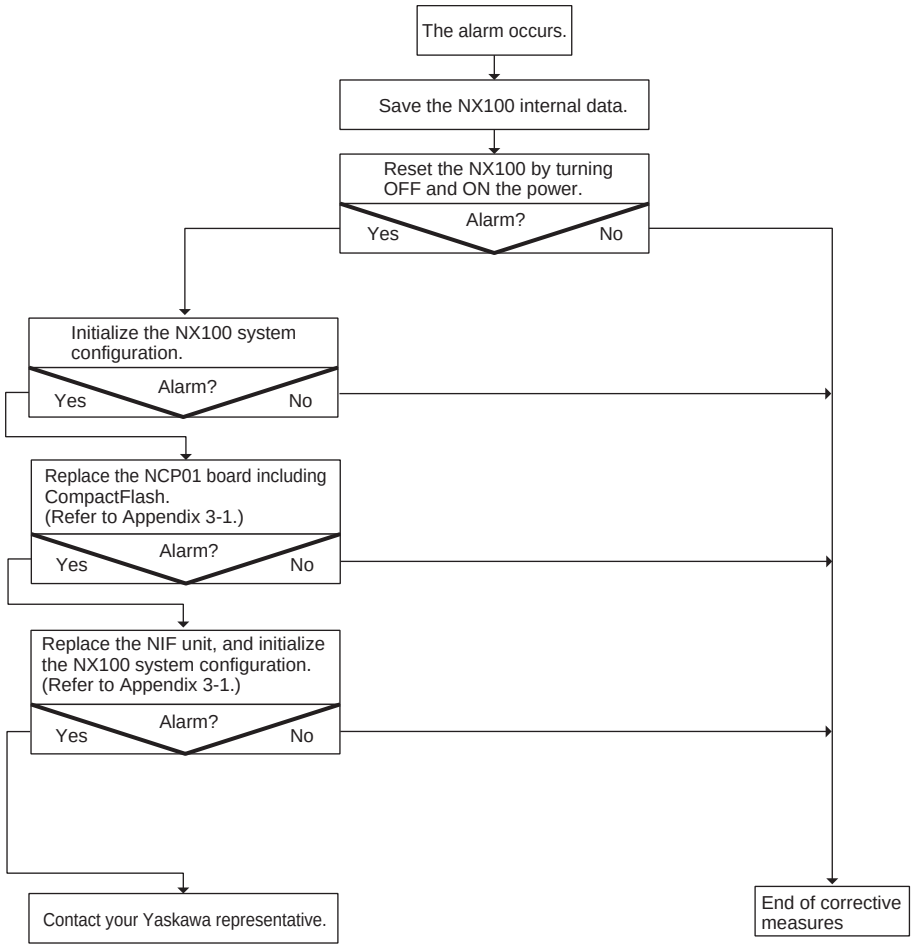


■ AL-0330

VERIFY ERROR (APPLICATION)

The NX100 verifies that the application parameters are correctly set during a startup process. This alarm occurs if the NX100 detects an error in the verification process.

Sub code: 0



■ AL-0350

VERIFY ERROR (Device Net allollcation FILE): [File No.]

The communicatin setting of the Device Net needs to be recorded when Device Net board is used as a communication master.

The file with the record is called the Device Net allocation file.

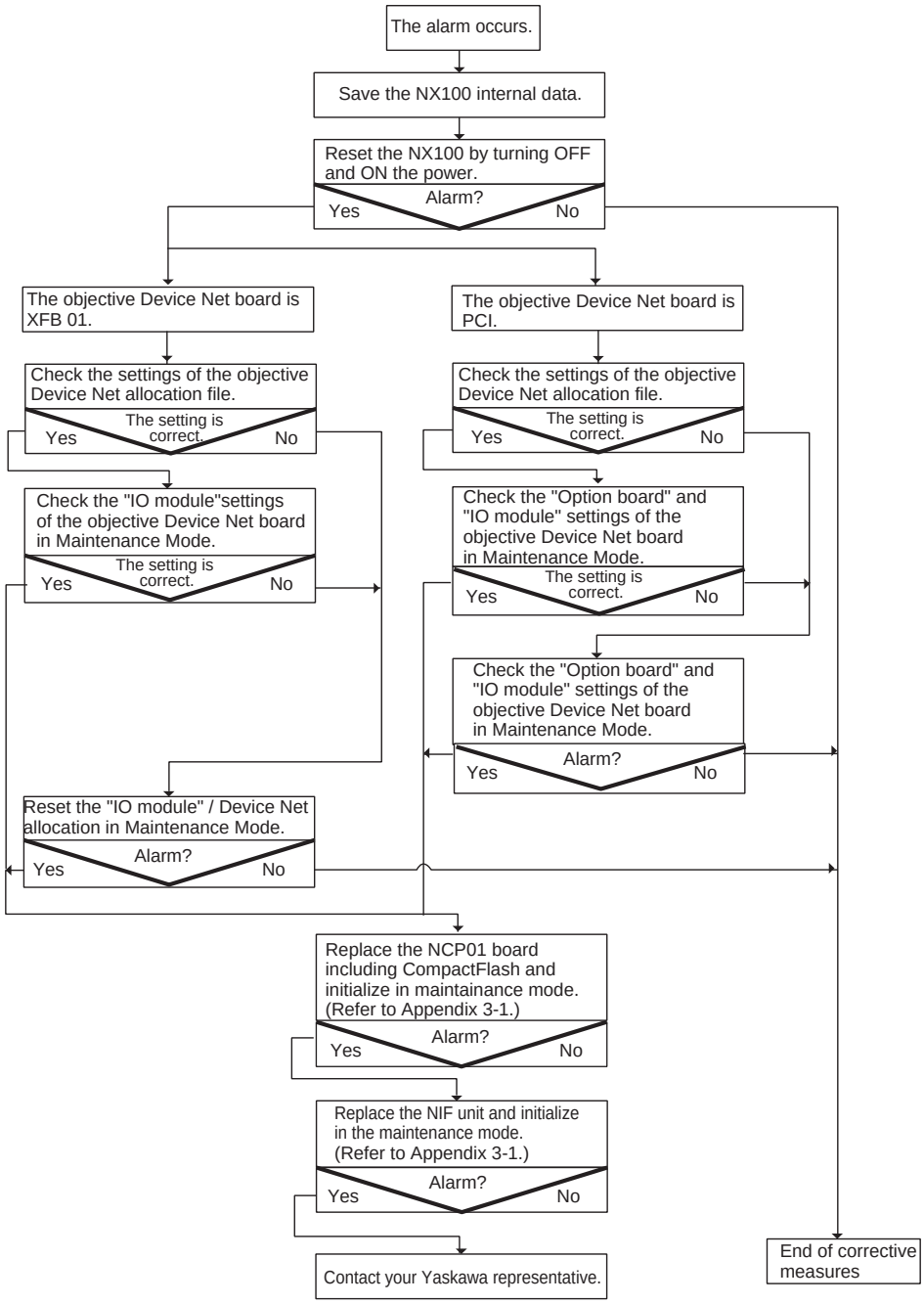
Check the correctness of this Device Net allocation file when turning on the power supply.

This alarm occurs if this file is found to be uncorrect.

Sub Code: File No. of DeviceNet allocation

0: Device Net allocation file 1

1: Device Net allocation file 2

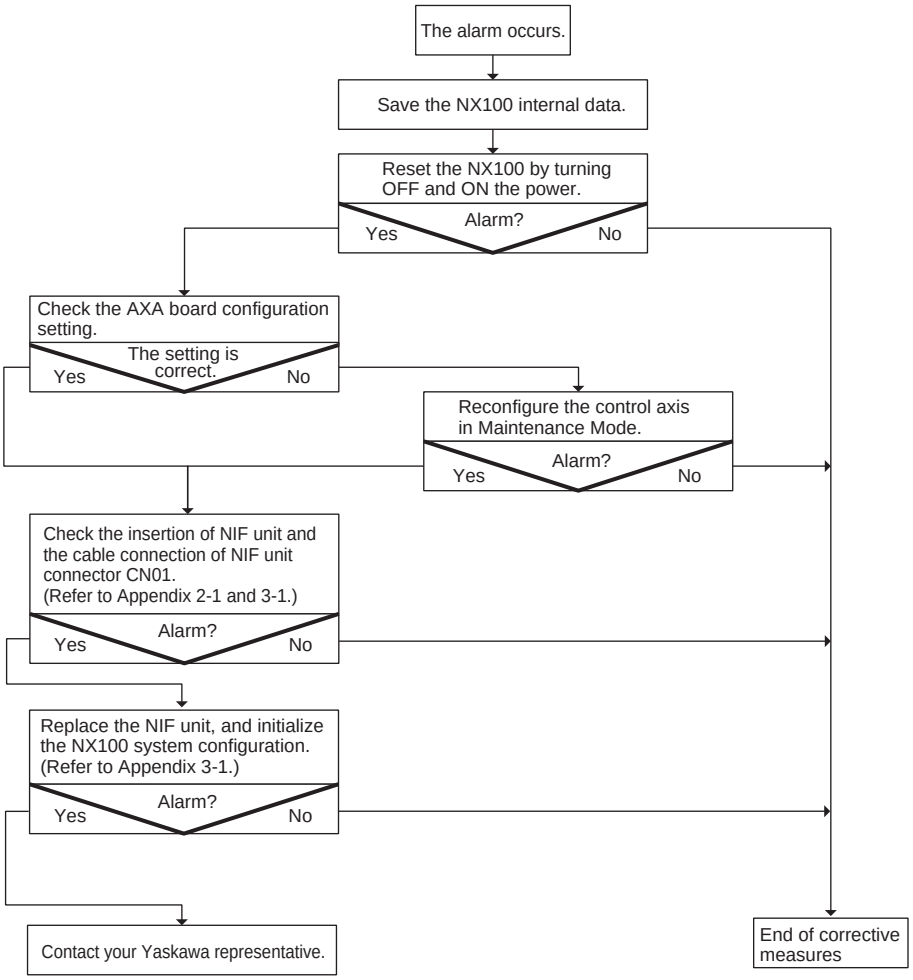


■ AL-0390

VERIFY ERROR (SEGMENT CLOCK)

The NX100 system uses a reference cycle called segment clock to control operation.
This alarm occurs if an incorrect value is set to this segment clock.
The segment clock is set with a parameter when the control group is set up.

Sub Code: None



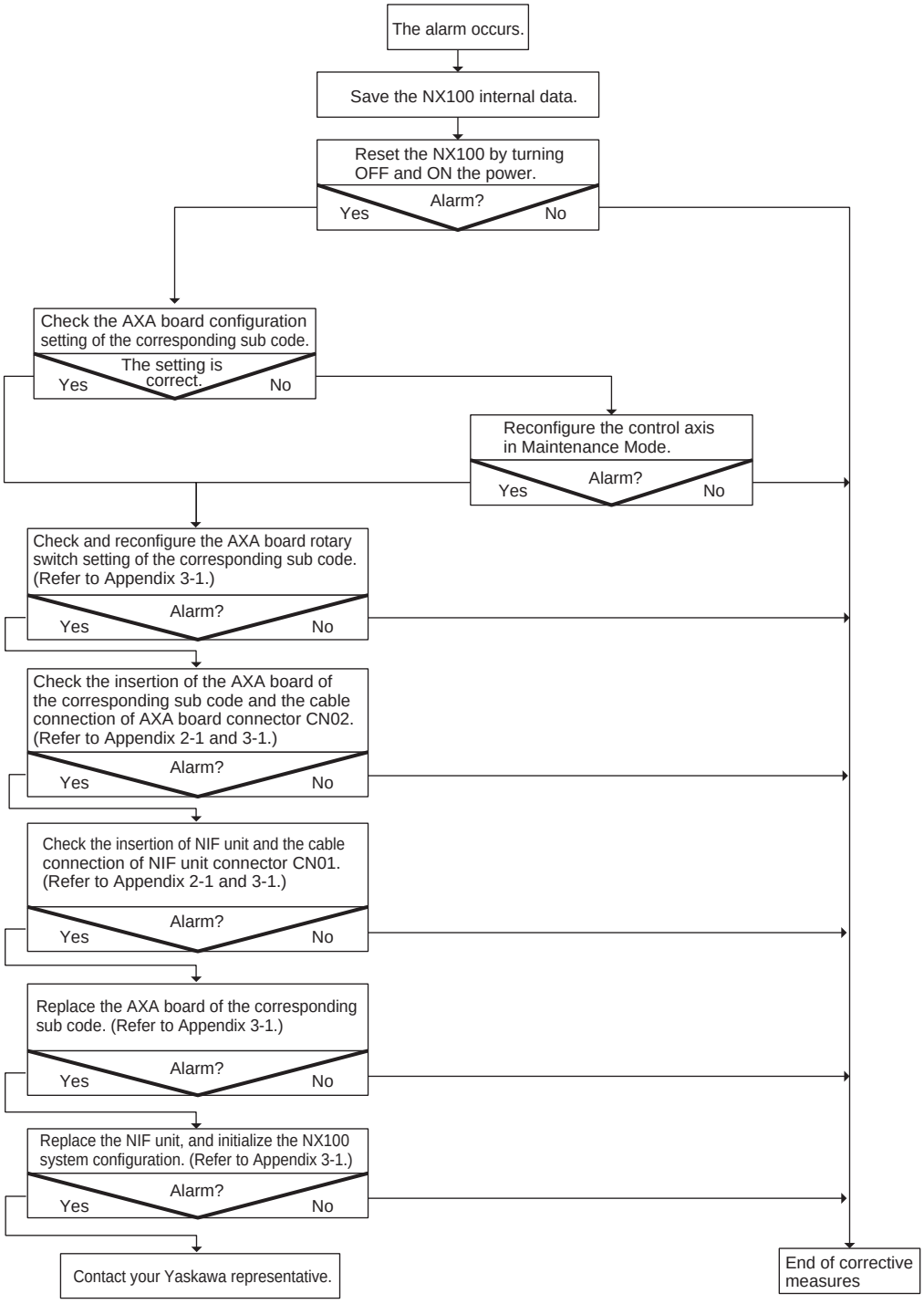
■ AL-0400

PARAMETER TRANSMISSION ERROR: [CPU Board]

The parameters required for the AXA board operation is transferred from the NCP01 board. This alarm occurs if the parameters are not successfully transferred.

Sub Code: CPU Board No.

50: AXA01/AXA02 (1st AXA board)
51: AXA01/AXA02 (2nd AXA board)
52: AXA01/AXA02 (3rd AXA board)
53: AXA01/AXA02 (4th AXA board)



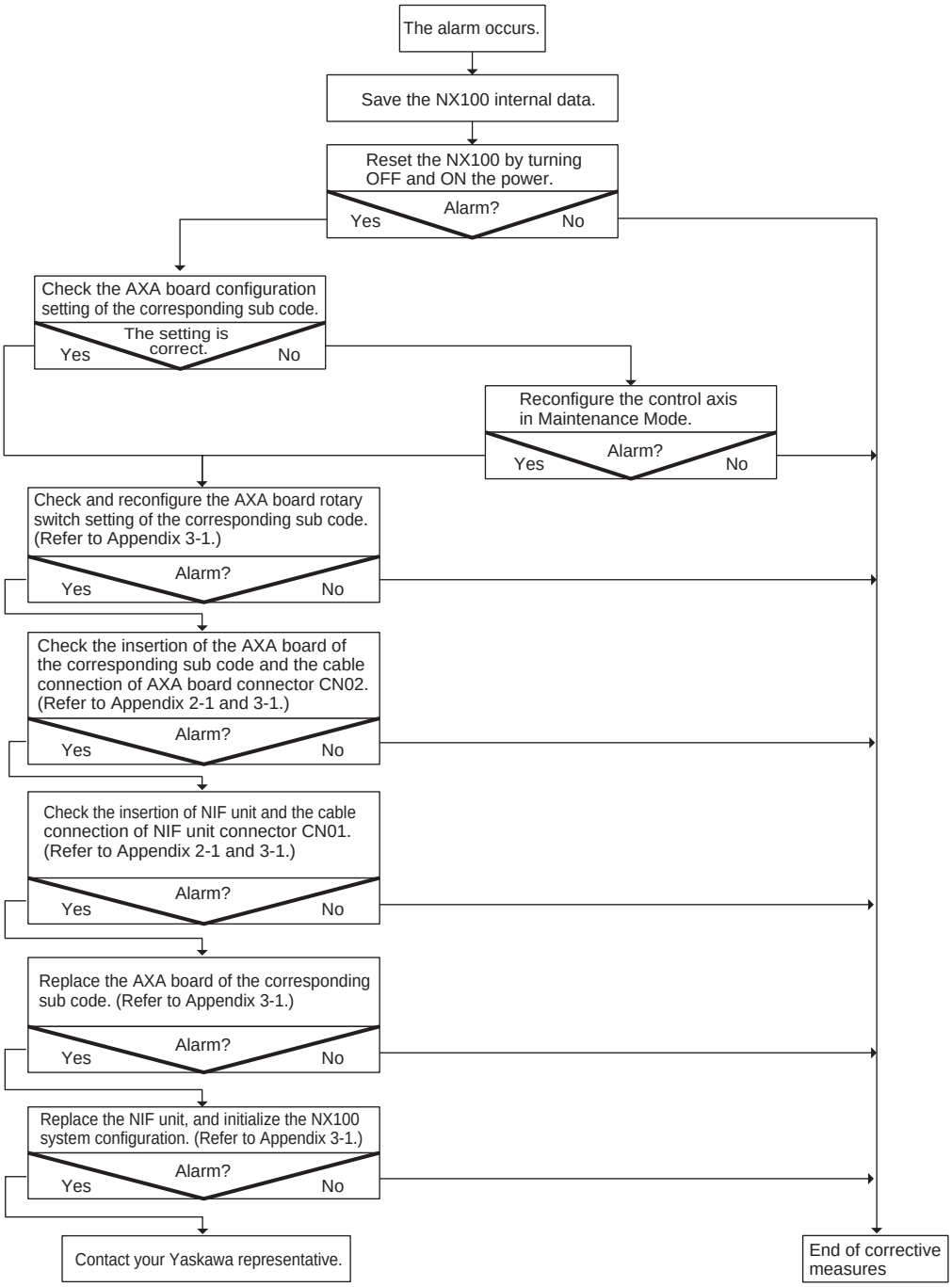
■ AL-0410

MODE CHANGE ERROR: [CPU Board]

The NX100 changes its operation modes during a startup process from the power-on operation until the startup process completion. Since mode change is required for the peripheral CPU boards as well as the main CPU board, the NX100 simultaneously performs a process as the mode change process. This alarm occurs if the mode change is not successfully performed.

Sub Code: CPU Board No.

50: AXA01/AXA02 (1st AXA board)
51: AXA01/AXA02 (2nd AXA board)
52: AXA01/AXA02 (3rd AXA board)
53: AXA01/AXA02 (4th AXA board)



■ AL-0420

DEVICE NET ALLOCATION FILE TRANSMISSION ERROR: [FILE No.]

The communication setting of the Device Net needs to be recorded when Device Net board is used as a communication master.

The file with the record is called the Device Net allocation file.

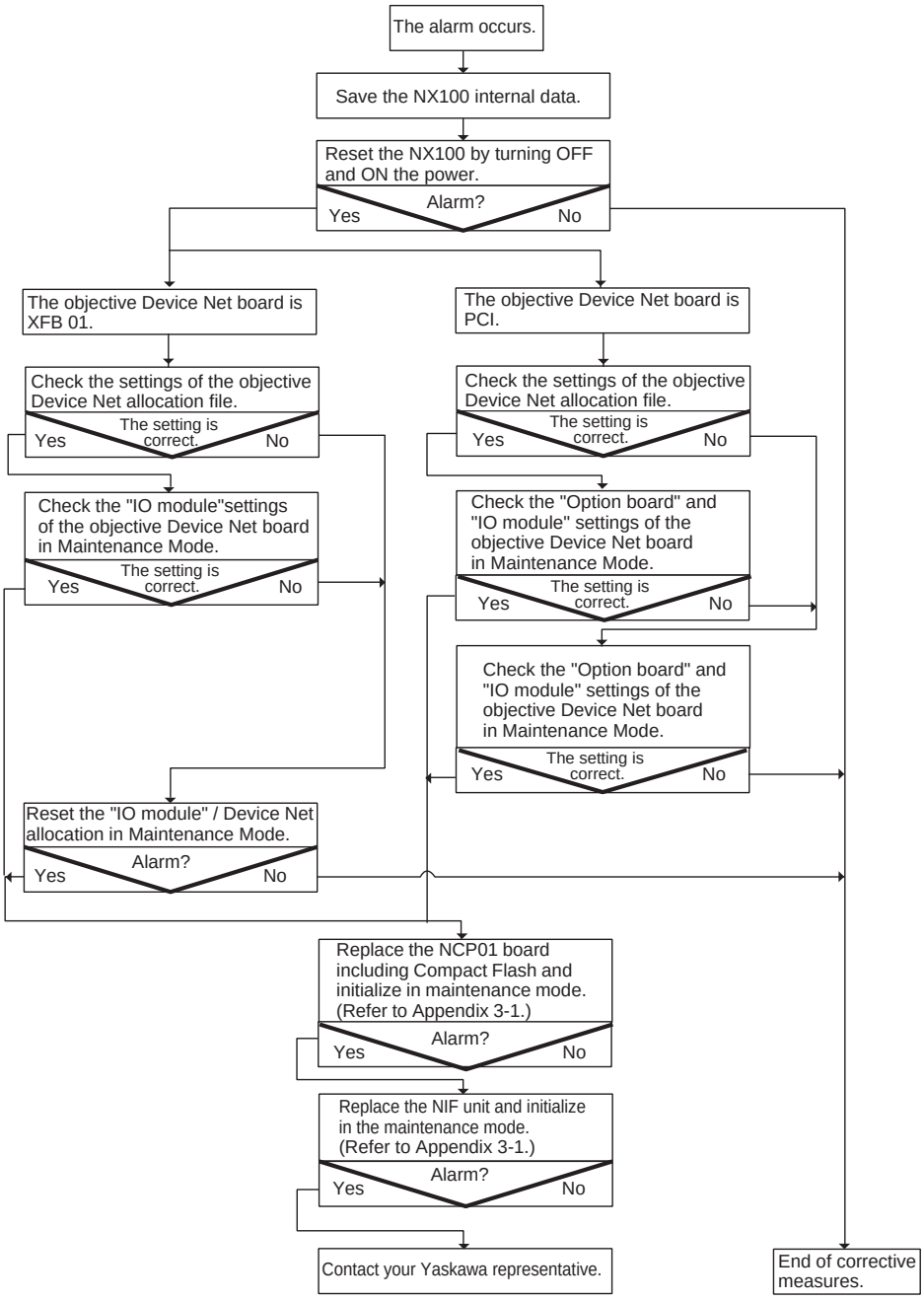
The Device Net allocation file is transmitted when turning on the power supply.

This alarm occurs if this file is found to be incorrect.

Sub Code: File No. of Device Net allocation

0: Device Net allocation file 1

1: Device Net allocation file 2

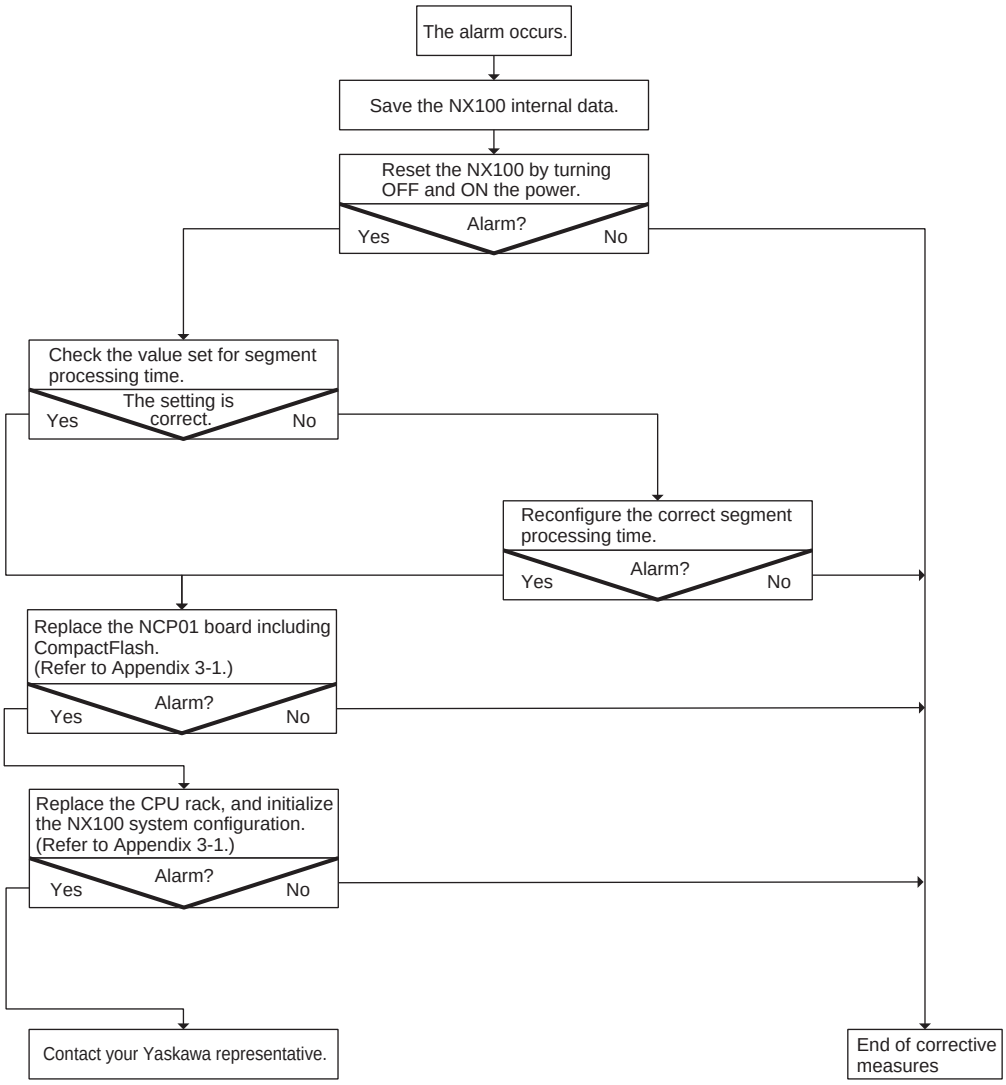


■ AL-0500

SEGMENT PROC NOT READY

To properly operate the manipulator, it is required to complete the processing of operation instructions within the specified time. This alarm occurs if the processing of operation instructions is not completed within the specified time.

Sub Code: None

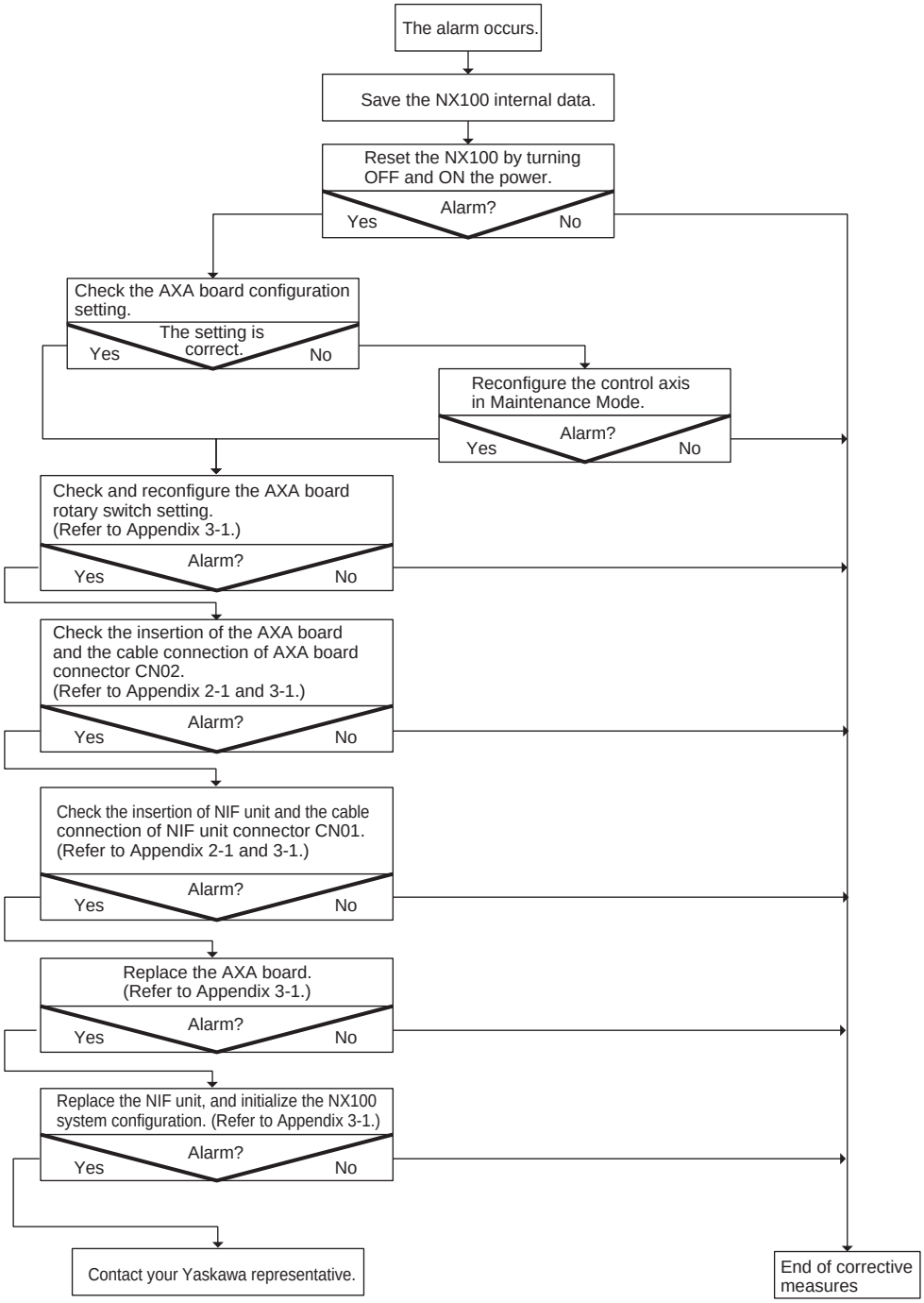


■ AL-0520

AXIS LIMIT OVER

In NX100, the maximum number of axes that can be controlled across the system is fixed.
This alarm occurs if the specified number of axes exceeds this upper limit.

Sub Code: None

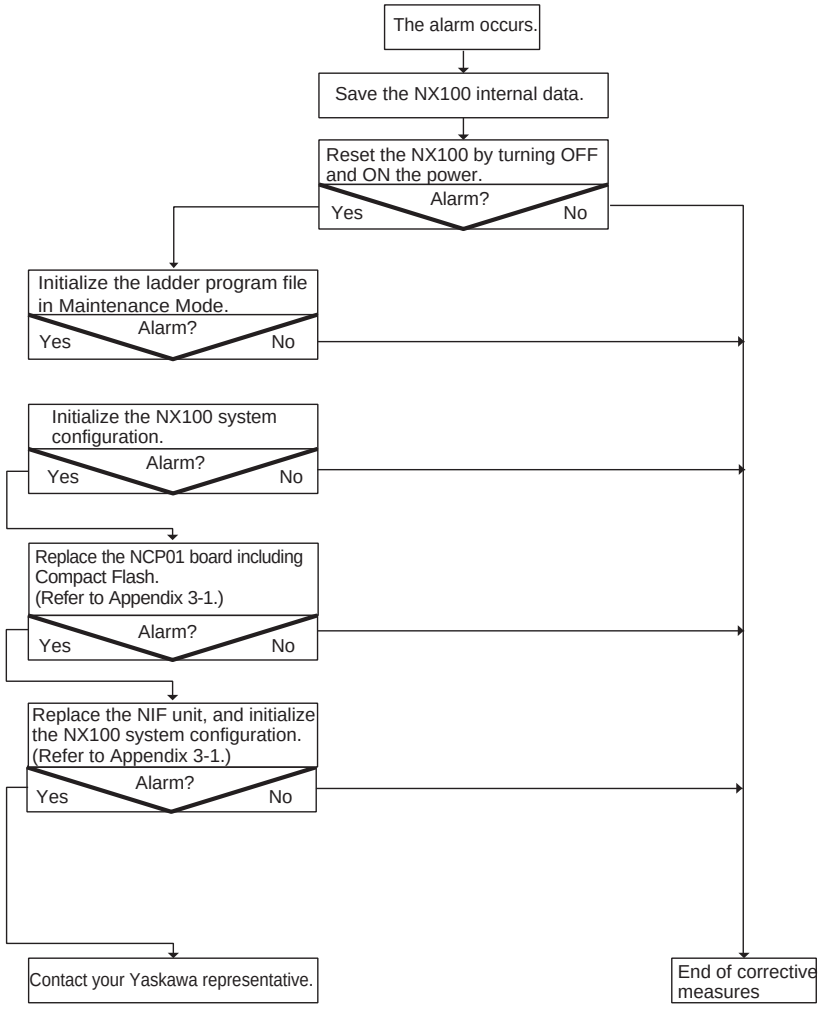


■ AL-0710

LADDER INITIALIZE ERROR

The ladder program was not initialized.

Sub Code: None

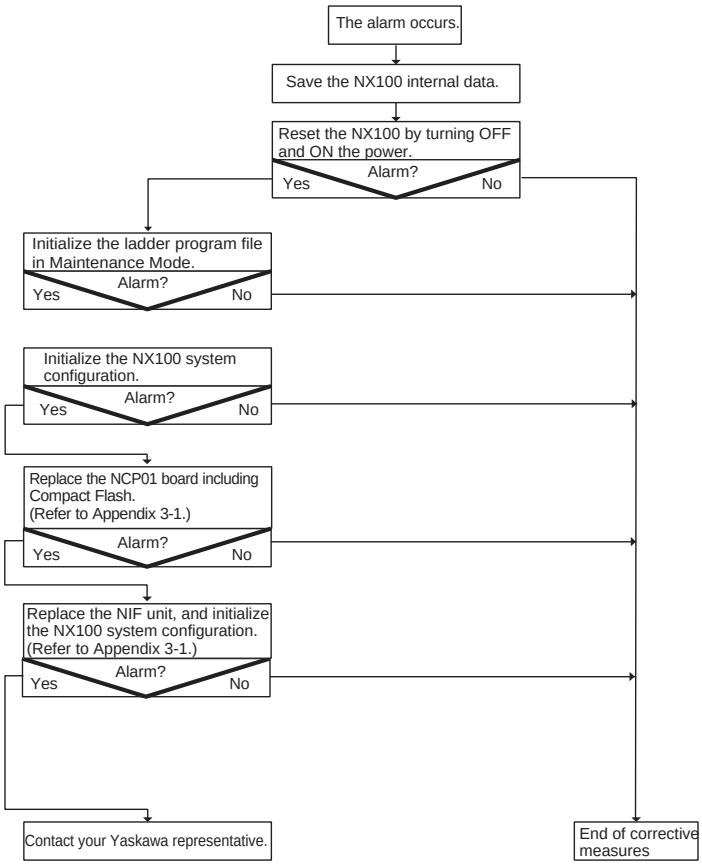


■ AL-0720

LADDER PROGRAM ERROR

This alarm occurs if the relay number specification is wrong.

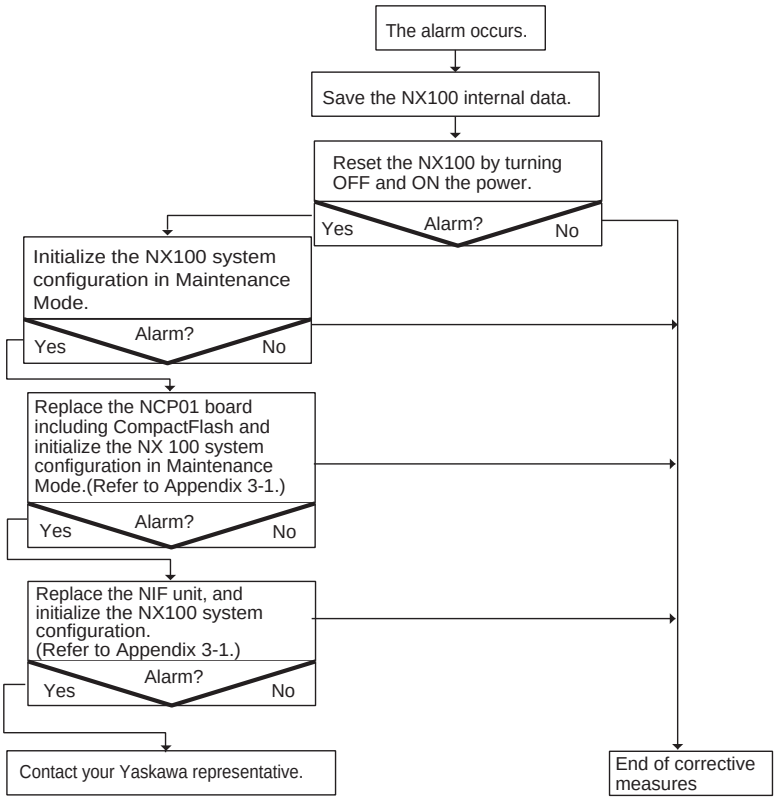
Sub Code: 1



■ AL-0800

FILE BACKUP ERROR (NCP01 CF)[FILE NO.]

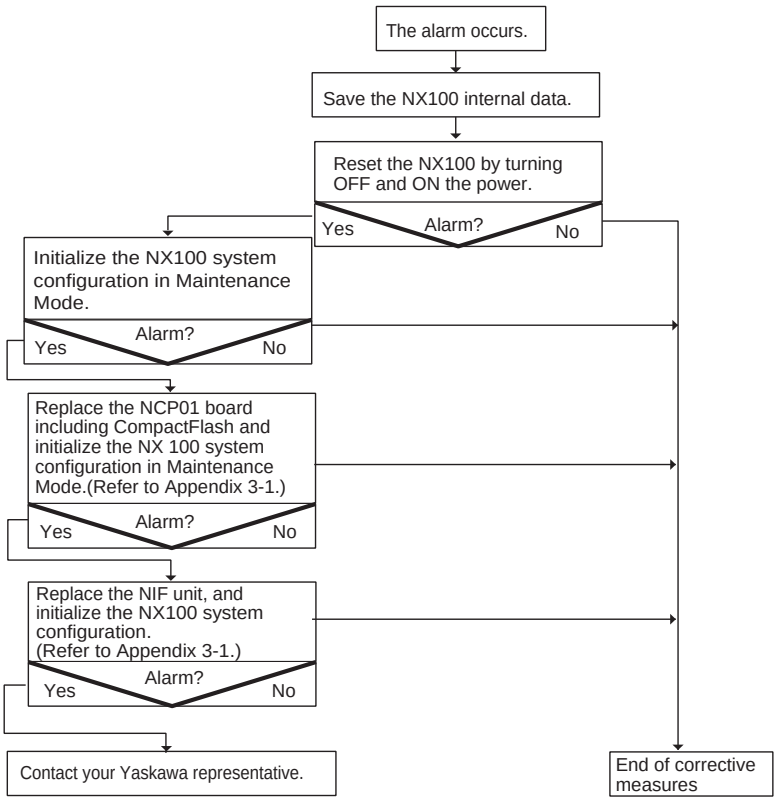
The NX100 saves a part of data needed for system operations on the Compact Flash in the NCP01 board.
When the data is changed, the new data is written in the on the Compact Flash.
This alarm occurs if the this writing is correctly done.
Sub Code: File No. of back up file in NX



■ AL-0801

FILE LOAD ERROR (NCP01 CF)[FILE NO.]

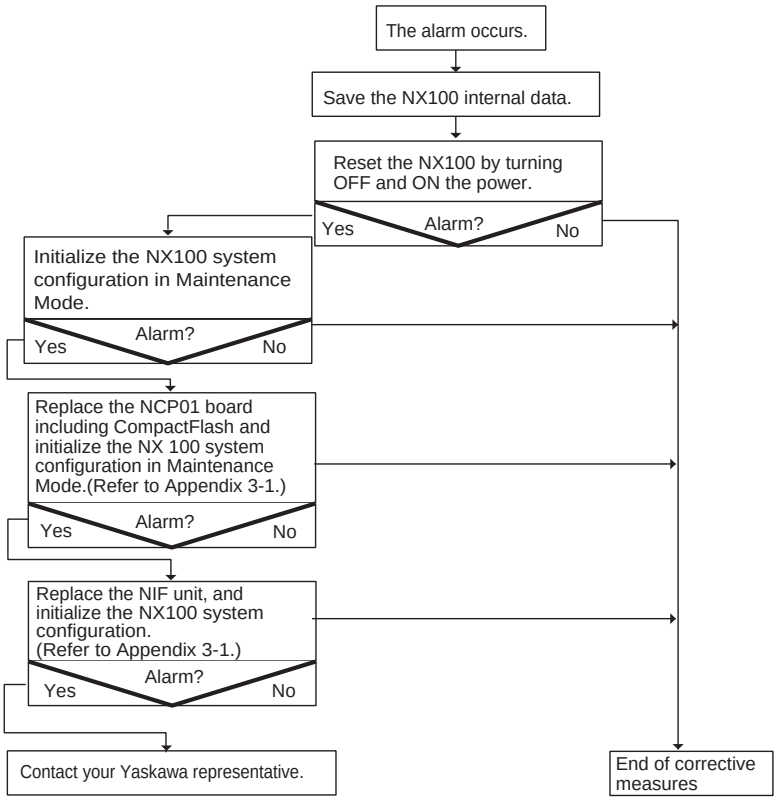
The NX100 saves a part of data needed for system operations on the Compact Flash in the NCP01 board.
When the data is changed, the new data is written in the on the Compact Flash.
This alarm occurs if the this writing is correctly done.
Sub Code: File No. of back up file in NX



■ AL-0802

FILE I/O ERROR (NCP01 CF)[INTERNAL ERROR NO.]

The NX100 saves a part of data needed for system operations on the Compact Flash in the NCP01 board. This alarm occurs if it cannot access to the NCP01correctly.
Usually, AL-0800 and AL-0801 couure simultaneously.
Sub Code: The number of the NX100 internal error code to the Compact Flash

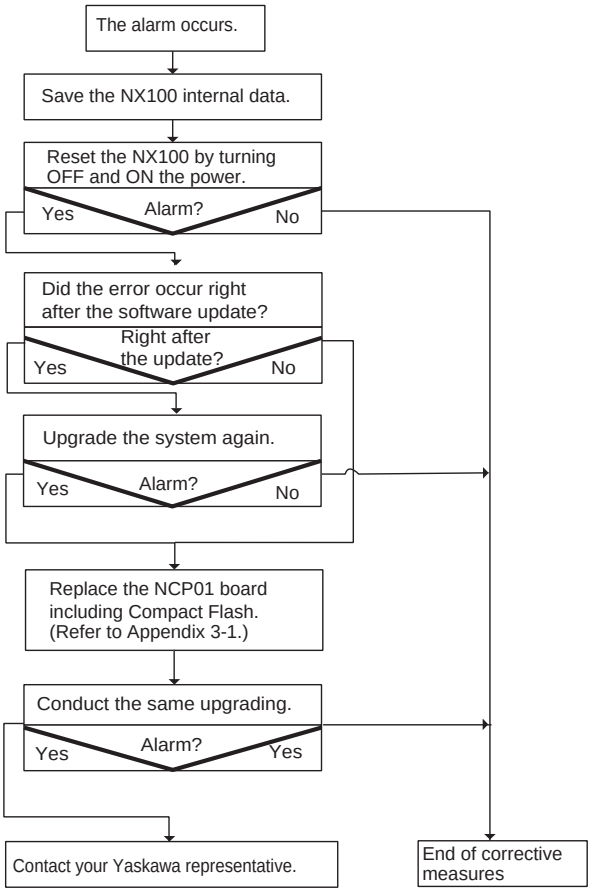


■ AL-0803

FILE ERROR

The NX100 new software controls the detailed data of robot or motor at the extra file called [MECHA.ROM]. This alarm occurs if this file cannot be read correctly.

Sub Code: None

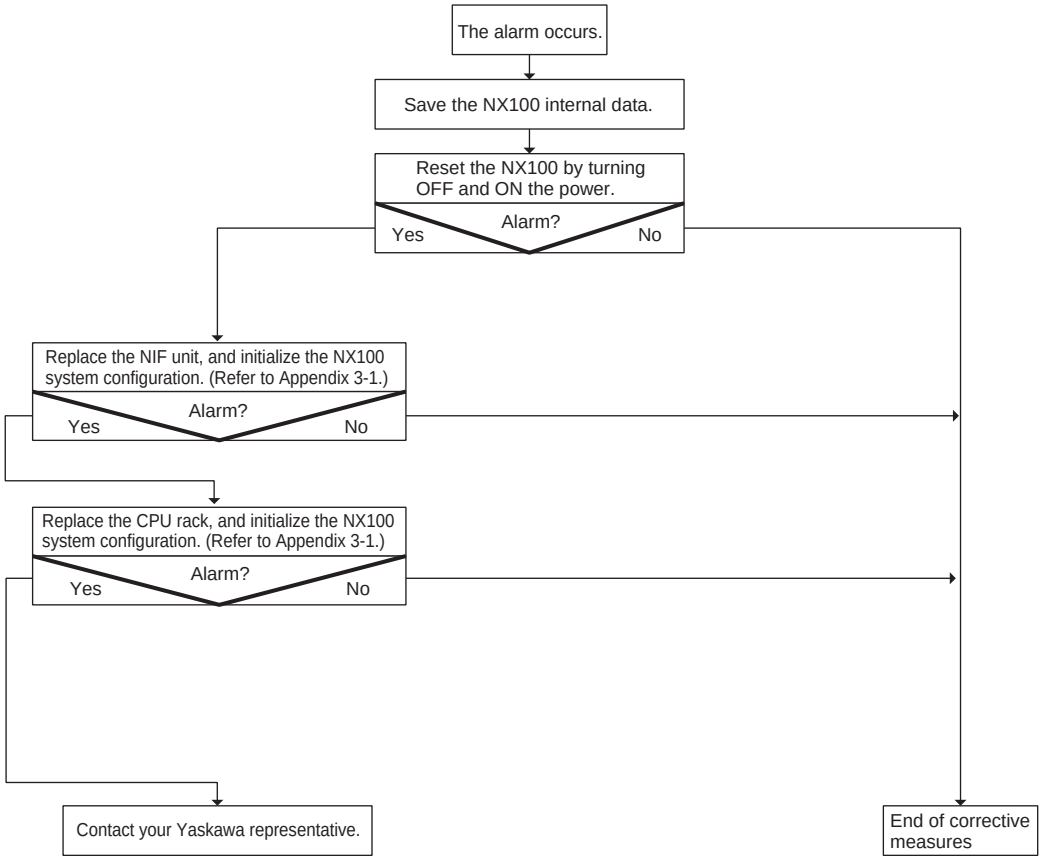


■ AL-0900

WATCHDOG TIMER ERROR (NIF BOARD)

The NX100 can safely stop the system with a watchdog function when an error occurred. This alarm occurs if a watchdog timeout is detected in the NIF board.

Sub Code: None

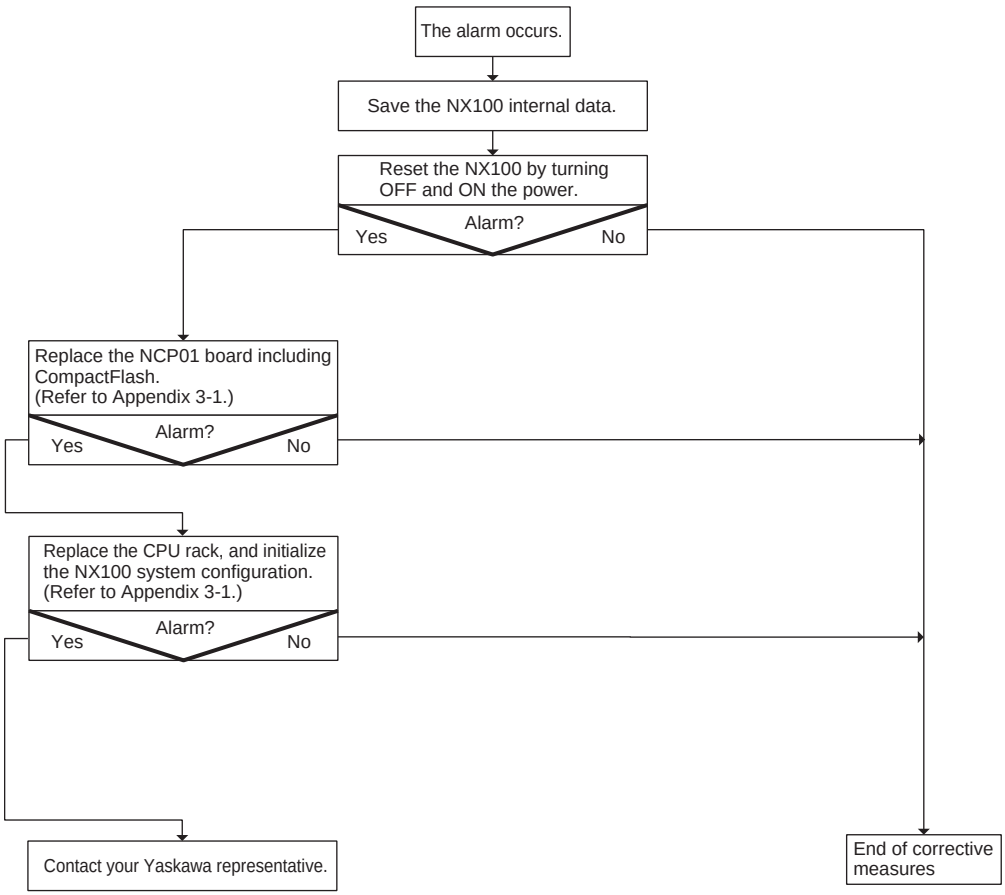


■ AL-0910

CPU ERROR (NCP01): [100]

The NX100 can safely stop the system even in case of an unexpected processing occurred in any board.
This alarm occurs if there is an unexpected processing in NCP01 (the main CPU board).

Sub Code: 100



■ AL-0920

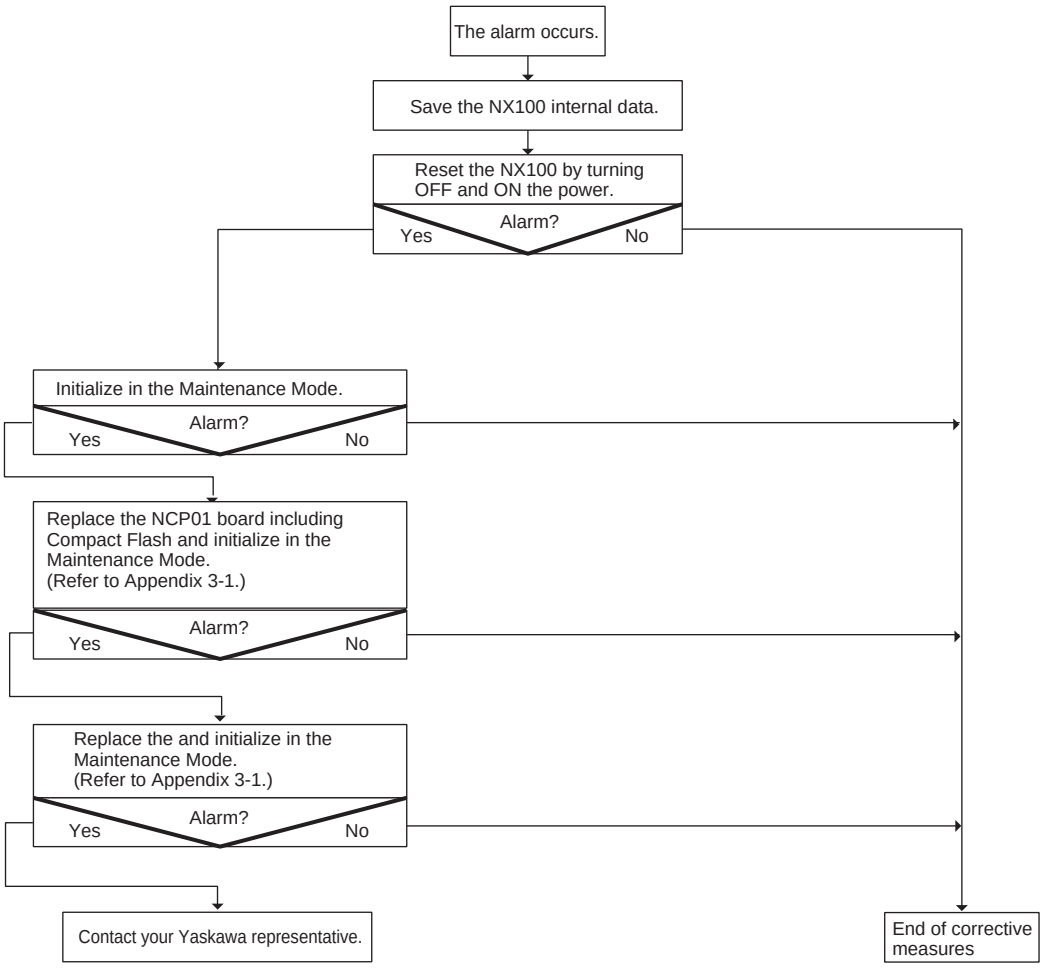
BUS ERROR (NCP01)

The NX100 uses the interruption signals which was lit by the internal IC as the standard signal of the system operation.

It alarms when this interruption signal is not appropriately input.

Sub Code: The type of the board is output

0: NCP01 board

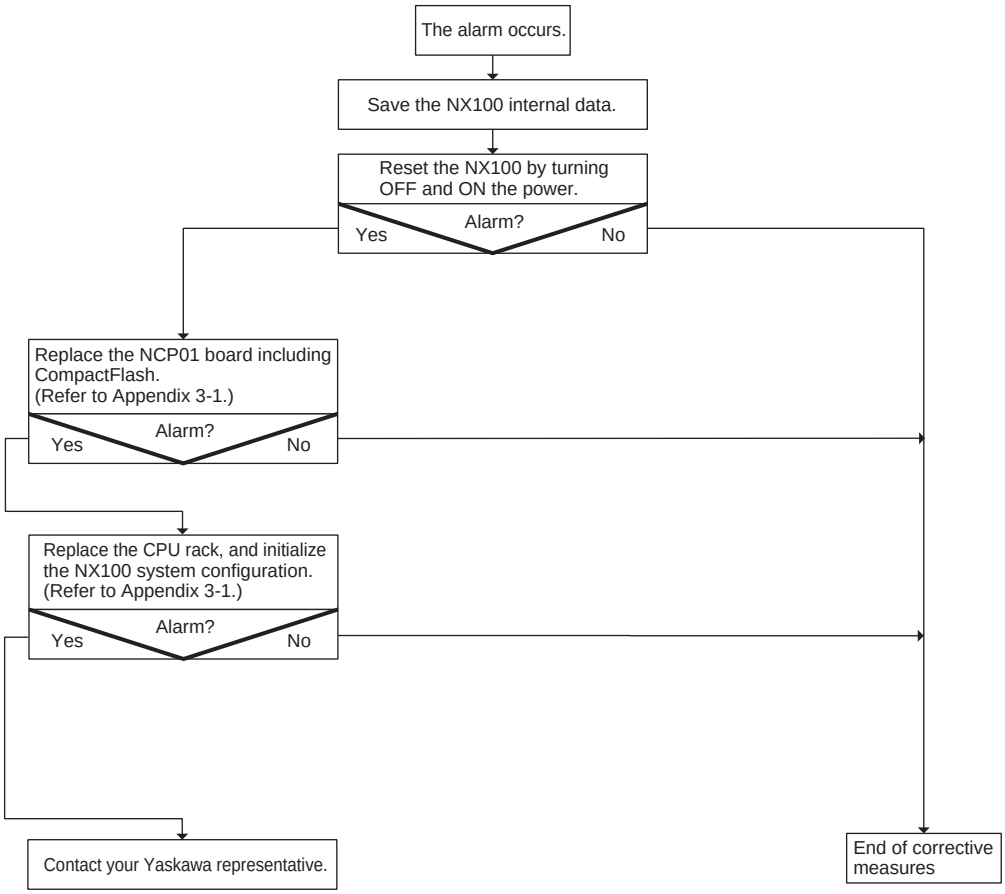


■ AL-0930

CPU HANG UP ERROR (NCP01)

The NX100 can safely stop the system even when the power supply voltage suddenly drops. This alarm occurs if a voltage drop is detected in NCP01 (the main CPU board).

Sub Code: None



■ AL-0950 / 0951 / 0952 / 0953

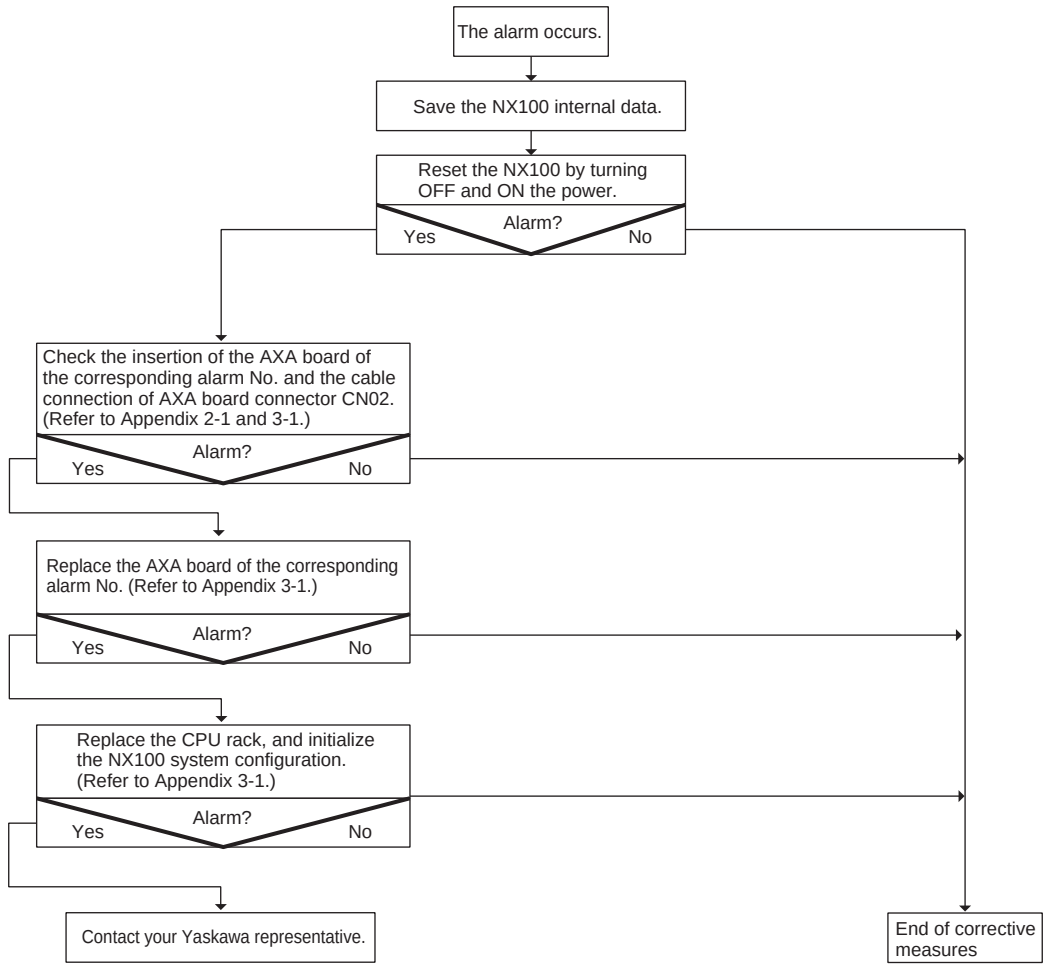
CPU ERROR (AXA#*): [100]

The NX100 can safely stop the system even when the power supply voltage suddenly drops. This alarm occurs if a voltage drop is detected in NCP01 (the main CPU board).
The least significant digit of the alarm signifies the AXA board No. of serial communications

Alarm

0950: 1st AXA board
0951: 2nd AXA board
0952: 3rd AXA board
0953: 4th AXA board

Sub Code: 100



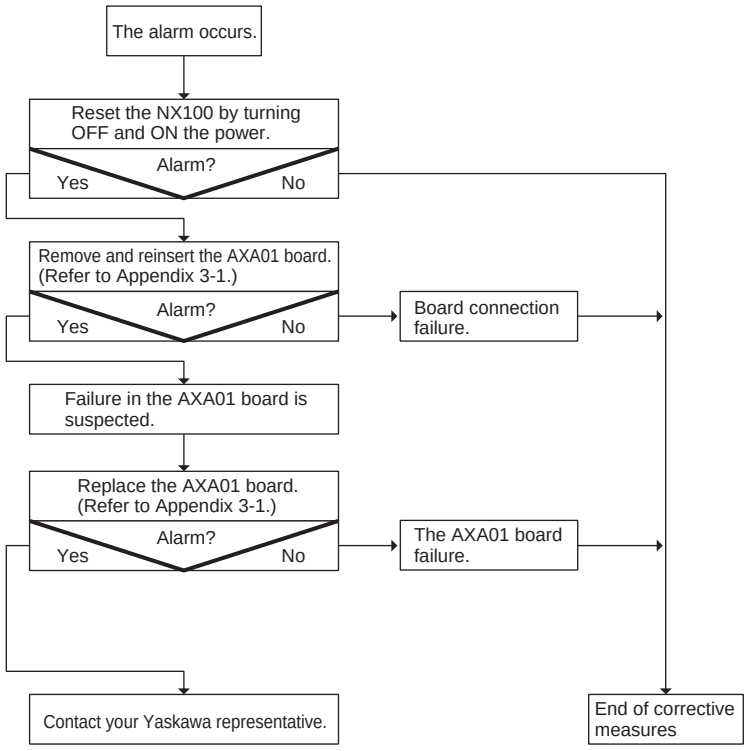
■ AL-1001

ROM ERROR (AXA01)

This alarm is caused by faulty data in ROM.
A hardware error in the board is considered as a major cause of this alarm. In rare cases, however, the cause is attributed to a loose connection or noise interference.

Sub Code

1*: EEPROM checksum error (*: axis No.)
20: EEPROM WRITE ENABLE error
21: EEPROM WRITE PROTECT error
22: EEPROM ERASE error
23: EEPROM CLEAR error
24: EEPROM writing error
25: EEPROM reading error
26: EEPROM verify error



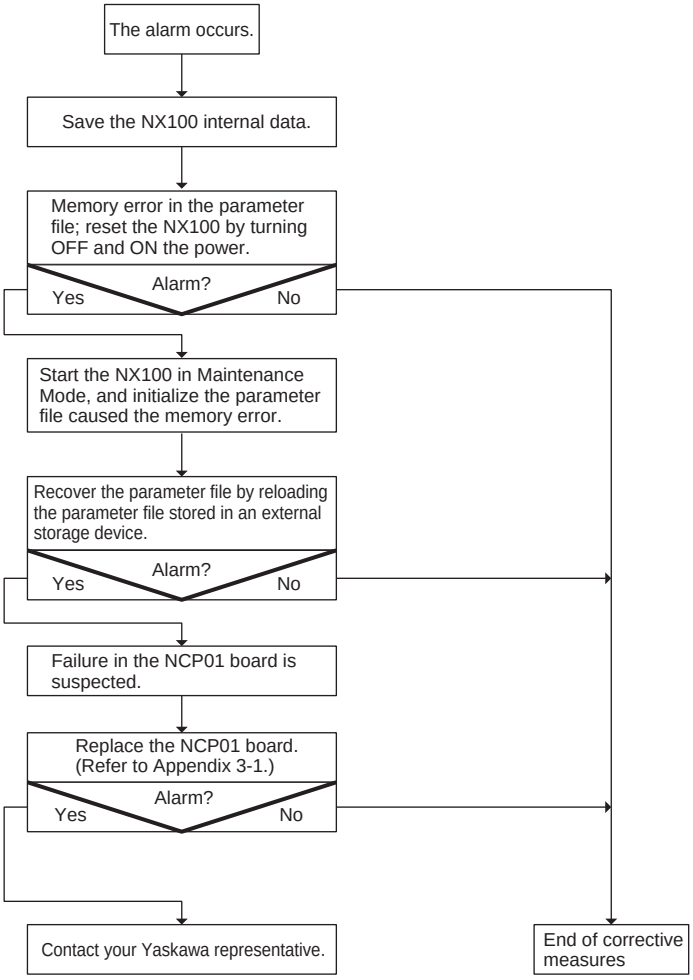
■ AL-1030

MEMORY ERROR (PARAMETER FILE)

This alarm occurs when an error is detected during total check of parameters.
For a general error recovery, start the NX100 in Maintenance Mode and initialize the parameter file which caused the memory error, then reload the parameter file stored in an external storage device.
As another possibility, this alarm may have been caused by a failure in CompactFlash card or memory error in the NCP01 board. In this case, replace the NCP01 board.

Sub Code

00:	RCD, RCxG parameter error	08:	A1P parameter error
01:	ROxG parameter error	09:	RS parameter error
02:	SVD, SVxG parameter error	10:	S1E parameter error
03:	SVMxG parameter error	11:	SVxB parameter error
04:	S1CxG, S2C, S3C, S4C parameter error	12:	AMCxG parameter error
05:	S1D, S2D, S3D, S4D parameter error	13:	SVPxG parameter error
06:	CIO parameter error	14:	MFxG parameter error
07:	FD parameter error	15:	SVxS parameter error



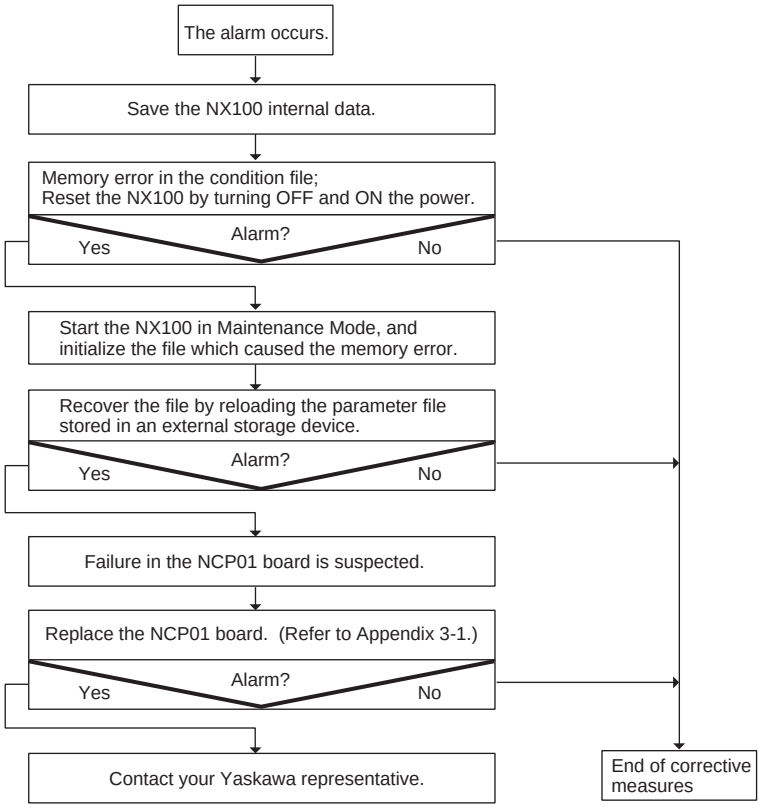
■ AL-1031

MEMORY ERROR (MOTION1): [File No. in CMOS]

This alarm occurs when an error is detected in the CMOS files used by MOTION.
For a general error recovery, start the NX100 in Maintenance Mode and initialize the file which caused the memory error, then reload the file stored in an external storage device.
As another possibility, this alarm may have been caused by a failure in CompactFlash card or memory error in the NCP01 board. In this case, replace the NCP01 board.

Sub Code

01: Home position calibration file
02: Tool file
03: User coordinates file
04: Robot calibration file
05: Tool calibration file
06: Weaving amplitude condition file
07: Home position correction data file
08: Conveyor calibration file
09: Arm and tool interference prevention file
20: Weaving file
21: Power Source condition data file
22: Welding condition auxiliary file
23: Arc start condition file
24: Arc end condition file
25: COMARC condition file
26: COMARC data file
27: Path correction condition file
28: Painting characteristics file
29: Painting condition file
30: Multi-layer index file
31: Multi-layer condition file
32: Sensor monitoring condition file
33: Name position file
34: Conveyor condition file
35: Press characteristics file
36: Servo float condition file
37: Spot welding Power Source condition data file
38: Air-gun condition file
39: Motor-gun condition file
40: Gun pressure file
41: Dry-spotting gun pressure file
42: Anticipation OT# output file
43: Anticipation OG# output file
44: Handling condition file
45: Form cut file
46: Spot (user) I/O allocation file
47: Linear servo float condition file
48: Macro definition file
49: Seal amount correction condition file (spray)
50: Seal amount correction condition file (undercoat)
51: Arc monitor file
52: Motor-gun condition auxiliary file
53: Job registration table
54: Painting device condition file
55: Painting system file
56: Painting condition file
57: Paint characteristics file
58: EVB gun file
59: Paint filling file
60: Welding pulse condition file
61: Clearance file
62: Linear scale condition file
63: Gauging sensor condition file
64: Conveyor condition auxiliary file


The alarm occurs.
Save the NX100 internal data.
Memory error in the condition file;
Reset the NX100 by turning OFF and ON the power.
Alarm?
Yes
No
Start the NX100 in Maintenance Mode, and
initialize the file which caused the memory error.
Recover the file by reloading the parameter file
stored in an external storage device.
Alarm?
Yes
No
Failure in the NCP01 board is suspected.
Replace the NCP01 board. (Refer to Appendix 3-1.)
Alarm?
Yes
No
Contact your Yaskawa representative.
End of corrective
measures

■ AL-1050

SET-UP PROCESS ERROR (SYSCON): [Cause No.]

The system software monitors if the setup of MOTION software is properly complete when the power is ON.

This alarm occurs if the MOTION software fails to properly complete the setup.

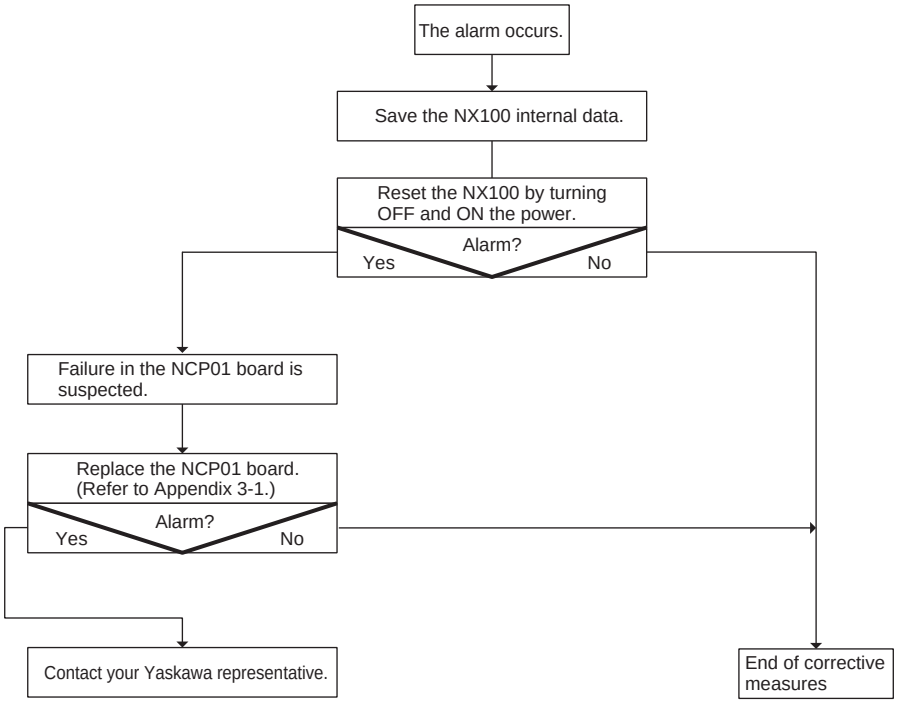
Note that the alarm AL-1051 (SETUP INITIALIZE ERROR (MOTION)) occurs in conjunction with this alarm.

(For details, refer to AL-1051.)

Sub Code: Error Cause Data No.

01: Incomplete setup of MOTION

02: Online error



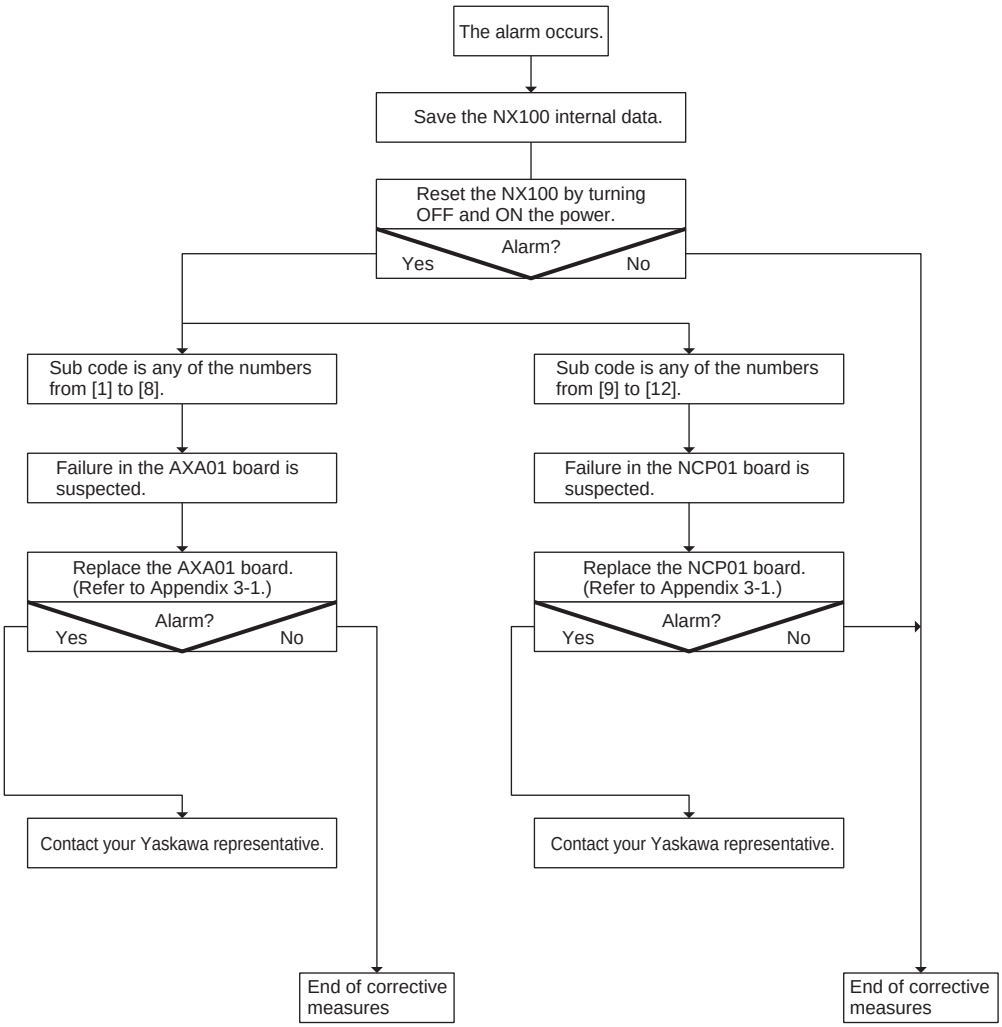
■ AL-1051

SETUP INITIALIZE ERROR (MOTION)

This alarm occurs if setup process of MOTION is not properly complete when the power is ON.

Sub Code: Error Cause Data No.

01: Unable to properly activate the servo control
02: The position data of when the power supply was turned OFF cannot be transmitted to the servo control section
03: The servo control section cannot receive the position data of when the power supply was turned OFF
05: Unable to send a request to turn ON the PG power supply for the mounted axis
06: Unable to turn ON the PG power supply for the mounted axis
07: Unable to send a request to prepare a feedback pulse
08: Unable to prepare a feedback pulse
09: Unable to send a request to initialize the arithmetic section (ARITH)
10: Unable to initialize ARITH
11: Unable to send a request to prepare the current position
12: Unable to prepare the current position



■ AL-1100

SYSTEM ERROR

This alarm is likely to be caused by failure in the system software, and it occurs when an alarm unregistered in the system software is detected.
As another possibility, noise interference causes RAM data failure in the NCP01 board, leading to this alarm.

Sub Code

C: * B: * F: *: Unregistered sub code (*: numbers)

■ AL-1101

SYSTEM ERROR (SYSTEM 1)

This alarm is likely to be caused by failure in the system software. The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

0 to 6: Internal control error in software

■ AL-1102

SYSTEM ERROR (SYSTEM 2)

This alarm is likely to be caused by failure in the system software. The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

0 to 524: Internal control error in software

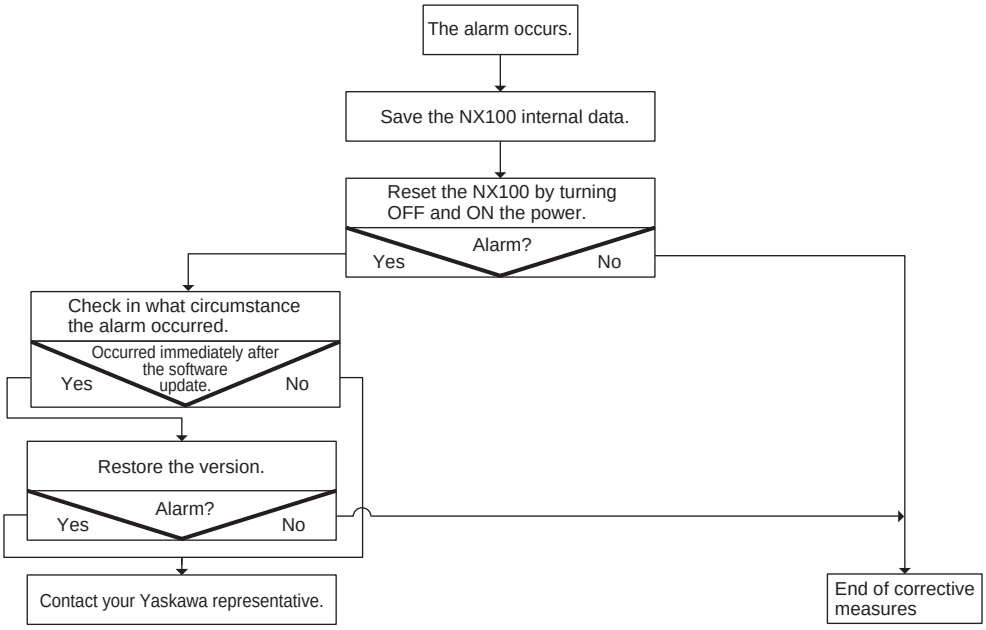
■ AL-1103

SYSTEM ERROR (EVENT)

This alarm is likely to be caused by failure in the system software. The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

0 to 8: Internal control error in software



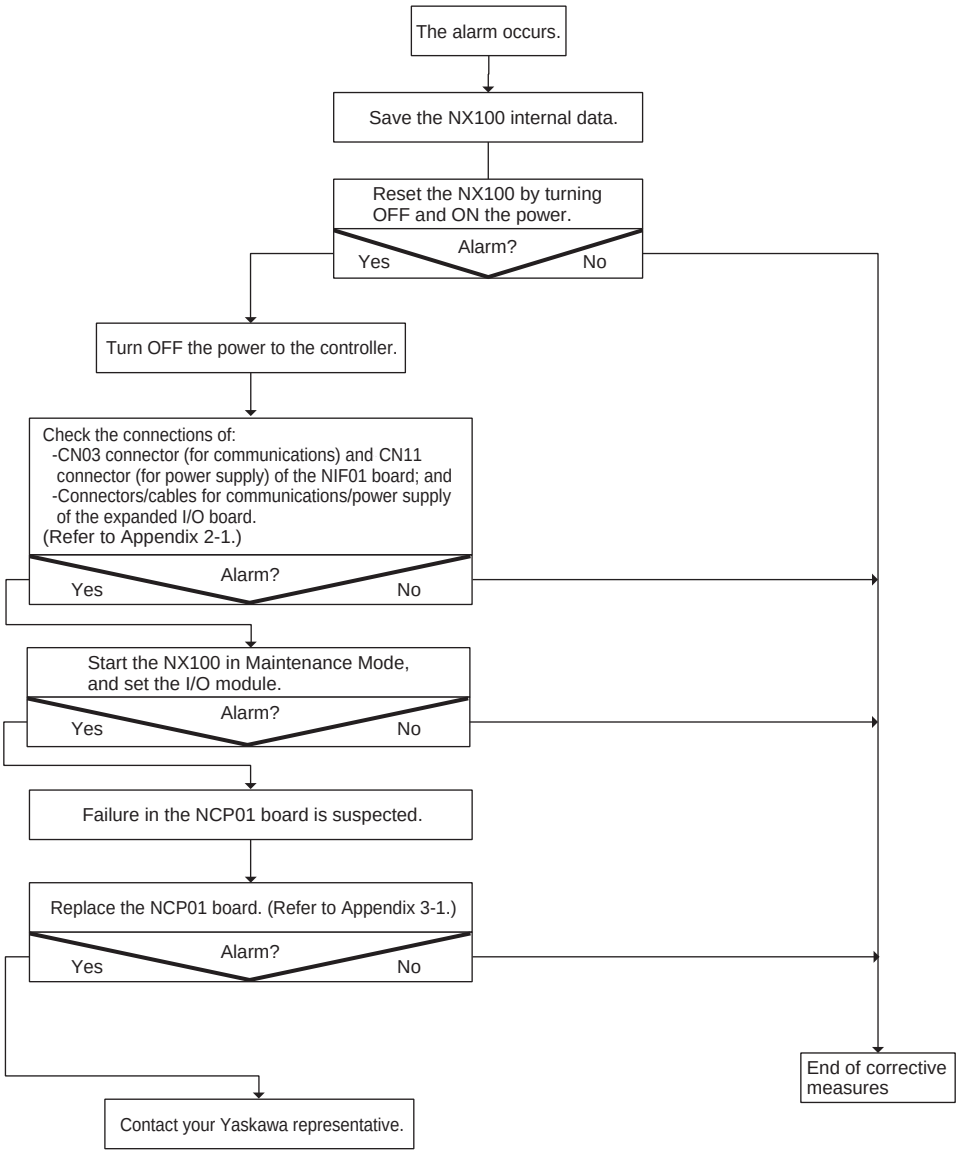
■ AL-1104

SYSTEM ERROR (CIO)

As a cause of this alarm, I/O module or CIO parameter setting error is suspected.
The alarm condition may be recovered by reconfiguring the I/O module in Maintenance Mode.
As another possibility, the alarm may have been caused by a failure in CompactFlash card or memory error in the NCP01 board. In this case, replace the NCP01 board.

Sub Code

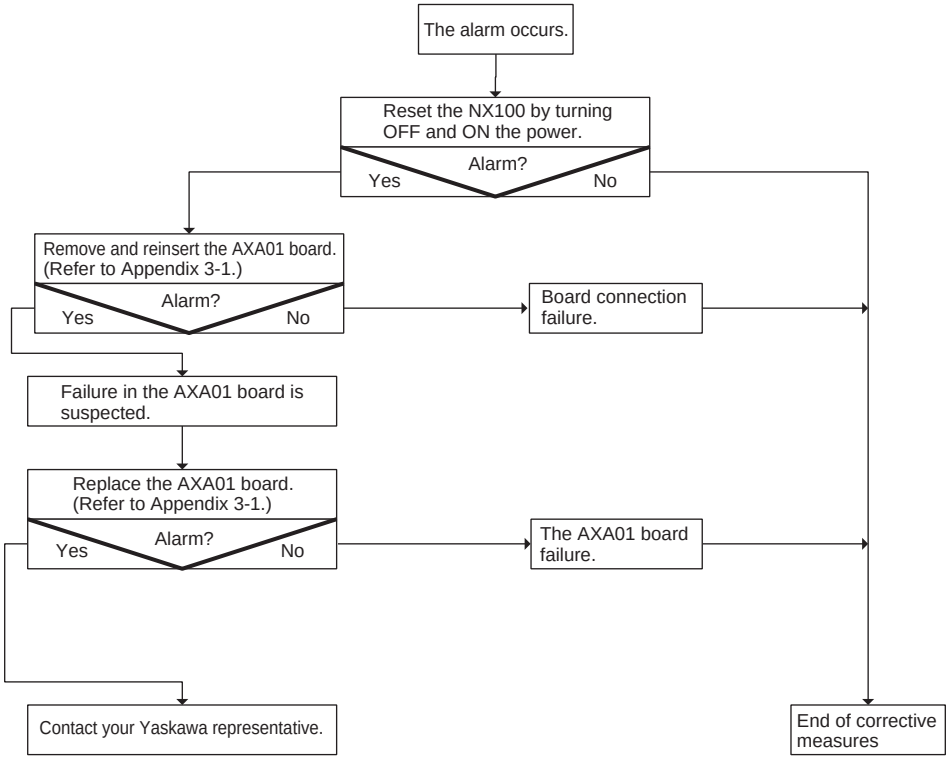
1000_0000: I/O module setting error



■ AL-1105

SYSTEM ERROR (SERVO)

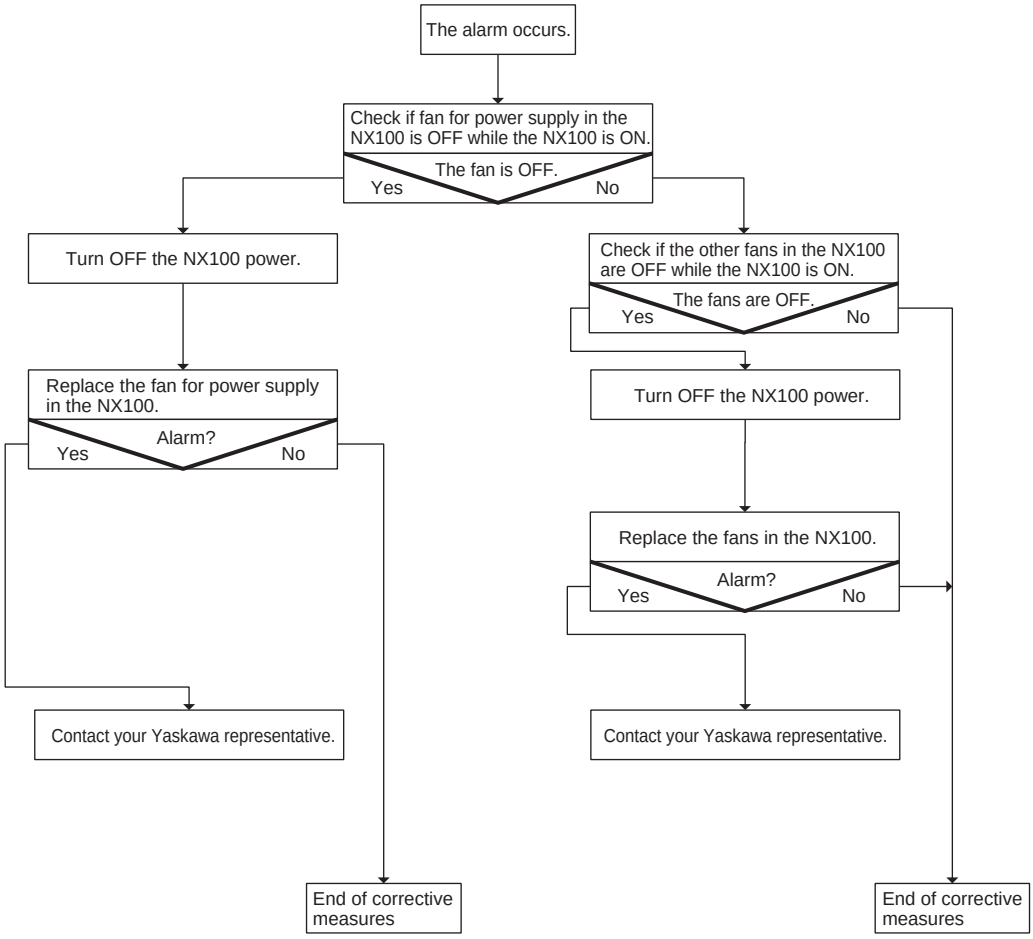
This alarm is caused by noise and control error. Contact your Yaskawa representative if the error persists even after restarting the NX100. (For each sub code, refer to "NX100 MAINTENANCE MANUAL (RE-CHO-A104)".



■ AL-1200

HIGH TEMPERATURE

This alarm occurs by high temperature inside the controller (CPS power supply unit), which is possibly caused by cooling fan error. If the error is due to the fan for power supply, the alarm occurs 10 minutes after turning on the power.
If the error is due to the other fans, the alarm AL-4119 may occur. AL-4119 FAN ERROR (IN CONTROL BOX) puts out a signal #50017 to activate the alarm 10 hours after the fan malfunction.
Sub Code: None



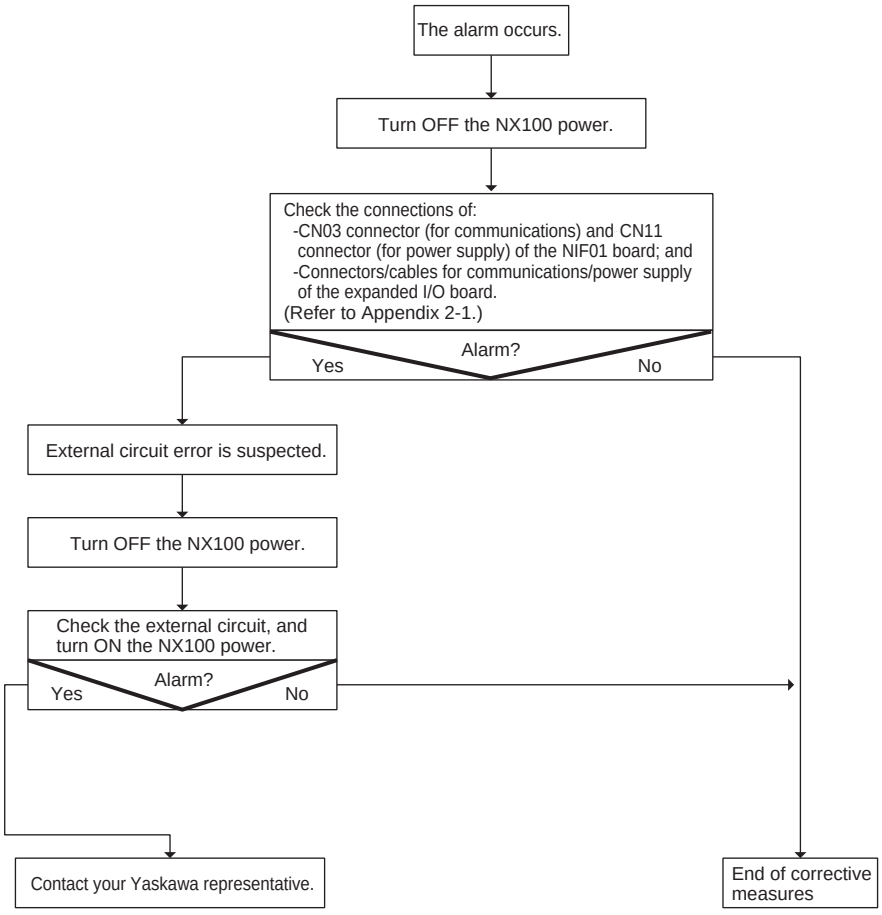
■ AL-1204

COMMUNICATION ERROR (IO MODULE)

As a cause of this alarm, I/O board communication error, loose connection of power connector, or I/O module setting error is suspected.
The alarm condition may be recovered by reconfiguring the I/O module in Maintenance Mode.
As another possibility, the alarm may have been caused by a failure in CompactFlash card or memory error in the NCP01 board. In this case, replace the controller unit.

Sub Code

0000_0000_0000_0001: NIF board communication error
0000_0000_0000_0010: Communication error in expanded I/O board station 1
0000_0000_0000_0100: Communication error in expanded I/O board station 2
...
1000_0000_0000_0000: Communication error in expanded I/O board station 15



■ AL-1207

BROKEN B_ON RELAY FUSE (NIF01)

This alarm occurs due to malfunction or blown fuse caused when electrical noise from the power supply interferes in the BROKEN B_ON RELAY FUSE detection circuit in the NIF01 board. Since the fuse is mounted to the NIF01 board, replace the controller unit if the alarm is caused by a blown fuse.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2

■ AL-1208

BROKEN S_ON RELAY FUSE (NIF01)

This alarm occurs due to malfunction or blown fuse caused when electrical noise from the power supply interferes in the BROKEN S_ON RELAY FUSE detection circuit in the NIF01 board. Since the fuse is mounted to the NIF01 board, replace the controller unit if the alarm is caused by a blown fuse.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2

■ AL-1209

EXTERNAL WDT BROKEN (NIF01)

This alarm occurs due to malfunction or failure caused when electrical noise from the power supply interferes in the EXTERNAL WDT BROKEN detection circuit in the NIF01 board. Replace the NIF01 board in case of failure.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2

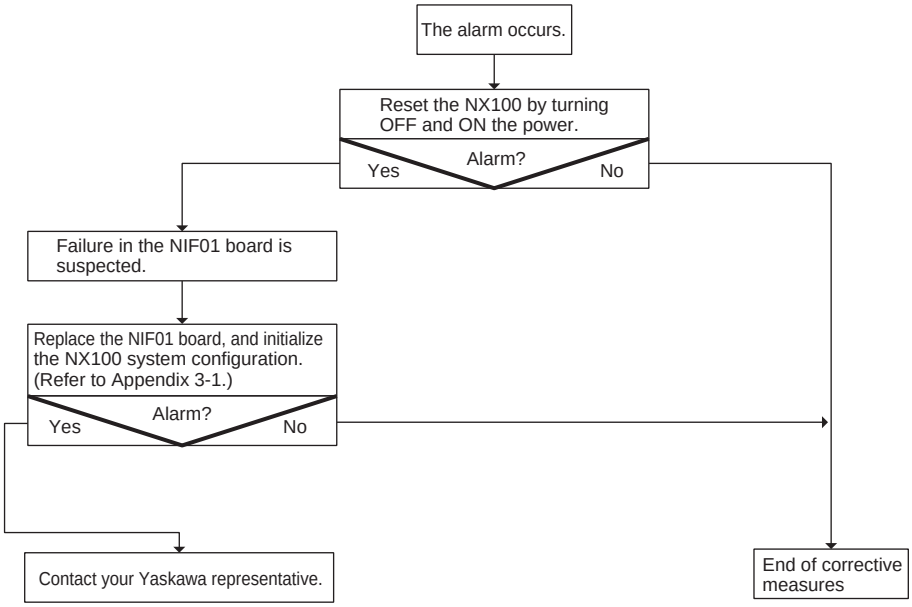
■ AL-1210

COMMUNICATION ERROR (PLDs) (NIF01)

This alarm occurs due to malfunction or failure caused when electrical noise from the power supply interferes in the SERIAL COMMUNICATION TOGGLE CHECK ERROR detection circuit in the NIF01 board. Replace the NIF01 board in case of failure.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2

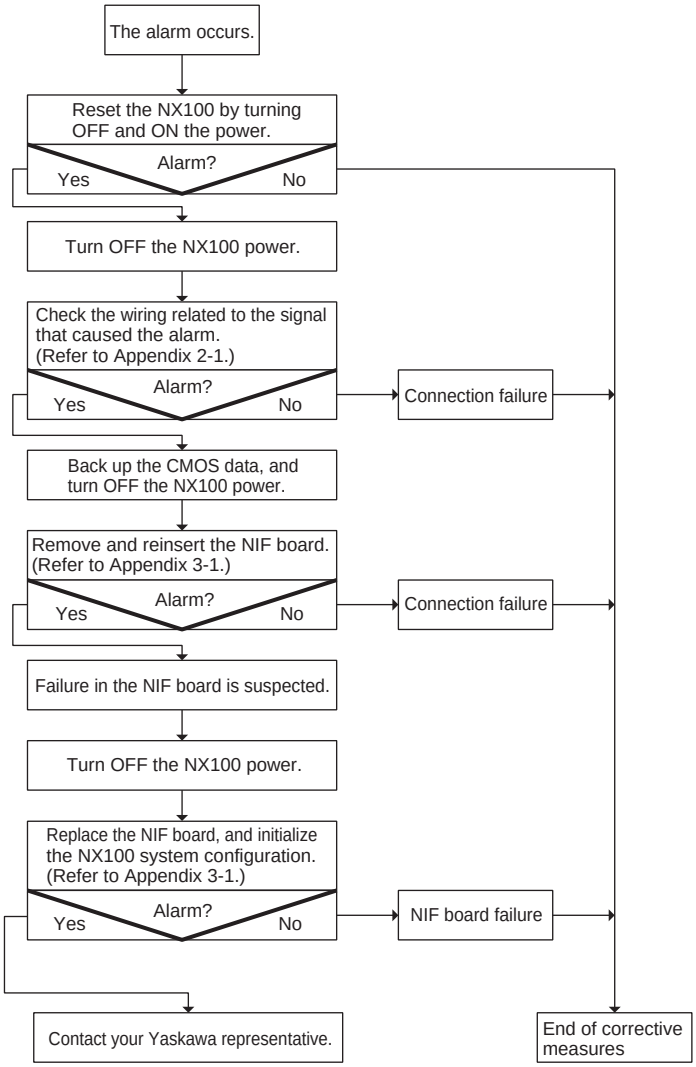

The alarm occurs.
Reset the NX100 by turning OFF and ON the power.
Alarm?
Yes
No
Failure in the NIF01 board is suspected.
Replace the NIF01 board, and initialize the NX100 system configuration. (Refer to Appendix 3-1.)
Alarm?
Yes
No
Contact your Yaskawa representative.
End of corrective measures

■ AL-1211

SAFE CIRCUIT INPUT FAULT (NIF01)

This alarm occurs if the user wiring of the signal corresponding to the sub code is false, or if there is a malfunction/failure caused by noise from the power supply in the NIF01 board.
If the sub code is not on the following list, malfunction/failure by noise from the power supply in the NIF01 board is suspected.
If the alarm is due to failure in the NIF01 board, replace the controller unit.
Sub Code: Error Cause Data No.

0000_0000_0000_0001: PBESP signal
0000_0000_0000_0010: PPESP signal
0000_0000_0000_0100: EXESP signal
0000_0000_0000_1000: ERRCPU signal
0000_0000_0001_0000: SYSRDY signal
0000_0000_0010_0000: SAF_F signal
0000_0000_1000_0000: EXSVON signal
0000_0010_0000_0000: MAINTE signal
0000_0100_0000_0000: DSW signal
0000_1000_0000_0000: EXDSW signal
0100_0000_0000_0000: SSP signal
1000_0000_0000_0000: FST signal



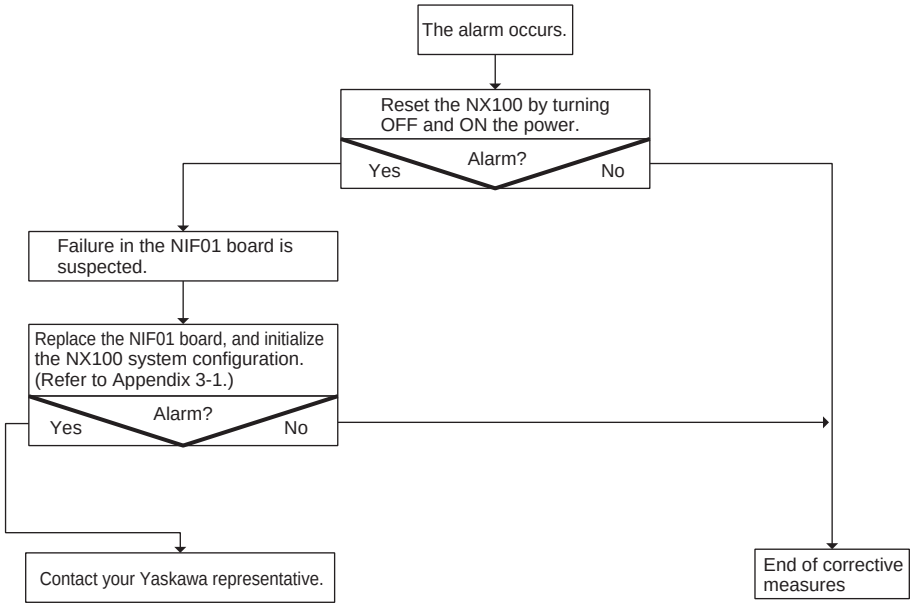
■ AL-1212

PLD MUTUAL MONITOR ERROR (NIF01)

This alarm occurs due to malfunction or failure caused when electrical noise from the power supply interferes in the PLD MUTUAL MONITOR ERROR detection circuit of the NIF01 board.
In case of failure, replace the controller unit.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2



■ AL-1213

MUTUAL WDT ERROR (NIF01)

If this alarm occurs in conjunction with AL-1211 (SAFE CIRCUIT INPUT FAULT (NIF01)), refer to the AL-1211 for troubleshooting.
When this alarm occurs independently, malfunction or failure caused by noise from the power supply in the NIF01 board is suspected.
In case of failure, replace the controller unit.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2

■ AL-1214

PBESP RELAY STICKING

This alarm occurs when the PBESP relay sticking detection circuit in the NIF01 board malfunctions due to electrical noise from the power supply, or when sticking/melting of overheated relay circuit wire is detected. Replace the NIF01 board if the alarm is caused by sticking/melting of the circuit wire.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2

■ AL-1215

PPESP STICKING

This alarm occurs when the PPESP relay sticking detection circuit in the NIF01 board malfunctions due to electrical noise from the power supply, or when sticking/melting of overheated relay circuit wire is detected. Replace the NIF01 board if the alarm is caused by sticking/melting of the circuit wire.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2

■ AL-1216

EXESP STICKING

This alarm occurs when the EXESP relay sticking detection circuit in the NIF01 board malfunctions due to electrical noise from the power supply, or when sticking/melting of overheated relay circuit wire is detected. Replace the NIF01 board if the alarm is caused by sticking/melting of the circuit wire.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2

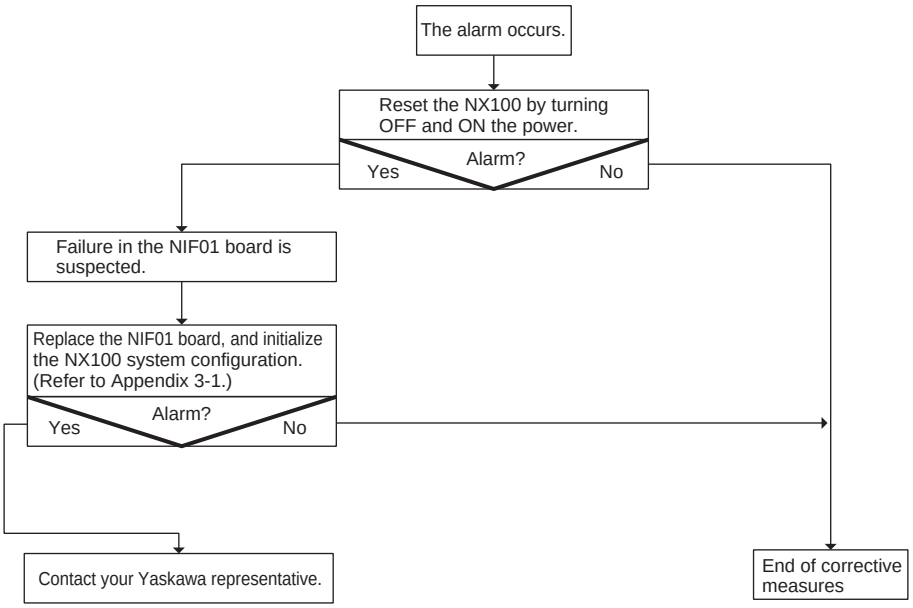
■ AL-1217

S_ON RELAY STICKING

This alarm occurs when the S_ON relay sticking detection circuit in the NIF01 board malfunctions due to electrical noise from the power supply, or when sticking/melting of overheated relay circuit wire is detected. Replace the NIF01 board if the alarm is caused by sticking/melting of the circuit wire.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2



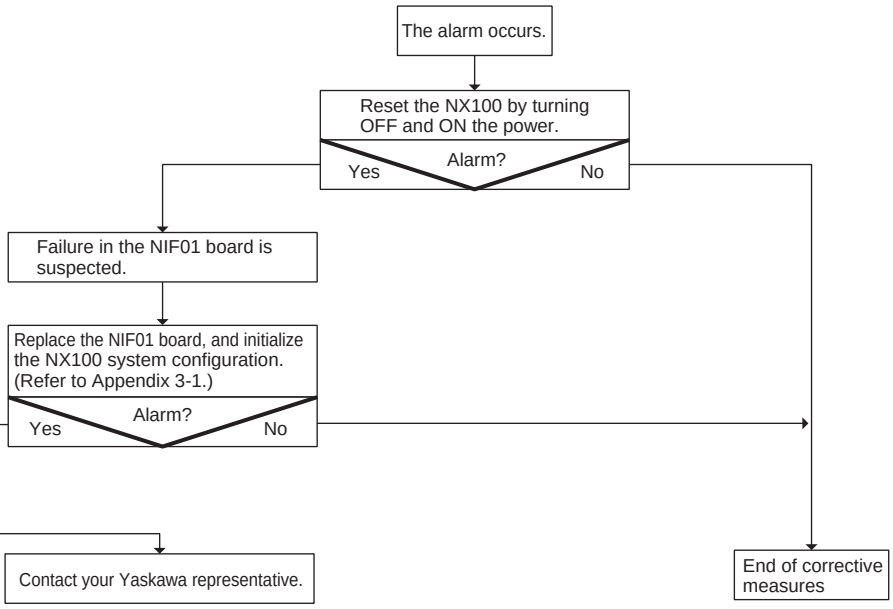
■ AL-1218

B_ON RELAY STICKING

This alarm occurs when the B_ON relay sticking detection circuit in the NIF01 board malfunctions due to electrical noise from the power supply, or when sticking/melting of overheated relay circuit wire is detected. Replace the NIF01 board if the alarm is caused by sticking/melting of the circuit wire.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2



■ AL-1219

ANOTHER PLD EXT WDT ERROR (NIF01)

If this alarm occurs in conjunction with AL-1211(SAFE CIRCUIT INPUT FAULT (NIF01)), refer to the AL-1211 troubleshooting.

When this alarm occurs independently, malfunction or failure caused by noise from the power supply in the NIF01 board is suspected.

In case of failure, replace the controller unit.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2

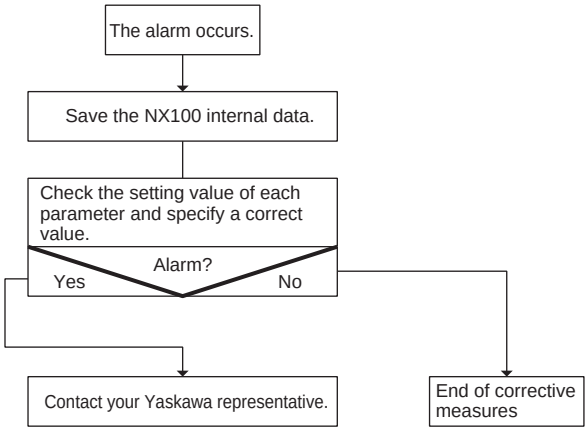
■ AL-1220

LAN COMMUNICATION PARAMETER ERROR

A parameter used by the Ethernet function contains an error. Specify a correct value.

Sub Code: The type of the abnormal address is output

1: IP address (RS70 to RS73)
2: Subnet mask (RS74 to RS77)
3: Default gateway (RS70 to RS73)
4: Server (host) address (RS70 to RS73)



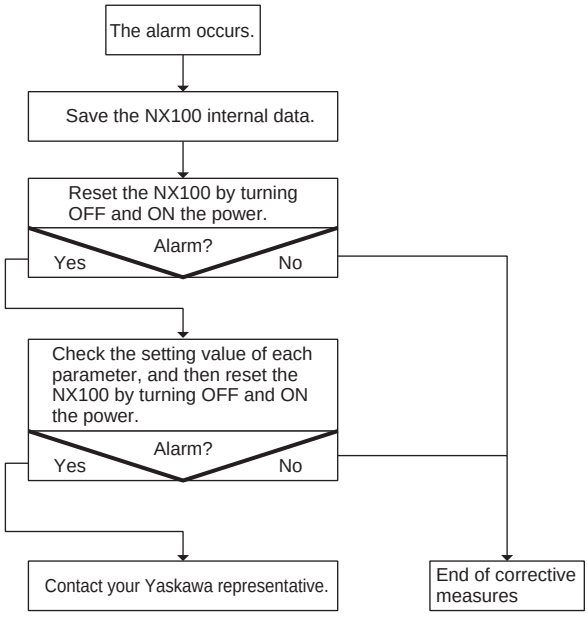
■ AL-1221

ETHERNET INITIAL PROCESS ERROR

This alarm occurs if an error is detected at the initialization of the Ethernet function.

Sub Code: The type of the abnormal process is output

1: Ethernet device setting
2: IP address setting
3: Subnet mask setting
4: Default gateway setting



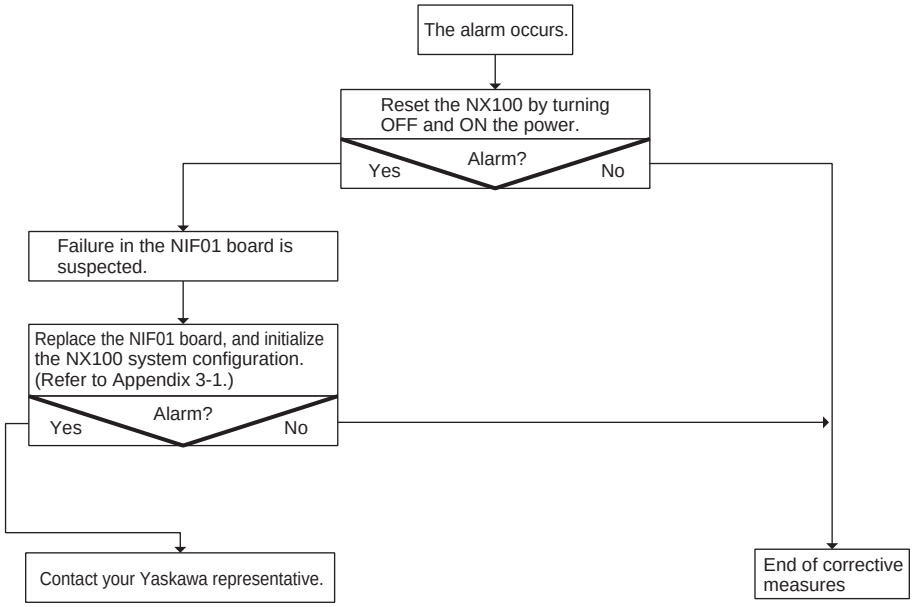
■ AL-1223

MONITOR PLD ERROR 1 (NIF01)

This alarm occurs due to malfunction or failure caused when electrical noise from the power supply interferes in the PLD MUTUAL MONITOR ERROR detection circuit of the NIF01 board.
In case of failure, replace the NIF01 board.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2



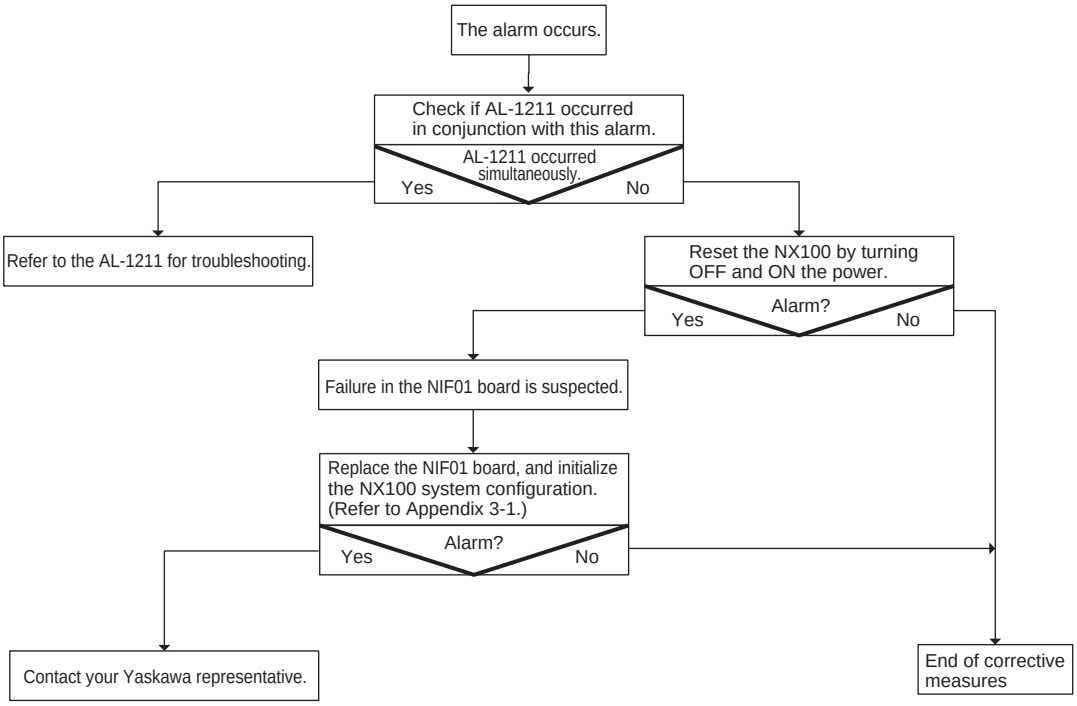
■ AL-1224

MONITOR PLD ERROR 2 (NIF01)

If this alarm occurs in conjunction with AL-1211 (SAFE CIRCUIT INPUT FAULT (NIF01)), refer to the AL-1211 troubleshooting.
When this alarm occurs independently, malfunction or failure caused by noise from the power supply in the NIF01 board is suspected.
In case of failure, replace the controller unit.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2



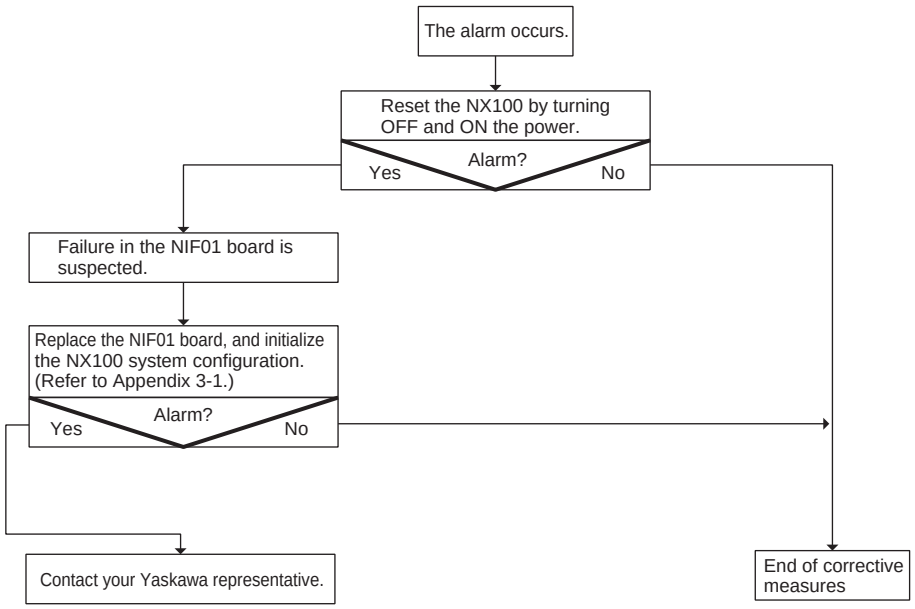
■ AL-1225

MONITOR PLD ERROR 3 (NIF01)

This alarm occurs due to malfunction or failure caused when electrical noise from the power supply is in the external WDT broken detection circuit in the NIF01 board.
In case of failure, replace the NIF01 board.

Sub Code

0000_0001: Error detected in PLD circuit 1
0000_0010: Error detected in PLD circuit 2
0000_0011: Error detected in PLD circuits 1 and 2



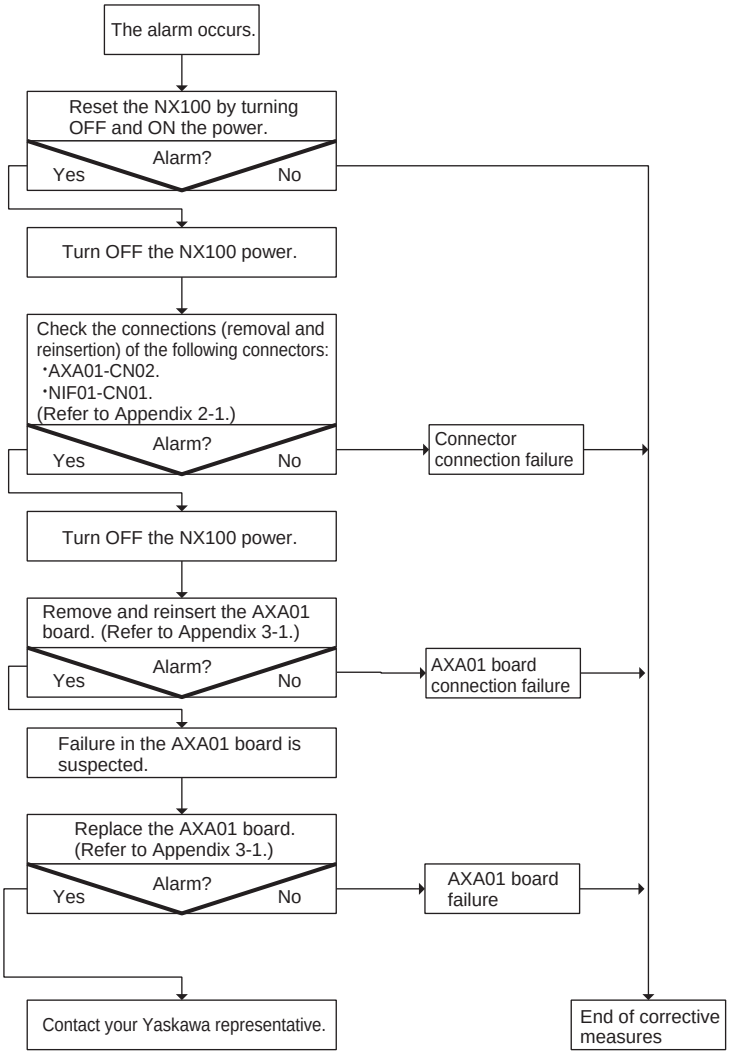
■ AL-1300

SERVO CPU SYNCHRONIZING ERROR

The NX100 system performs serial communications between the main CPU board and the servo control board (AXA01), checking the synchronization of the servo regular process and the communication process. This alarm occurs if any synchronization error is detected.

As a major cause of this alarm, a hardware error in the servo control board is suspected. In rare cases, however, the cause is attributed to a loose connection, noise interference, or internal data inconsistency.

Sub Code: None



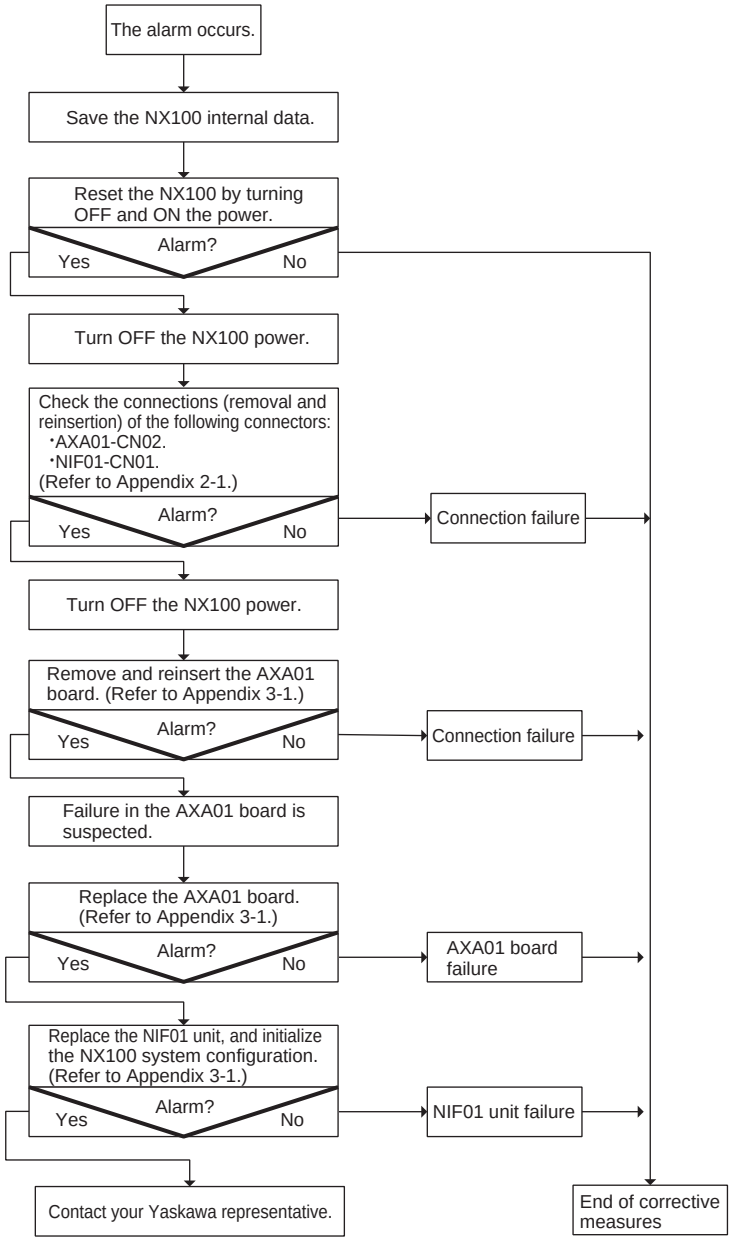
■ AL-1301

COMMUNICATION ERROR (SERVO)

This alarm occurs if there is a communication error between the robot interface board (NIF01) and the servo control board (AXA01).
As a major cause of this alarm, a loose connection is suspected. In rare cases, however, the cause is attributed to a noise interference, defective board/cable, or internal data inconsistency.

Sub Code

0: Communication status error
1: Watchdog timer error
2: JL040 alarm
3: Communication status error
4: Data consistency error



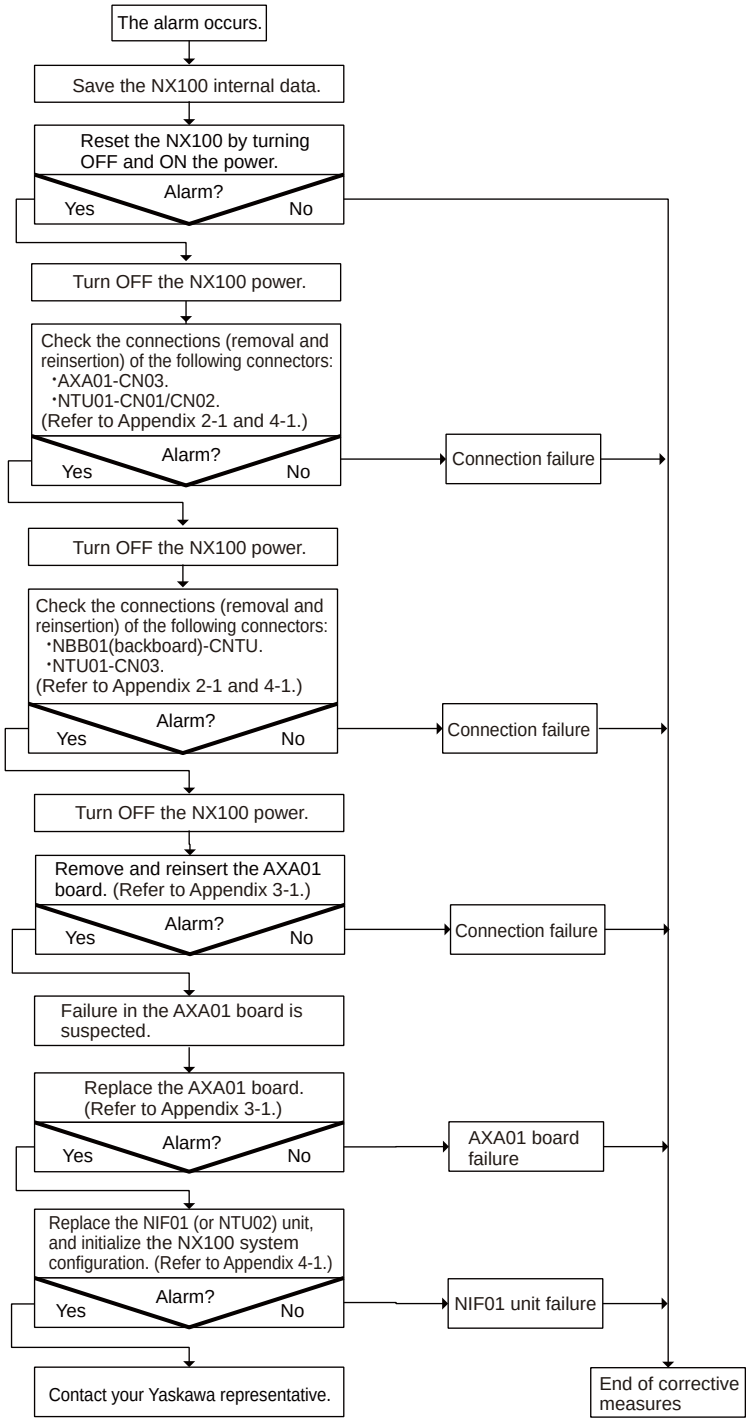
■ AL-1302

COMMUNICATION ERROR (SERVO I/O)

This alarm occurs if there is a communication error between the servo control board (AXA01) and the power-on unit (NTU01).
As a major cause of this alarm, a loose connection is suspected. In rare cases, however, the cause is attributed to a noise interference, defective board/cable, or internal data inconsistency.

Sub Code

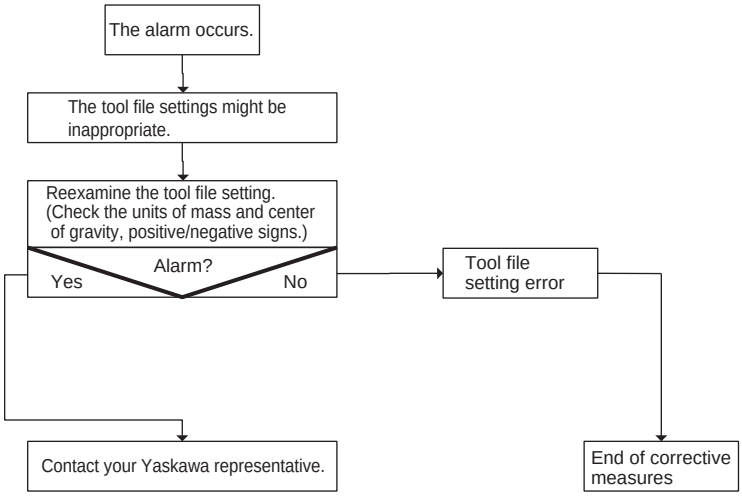
1: Communication loop back
2: No interrupt
4: Watchdog timer error
5: Command timeout
10: Communication loop back (: AXA board number of the connected unit)
**11: Address matching error
12: Servo I/O communication buffer error (: AXA board number of the connected unit)



■ AL-1303

ARITHMETIC ERROR (SERVO)

This alarm occurs if a control-related parameter calculated from the input parameter (tool file) is not within the specified range.

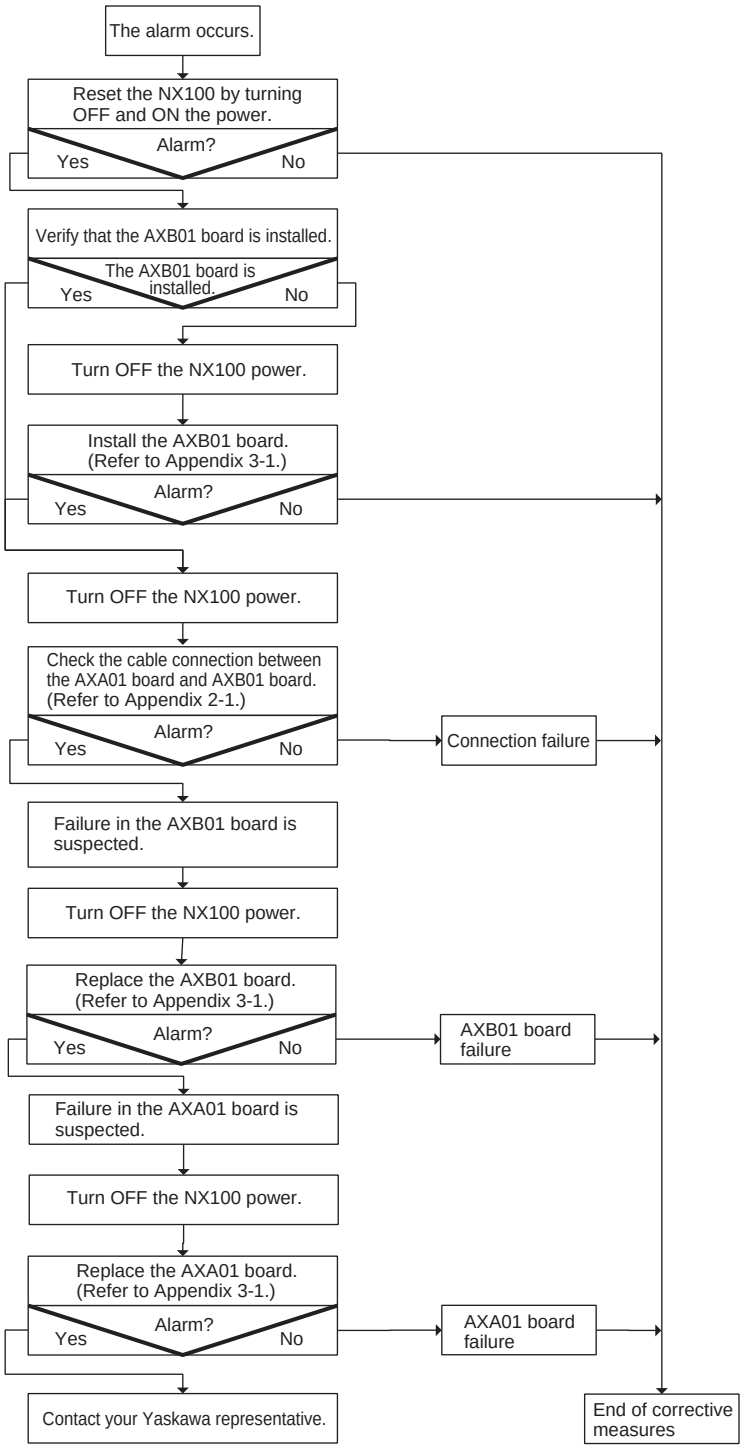


■ AL-1304

EX-AXIS BOARD NOT INSTALLED

This alarm occurs during operation if the external axis board which has not been installed is assigned, or if the board is defective. In rare cases, the cause is attributed to a loose connection or noise interference.

Sub Code: None



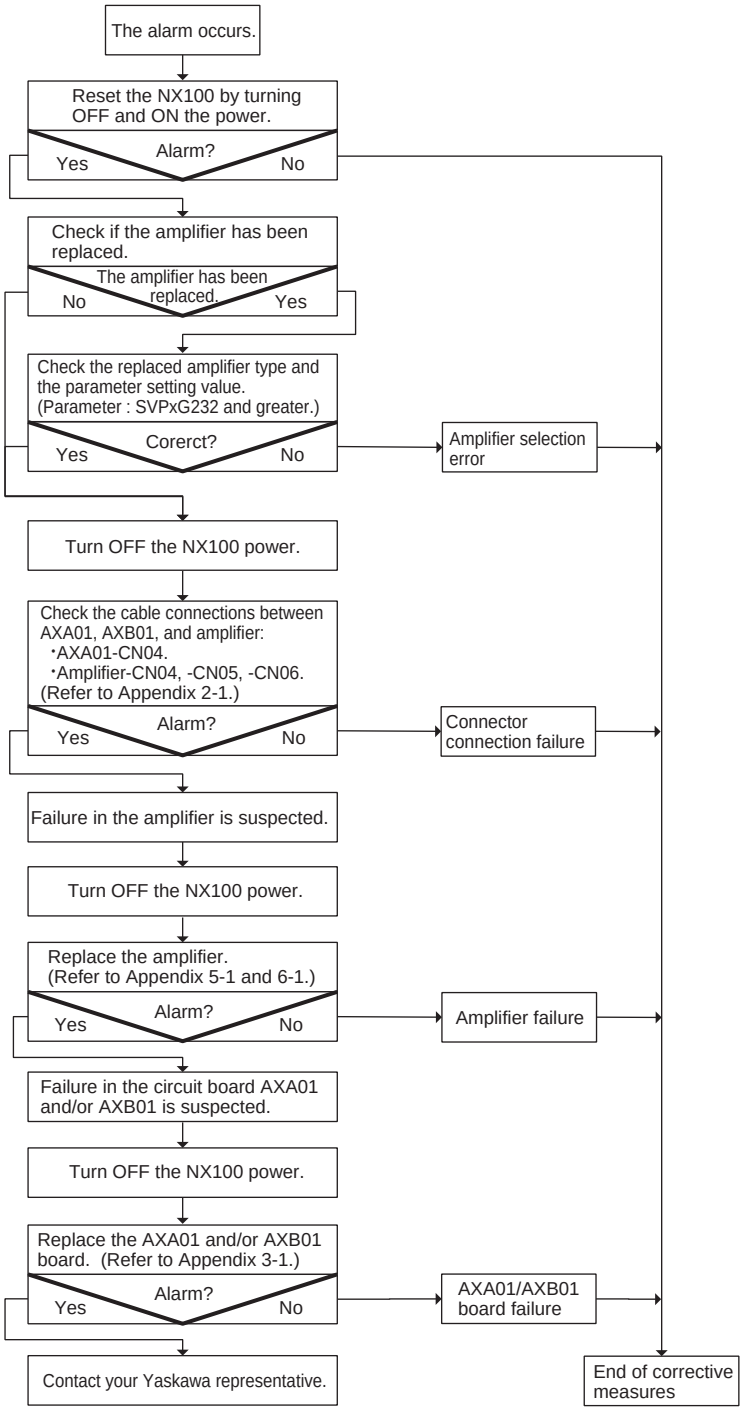
■ AL-1306

AMPLIFIER TYPE MISMATCH: [Physical Axis Bit]

When the controller power is ON, the NX100 system checks the current capacity of the SERVOPACK amplifier in the servo control circuit board (AXA01, AXB01). This alarm occurs if there is a difference in the capacity between the set value and the mounted amplifier.

If the alarm occurs during operation, a parameter setting error or board failure is suspected. In rare cases, the cause is attributed to a loose connection or noise interference.

Sub Code: Signifies the axis in which the alarm occurred



■ AL-1307

ENCODER TYPE MISMATCH: [Physical Axis Bit]

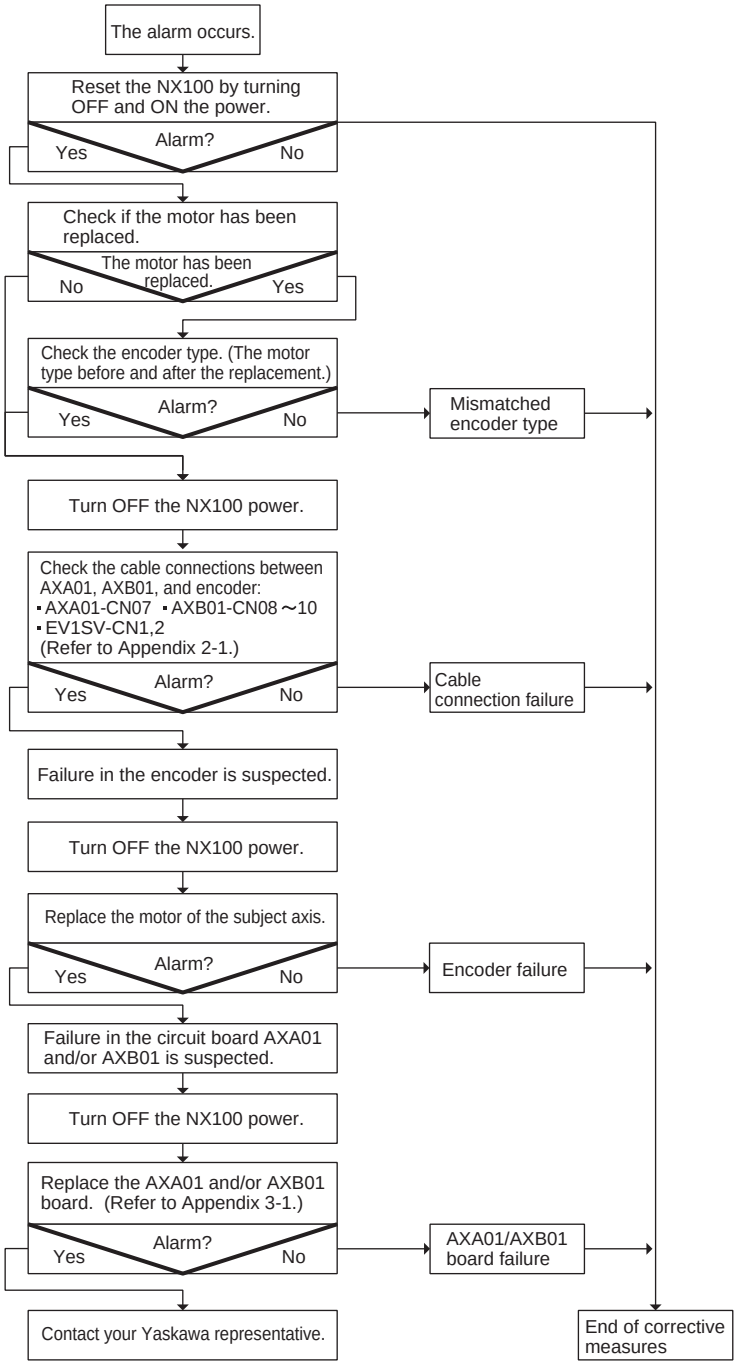
When the controller power is ON, the NX100 system checks the encoder type of the manipulator in the servo control circuit board (AXA01, AXB01). This alarm occurs if there is a difference between the set value and the mounted encoder.

Since the proper encoder has been configured before shipment, an encoder error is suspected as the cause of this alarm occurred during operation. If the encoder has been replaced, type mismatch is suspected.

In rare cases, the cause is attributed to a loose connection, defective board, or noise interference.

The NX100 system identifies the 16-bit and 17-bit serial encoder types; the resolutions of absolute encoders of which motor types are SGM☐☐-☐☐A1A and SGM☐☐-☐☐A2A are respectively 16 bits and 17 bits.

Sub Code: Signifies the axis in which the alarm occurred



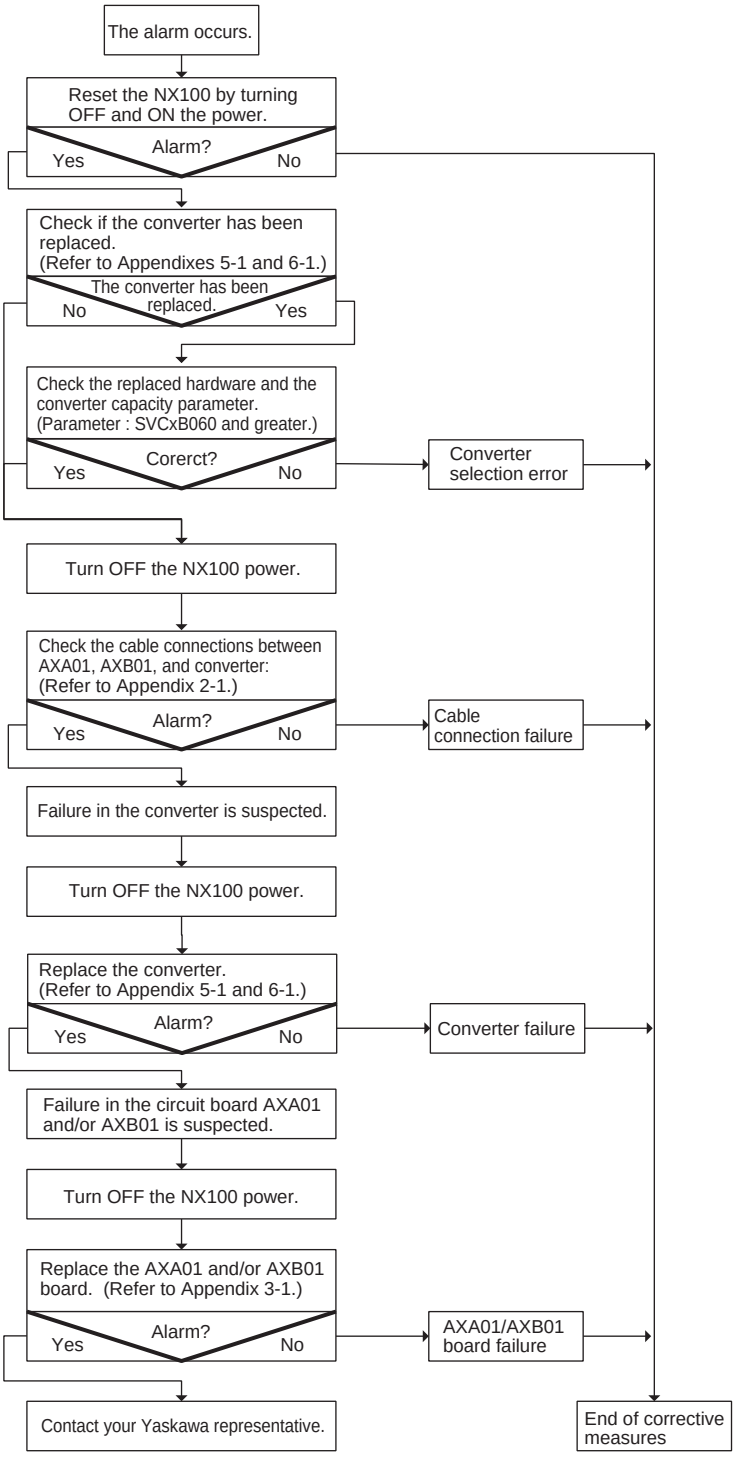
■ AL-1308

CONVERTER TYPE MISMATCH: [Physical Converter Bit]

When the controller power is ON, the NX100 system checks the converter type. This alarm occurs if there is a difference between the set value and the mounted converter type.

If the alarm occurs during operation, a servo control board (AXA01, AXB01) or converter failure is suspected. In rare cases, the cause is attributed to a loose connection or noise interference.

Sub Code: Signifies the converter in which the alarm occurred



■ AL-1309

HARDWARE ERROR (CONVERTER): [Physical Converter Bit]

■ AL-1310

CHARGE ERROR (CONVERTER): [Physical Converter Bit]

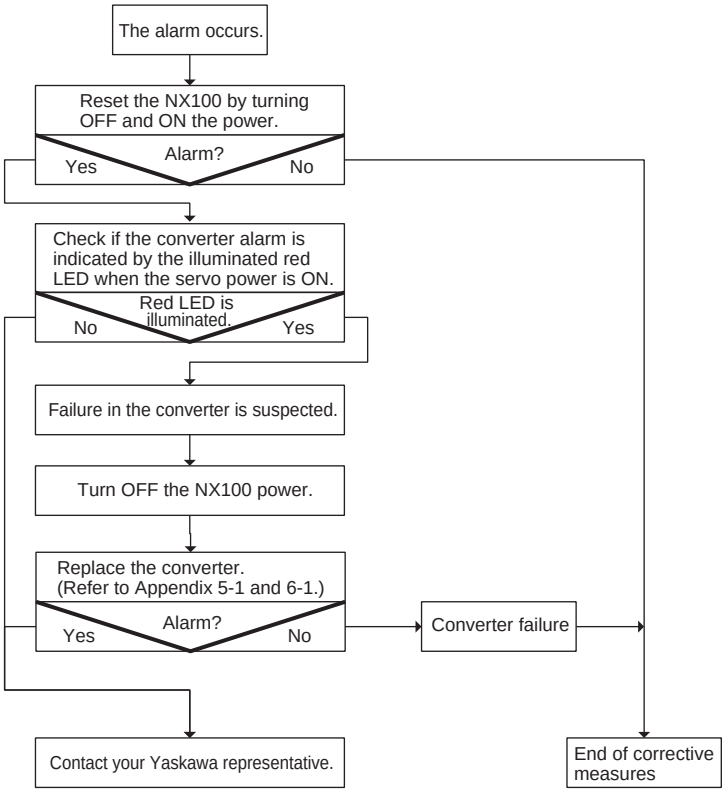
■ AL-1311

A/D DETECTION ERROR (CONVERTER): [Physical Converter Bit]

■ AL-1312

ID ERROR (CONVERTER): [Physical Converter Bit]

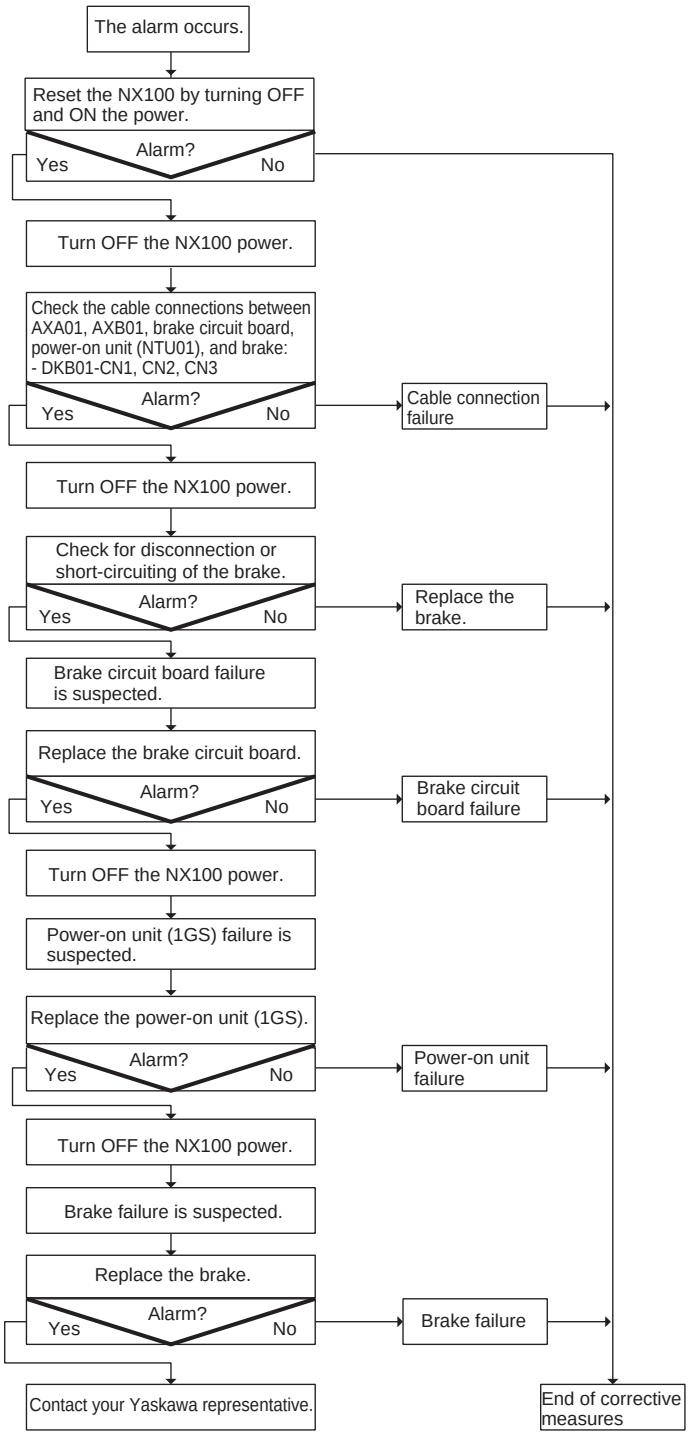
The NX100 system checks internal state of converter in the converter board.
This alarm occurs if any malfunction is found.
As a major cause of this alarm, converter defect is suspected.
Sub Code: Signifies the converter in which the alarm occurred



■ AL-1321

BRAKE BOARD ERROR

This alarm occurs if an error is detected in the brake circuit board.
The alarm cause is attributed to a connection error of cable, or defective brake circuit board/power-on unit (NTU01)/brake.



■ AL-1322

BRAKE BOARD STICKING

This alarm occurs if sticking/melting is detected at the brake circuit board.
Replace the brake relay (CRBRK1, 2).

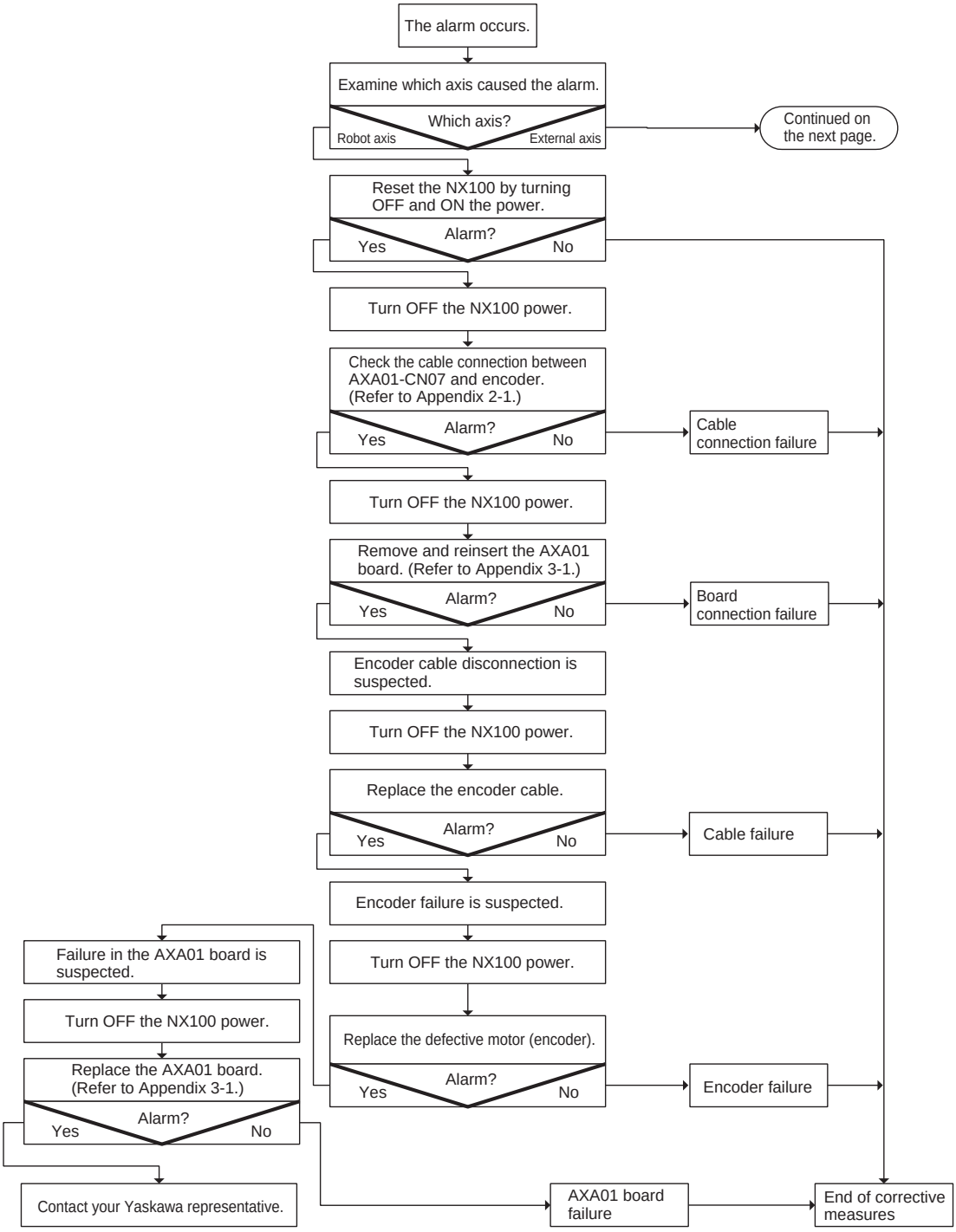
■ AL-1325

COMMUNICATION ERROR (ENCODER)

The NX100 system performs serial communications between controller and encoder. This alarm occurs if the system fails to establish the communications.

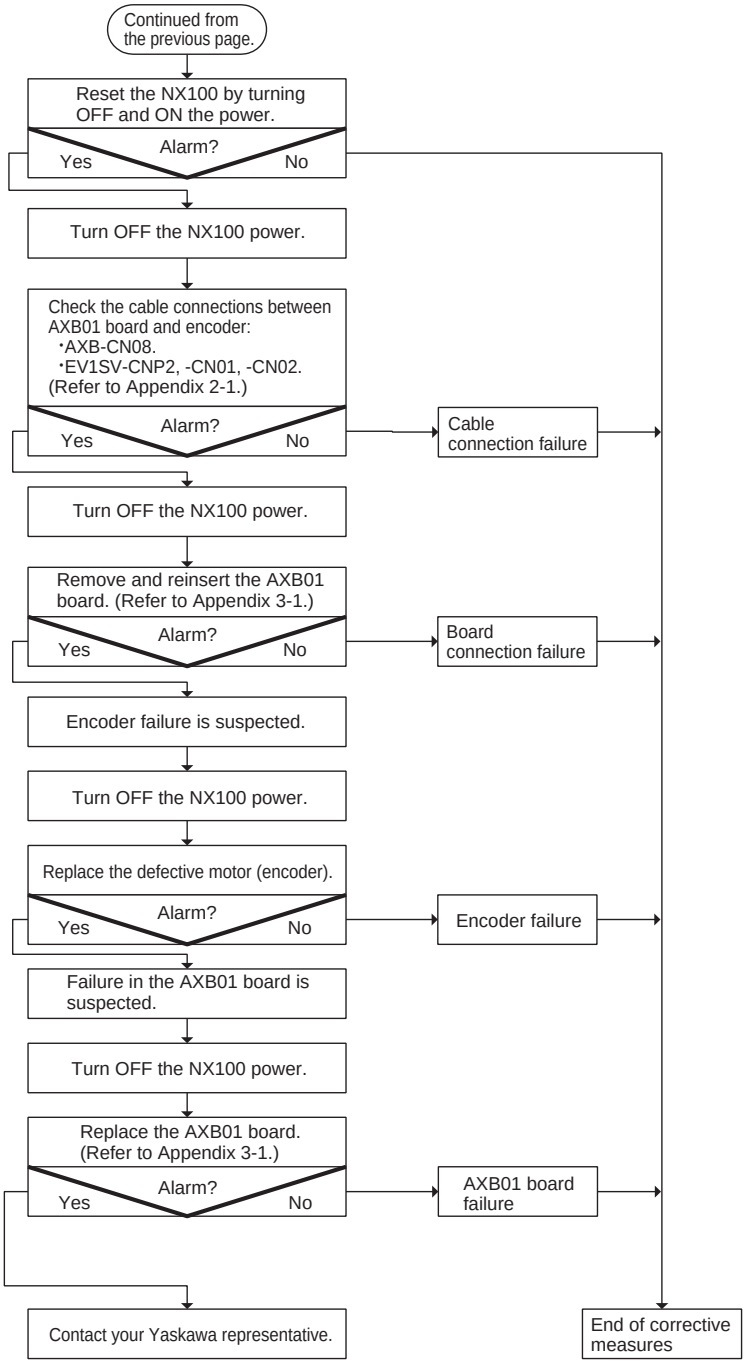
If the alarm occurs during operation, a communications disconnection due to a connection error of encoder cable is suspected. In rare cases, the cause is attributed to a noise interference, encoder malfunction, or defective servo control board(AXA01, AXB01).

Sub Code: Signifies the axis in which the alarm occurred



■ AL-1325

COMMUNICATION ERROR (ENCODER)



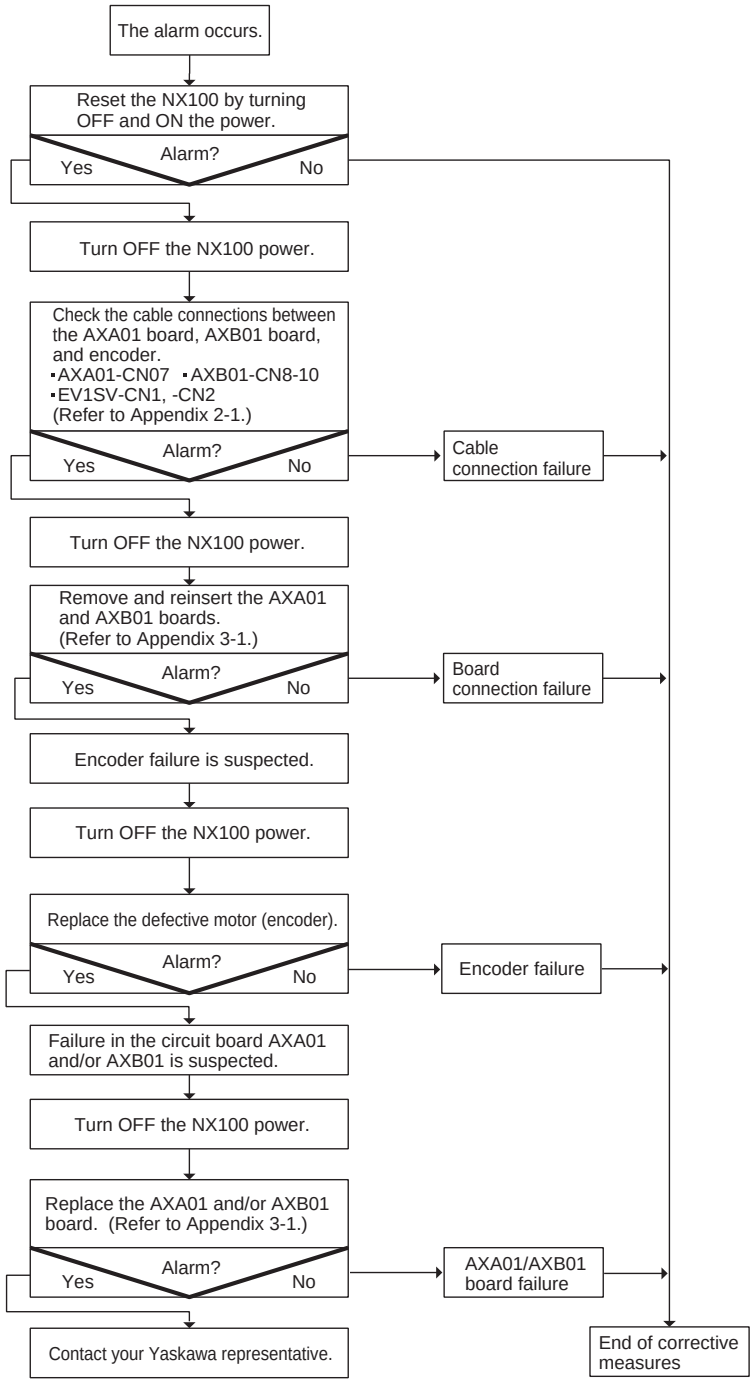
■ AL-1326

DEFECTIVE ENCODER ABSOLUTE DATA

When the controller power is ON, the NX100 system checks the encoder data. This alarm occurs if the encoder data exceeds the set value.

Under normal operating condition, the internal data does not exceed the set value; therefore, if the alarm occurs during operation, encoder error is suspected as an alarm cause. In rare cases, the cause is attributed to a loose connection, servo control board (AXA01, AXB01) defect, or noise interference.

Sub Code: Signifies the axis in which the alarm occurred



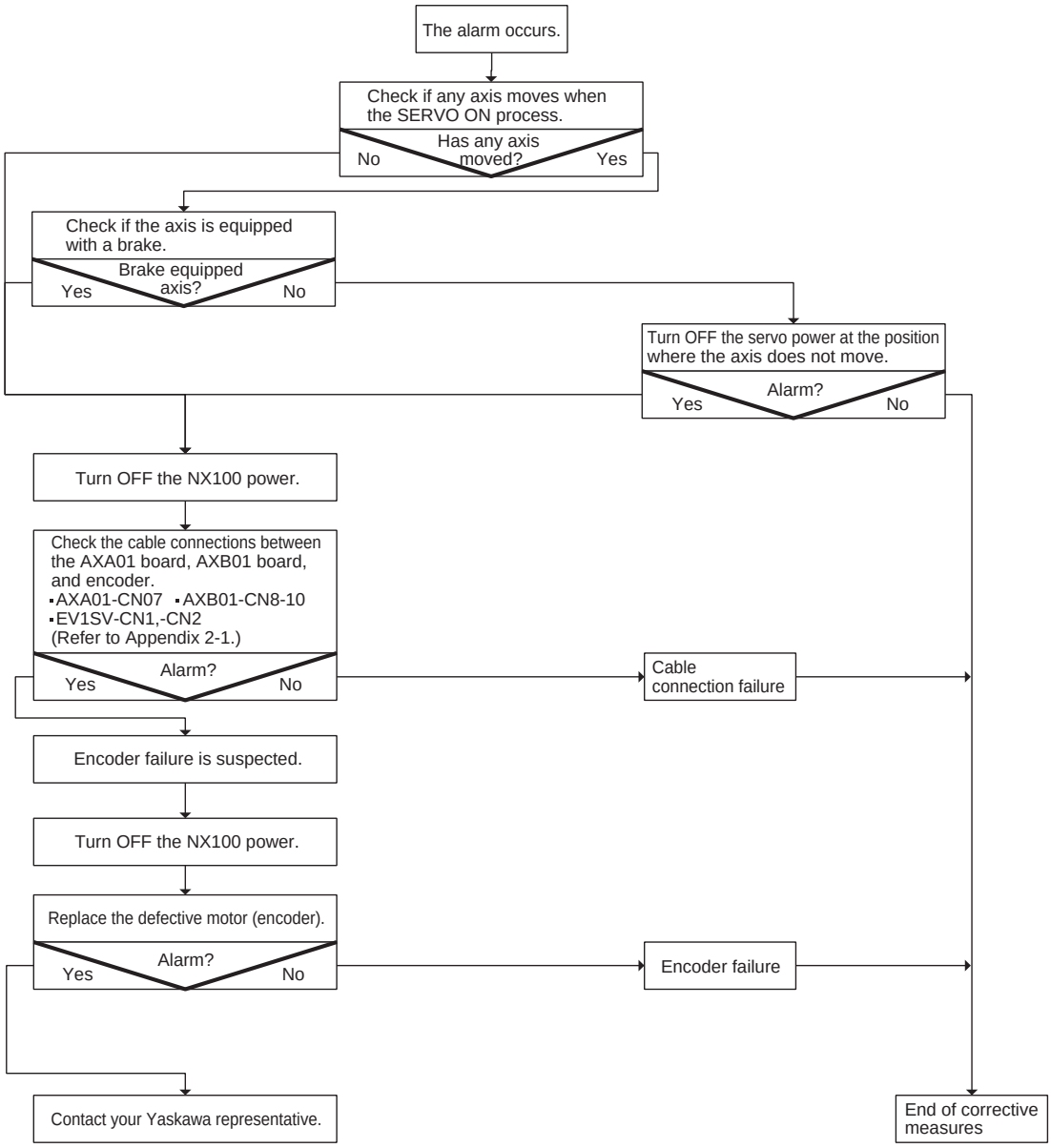
■ AL-1327

ENCODER OVER SPEED

When the controller power is ON, the NX100 system checks the encoder data. This alarm occurs if any axis moves (i.e. falls by its own weight), or an encoder rotation speed of 400 rpm or more is detected during the power-ON process.

A loose connection or encoder error is suspected if the alarm occurs when the manipulator is in halt condition.

Sub Code: Signifies the axis in which the alarm occurred



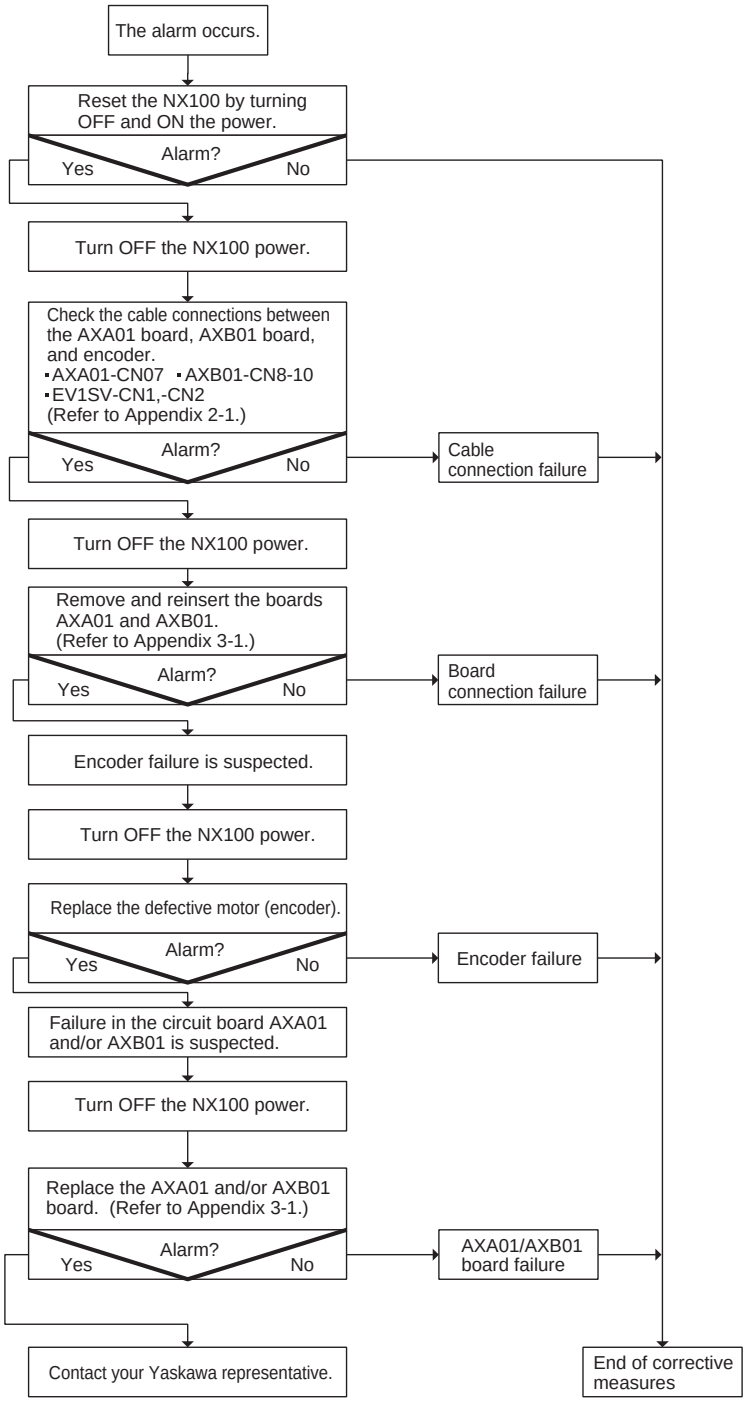
■ AL-1328

DEFECTIVE SERIAL ENCODER

The NX100 system performs serial communications between controller and encoder. This alarm occurs if there is an error in the internal data of the encoder.

If the alarm occurs during operation, a connection error of peripheral cable or a defective encoder is suspected. In rare cases, the alarm cause is attributed to a noise interference or defective servo control board (AXA01, AXB01).

Sub Code: Signifies the axis in which the alarm occurred



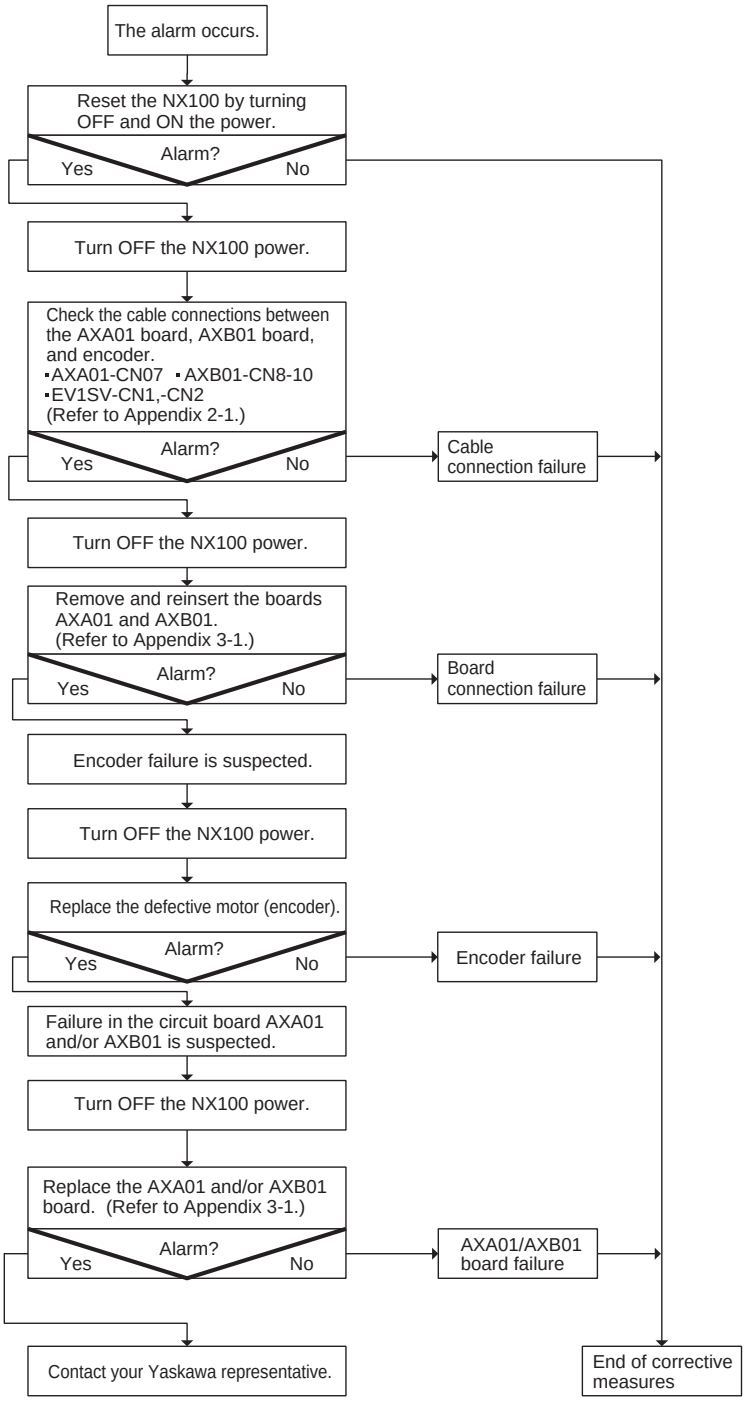
■ AL-1329

CANNOT EXECUTE SERIAL ENCODER COMMAND

The NX100 system performs serial communications between controller and encoder. This alarm occurs if the operation in response to the encoder command is a malfunction.

If the alarm occurs during operation, a connection error of peripheral cable or a defective encoder is suspected. In rare cases, the alarm cause is attributed to a noise interference or defective servo control board (AXA01, AXB01).

Sub Code: Signifies the axis in which the alarm occurred

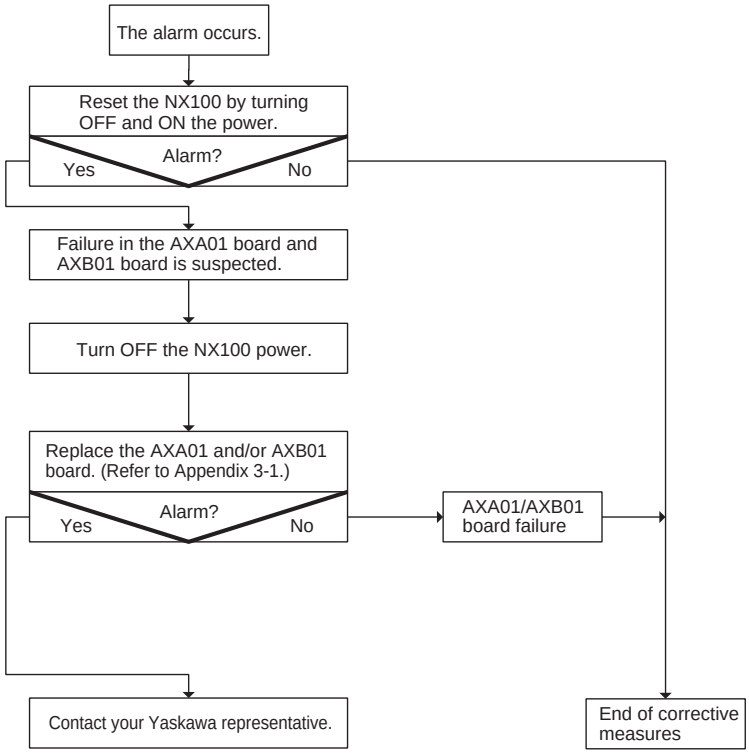


■ AL-1330

MICRO PROGRAM TRANSMIT ERROR

When the NX100 power is ON, the SERVO software transfers a microprogram which controls an electric current to JL056 (the current-limiting LSI). This alarm occurs if any inconsistency is found in verification performed after the transfer. The cause for the alarm is a defective servo control board (AXA01,AXB01).

Sub Code: Signifies the axis in which the alarm occurred



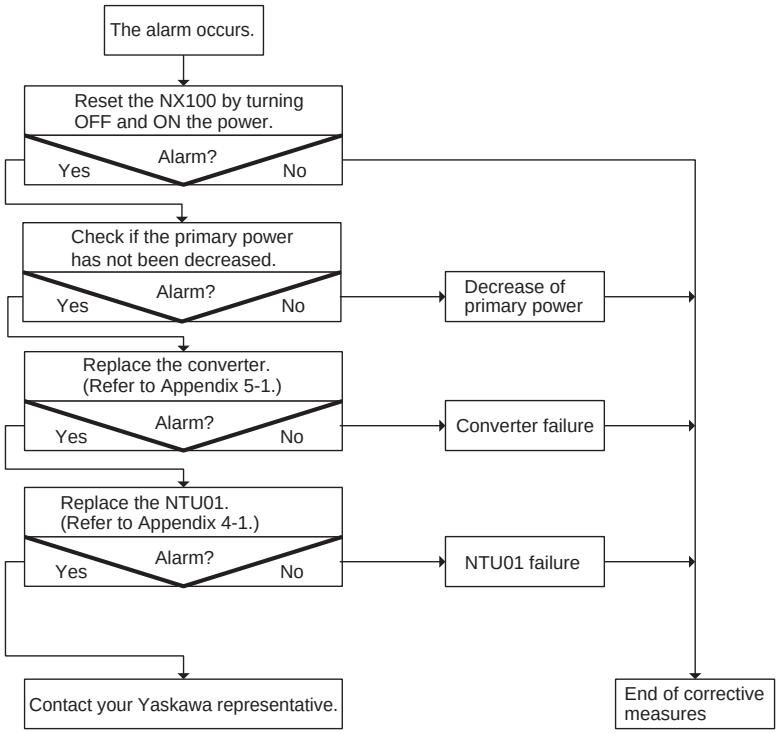
■ AL-1331

CONVERTER CHARGE ERR (CONVERTER)

This alarm occurs if charge error is sent from the converter.

The primary power may have been decreased.

In rare cases, the alarm cause is defective converter or defective power-ON unit (NTU01).

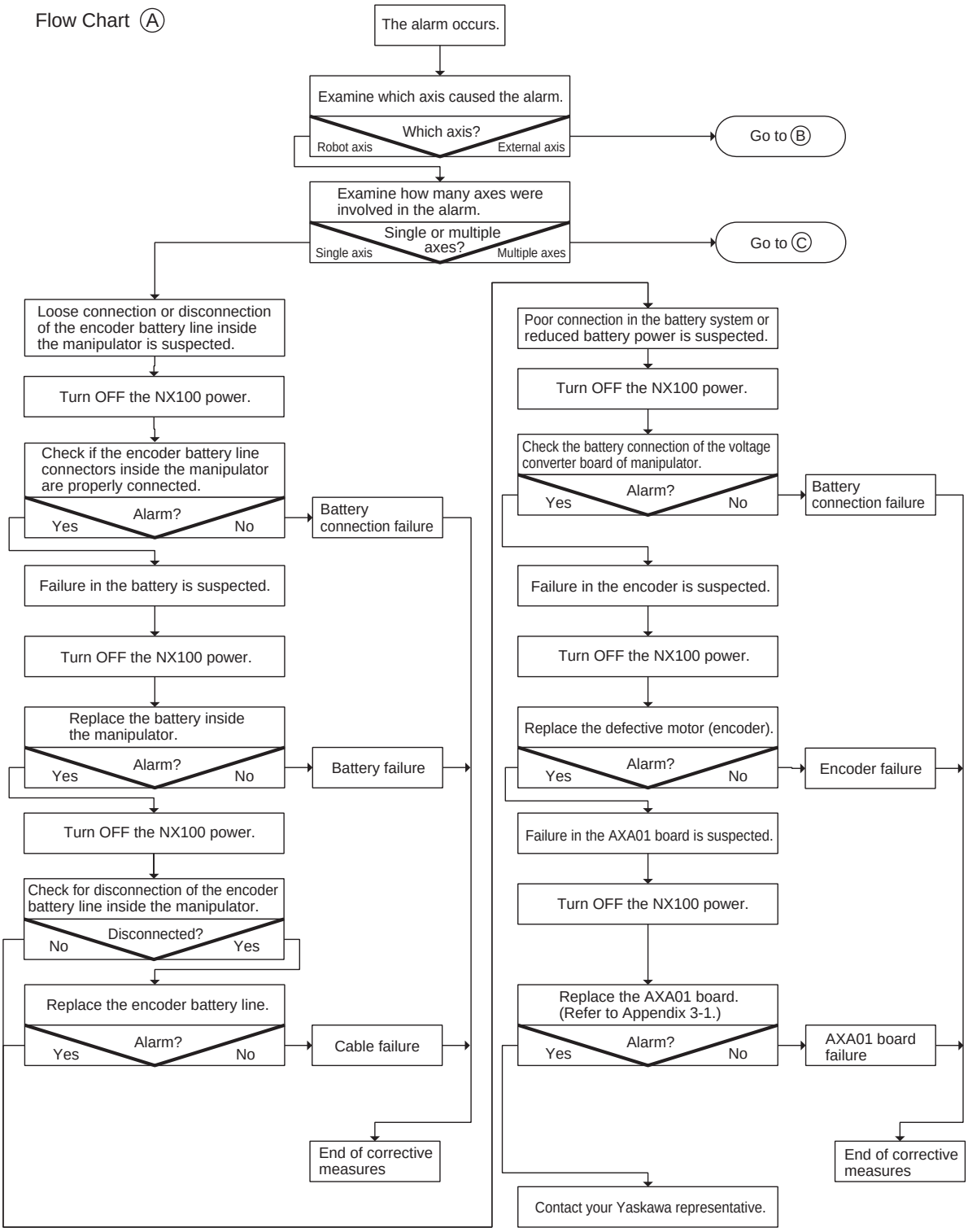


■ AL-1335

INCOMPLETE ENCODER RESET

This alarm occurs when reset operation to recover from the encoder backup error did not complete. As a cause of the alarm, a defective battery line (poor connection, voltage reduction) is suspected.

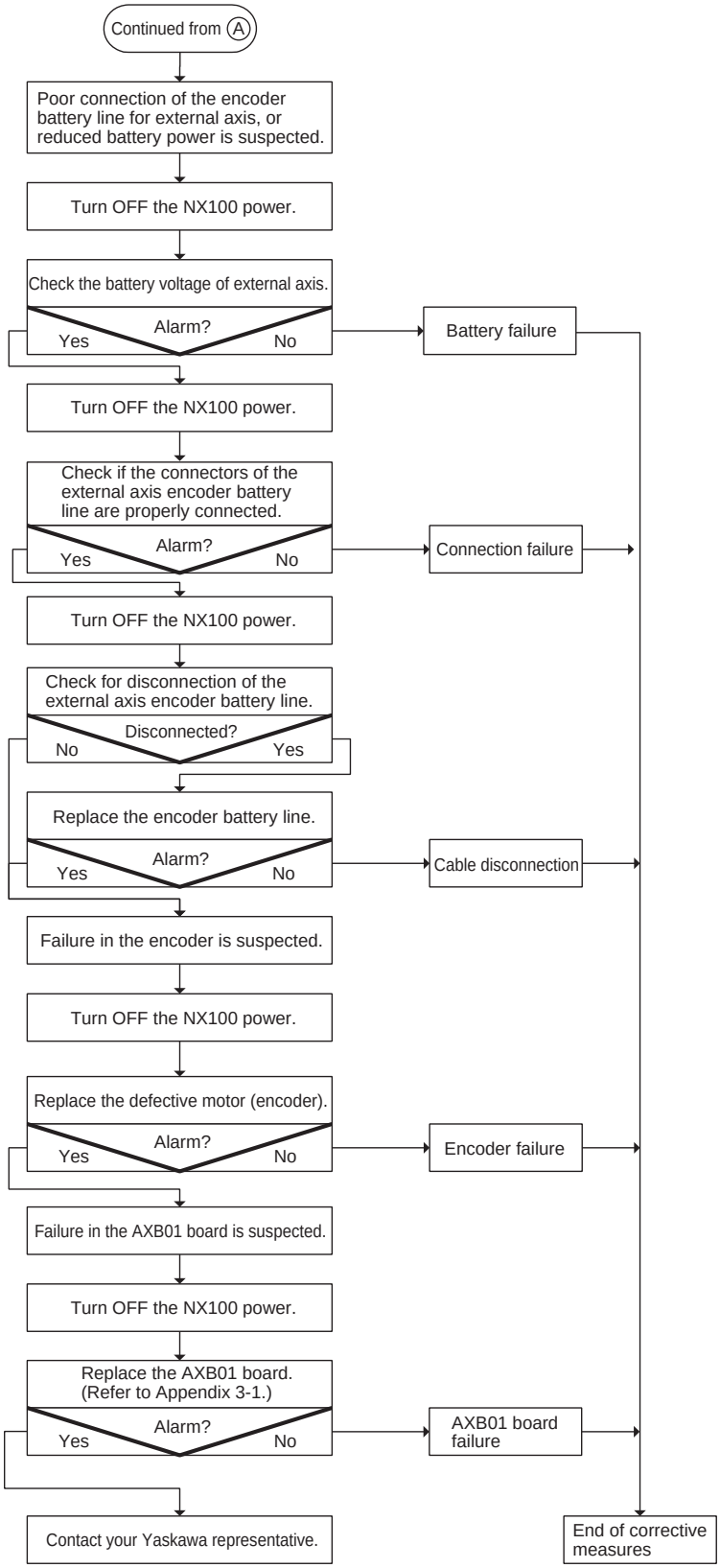
Sub Code: Signifies the axis in which the alarm occurred



■ AL-1335

INCOMPLETE ENCODER RESET

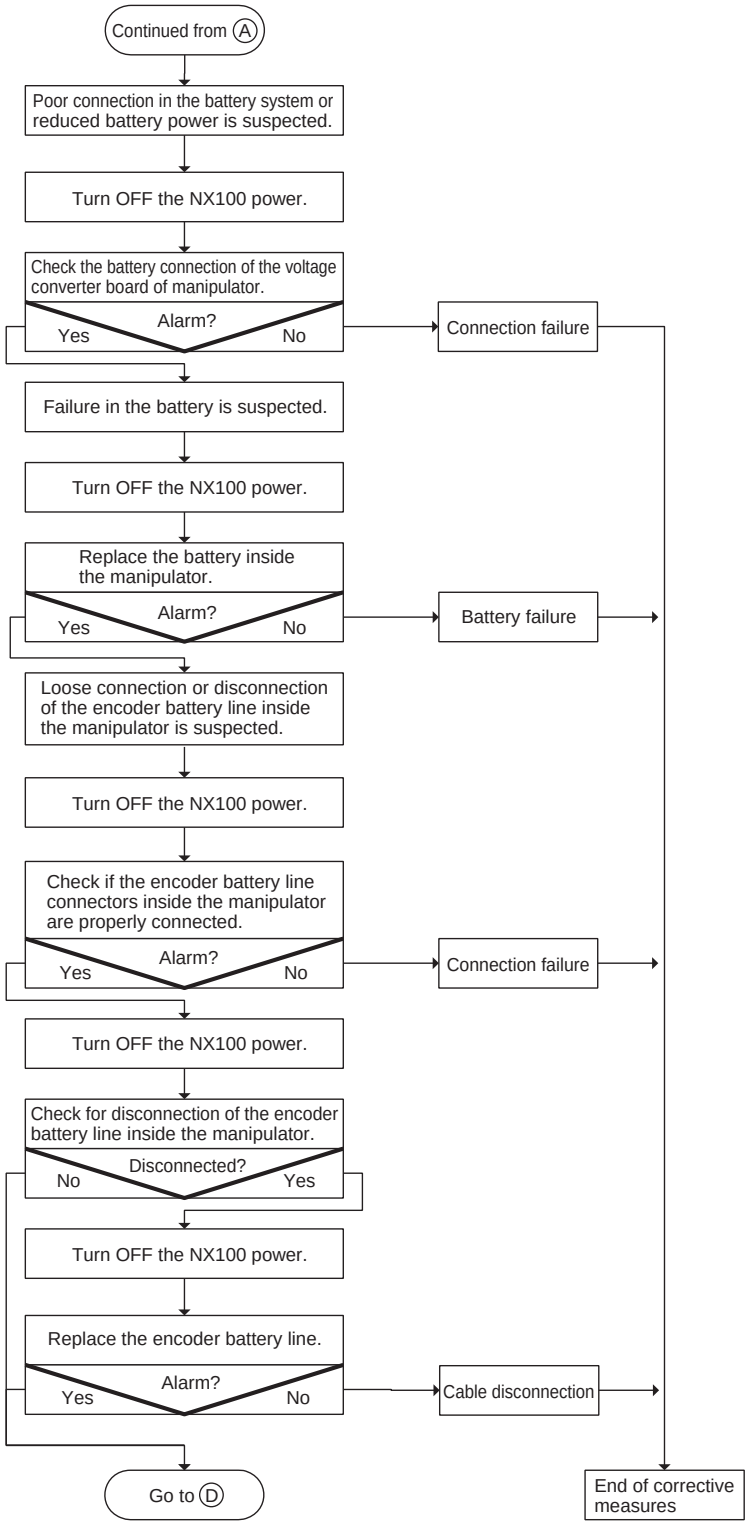
Flow Chart Ⓑ



■ AL-1335

INCOMPLETE ENCODER RESET

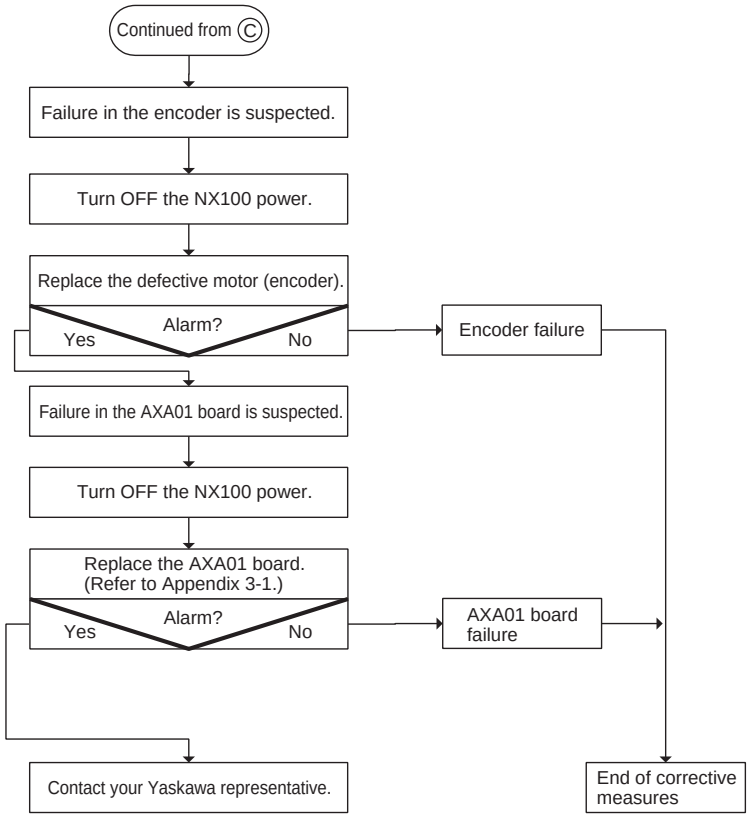
Flow Chart Ⓒ



■ AL-1335

INCOMPLETE ENCODER RESET

Flow Chart Ⓓ



■ AL-1343

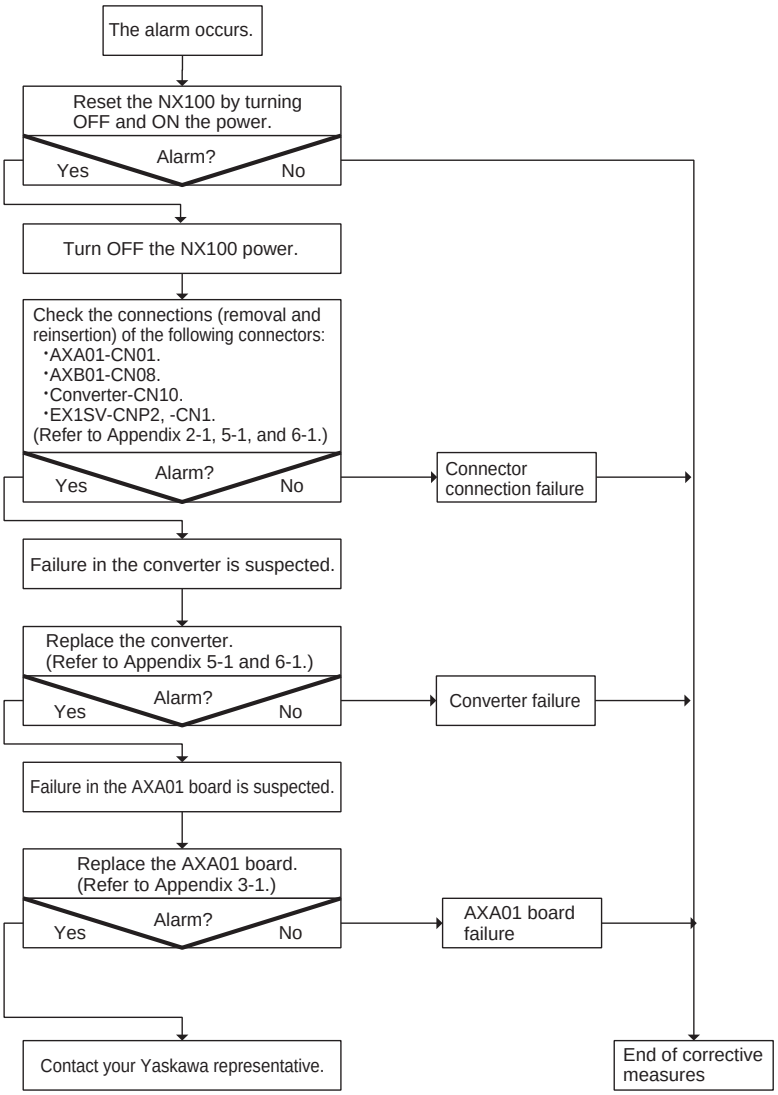
COMMUNICATION ERROR (CONVERTER)

The NX100 system performs serial communications between the servo control board and the converter. This alarm occurs if there is an error in the serial communications.

As a major cause of this alarm, a loose connection or defective converter is suspected. In rare cases, however, the cause is attributed to a noise interference or defective servo control board (AXA01, AXB01).

Sub Code

01: Communication status error (: converter No.)
02: Command timeout (: converter No.)
03: Sent buffer FULL (: converter No.)
04: CRC-16 error (: converter No.)
05: Error code received (: converter No.)
06: Received command error (: converter No.)



■ AL-1345

SAFETY CIRCUIT SIGNAL UNMATCH (SERVO): [XYY]

This alarm occurs if unmatched signal is detected in the double-check of the receive data to the power-ON unit.

Servo NO.: SV#X (Signifies the alarm occurrence servo in the multi-robot system)

X: Servo No. (AXA01 board rotary SW)

1: Servo 1 (AXA01 board rotary SW is "0")
2: Servo 2 (AXA01 board rotary SW is "1")
3: Servo 3 (AXA01 board rotary SW is "2")
4: Servo 4 (AXA01 board rotary SW is "3")

Sub Code: XYY (Signifies the power-ON unit No. and unmatched signal No.)

X: Power-ON unit

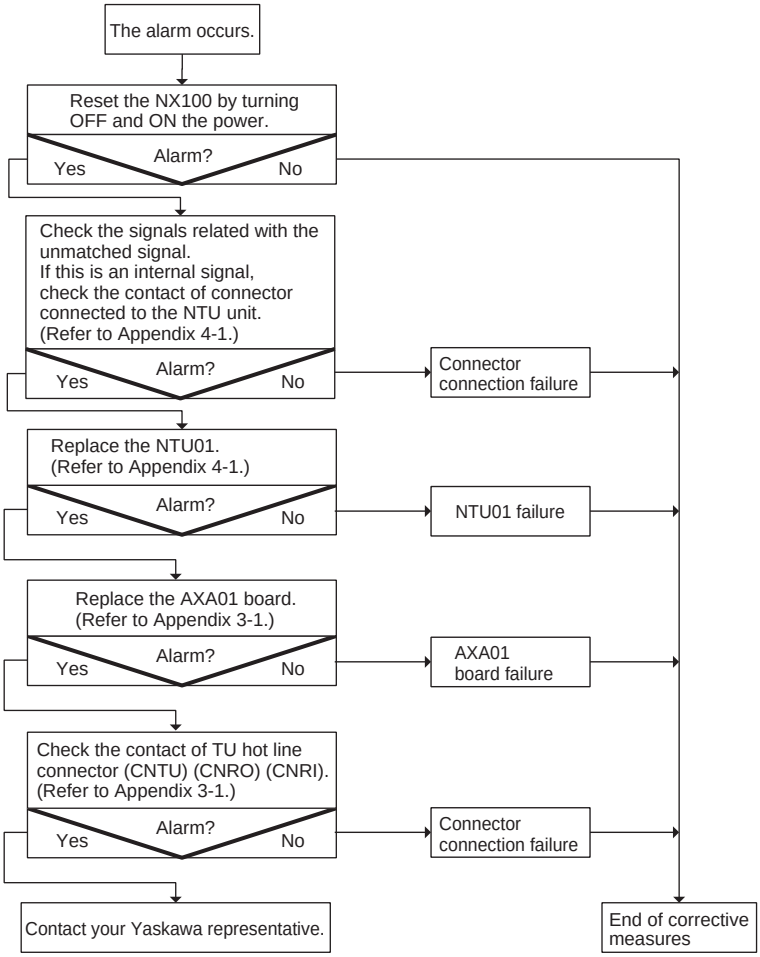
1: Power-ON unit1 (TU#1)
2: Power-ON unit2 (TU#2)
3: Power-ON unit3 (TU#3)
4: Power-ON unit4 (TU#4)
5: Power-ON unit5 (TU#5)
6: Power-ON unit6 (TU#6)

YY: Unmatched signal

01: KMMA signal unmatched error
02: SVMAIN signal unmatched error
03: SVMAIN1/2 signal unmatched error
04: IORDY signal unmatched error
05: ONEN signal unmatched error
06: FUCUT signal unmatched error
07: SHOCK1 signal unmatched error
08: EXOT signal unmatched error
09: OT signal unmatched error
10: TUSONER signal unmatched error
11: SVCMPER signal unmatched error
12: TCER signal unmatched error
13: SON_OUT signal unmatched error
14: BRRVER signal unmatched error

60: Error due to unmatched output signal for servo board failure
61: Error due to unmatched signal for the main contactor state (closed contact)
62: Error due to unmatched signal for the main contactor state (open contact)
63: Error due to unmatched input signal for the main contractor control relay
64: Error due to unmatched input signal for the OT recovery
65: Error due to unmatched input signal for the external WDT
66: Error due to unmatched 1FB input signal: the brake release control signal

The above signals are connected to CN06 and CN07 on the NTU unit.
Other signals are internal signals.
ex.) Servo No.: SV#1, Sub Code: 208
Unmatched EXOT signal error in the power-ON unit 2 (TU#2) connected to servo 1


The alarm occurs.
Reset the NX100 by turning OFF and ON the power.
Alarm?
Yes
No
Check the signals related with the unmatched signal.
If this is an internal signal, check the contact of connector connected to the NTU unit.
(Refer to Appendix 4-1.)
Alarm?
Yes
No
Connector connection failure
Replace the NTU01.
(Refer to Appendix 4-1.)
Alarm?
Yes
No
NTU01 failure
Replace the AXA01 board.
(Refer to Appendix 3-1.)
Alarm?
Yes
No
AXA01 board failure
Check the contact of TU hot line connector (CNTU) (CNRO) (CNRI).
(Refer to Appendix 3-1.)
Alarm?
Yes
No
Connector connection failure
Contact your Yaskawa representative.
End of corrective measures

■ AL-1349

POWER LOST DETECTION (AXA01/02)

The primary power may be insufficient.
Check the voltage of primary power.

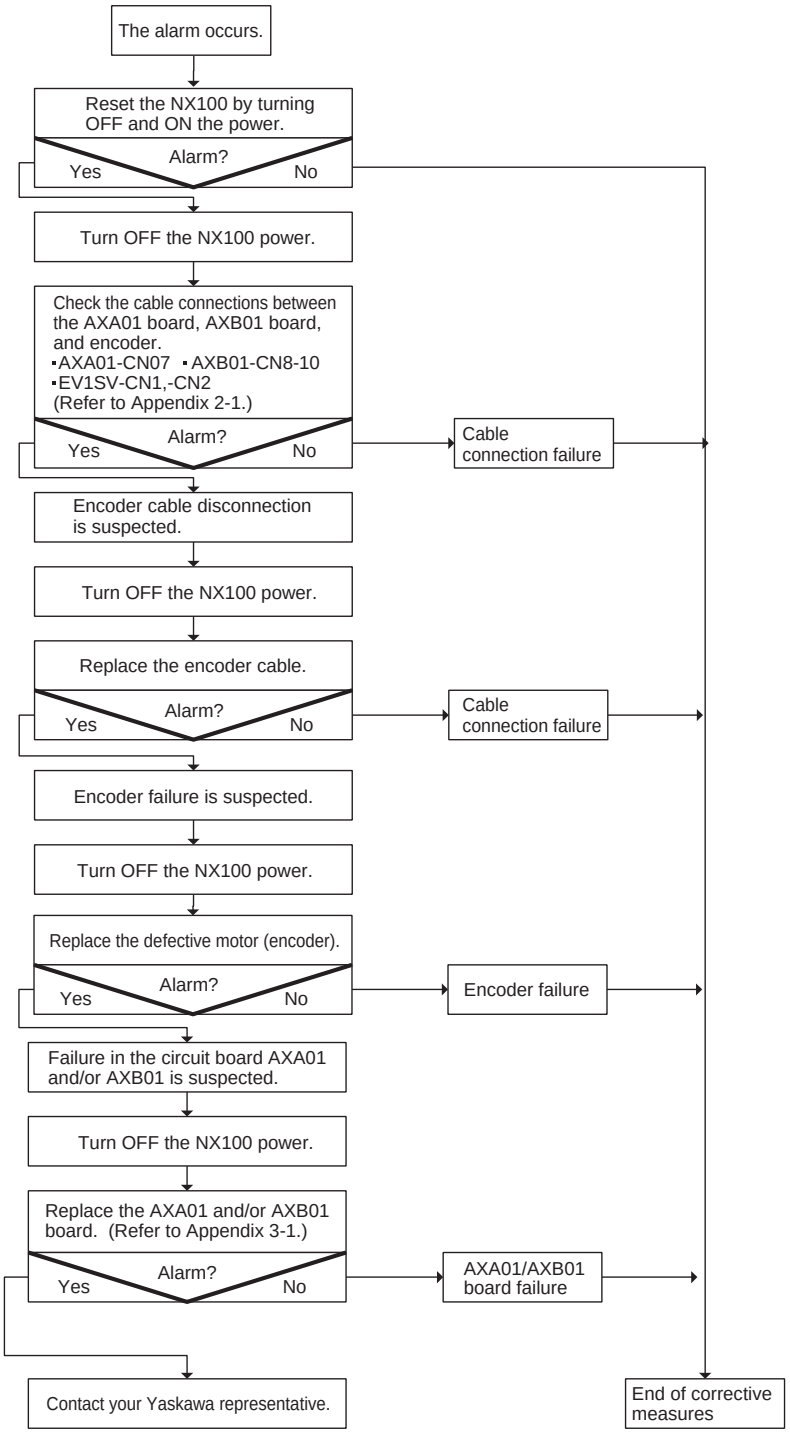
■ AL-1352

SERIAL ENCODER CORRECTION ERROR

The NX100 system controls the manipulator based on the position data from encoder. If a communication error occurs in a control cycle, the system controls the manipulator in accordance with the previous position data checking the compensation data. This alarm occurs if the compensation data is not within the specified value.

If the alarm occurs during operation, defect in the encoder is suspected as a cause. In rare cases, the alarm cause is attributed to a loose connection, defective servo control board (AXA01, AXB01), or noise interference.

Sub Code: Signifies the axis in which the alarm occurred



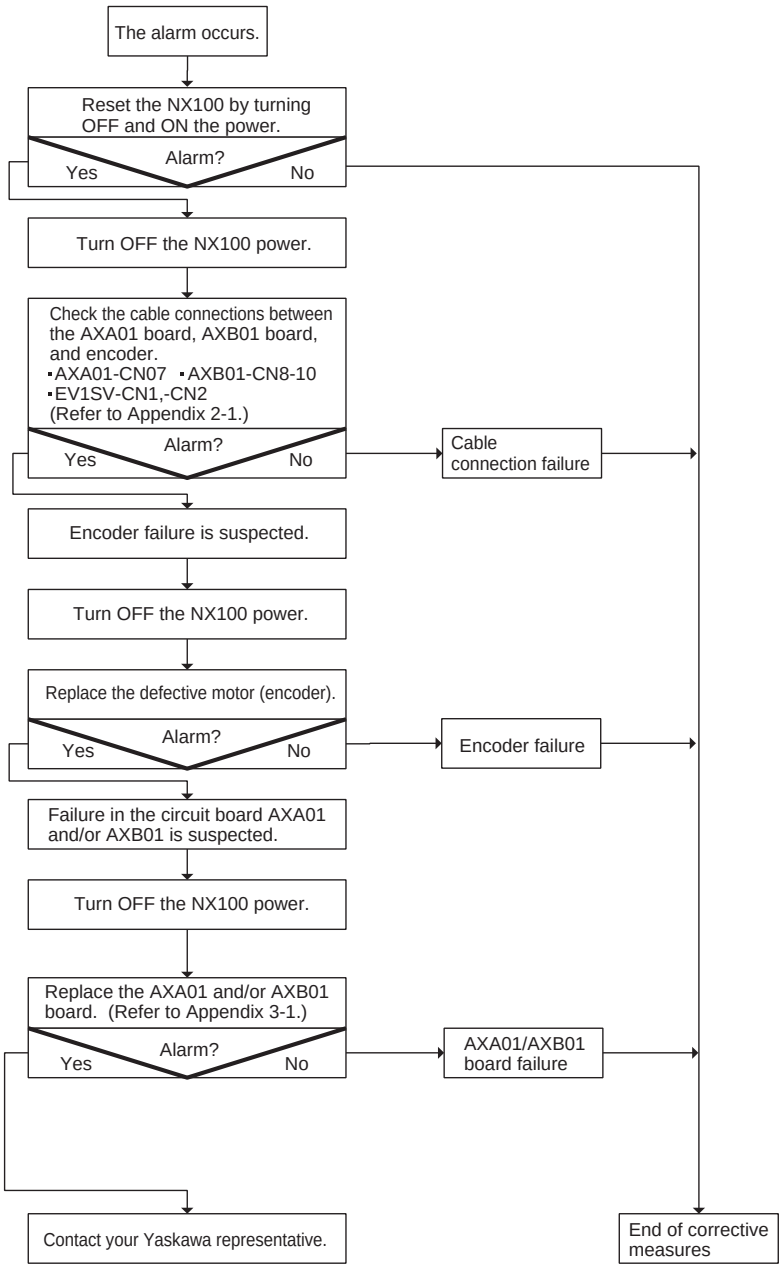
■ AL-1355

SERIAL ENC MULTITURN LIMIT ERR

When the controller power is ON, the NX100 system checks the encoder multi-turn quantity. This alarm occurs if the multi-turn quantity data is not within the normal range.

If the alarm occurs during operation, defect in the encoder is suspected as a cause. In rare cases, the alarm cause is attributed to a loose connection, defective servo control board (AXA01, AXB01), or noise interference.

Sub Code: Signifies the axis in which the alarm occurred



■ AL-1500

SAFETY CIRCUIT FAULT (SERVO I/O)
This alarm occurs if PLD internal mutual monitoring error is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1501

SVMX RELAY STICKING (SERVO I/O)
This alarm occurs if SVMX relay sticking is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1502

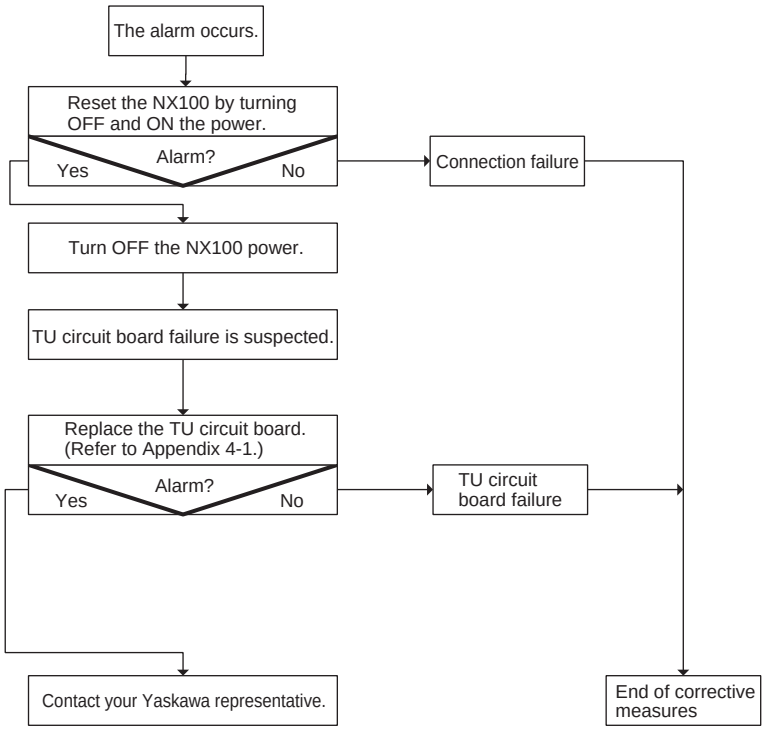
CONTACTOR STICKING (SERVO I/O)
This alarm occurs if contactor sticking is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1503

SAFETY CIRCUIT IN FAULT(SV I/O)

This alarm occurs if input comparison error is sent from the TU circuit board.

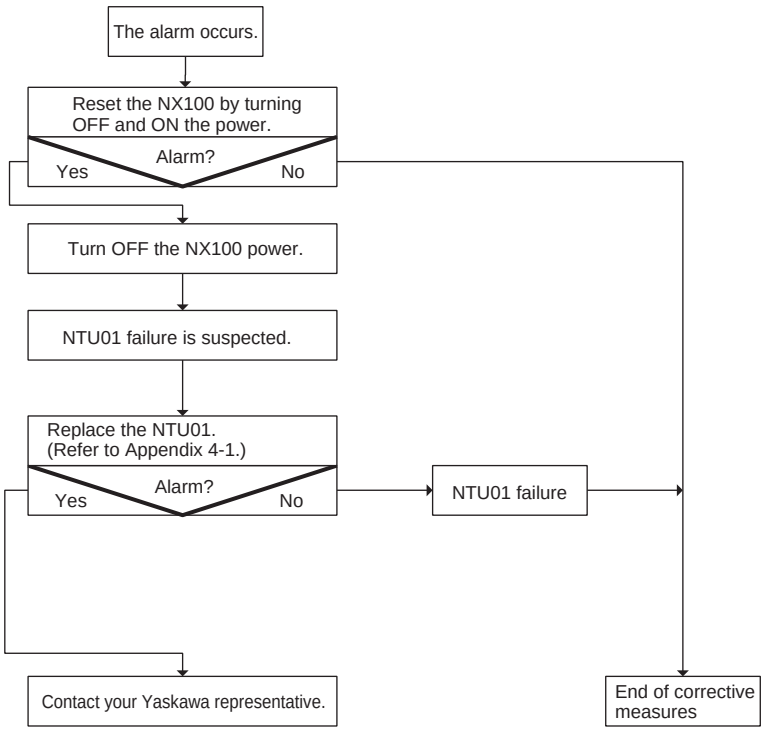
Sub Code: None



■ AL-1504

TUSON RELAY STICKING (SERVO I/O)

This alarm occurs if TUSON relay breakdown is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.



■ AL-1505

BROKEN B_ON FUSE (SERVO I/O)
This alarm occurs if B_ON relay fuse breakdown is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1506

BROKEN CONTACTOR FUSE (SERVO I/O)
This alarm occurs if main contactor relay fuse breakdown is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1507

BROKEN S_ON FUSE (SERVO I/O)
This alarm occurs if S_ON relay fuse breakdown is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1508

SAFETY CIRCUIT WDT ERROR (SV I/O)
This alarm occurs if mutual WDT error is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1509

EXTERNAL WDT OVER (SERVO I/O)
This alarm occurs if external WDT over is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

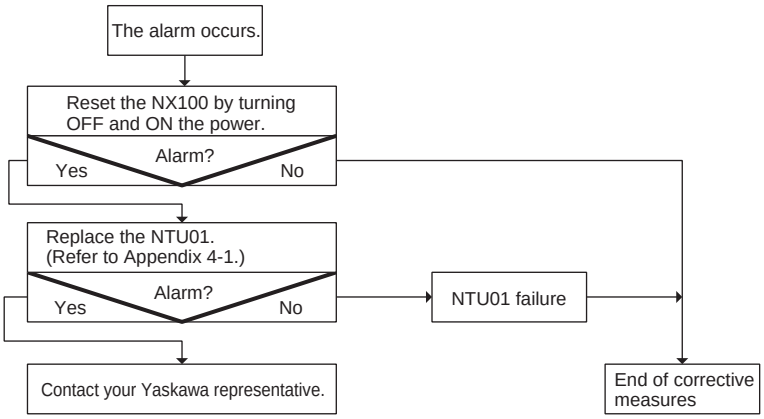
■ AL-1510

EXTERNAL WDT BROKEN (SERVO I/O)
This alarm occurs if external WDT breakdown is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1511

COMMUNICATION ERR(PLDs) (SERVO I/O)

This alarm occurs if serial communication toggle check error is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.



■ AL-1512

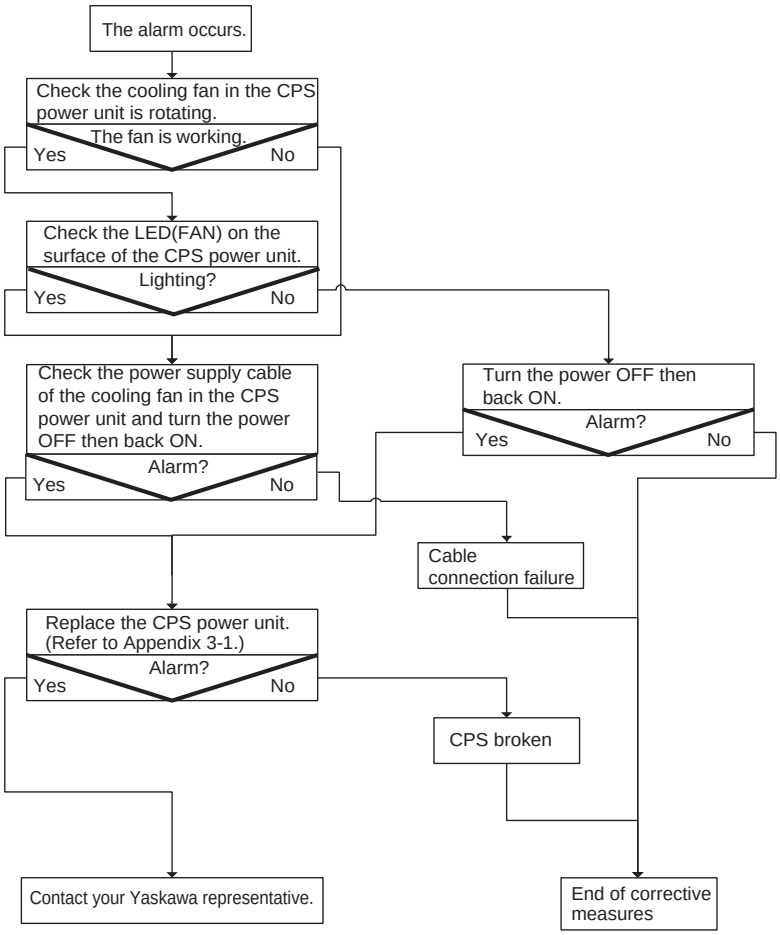
POWER SUPPLY FAN ERROR (SERVO)

This alarm occurs if the rotation speed of in-panel cooling fan decreased.

As a cause of the alarm, a loose connection is suspected.

As a cause of the alarm, the defective CPS power supply unit is suspected.

Sub Code: None

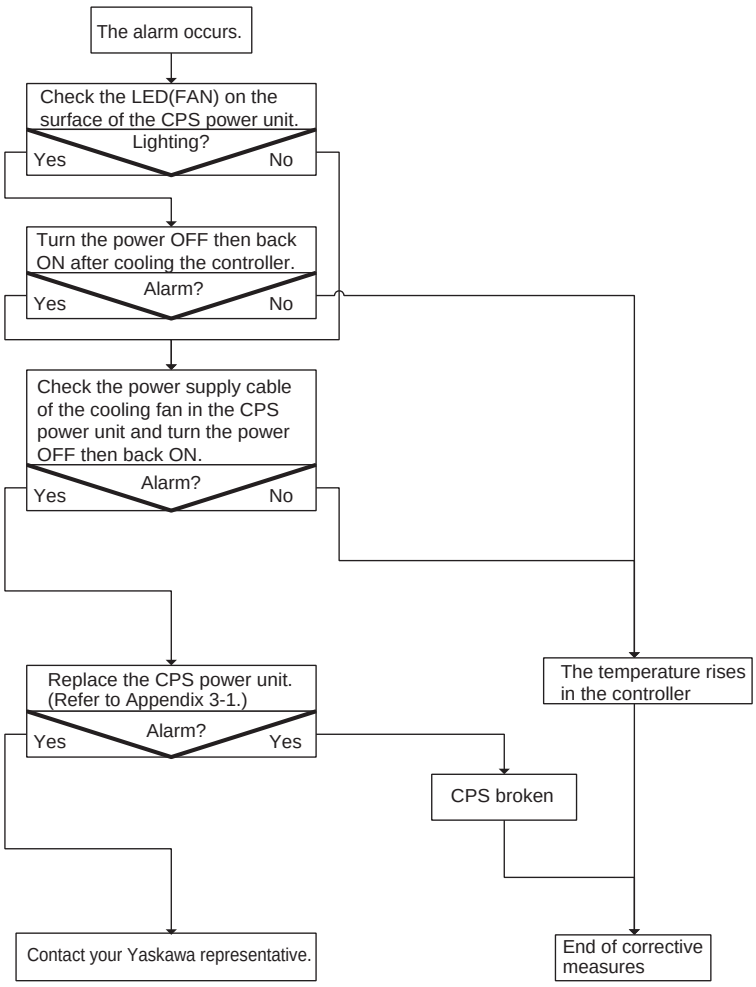


■ AL-1513

POWER SUPPLY OVERHEAT (SERVO)

This alarm occurs if the temperature sensor in the CPS power unit is activated.
It seems that the internal temperature of CPS power unit has risen due to a defective CPS cooling fan or some other reason.

Sub Code: None

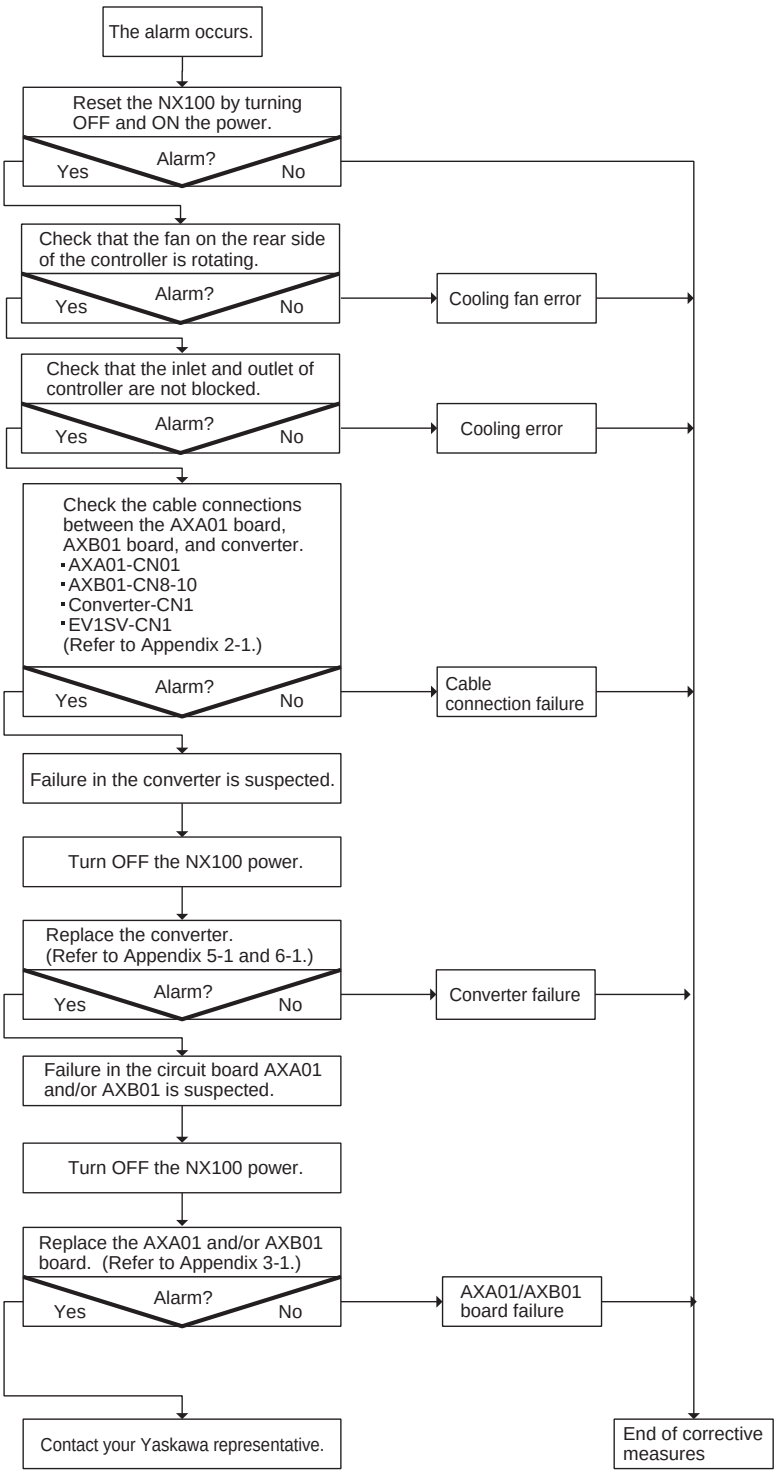


■ AL-1514

OVERHEAT (AMPLIFIER)

This alarm occurs if the temperature sensor in the servo amp is activated.
It seems that the temperature of servo amp has risen due to a defective cooling fan or some other reason.

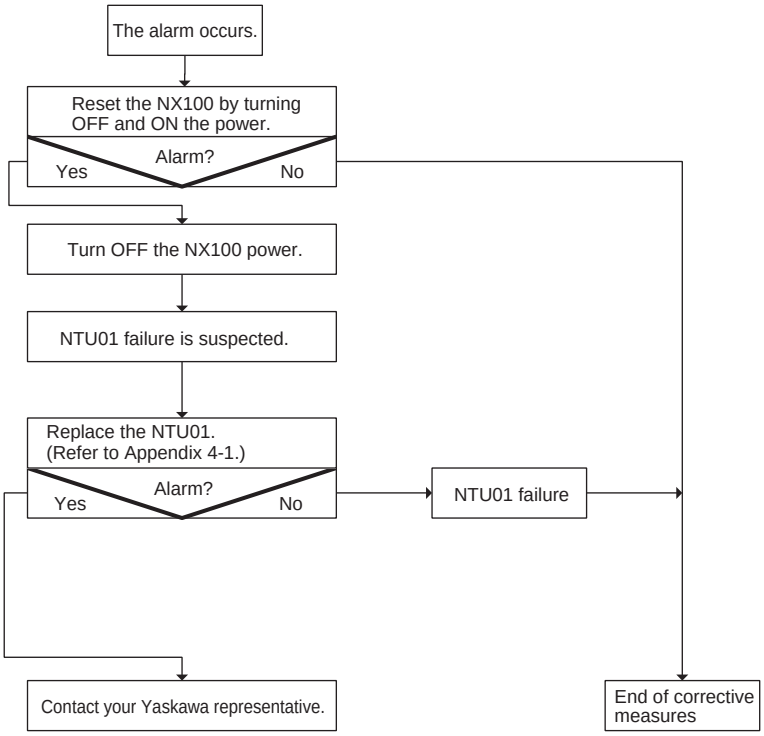
Sub Code: Signifies the axis in which the alarm occurred.



■ AL-1515

SON_OUT RELAY STICKING (SERVO)

This alarm occurs if SON_OUT relay sticking is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.



■ AL-1516

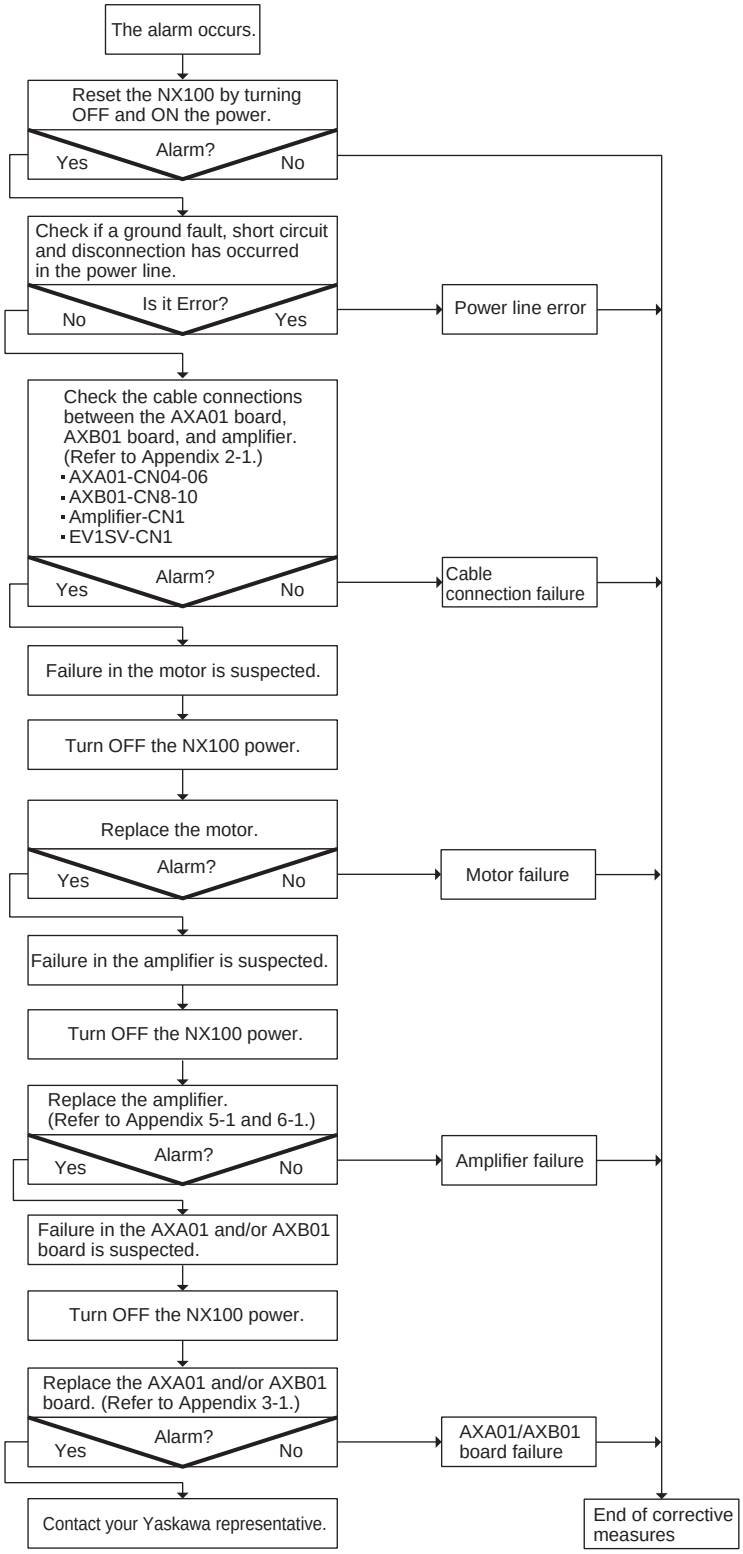
BRRCER RELAY STICKING (SERVO I/O)

This alarm occurs if BRRCER relay sticking is sent from the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1547

CURRENT FEEDBACK ERROR

This error occurs if an excessive current flows through the motor.
A ground fault of motor power line, a poor connector connection, or a defective motor is suspected.
This alarm occurs if a defective amplifier or defective servo control board (AXA01, AXB01).
Sub Code: Signifies the axis in which the alarm occurred.



■ AL-1592

MONITOR PLD ERROR 1 (SERVO I/O)
This alarm occurs if external WDT failure was detected in the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1593

MONITOR PLD ERROR 2 (SERVO I/O)
This alarm occurs if external WDT over was detected in the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

■ AL-1594

MONITOR PLD ERROR 3 (SERVO I/O)
This alarm occurs if mutual WDT error was detected in the power-ON unit (NTU01).
The alarm cause is attributed to a defective power-ON unit (NTU01).
(1) Reset the alarm.
(2) If the alarm occurs again, replace the power-ON unit (NTU01).
Sub Code: Signifies the power-ON unit (NTU01) number in which the alarm occurred.

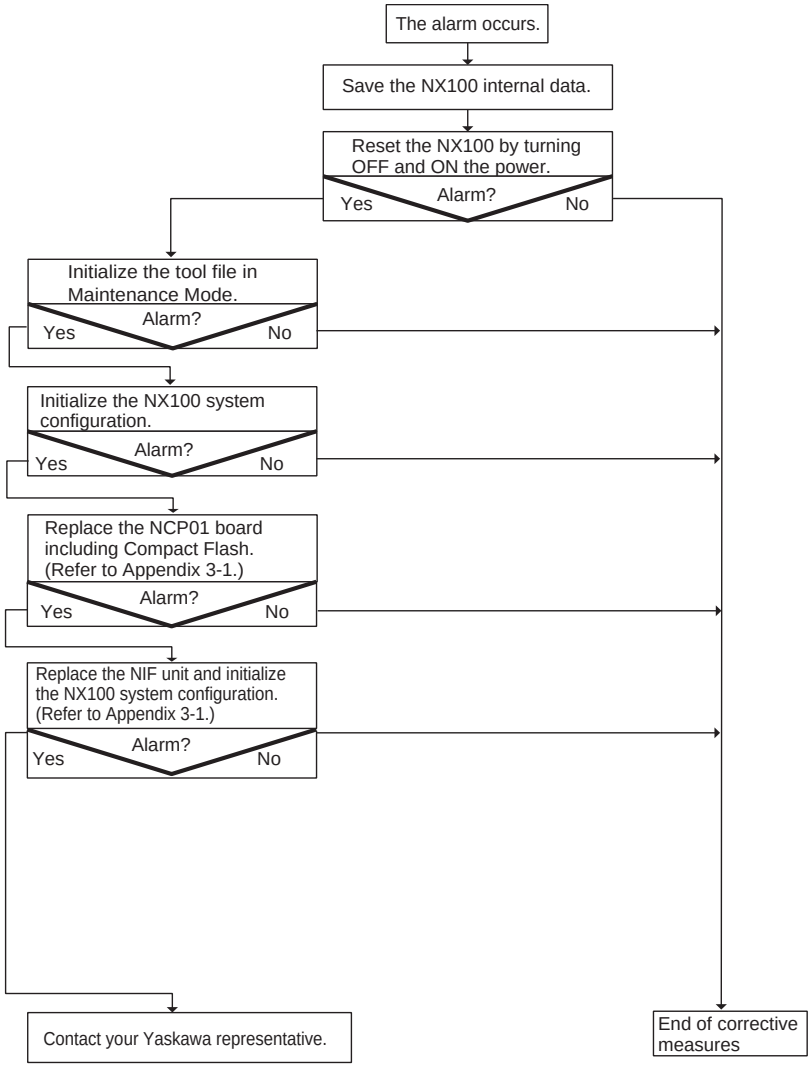
■ AL-4000

MEMORY ERROR (TOOL FILE)

An error was detected at memory check.

The memory for the tool file is damaged.

Sub Code: Signifies the tool number in which the alarm occurred.



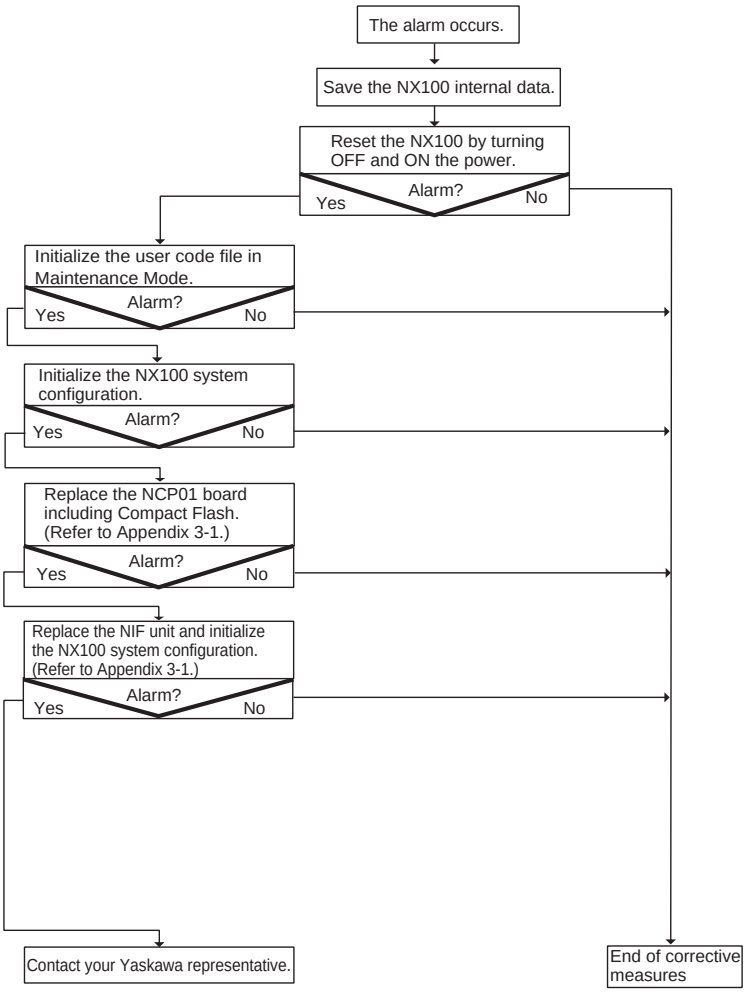
■ AL-4001

MEMORY ERROR (USER COORD FILE)

An error was detected at memory check.

The memory for the user code file is damaged.

Sub Code: Signifies the user coordinate number in which the alarm occurred.



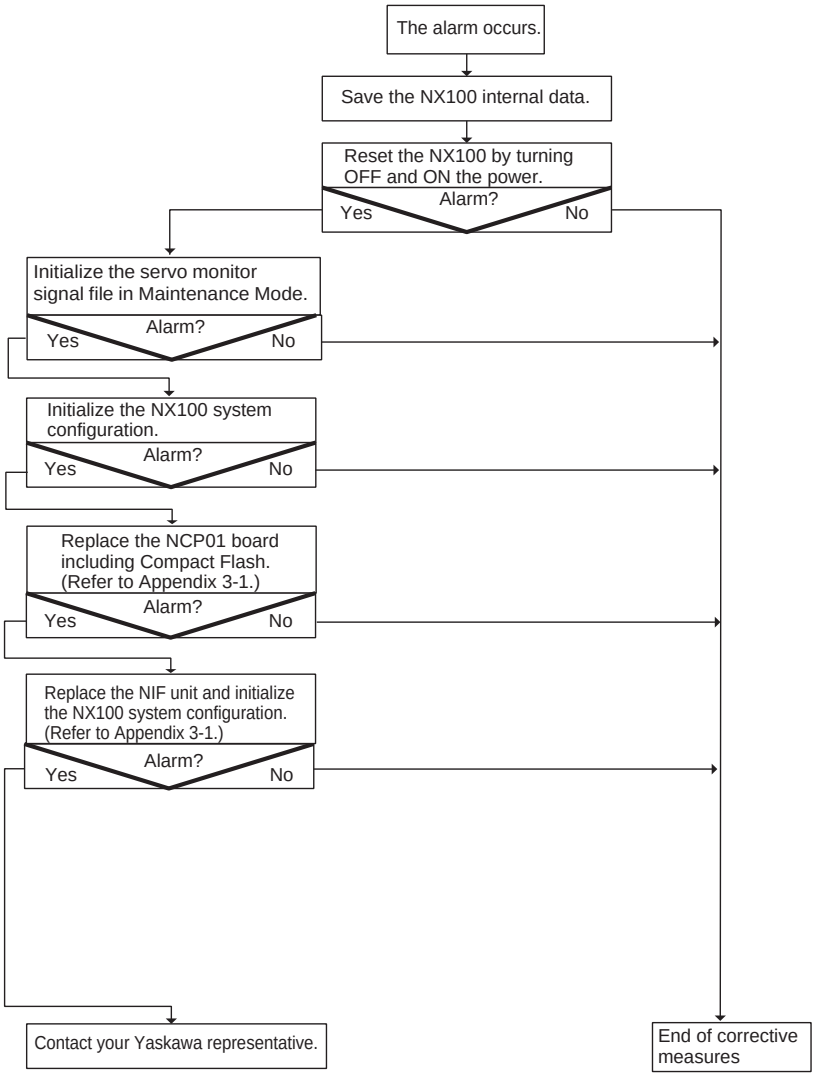
■ AL-4002

MEMORY ERROR (SV MON SIGNAL FILE)

An error was detected at memory check.

The memory for the servo monitor signal file is damaged.

Sub Code: None



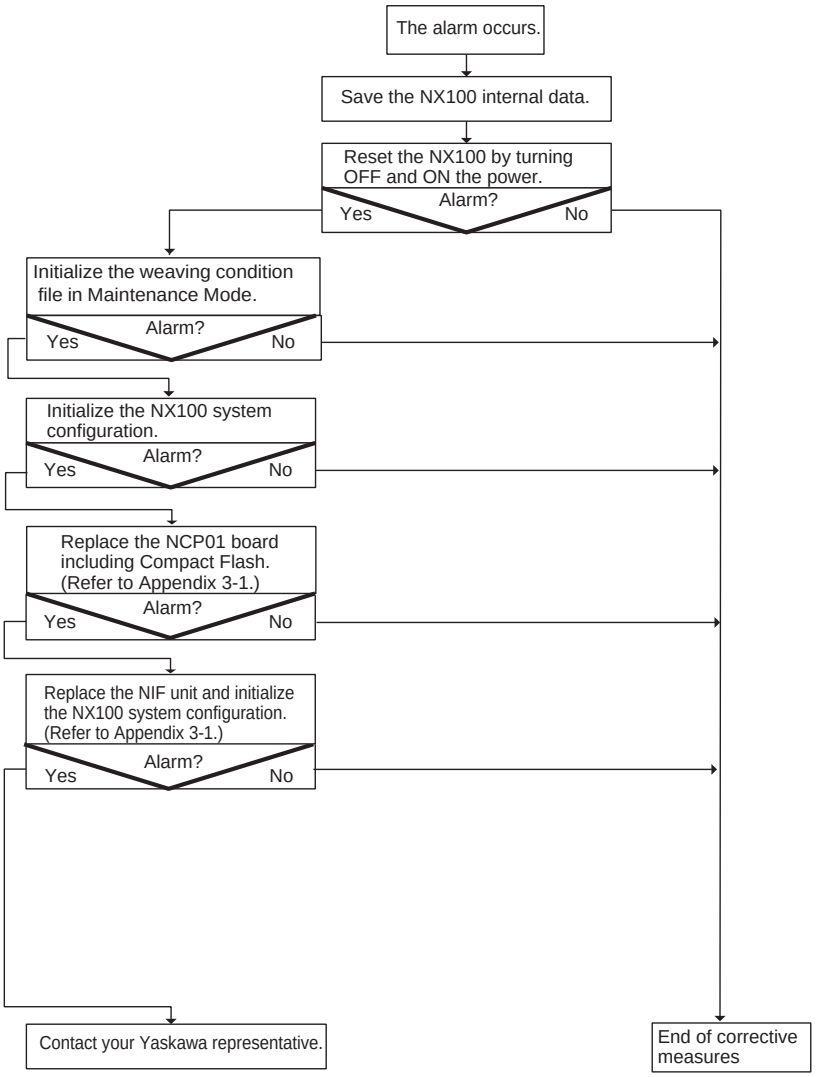
■ AL-4003

MEMORY ERROR (WEAVING FILE)

An error was detected at memory check.

The memory for the weaving condition file is damaged.

Sub Code: Signifies the page number in which the alarm occurred.



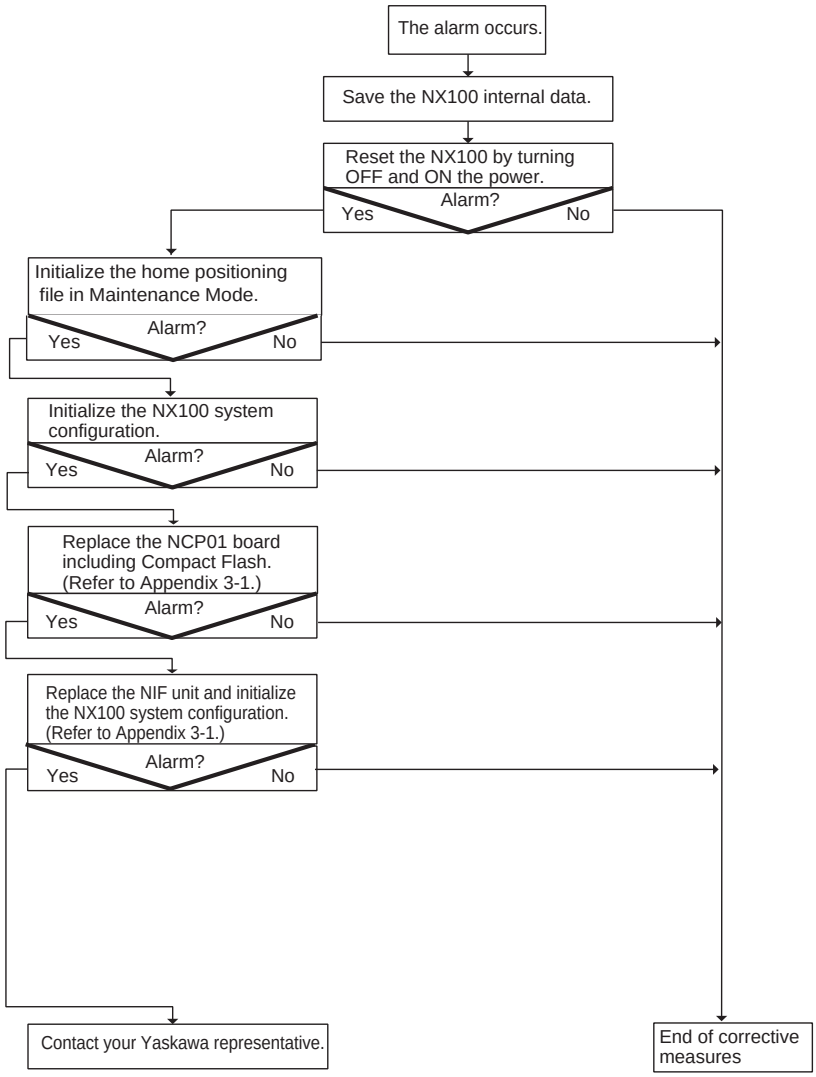
■ AL-4004

MEMORY ERROR (HOME POS FILE)

An error was detected at memory check.

The memory for the home positioning file is damaged.

Sub Code: None



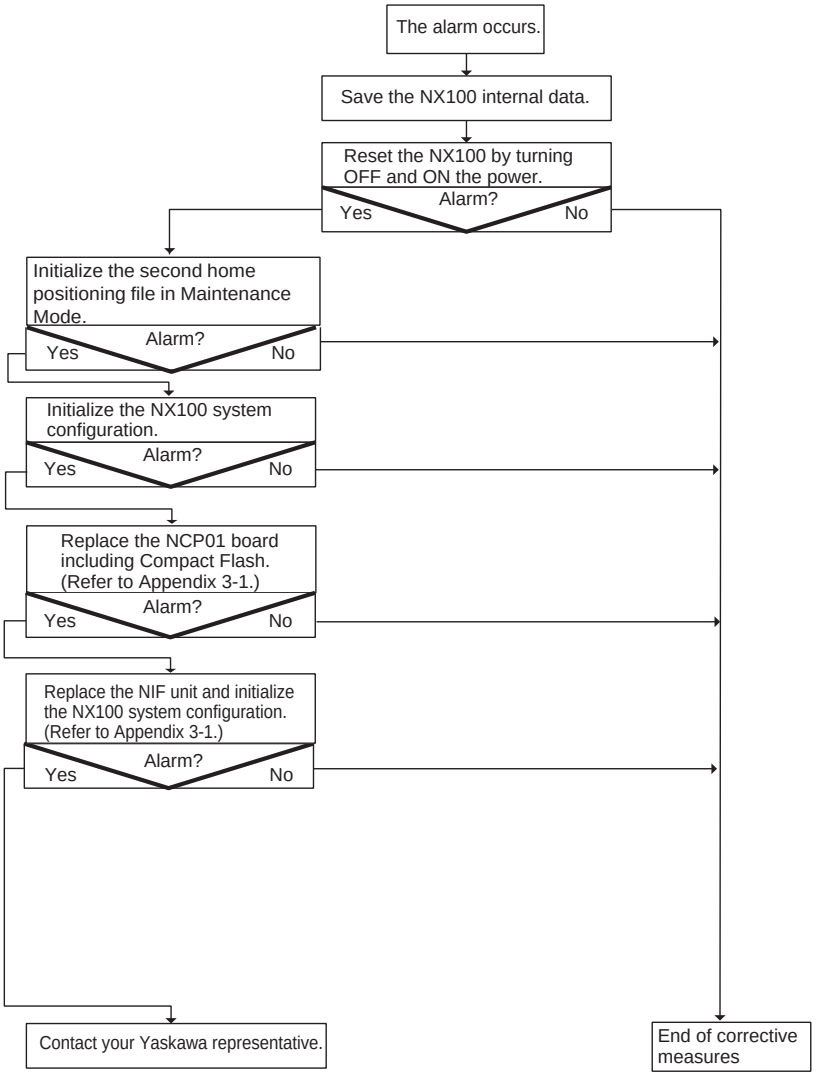
■ AL-4005

MEMORY ERROR (SECOND HOME POS FILE)

An error was detected at memory check.

The memory for the second home positioning file is damaged.

Sub Code: None



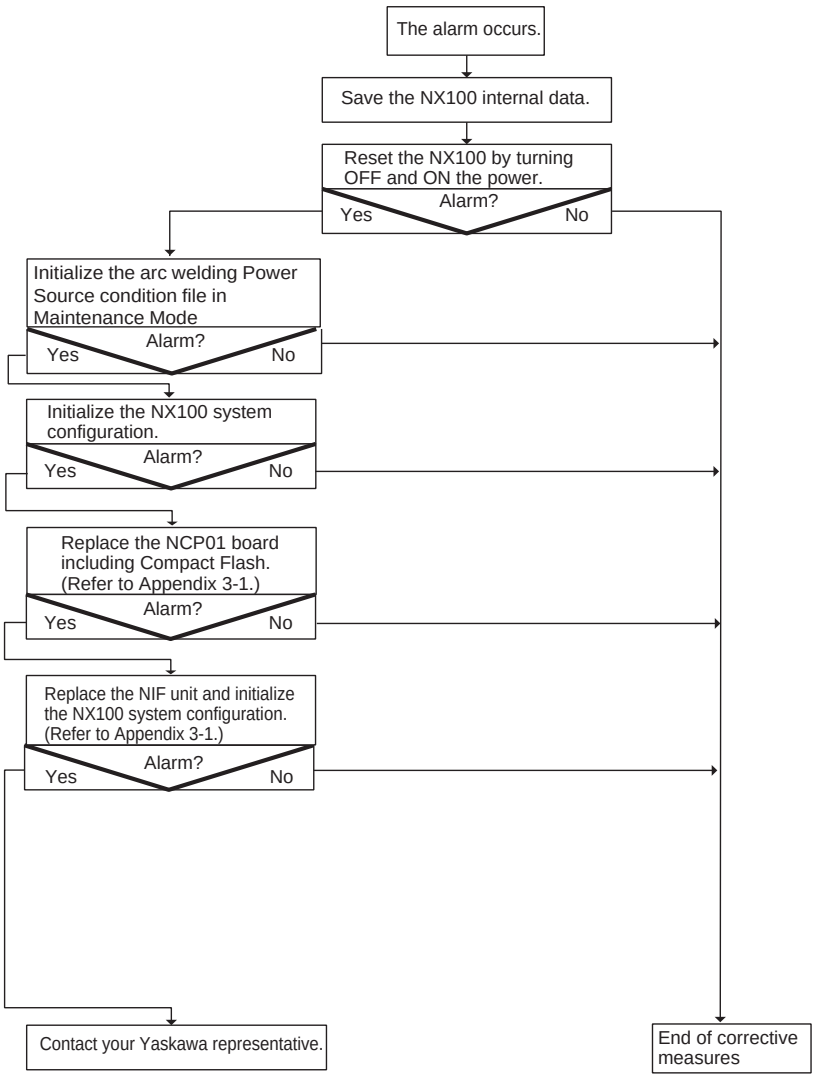
■ AL-4006

MEMORY ERROR (POWER SOURCE COND)

An error was detected at memory check.

The memory for the arc welding Power Source condition file is damaged.

Sub Code: Signifies the page number in which the alarm occurred.



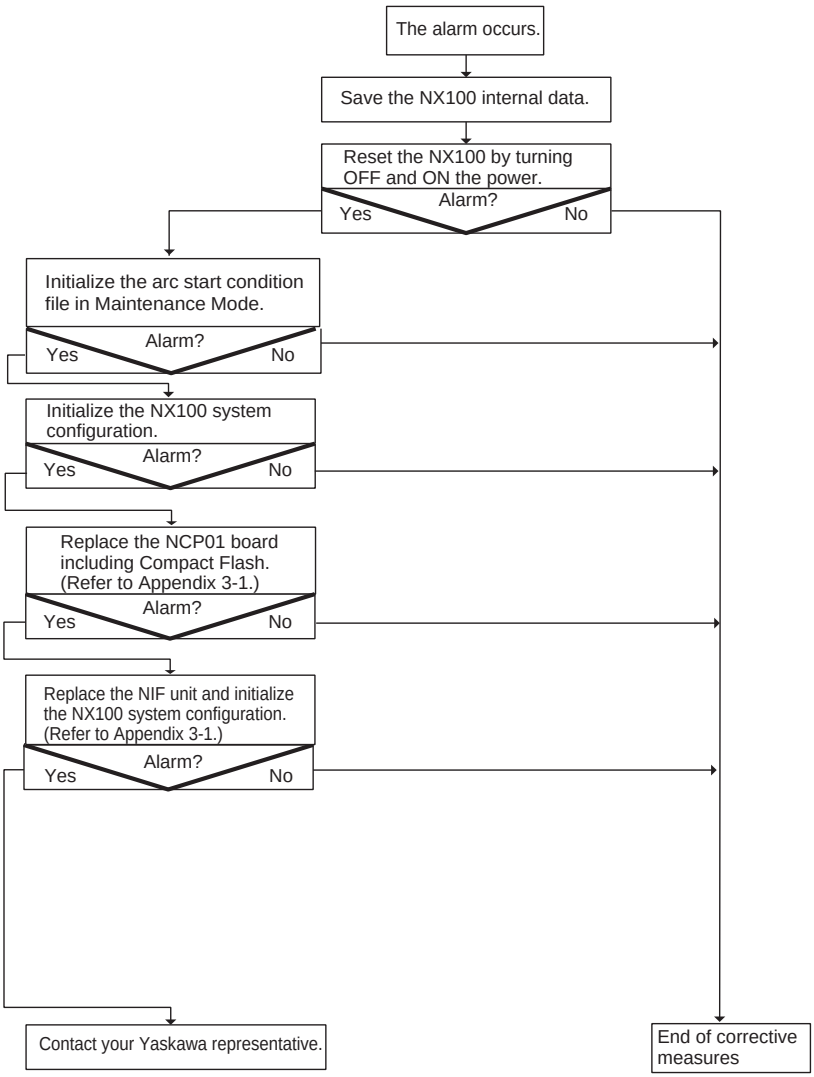
■ AL-4007

MEMORY ERROR (ARC START COND FILE)

An error was detected at memory check.

The memory for the arc start condition file is damaged.

Sub Code: Signifies the page number in which the alarm occurred.



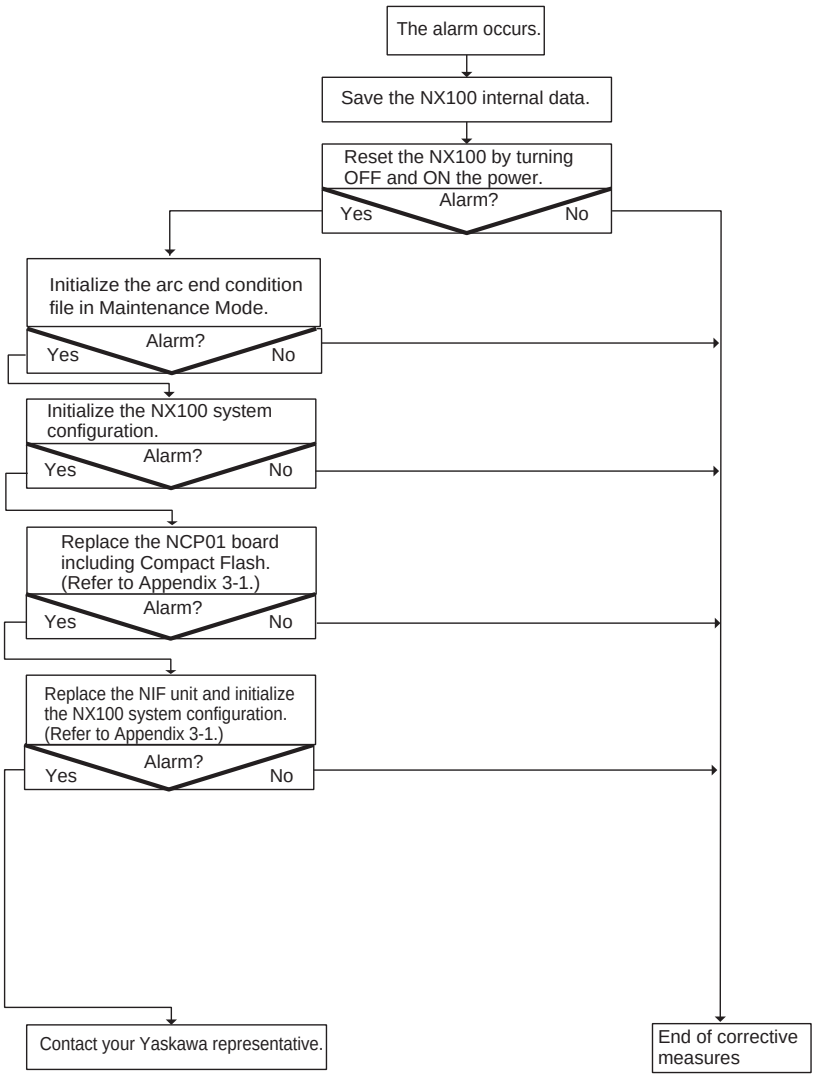
■ AL-4008

MEMORY ERROR (ARC END COND FILE)

An error was detected at memory check.

The memory for the arc end condition file is damaged.

Sub Code: Signifies the page number in which the alarm occurred.



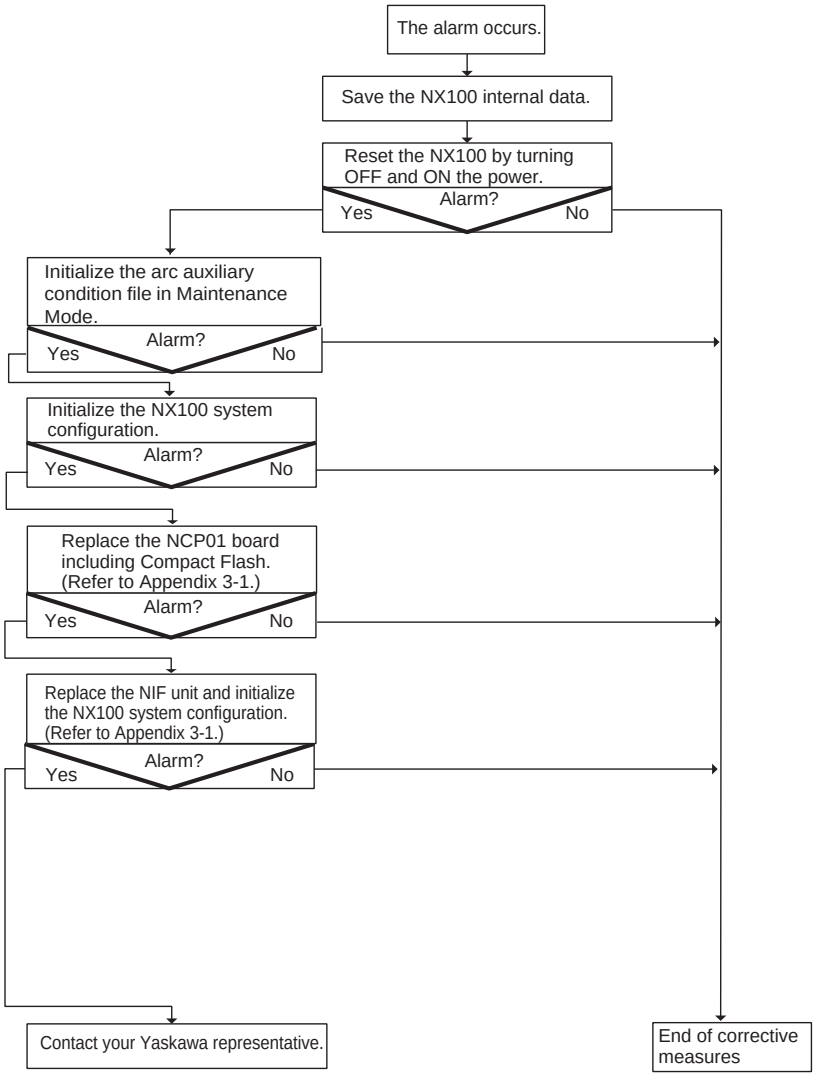
■ AL-4009

MEMORY ERROR (ARC AUX COND FILE)

An error was detected at memory check.

The memory for the arc auxiliary condition file is damaged.

Sub Code: Signifies the page number in which the alarm occurred.



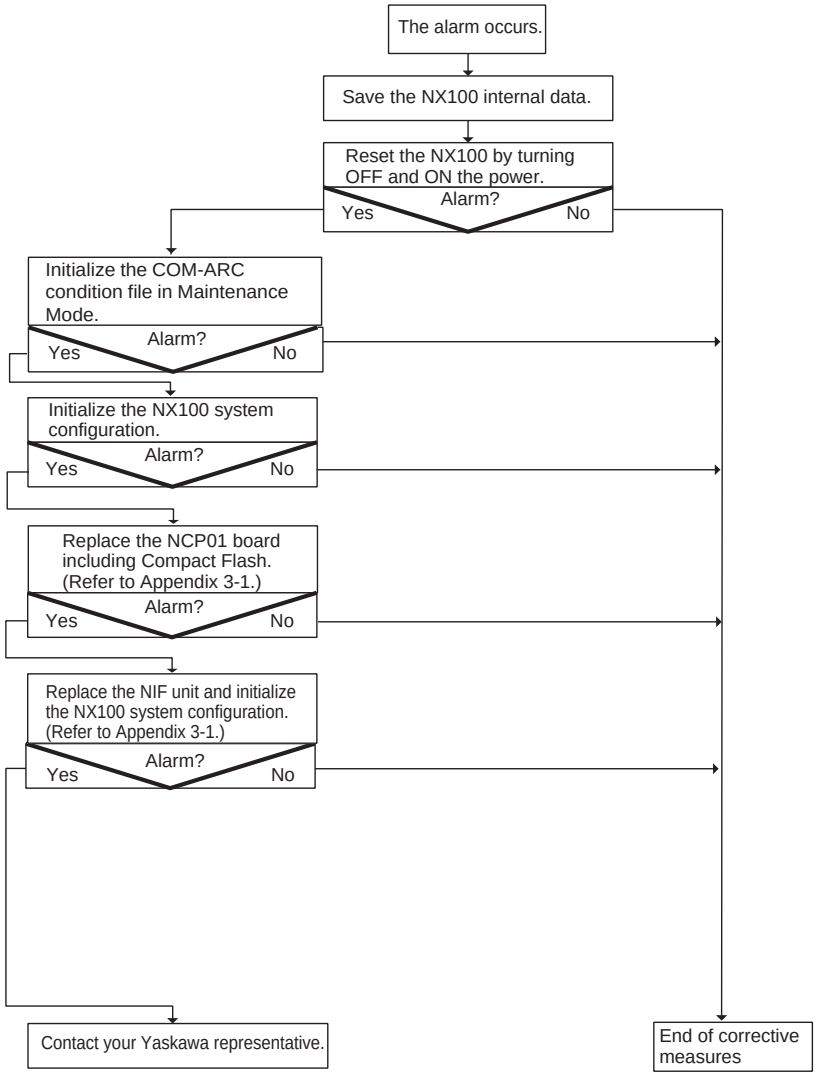
■ AL-4010

MEMORY ERROR (COM-ARC COND FILE)

An error was detected at memory check.

The memory for COM-ARC condition file is damaged.

Sub Code: Signifies the page number in which the alarm occurred.



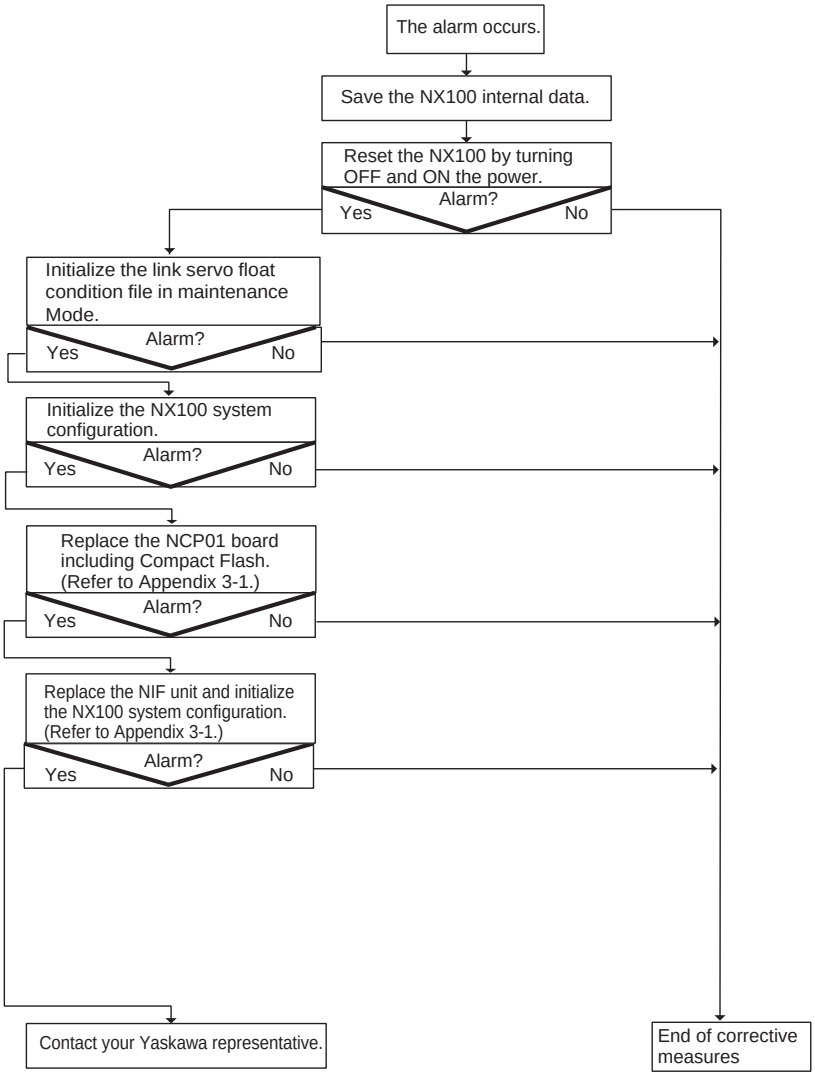
■ AL-4012

MEMORY ERROR (LINK SERVOFLOAT FILE)

An error was detected at memory check.

The memory for the link servo float condition file is damaged.

Sub Code: Signifies the condition file number in which the alarm occurred.



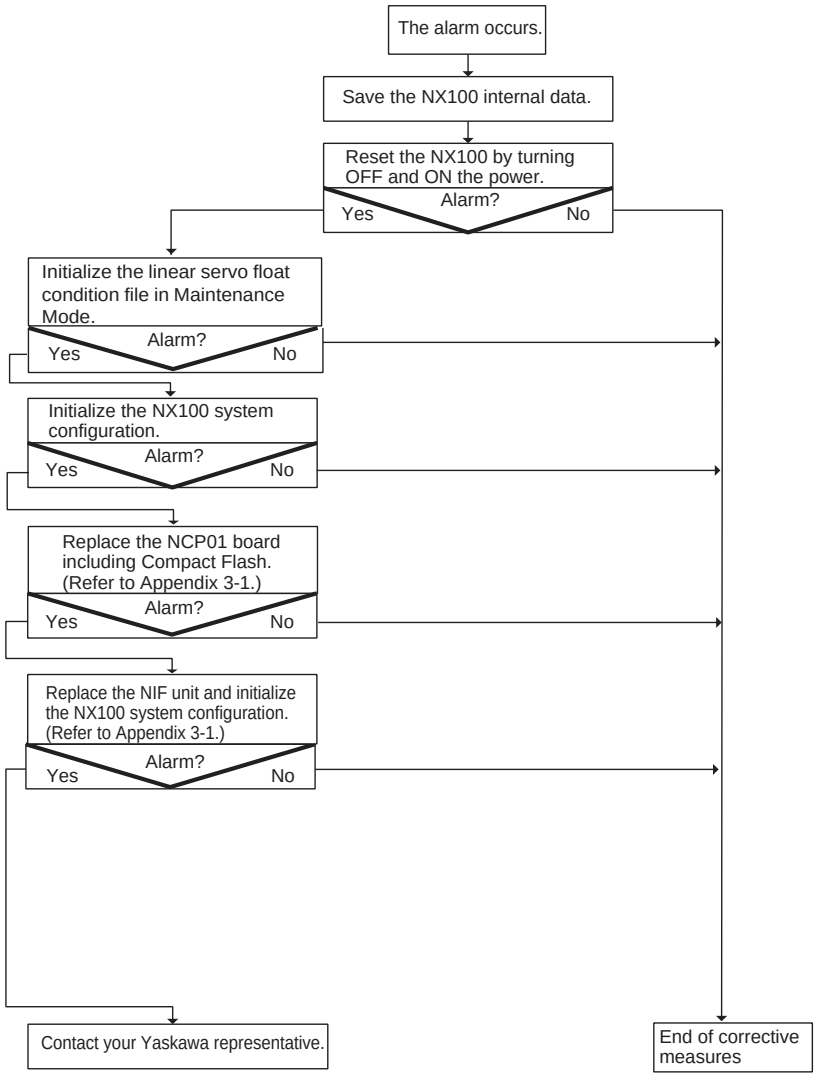
■ AL-4013

MEMORY ERROR (LINEAR SERVOFLOAT FILE)

An error was detected at memory check.

The memory for the linear servo float condition file is damaged.

Sub Code: Signifies the condition file number in which the alarm occurred.



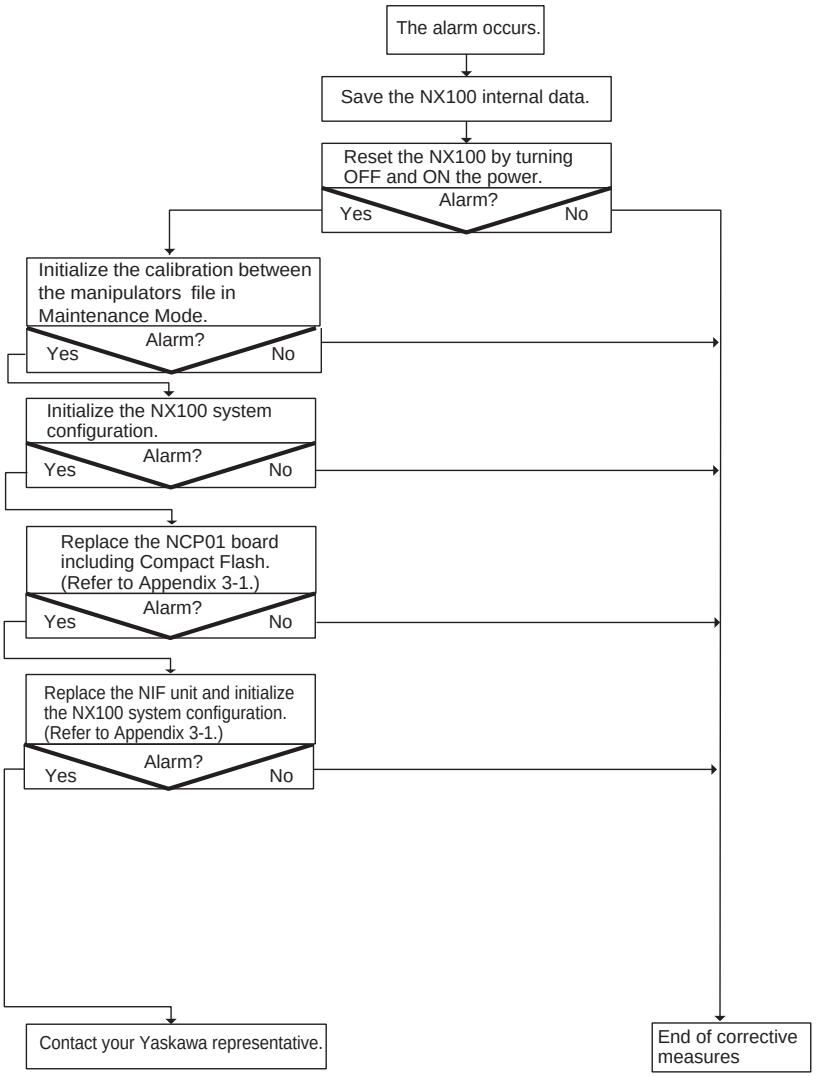
■ AL-4014

MEMORY ERROR (ROBOT CALIB FILE)

An error was detected at memory check.

The memory for the file for calibration between manipulators is damaged.

Sub Code: Signifies the page number in which the alarm occurred.



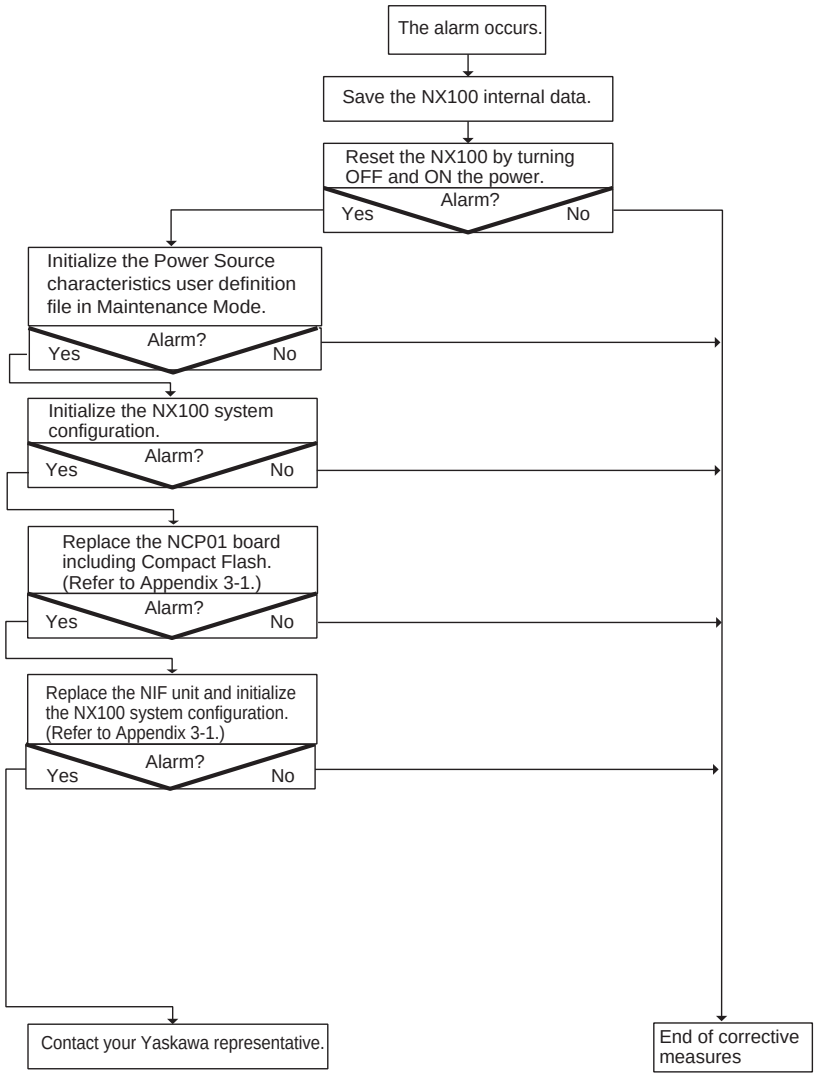
■ AL-4017

MEMORY ERROR (POWER SRC USER FILE)

An error was detected at memory check.

The memory for the power source characteristics user definition file is damaged.

Sub Code: None



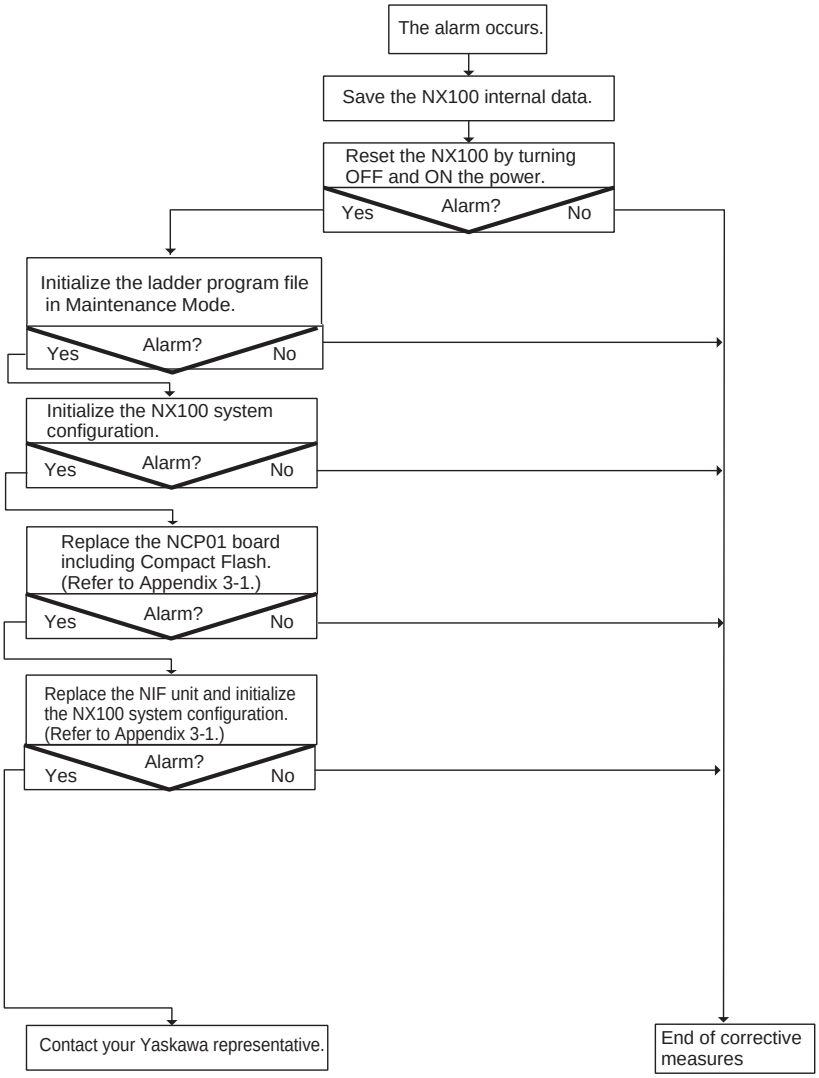
■ AL-4018

MEMORY ERROR (LADDER PRG FILE)

An error was detected at memory check.

The memory for the ladder program file is damaged.

Sub Code: None



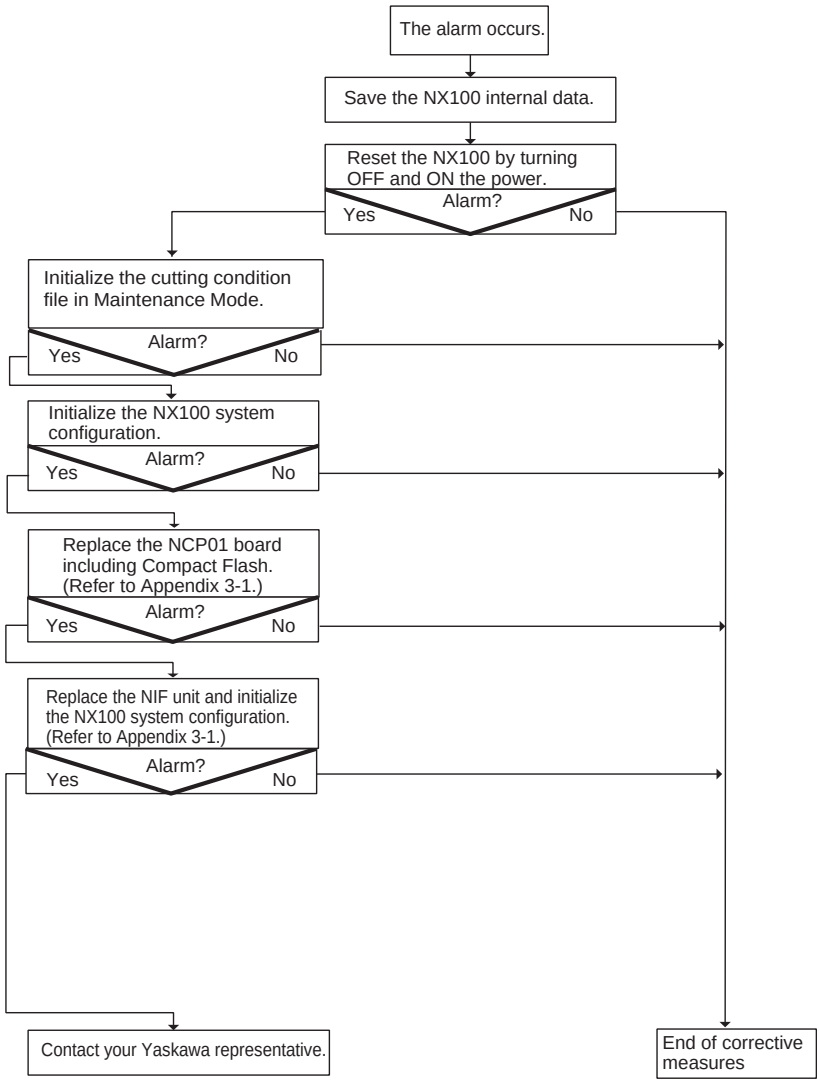
■ AL-4019

MEMORY ERROR (CUTTING COND FILE)

An error was detected at memory check.

The memory for the user coordinates file is damaged.

Sub Code: Signifies the page number in which the alarm occurred.



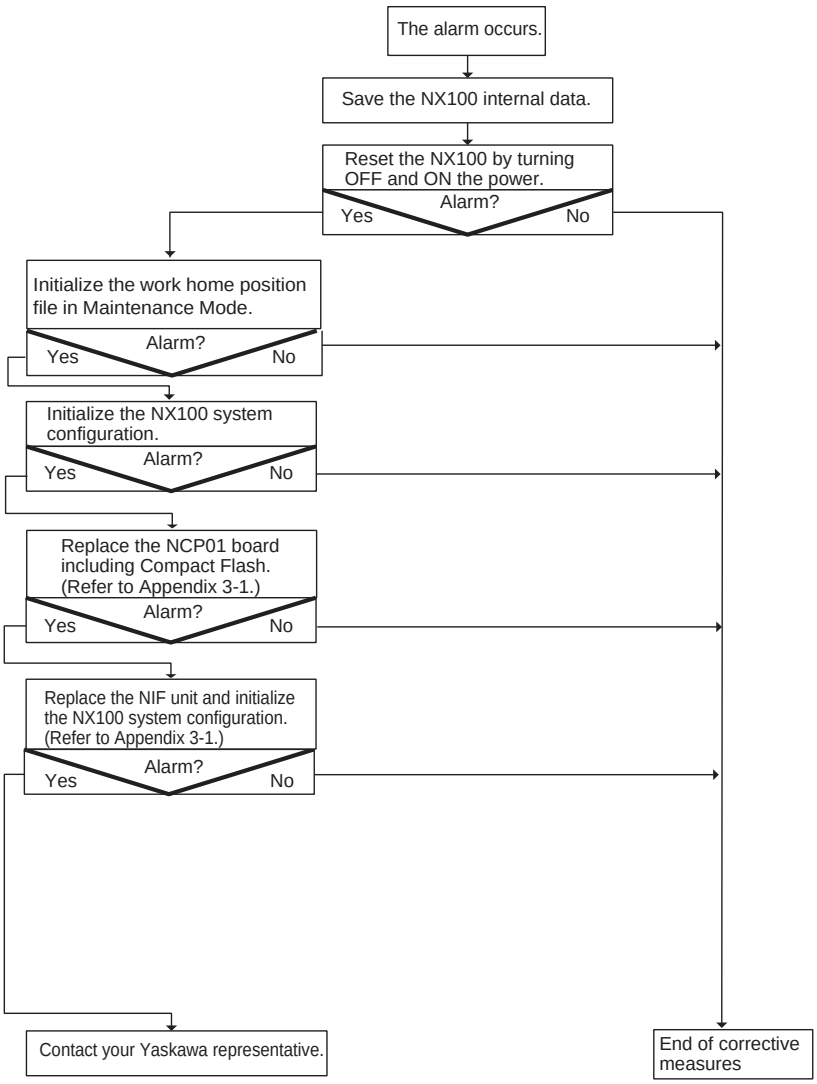
■ AL-4020

MEMORY ERROR (OPERATION ORIGIN)

An error was detected at memory check.

The memory for the work home position file is damaged.

Sub Code: None



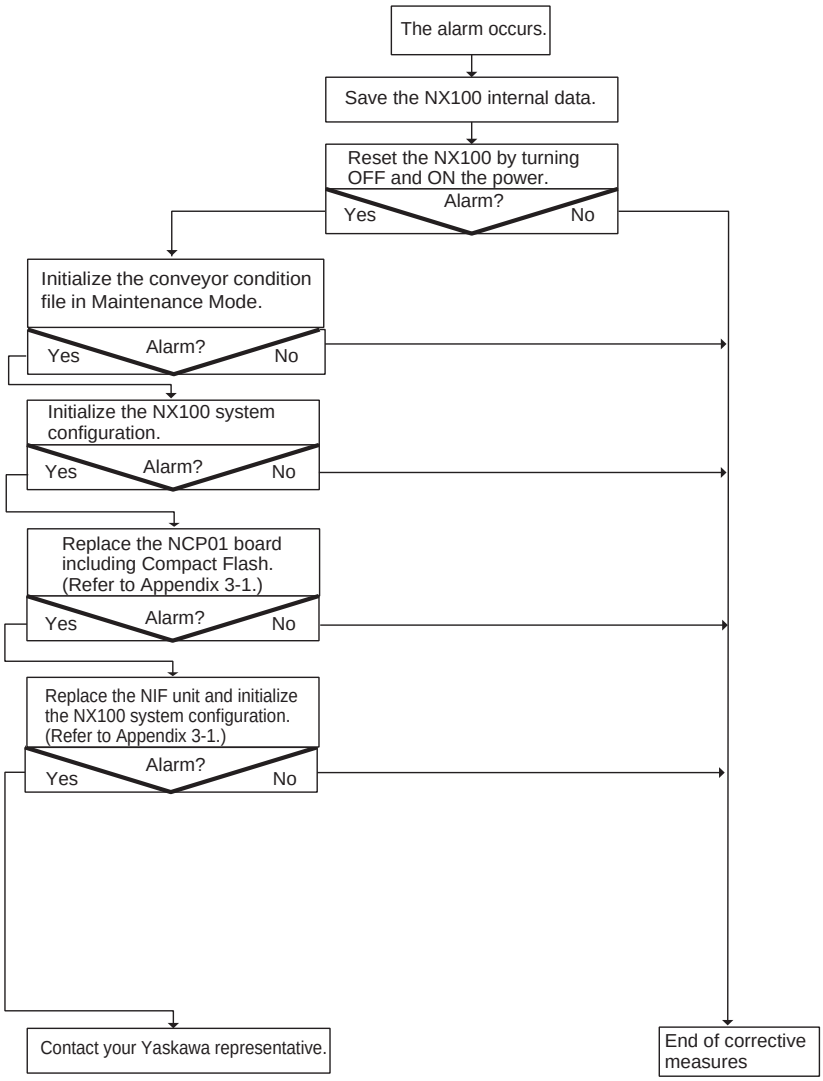
■ AL-4021

MEMORY ERROR (CONVEYOR COND FILE)

An error was detected at memory check.

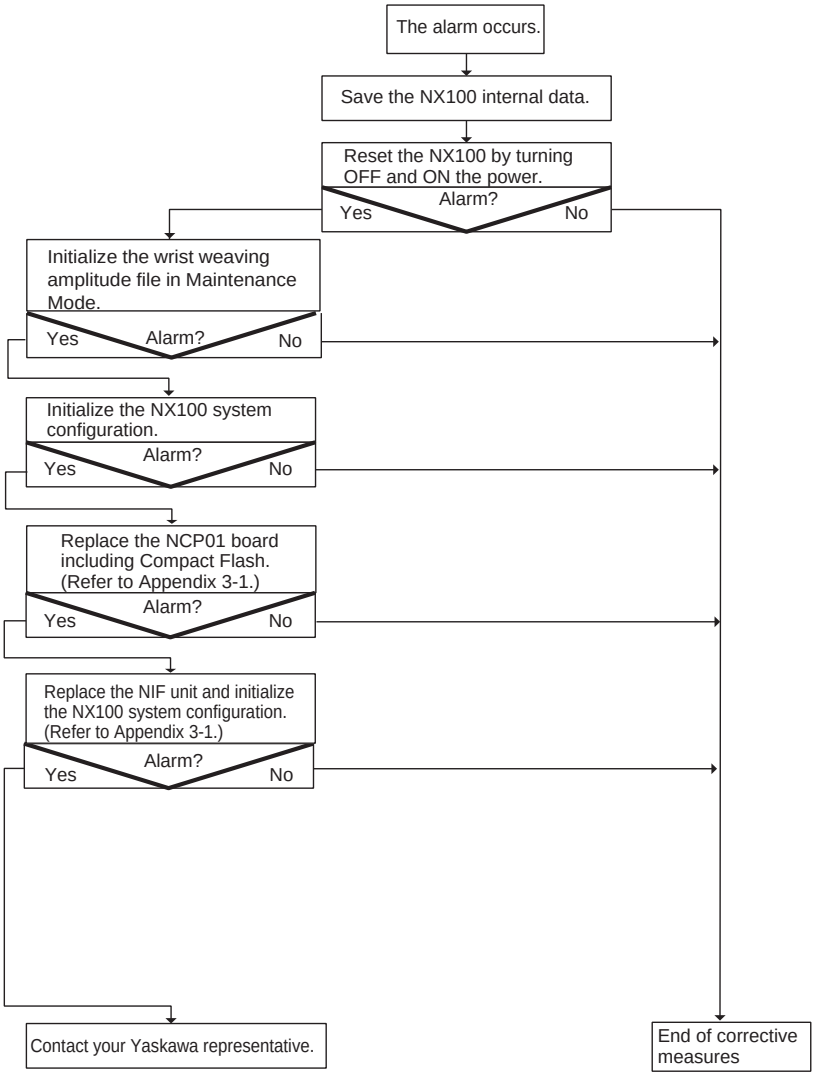
The memory for the conveyor condition file is damaged.

Sub Code: None



■ AL-4024

MEMORY ERROR (WRIST WEAV AMP FILE)
An error was detected at memory check.
The memory for the wrist weaving amplitude file is damaged.
Sub Code: None



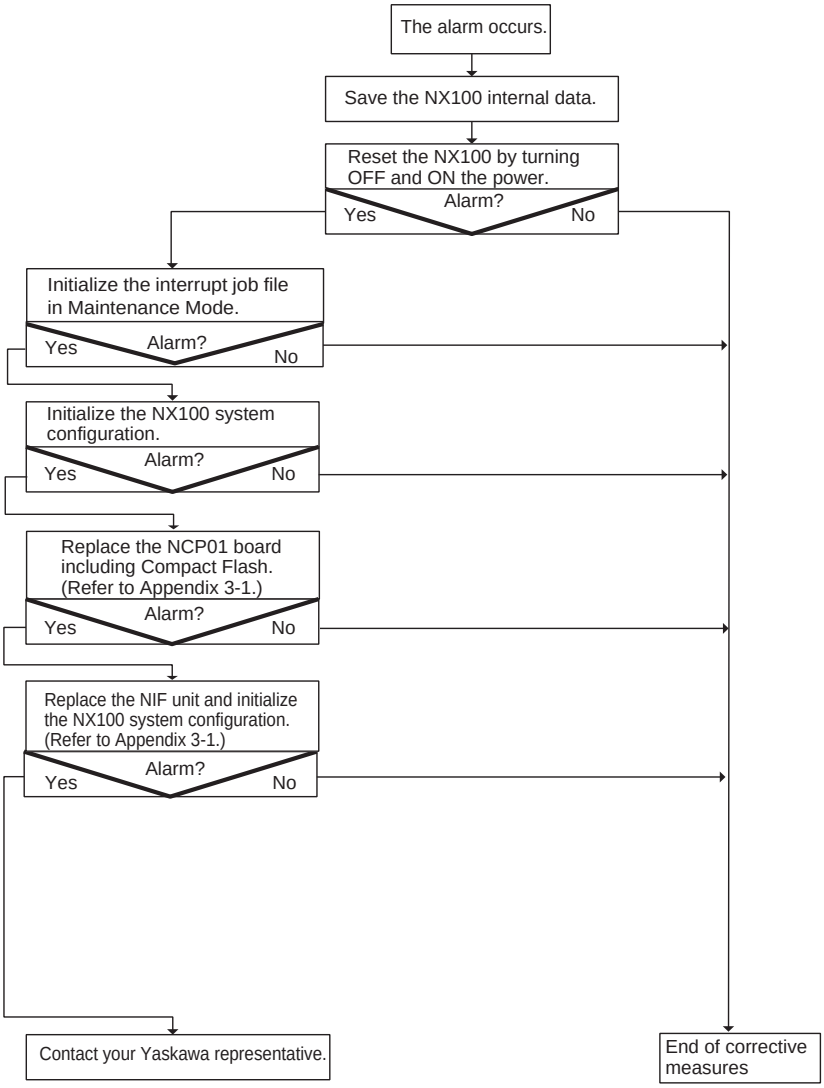
■ AL-4025

MEMORY ERROR (INTERRUPT JOB FILE)

An error was detected at memory check.

The memory for the interrupt job file is damaged.

Sub Code: None



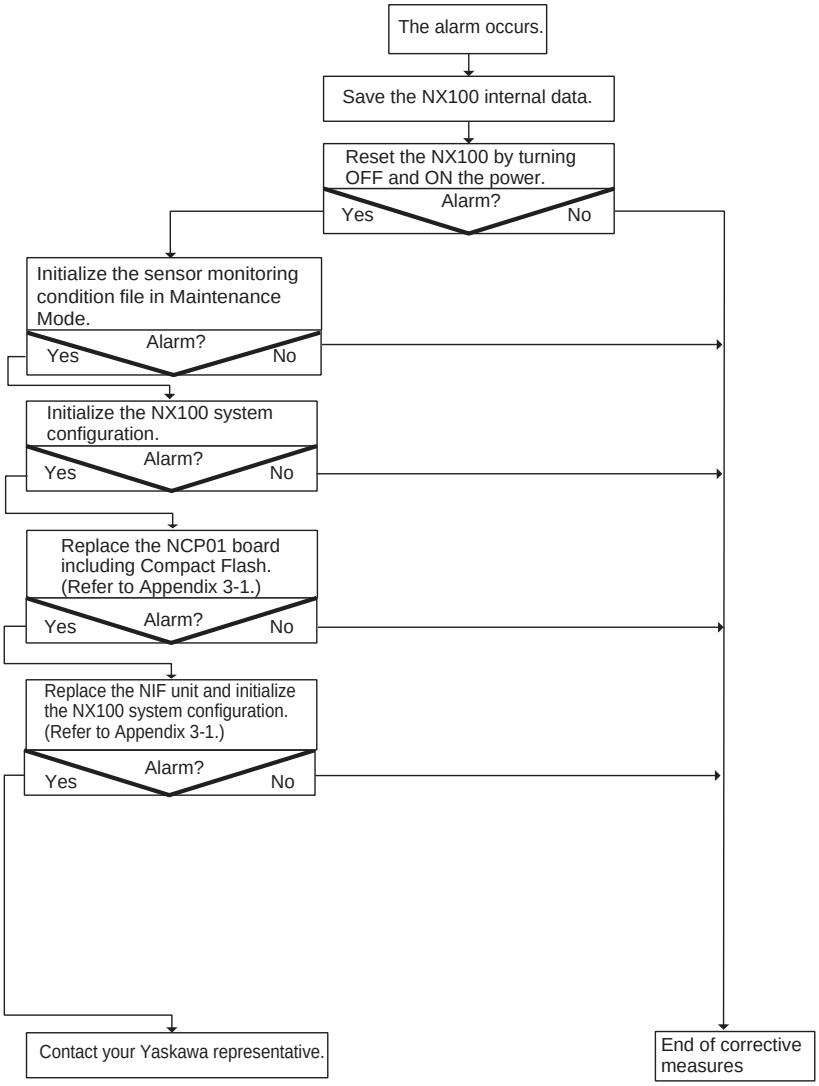
■ AL-4028

MEMORY ERROR (SENSOR MON COND FILE)

An error was detected at memory check.

The memory for the sensor monitoring condition file is damaged.

Sub Code: None



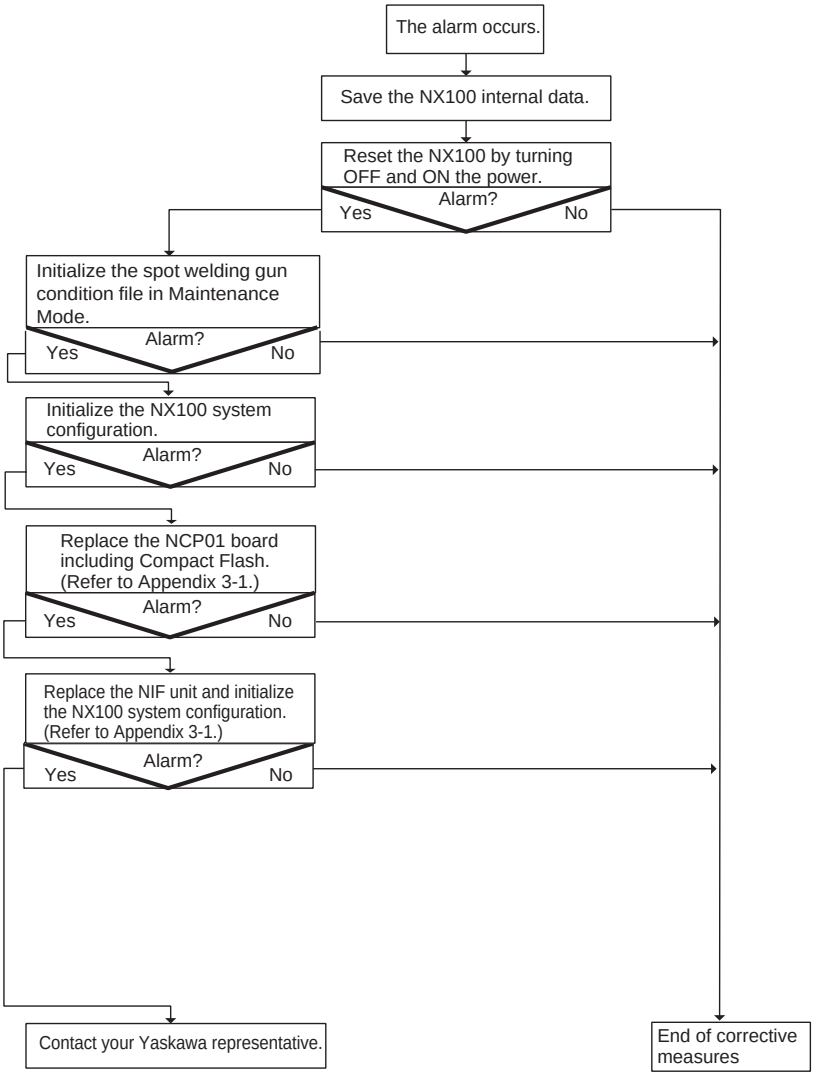
■ AL-4031

MEMORY ERROR (GUN COND FILE)

An error was detected at memory check.

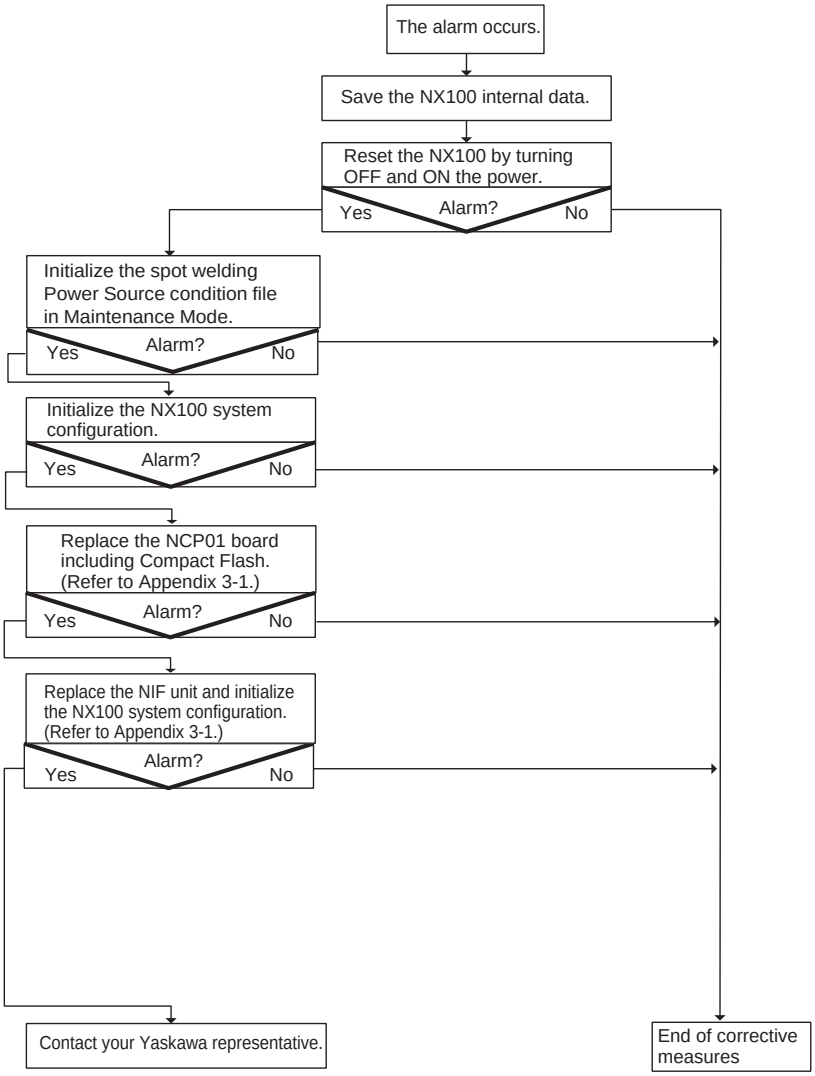
The memory for the spot welding gun condition file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



■ AL-4032

MEMORY ERROR (SPOT WELD COND FILE)
An error was detected at memory check.
The memory for the spot welding gun condition file is damaged.
Sub Code: Signifies the file number in which the alarm occurred.



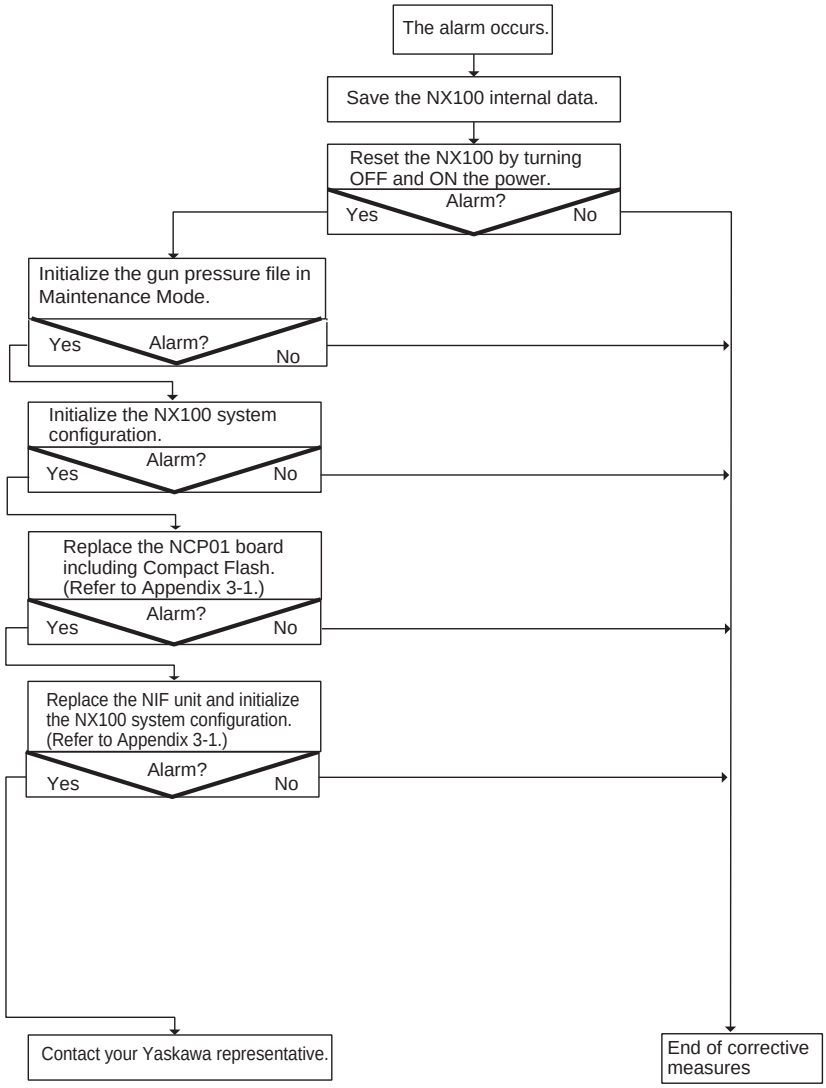
■ AL-4033

MEMORY ERROR (GUN PRESSURE FILE)

An error was detected at memory check.

The memory for the gun pressure file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



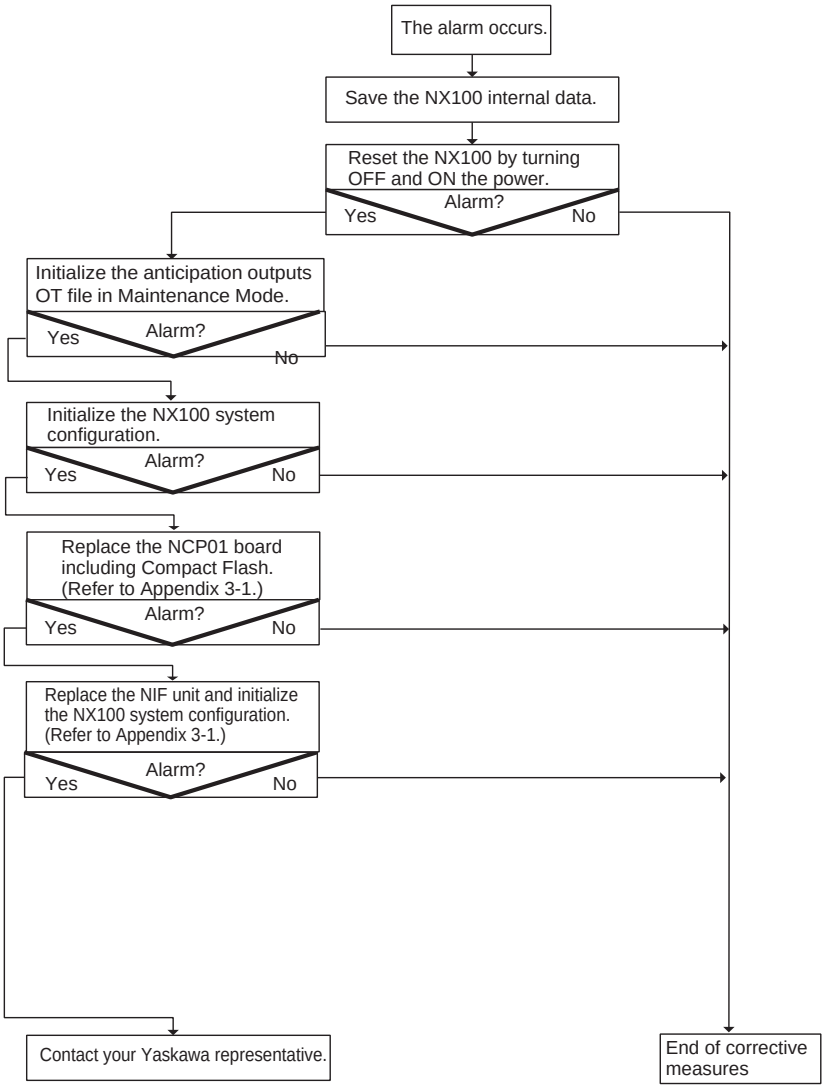
■ AL-4034

MEMORY ERROR (ANTICIPATION OT FILE)

An error was detected at memory check.

The memory for the anticipation outputs (OT) file is damaged.

Sub Code: None



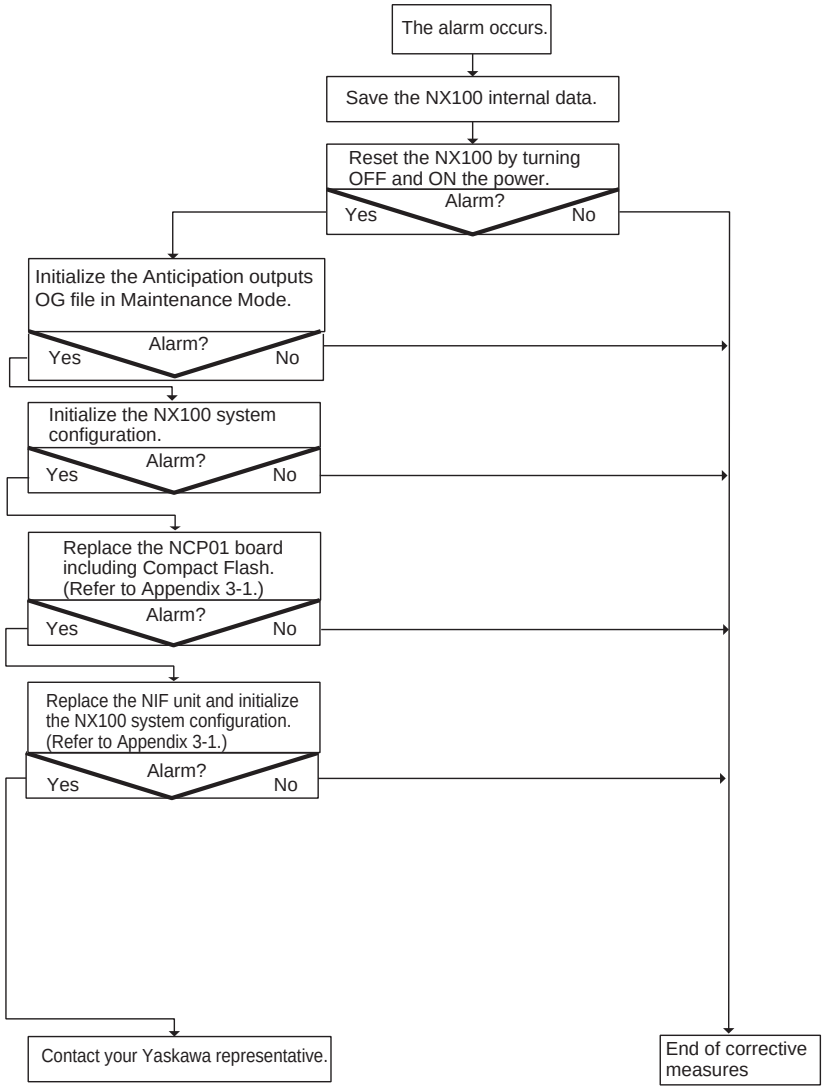
■ AL-4035

MEMORY ERROR (ANTICIPATION OG FILE)

An error was detected at memory check.

The memory for the anticipation outputs (OG) file is damaged.

Sub Code: None



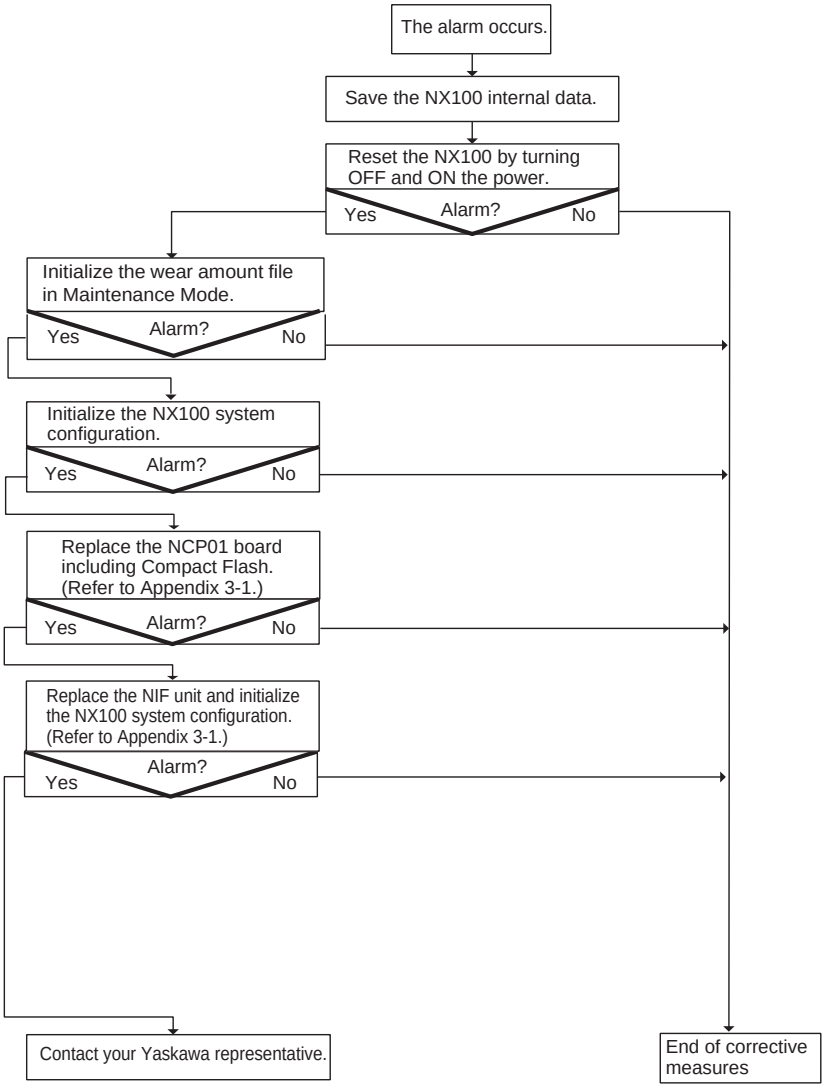
■ AL-4036

MEMORY ERROR (WEARING FILE)

An error was detected at memory check.

The memory for the wear amount file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



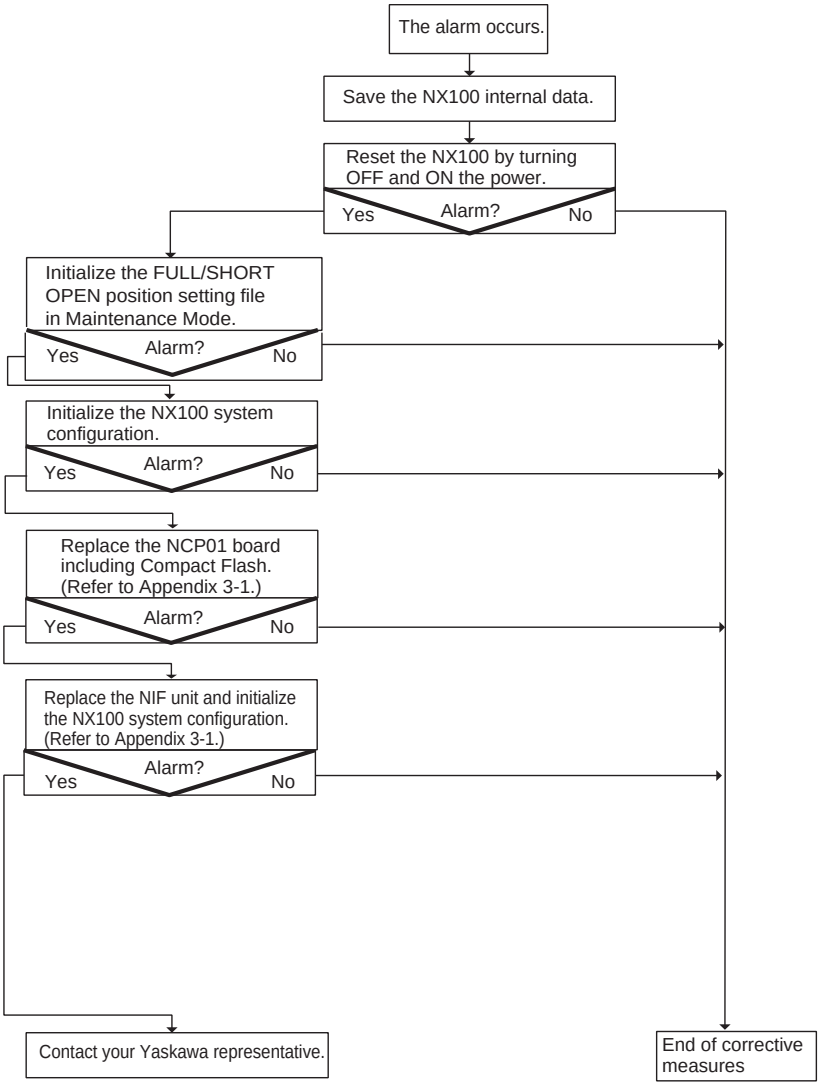
■ AL-4037

MEMORY ERROR (FULL-SHORT POSITION STROKE FILE)

An error was detected at memory check.

The memory for the FULL/SHORT OPEN position file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



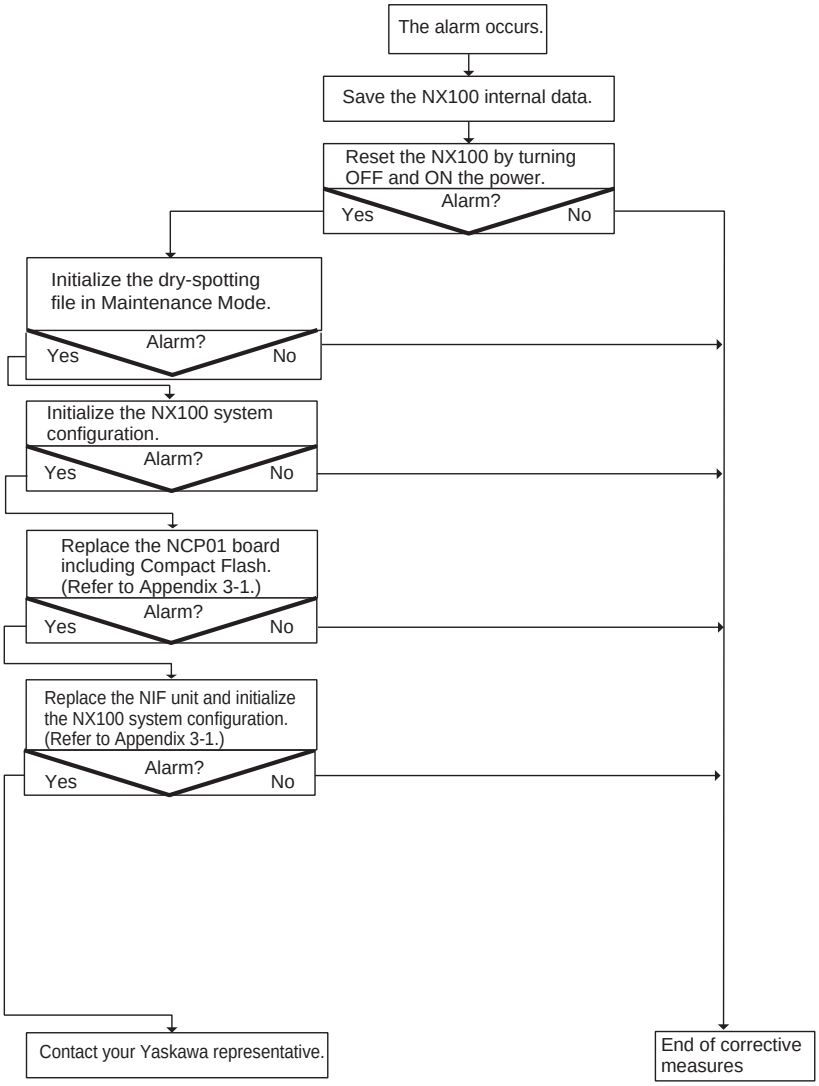
■ AL-4038

MEMORY ERROR (PRESSURE FILE)

An error was detected at memory check.

The memory for the dry-spotting pressure file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



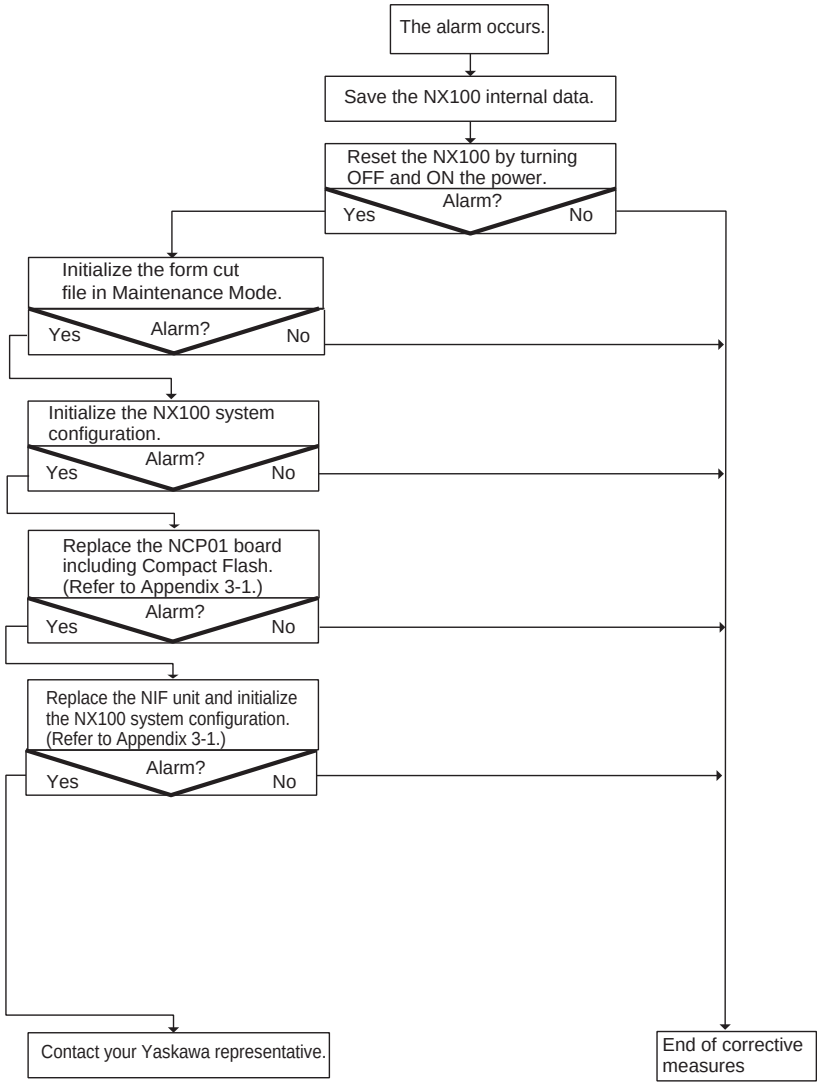
■ AL-4039

MEMORY ERROR (FORM CUT FILE)

An error was detected at memory check.

The memory for the form cut file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



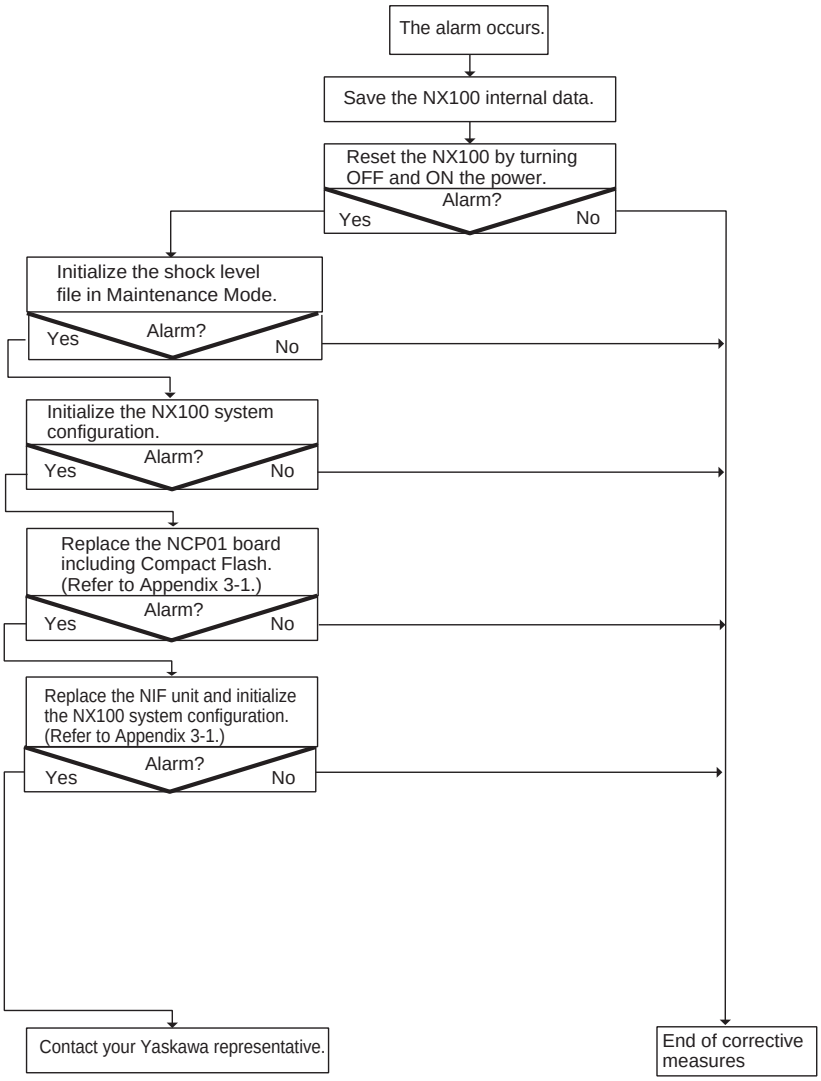
■ AL-4040

MEMORY ERROR (SHOCK LEVEL FILE)

An error was detected at memory check.

The memory for the shock level file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



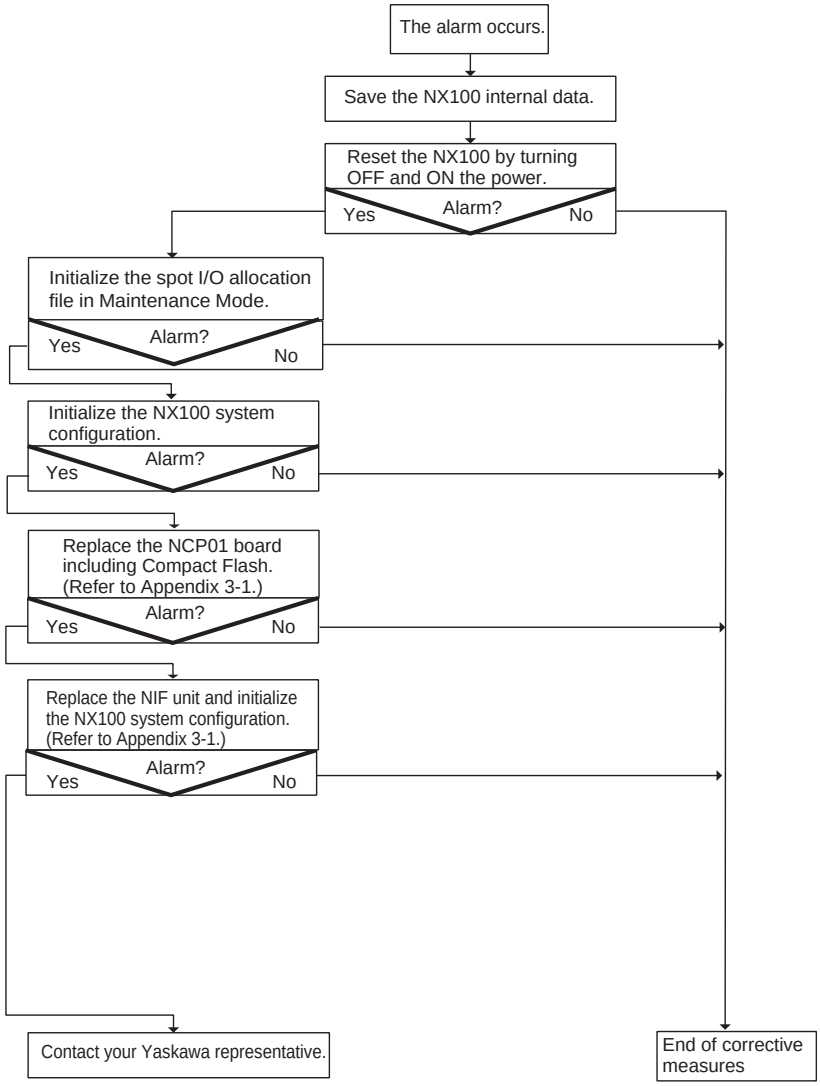
■ AL-4041

MEMORY ERROR (SPOT I/O ALLOCATE FL)

An error was detected at memory check.

The memory for the spot I/O allocation file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



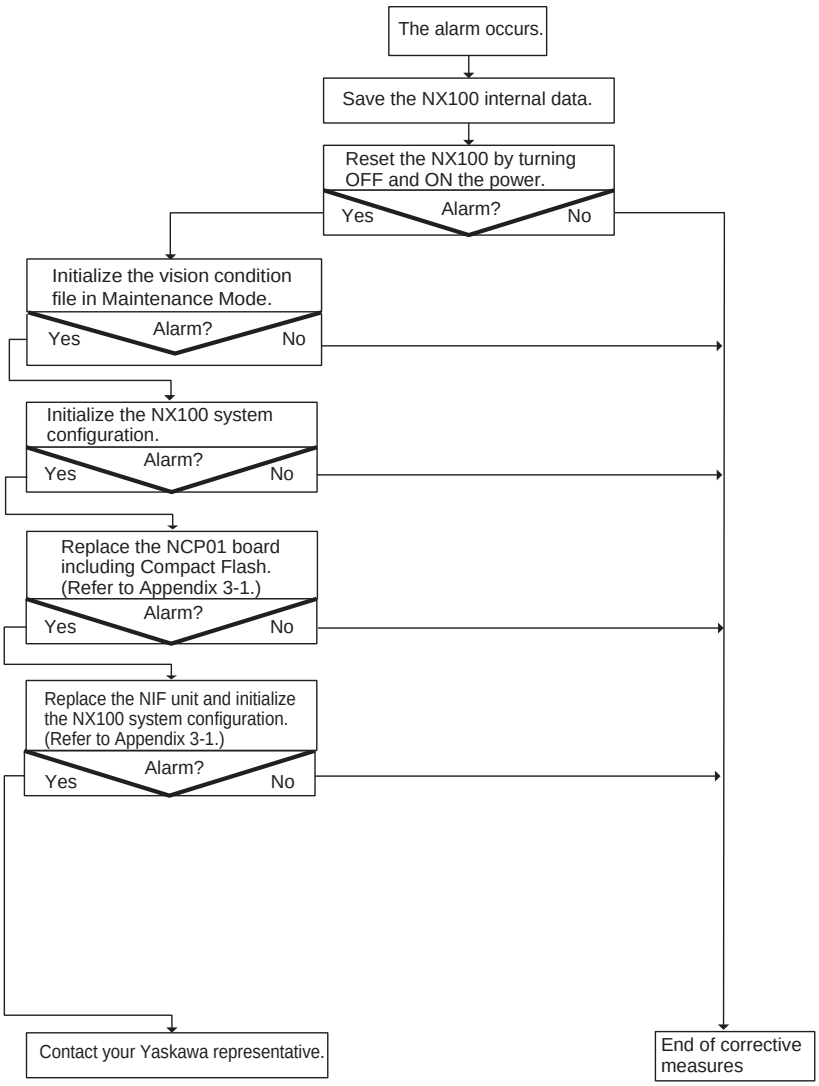
■ AL-4042

MEMORY ERROR (VISION FILE)

An error was detected at memory check.

The memory for the vision condition file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



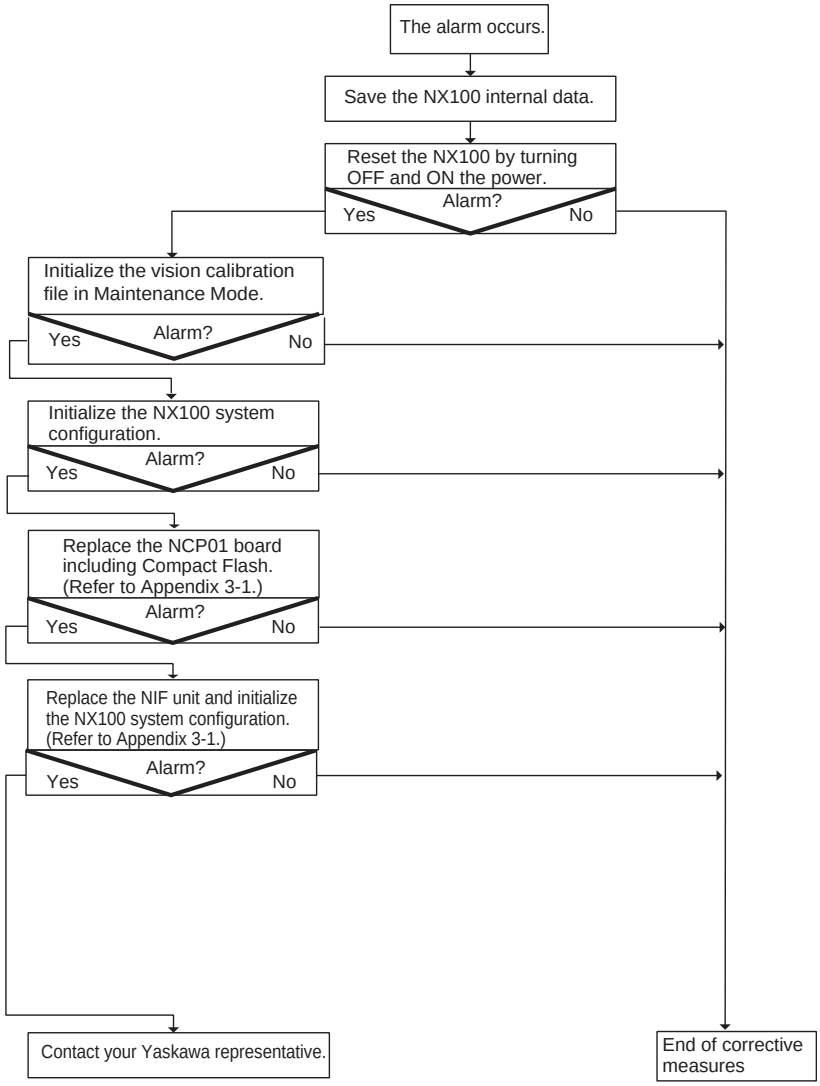
■ AL-4043

MEMORY ERROR (VISION CALIBRATION)

An error was detected at memory check.

The memory for the vision calibration file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



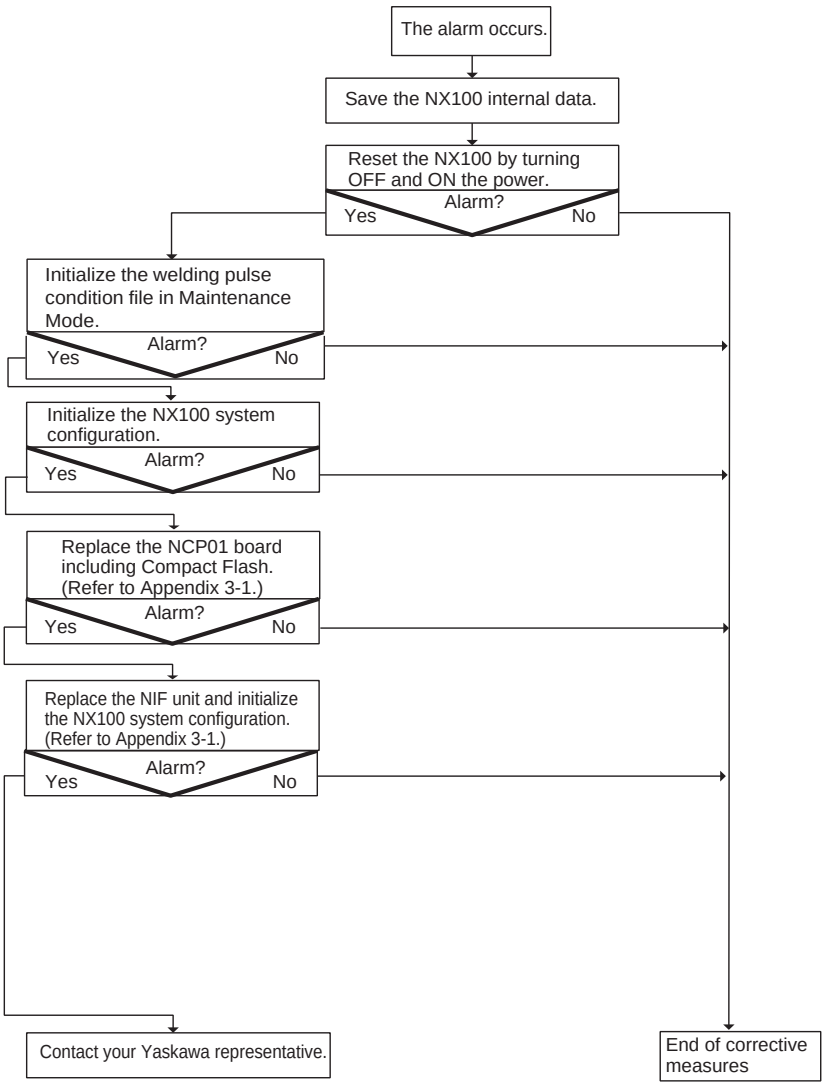
■ AL-4044

MEMORY ERROR (WELDING PULSE COND FILE)

An error was detected at memory check.

The memory for the welding pulse condition file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



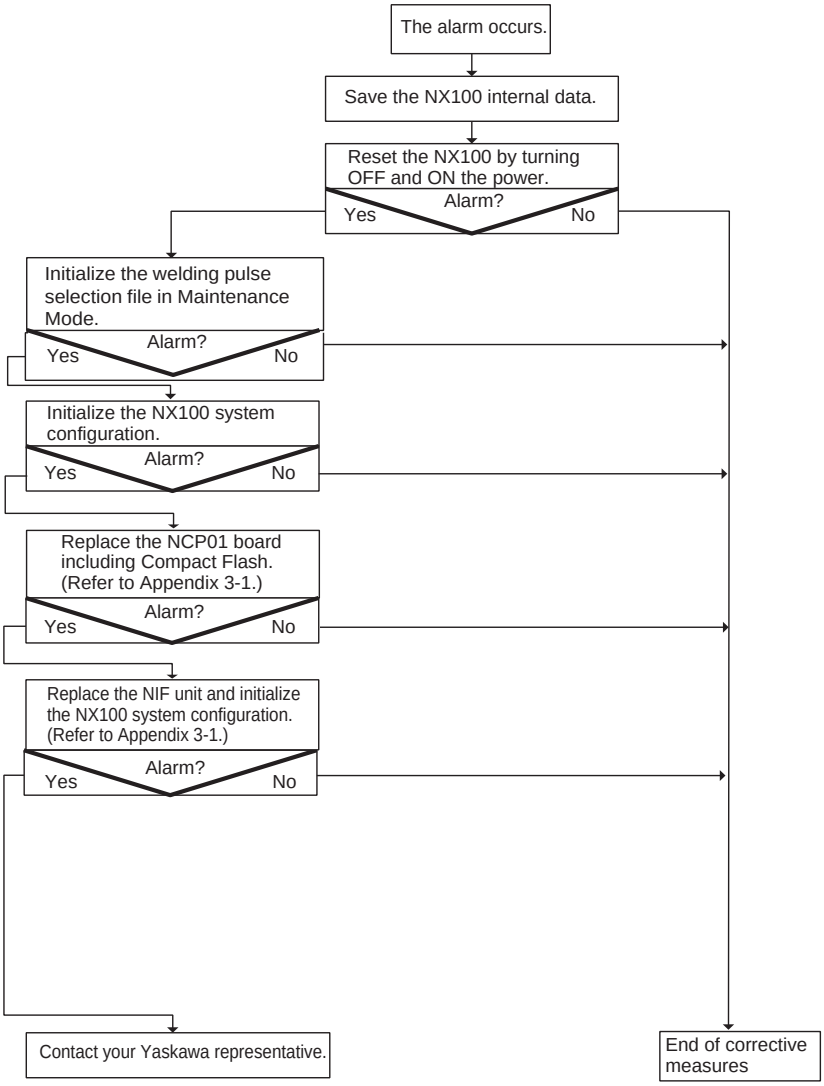
■ AL-4045

MEMORY ERROR (WELDING PULSE SELECTION FILE)

An error was detected at memory check.

The memory for the welding pulse selection file is damaged.

Sub Code: None



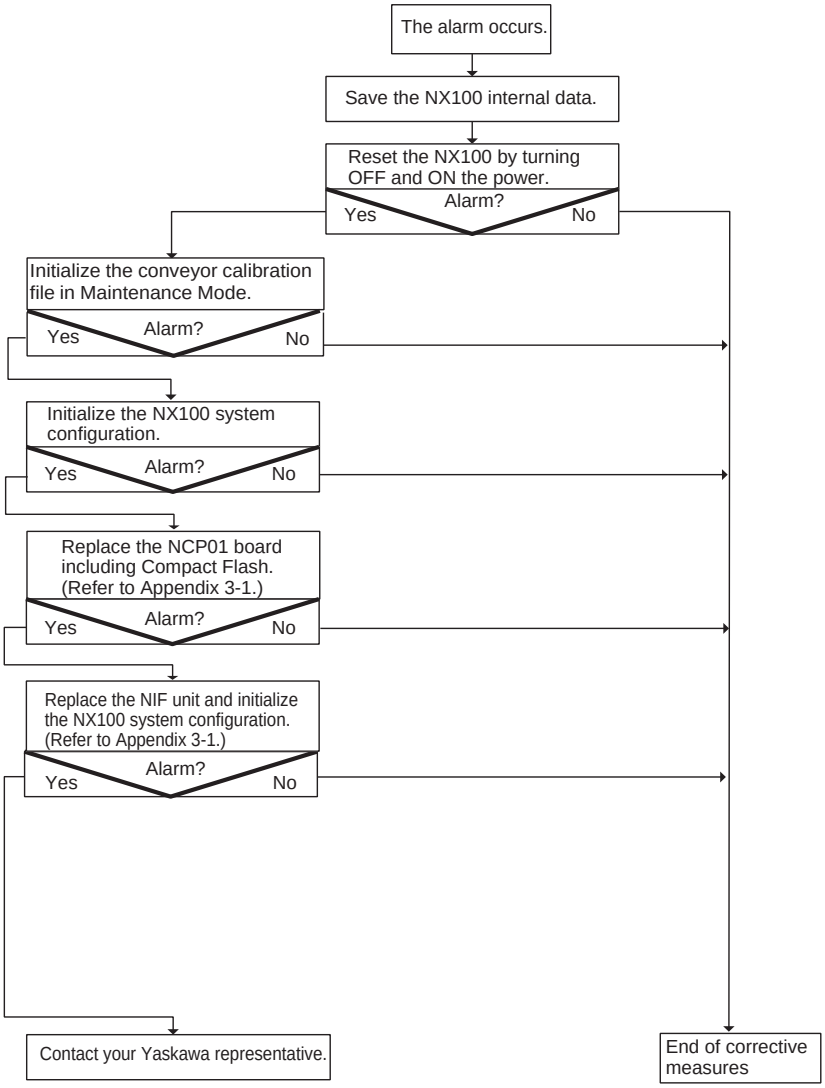
■ AL-4046

MEMORY ERROR (CONVEYOR CALIBRATION)

An error was detected at memory check.

The memory for the conveyor calibration file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



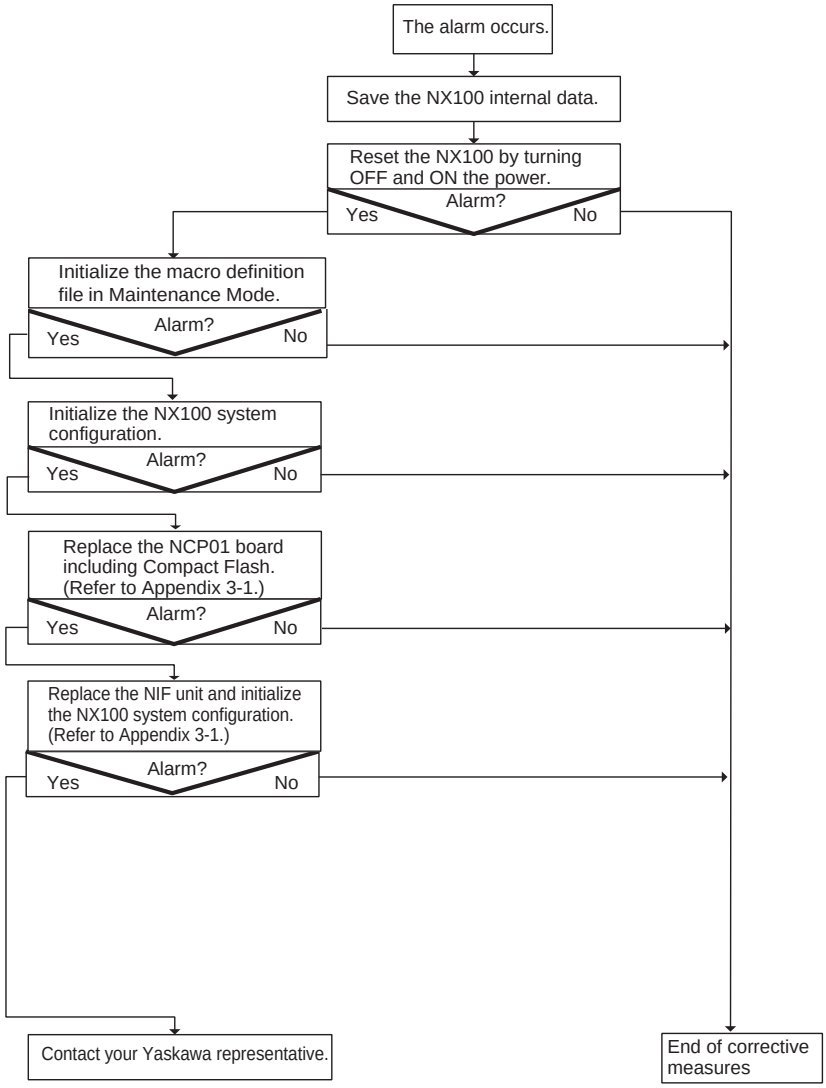
■ AL-4047

MEMORY ERROR (MACRO DEFINITION FILE)

An error was detected at memory check.

The memory for the macro definition file is damaged.

Sub Code: None



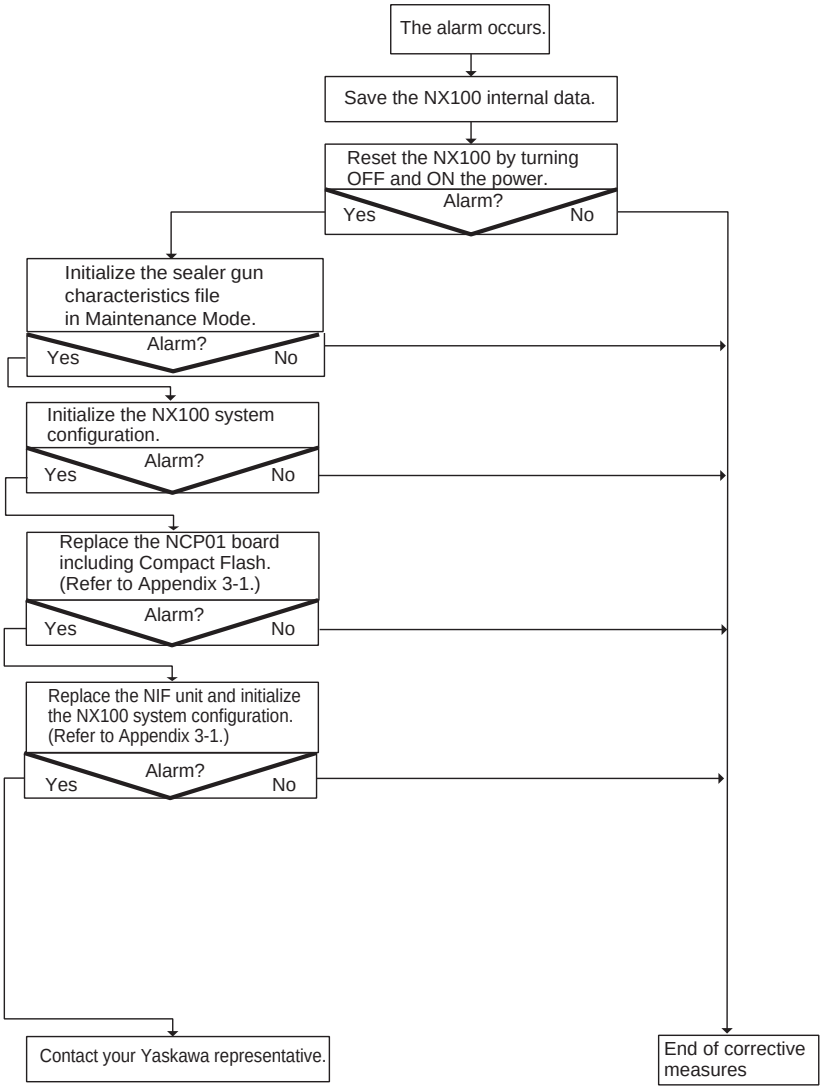
■ AL-4048

MEMORY ERROR (SERVO S-GUN FILE)

An error was detected at memory check.

The memory for the sealer gun characteristics file is damaged.

Sub Code: None



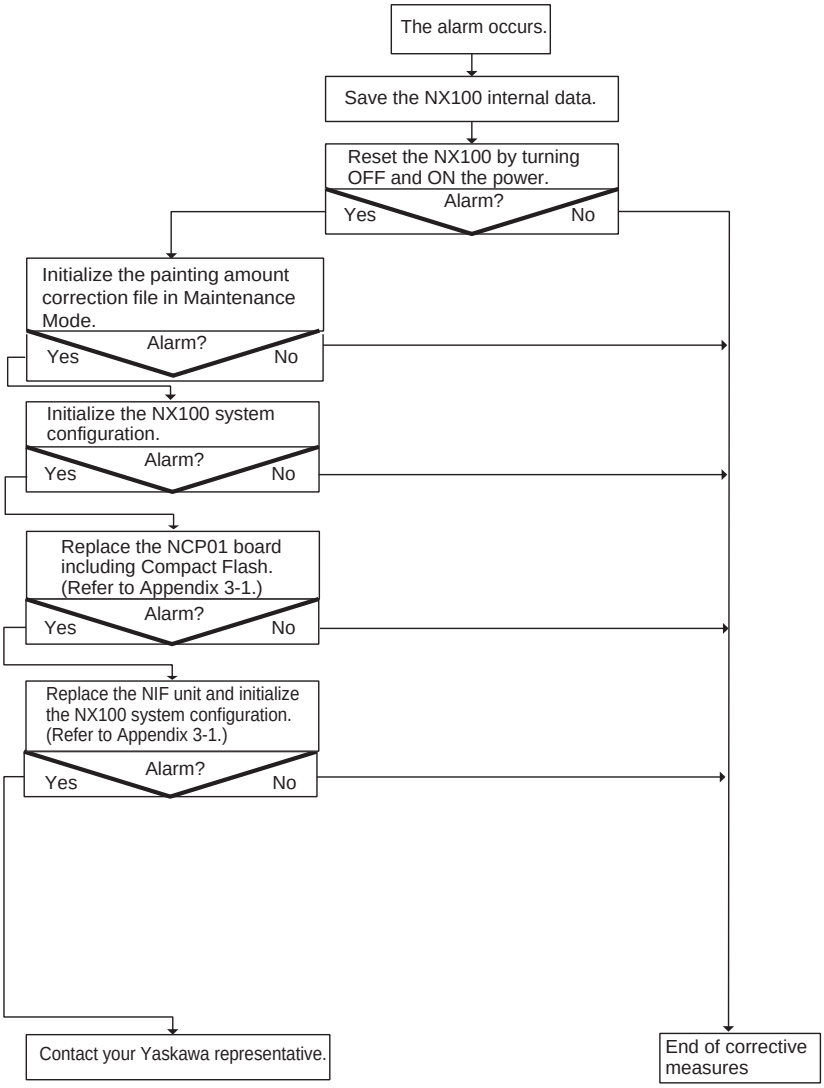
■ AL-4049

MEMORY ERROR (PASTE GUAN COMPENSATION FILE)

An error was detected at memory check.

The memory for the painting amount correction file is damaged.

Sub Code: None



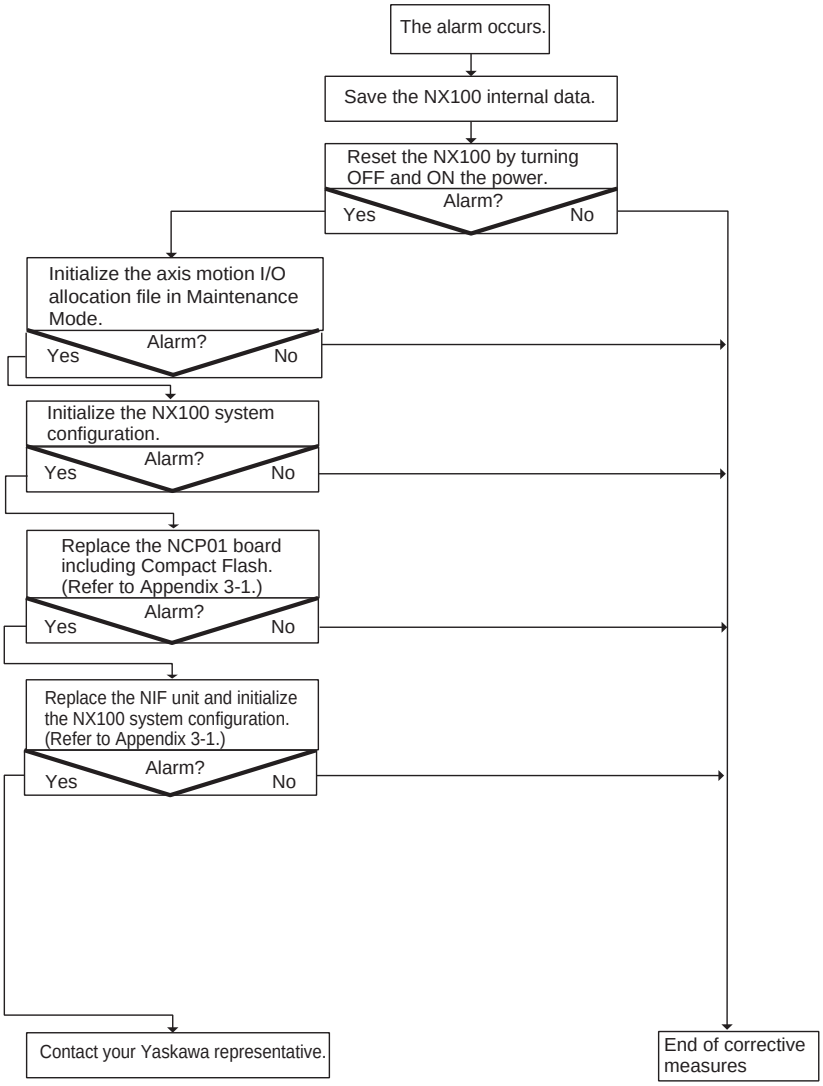
■ AL-4050

MEMORY ERROR (AXIS I/O ALLOCATION FILE)

An error was detected at memory check.

The memory for the axis motion I/O allocation file is damaged.

Sub Code: None



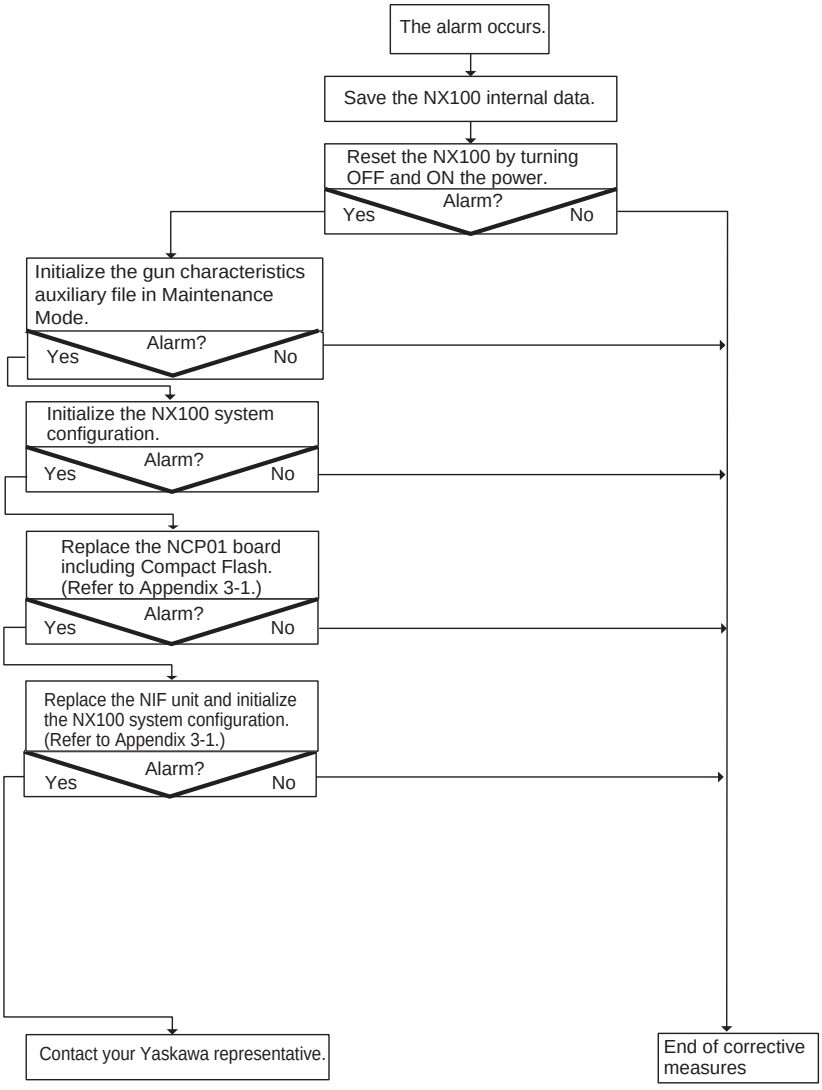
■ AL-4051

MEMORY ERROR (GUN COND AUX FILE)

An error was detected at memory check.

The memory for the gun characteristics auxiliary file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



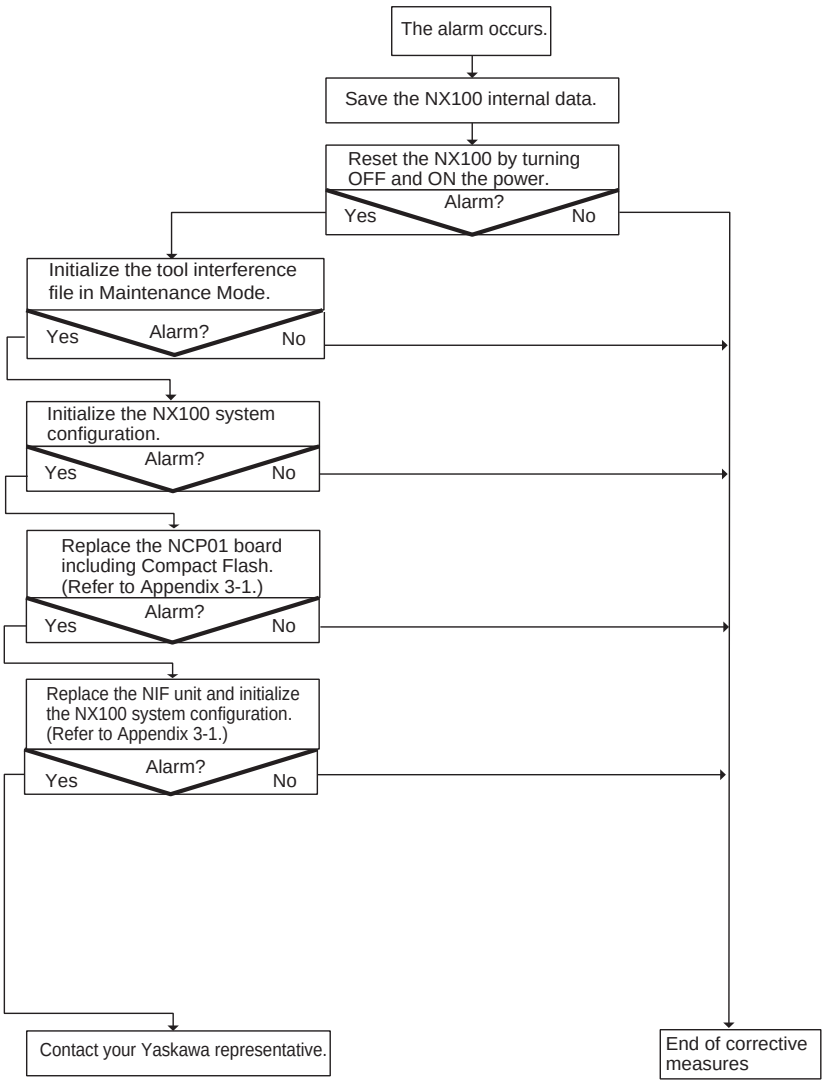
■ AL-4052

MEMORY ERROR (TOOL INTERFERENCE FILE)

An error was detected at memory check.

The memory for the tool interference file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



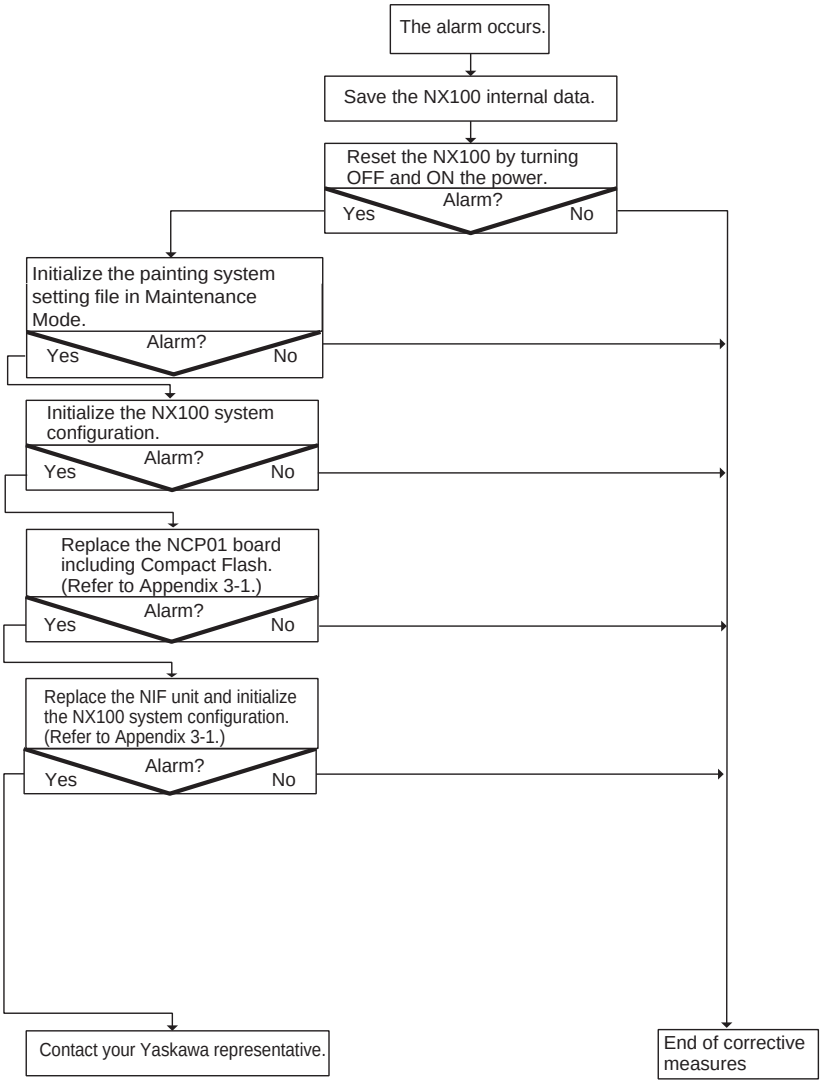
■ AL-4053

MEMORY ERROR (PAINTING SYSTEM CONFIGURATION FILE)

An error was detected at memory check.

The memory for the painting system setting file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



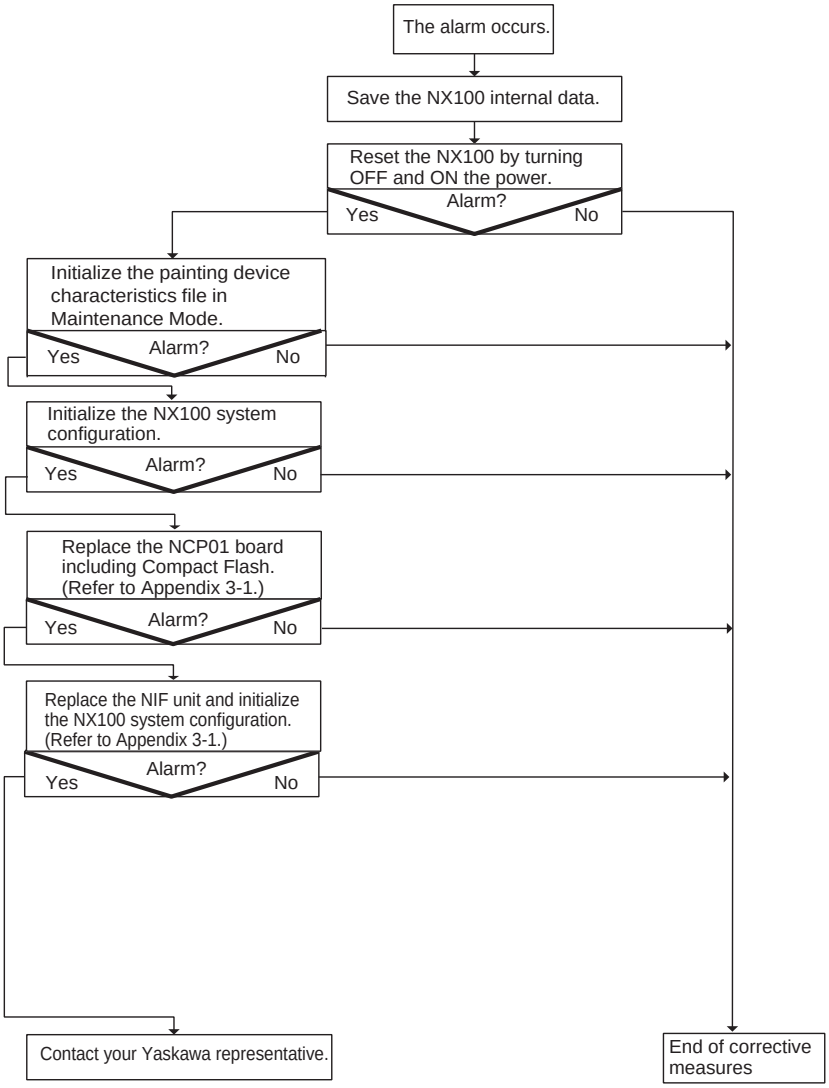
■ AL-4054

MEMORY ERROR (PAINTING SPECIAL FILE)

An error was detected at memory check.

The memory for the painting system setting file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



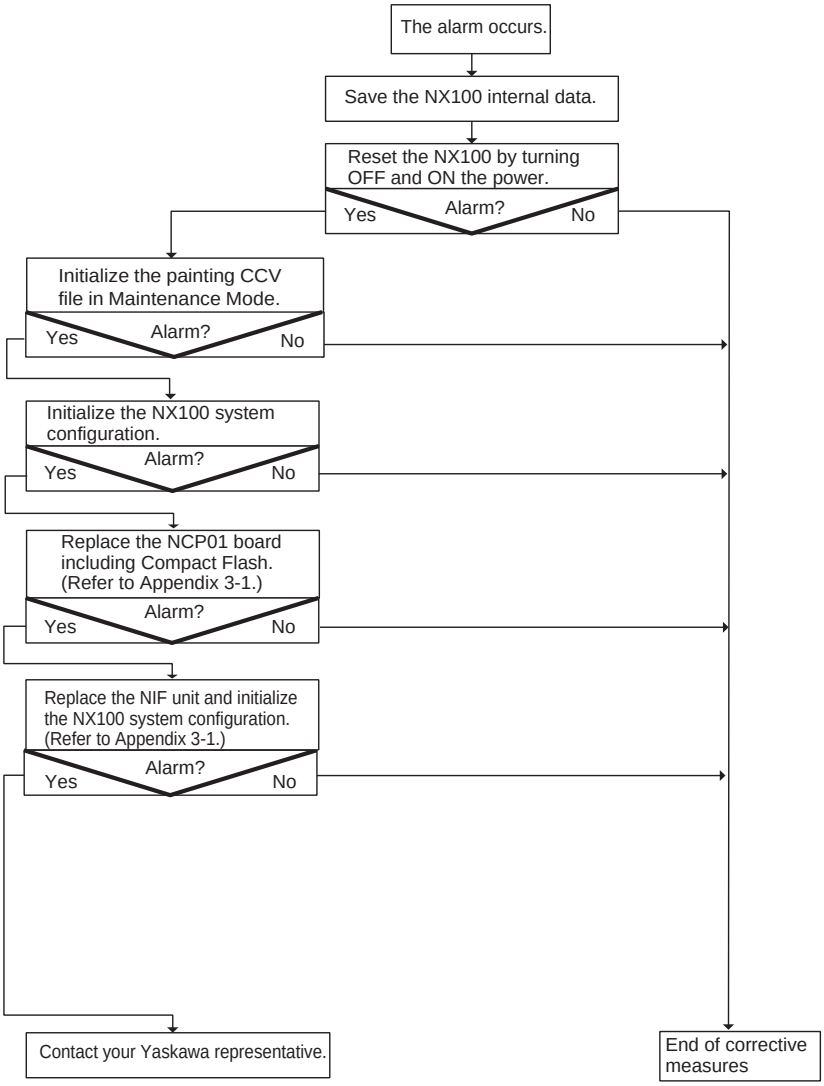
■ AL-4055

MEMORY ERROR (CCV-PAINT TABLE FILE)

An error was detected at memory check.

The memory for the painting CCV file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



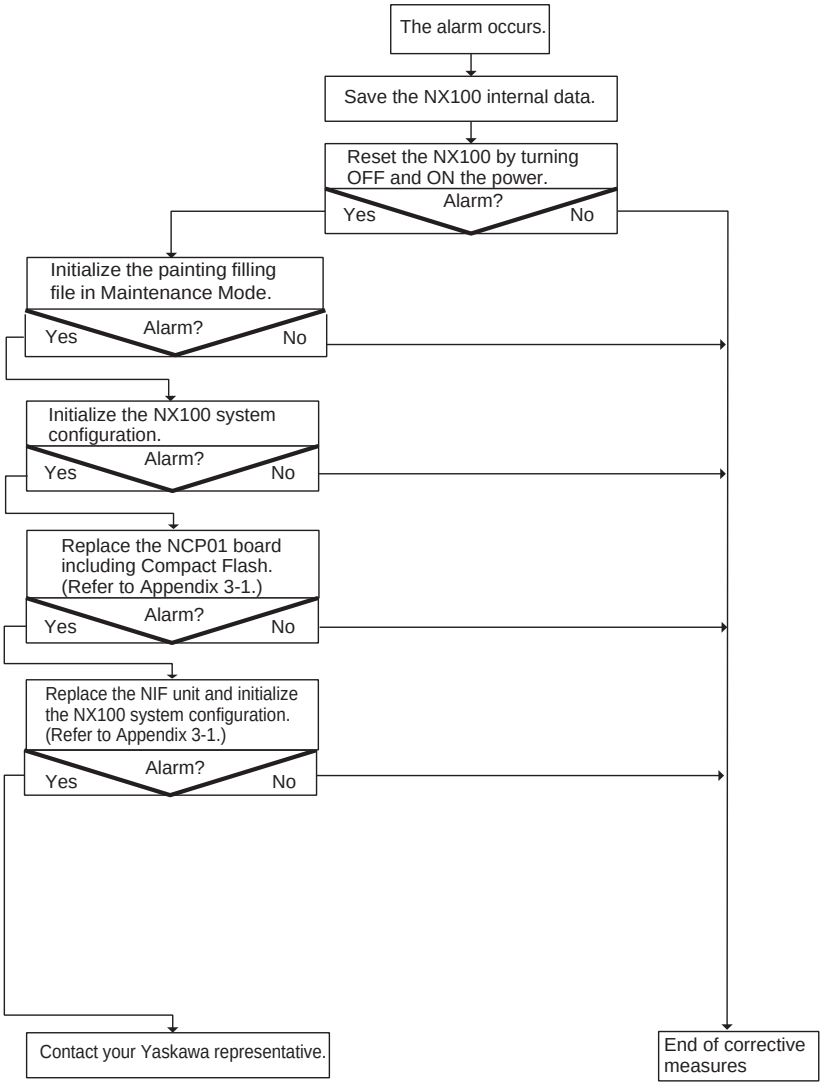
■ AL-4056

MEMORY ERROR (PLUG VOLUME FILE)

An error was detected at memory check.

The memory for the paint filling file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



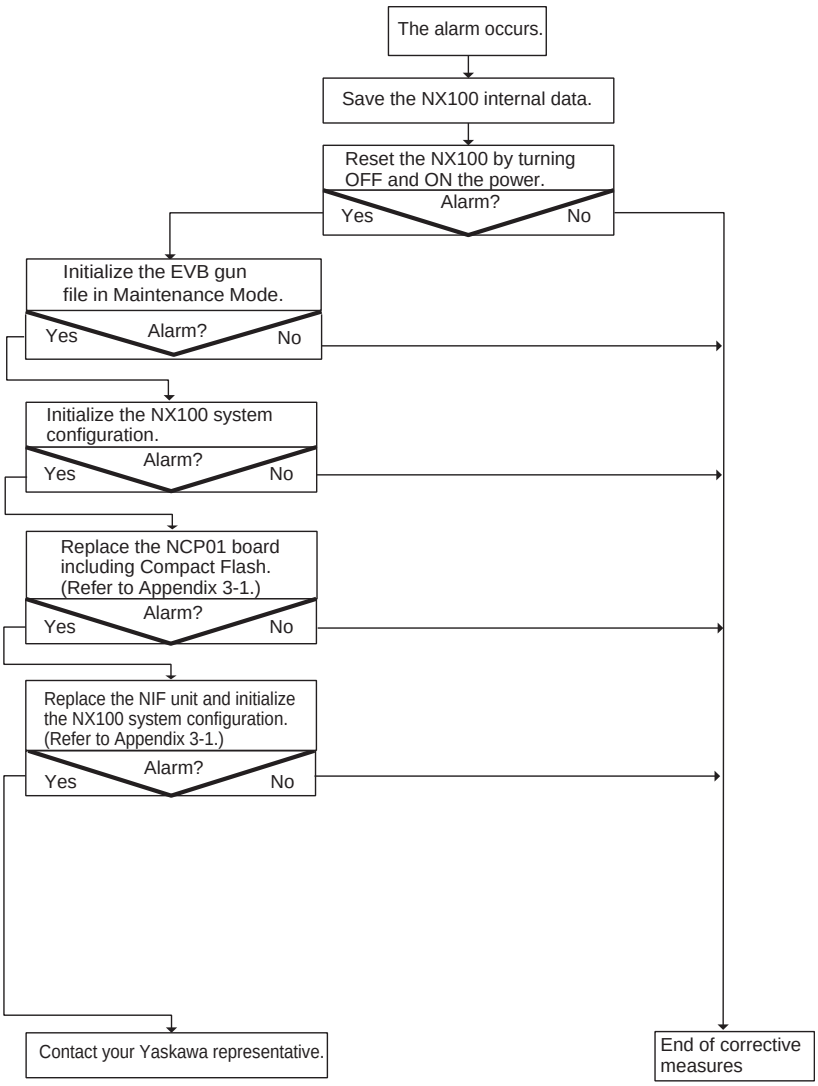
■ AL-4057

MEMORY ERROR (EVB GUN COND)

An error was detected at memory check.

The memory for the EVB gun file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



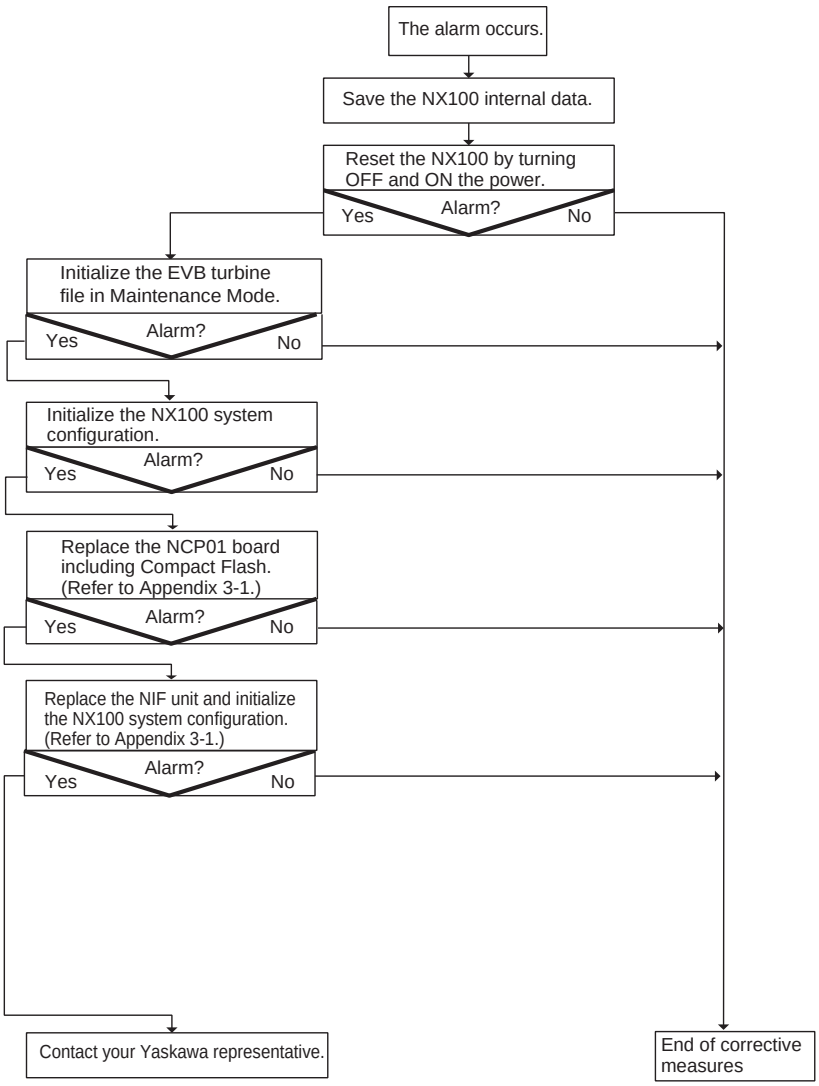
■ AL-4058

MEMORY ERROR (EVB TURBINE COND FILE)

An error was detected at memory check.

The memory for the EVB turbine condition file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



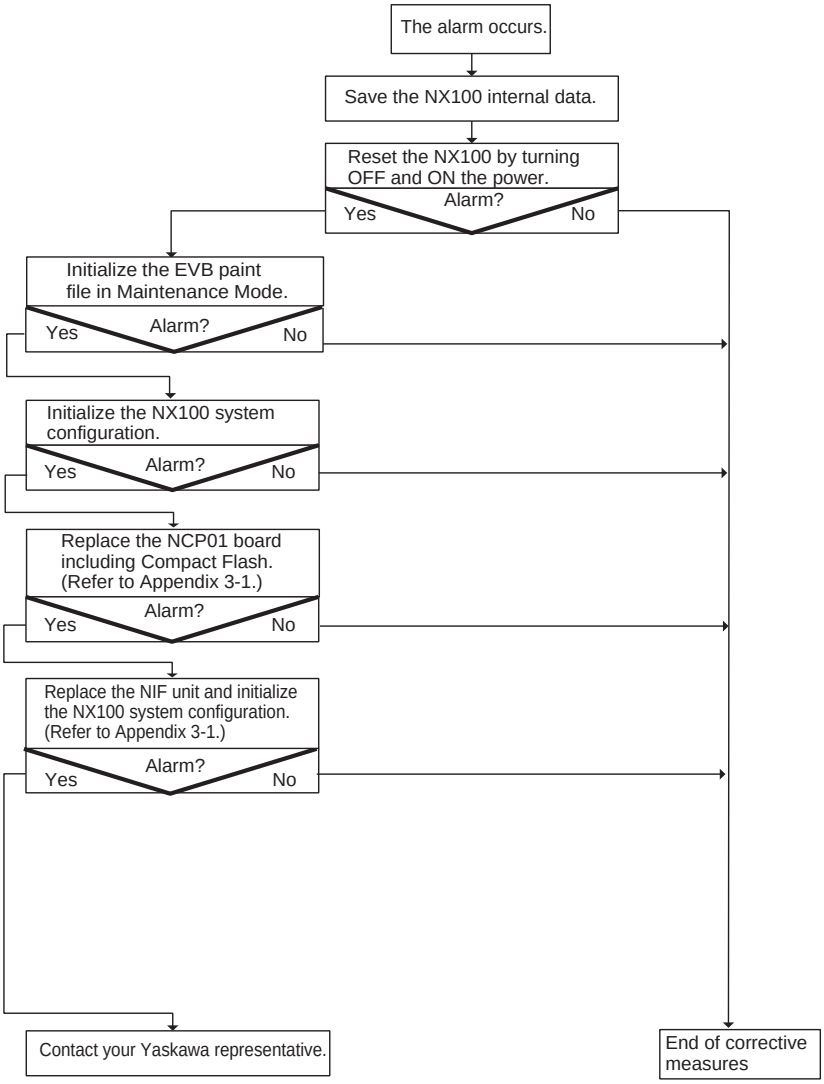
■ AL-4059

MEMORY ERROR (EVB PAINT COND FILE)

An error was detected at memory check.

The memory for the EVB paint condition file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



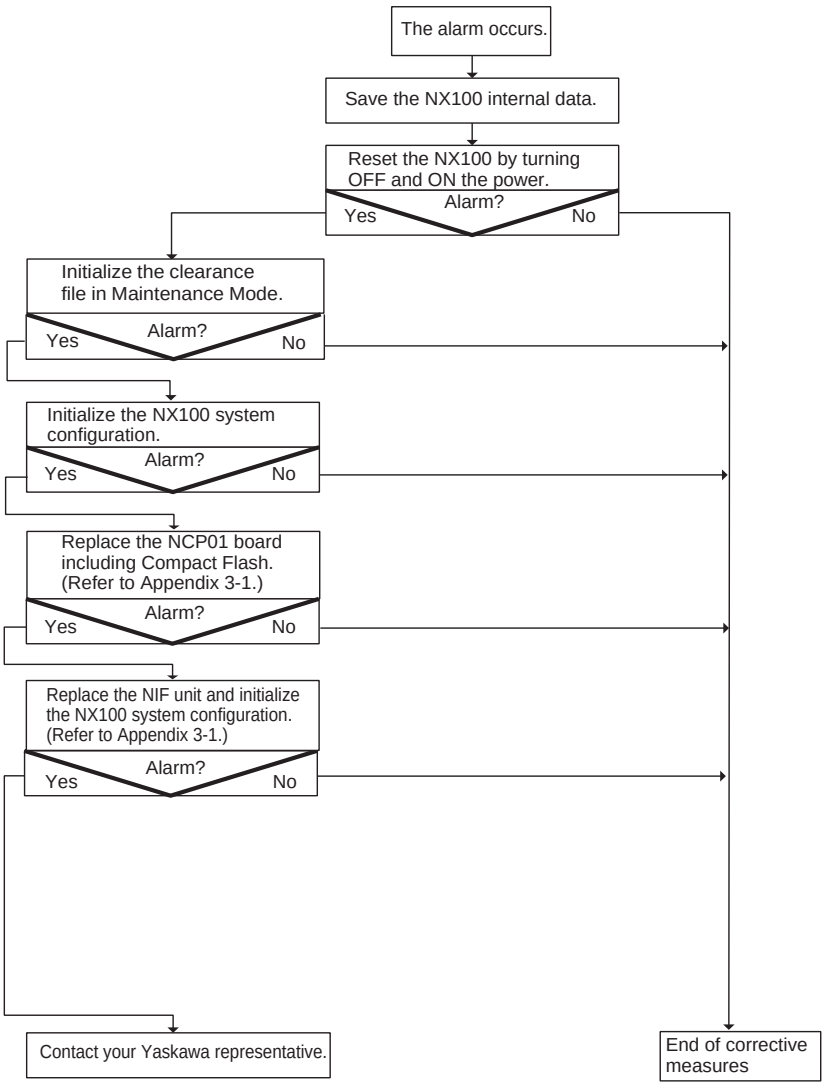
■ AL-4060

MEMORY ERROR (CLEARANCE FILE)

An error was detected at memory check.

The memory for the clearance file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



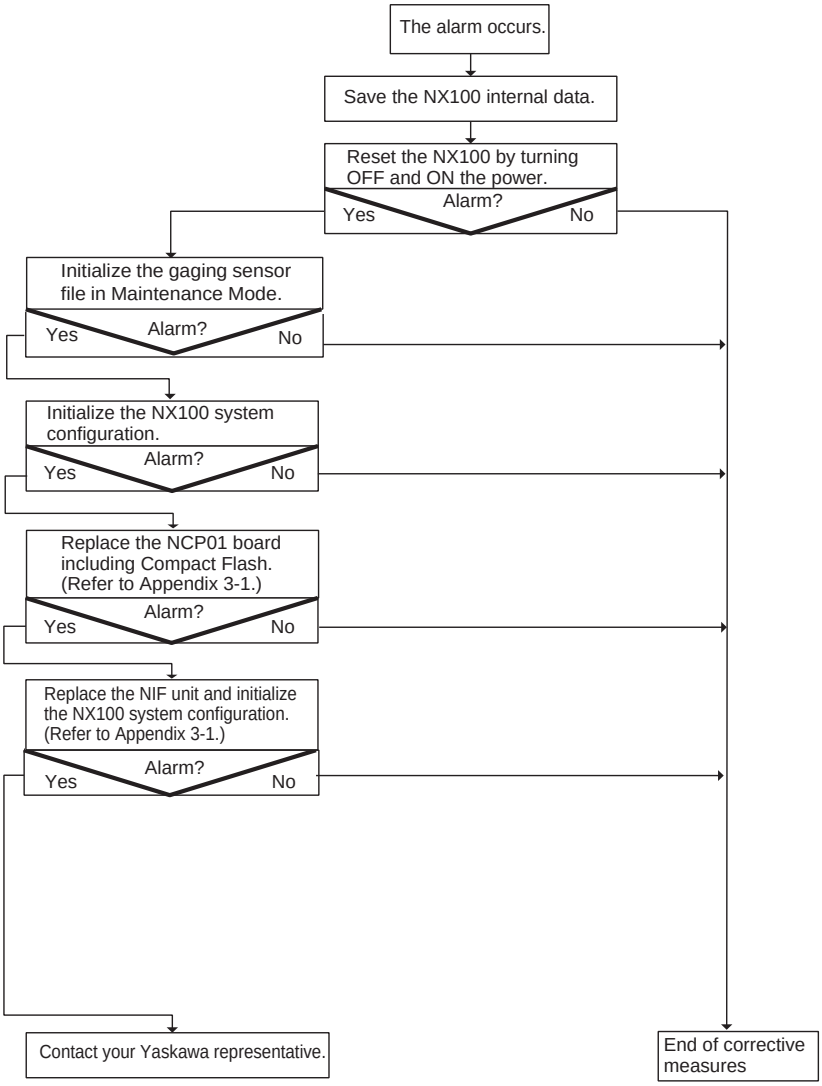
■ AL-4061

MEMORY ERROR (GAGING SONSOR FILE)

The memory for the gaging sensor file is damaged.

An error was detected at memory check.

Sub Code: Signifies the file number in which the alarm occurred.



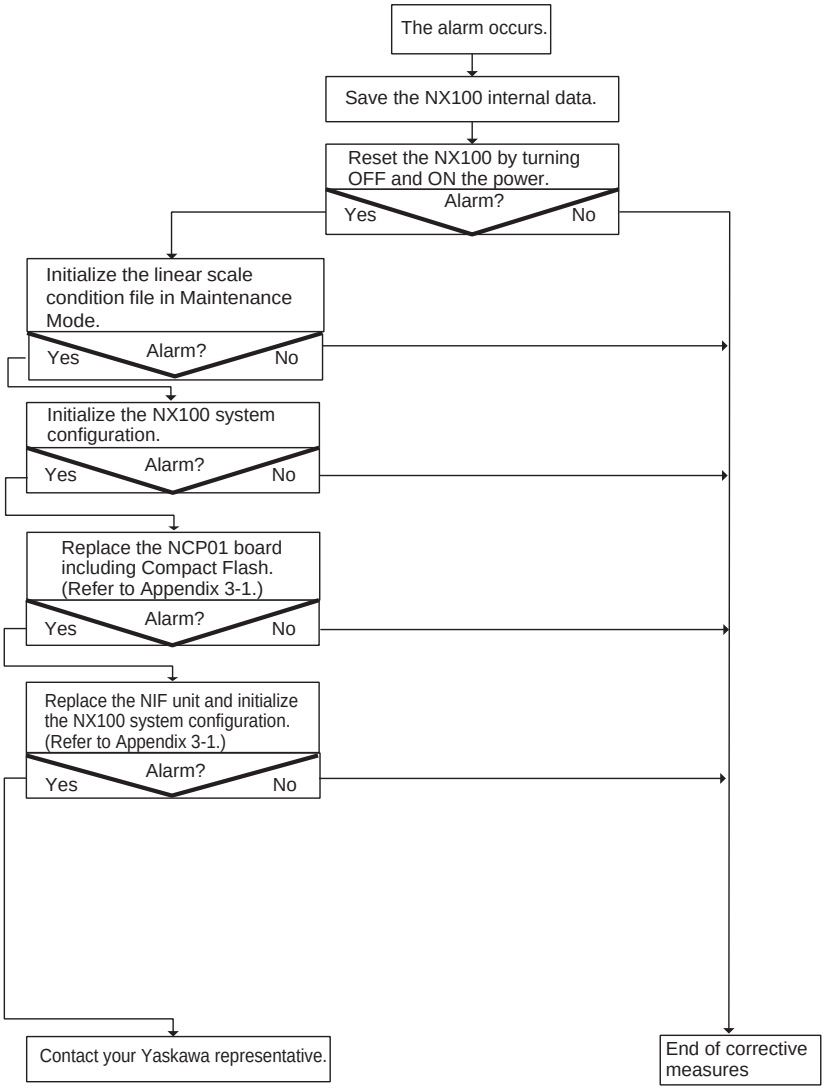
■ AL-4062

MEMORY ERROR (LINEAR SCALE FILE)

The memory for the linear scale condition file is damaged.

An error was detected at memory check.

Sub Code: Signifies the file number in which the alarm occurred.



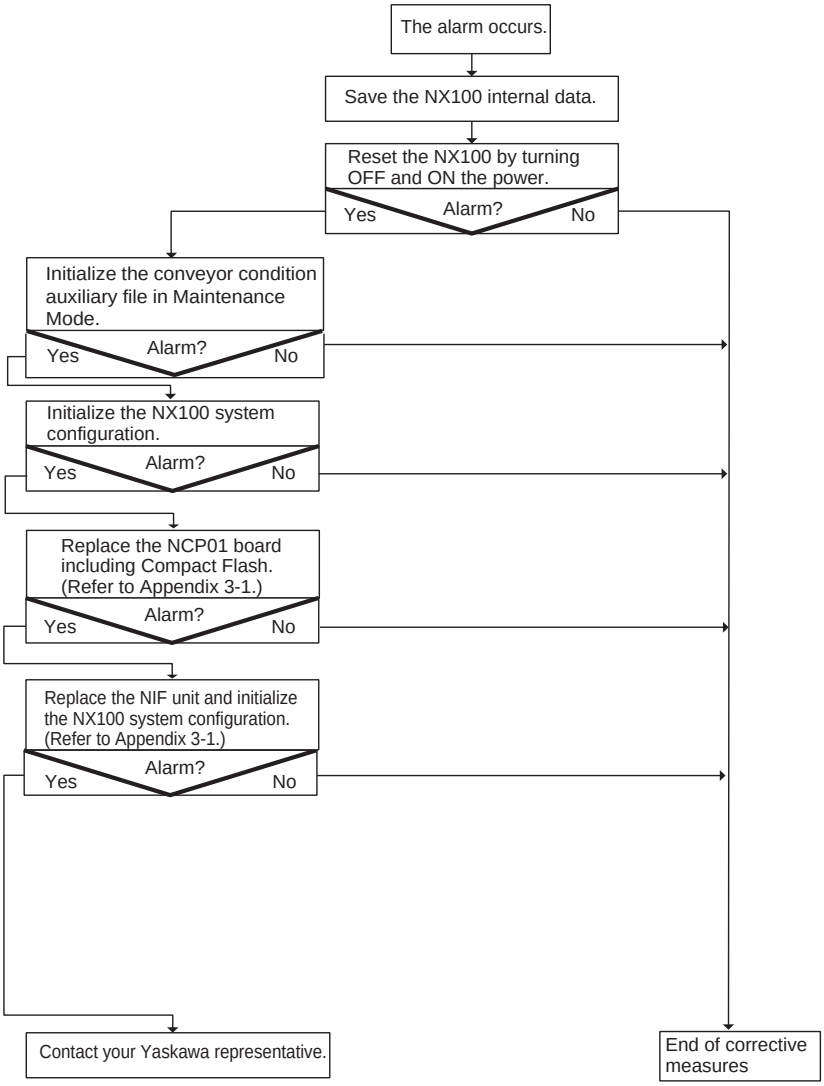
■ AL-4063

MEMORY ERROR (CONVEYOR COND SUPP.)

An error was detected at memory check.

The memory for the conveyor condition auxiliary file is damaged.

Sub Code: Signifies the file number in which the alarm occurred.



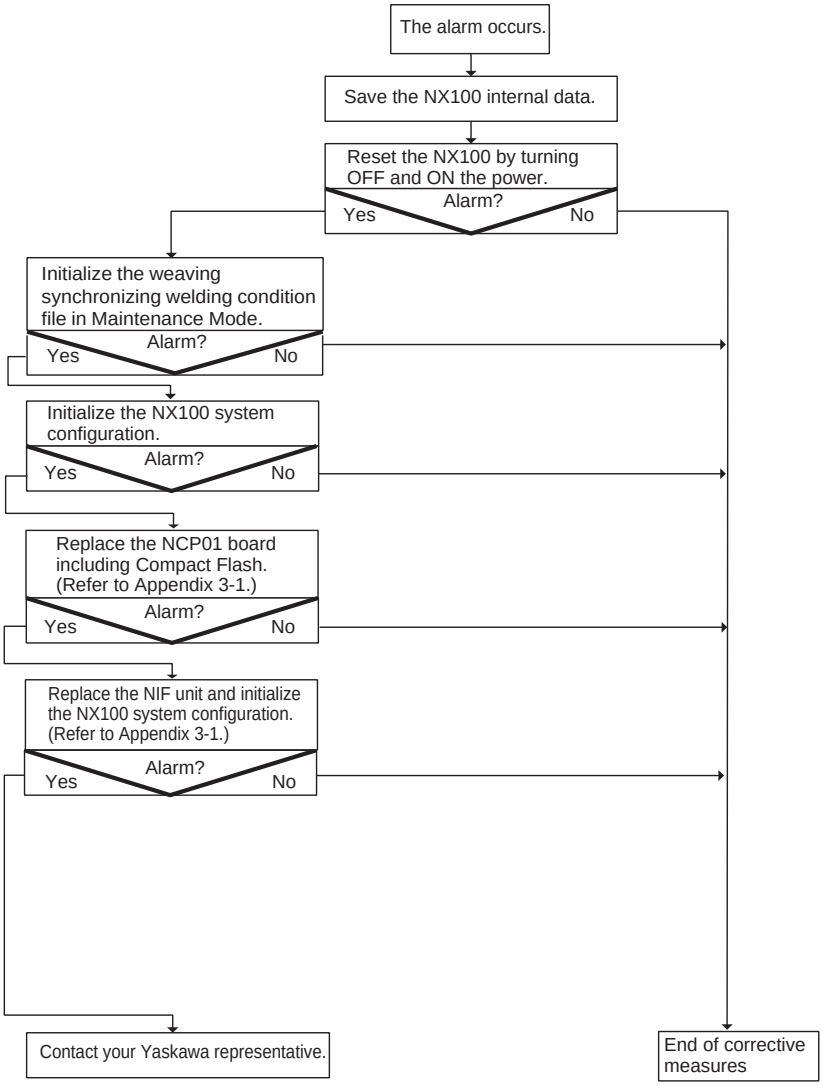
■ AL-4064

MEMORY ERROR (WEAVING SYNCHRONIZING WELD FILE)

An error was detected at memory check.

The memory for the weaving synchronizing welding condition file is damaged.

Sub Code: None



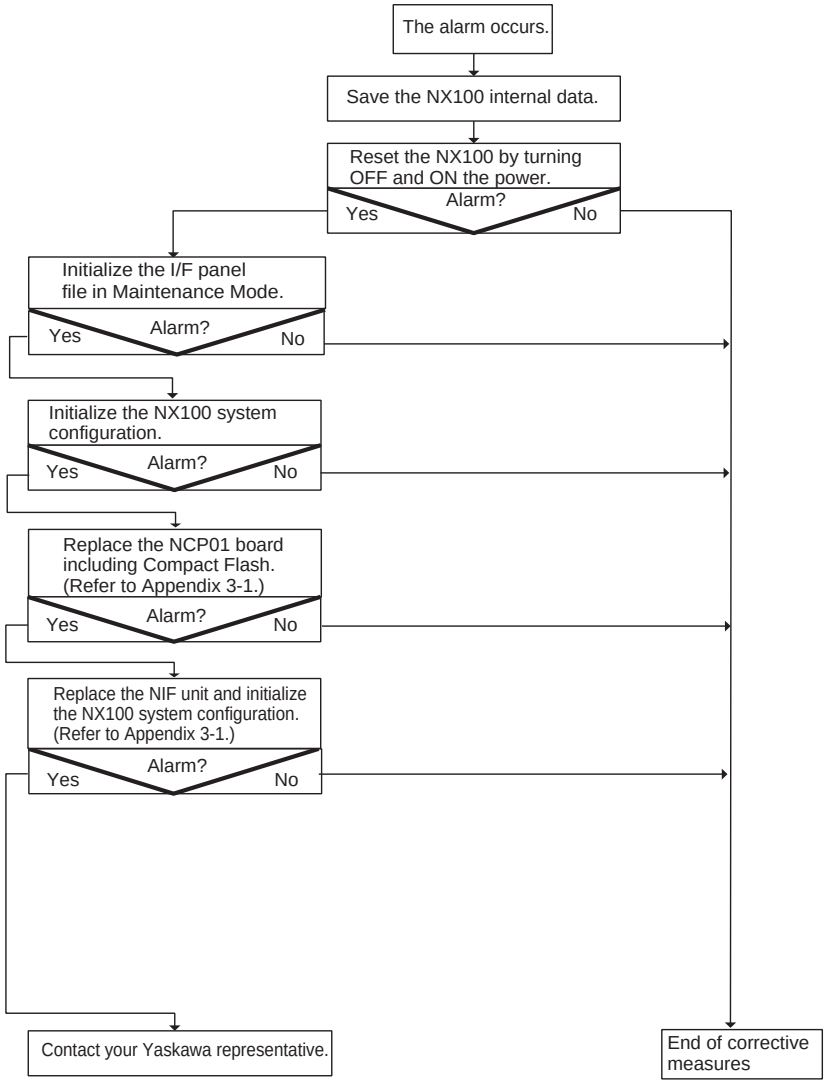
■ AL-4065

MEMORY ERROR (I/F PANEL FILE)

An error was detected at memory check.

The memory for the I/F panel file is damaged.

Sub Code: None



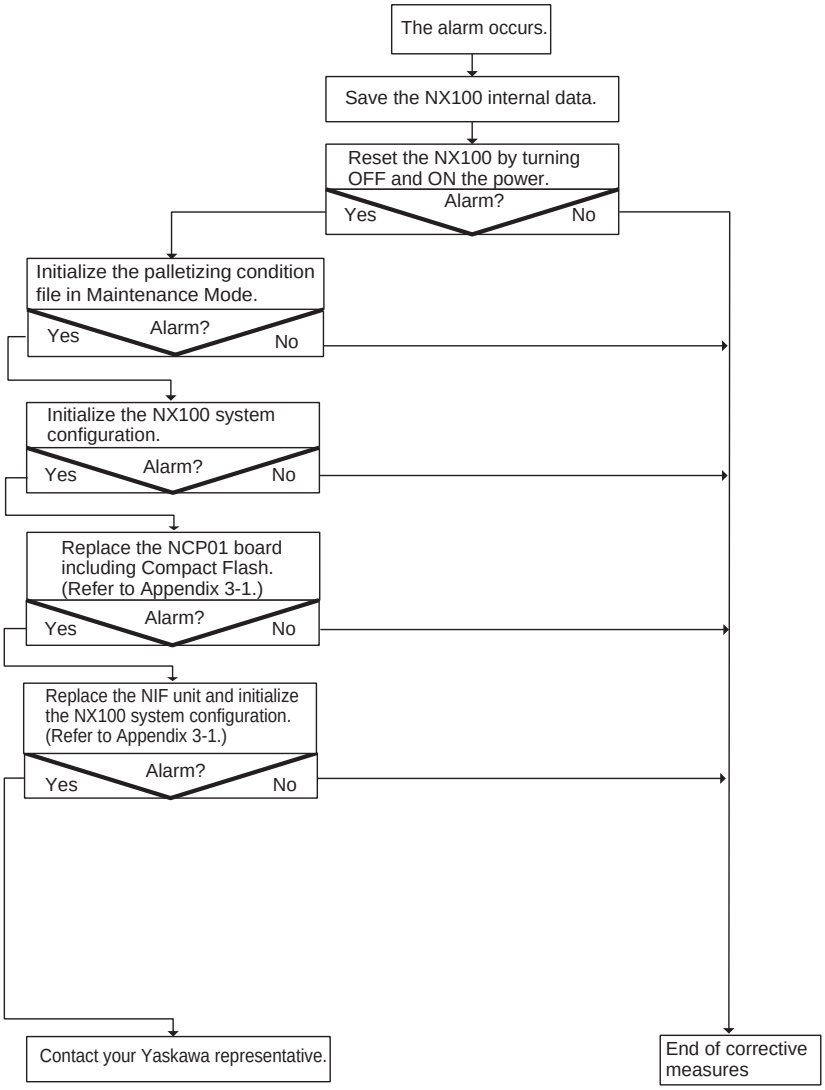
■ AL-4069

MEMORY ERROR (PALLETIZING COND FILE)

An error was detected at memory check.

The memory for the palletizing condition file is damaged.

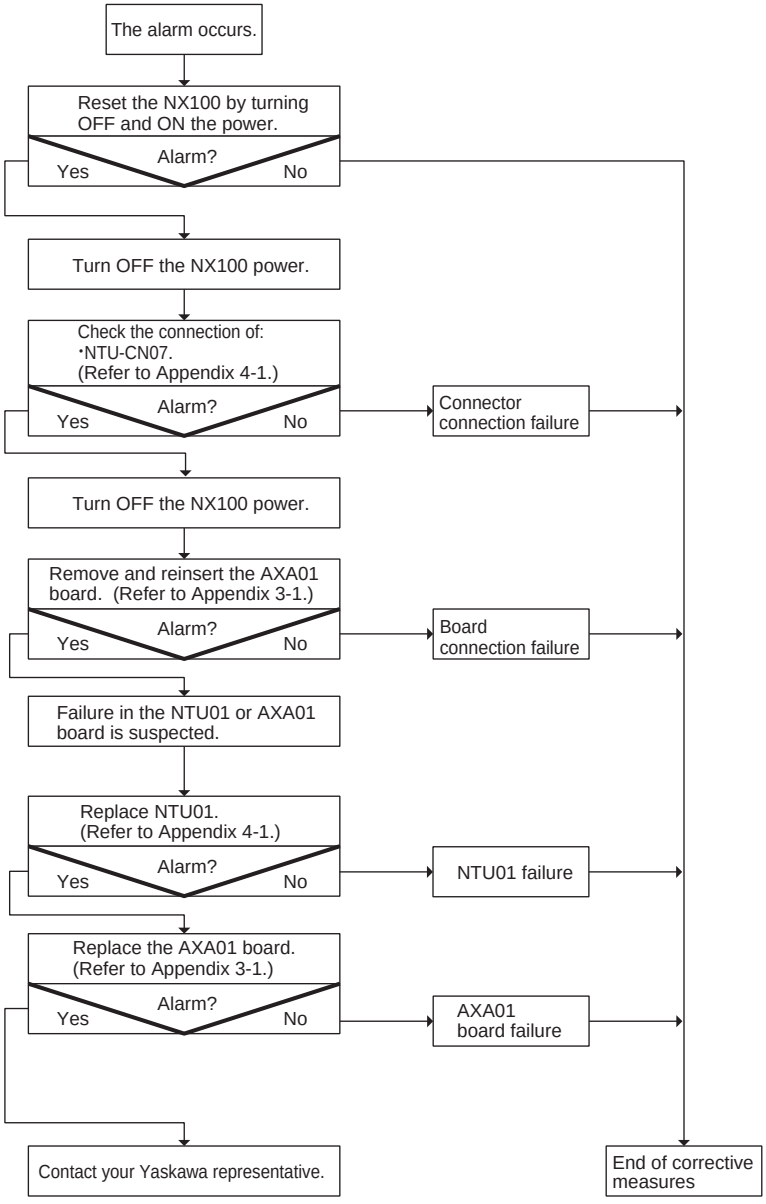
Sub Code: None



■ AL-4100

OVERRUN (ROBOT AXIS)

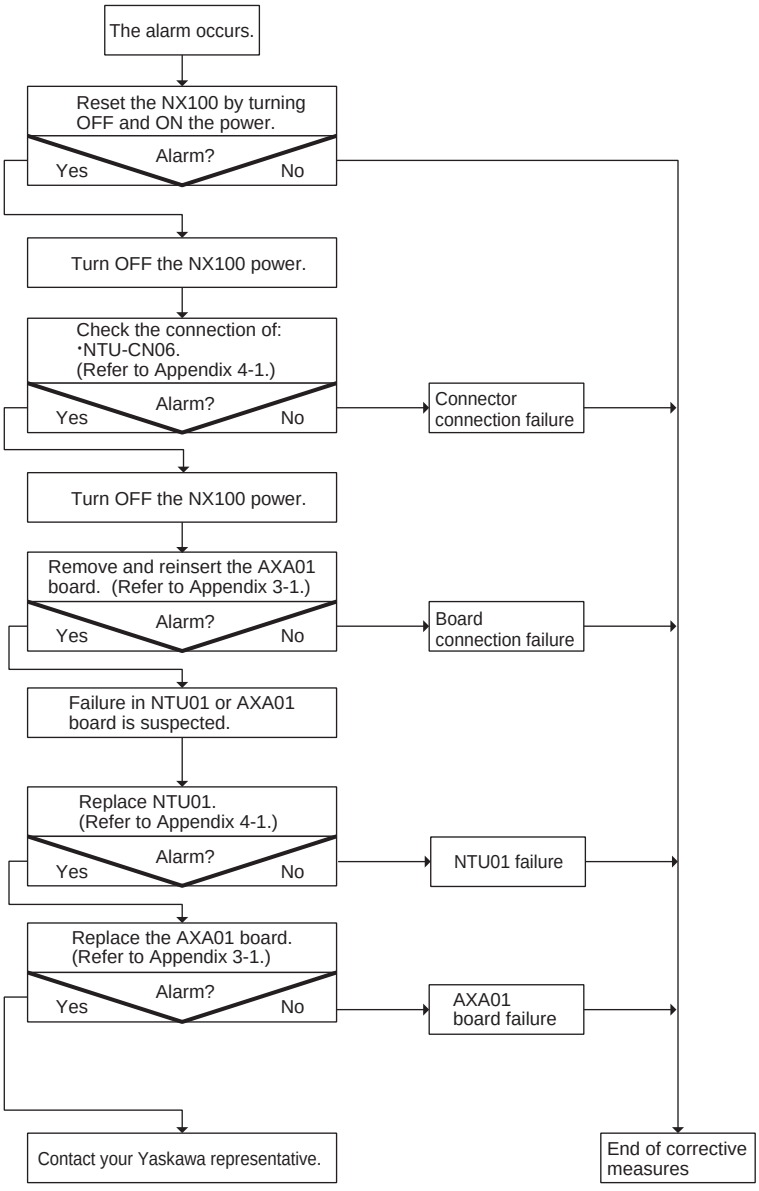
This alarm occurs if the manipulator overrun signal is detected.
Since this overrun signal is short-circuited to disable the alarm before shipment, a cable disconnection or defective board is suspected as the alarm cause.
Sub Code: Signifies the physical No. of contactor in which the alarm occurred



■ AL-4101

OVERRUN (EXTERNAL AXIS)

This alarm occurs if the overrun signal of external axis is detected.
Since this overrun signal is short-circuited to mask the alarm before shipment, a defective cable or board is suspected as the alarm cause.
Sub Code: Signifies the physical No. of contactor in which the alarm occurred



■ AL-4102

SYSTEM DATA HAS BEEN CHANGED
The NX100 detected that the servo power supply was turned ON after data (parameter) which affecting the system was changed.
Turn the NX100 controller ON again.
Sub Code: 1

■ AL-4103

PARALLEL START INSTRUCTION ERROR
An error was detected in the independent control startup operation.
As a cause of the alarm, the operation timing or setting failure of sub task in the independent control function is suspected.
Sub Code:

1: Sub task being executed
2: Group axis being used
3: Multiple start of same job
4: Unregistered master job
5: Synchronization instruction error
6: Stopped by an alarm

■ AL-4104

WRONG EXECUTION OF LOAD INST
An error was detected when a load instruction (LOADV, LOADJ) was executed by the DCI function.
* For details and countermeasures, refer to NX100 OPTIONS INSTRUCTIONS FOR DATA TRANSMISSION FUNCTION(RE-CKI-A445).
Sub Code:

1 to 245: Codes that indicate the details of error.

■ AL-4105

WRONG EXECUTION OF SAVE INST
An error was detected when a save instruction (SAVEV, SAVEJ) was executed by the DCI function.
* For details and countermeasures, refer to NX100 OPTIONS INSTRUCTIONS FOR DATA TRANSMISSION FUNCTION(RE-CKI-A445).
Sub Code:

1 to 245: Codes that indicate the details of error.

■ AL-4106

WRONG EXECUTION OF DELETE INST
An error was detected when a delete instruction (DELETEJ) was executed by the DCI function.
* For details and countermeasures, refer to NX100 OPTIONS INSTRUCTIONS FOR DATA TRANSMISSION FUNCTION(RE-CKI-A445).
Sub Code:

1 to 245: Codes that indicate the details of error.

■ AL-4107

OUT OF RANGE (ABSO DATA)

The position difference between when the power was turned OFF and when the power was turned ON again exceeded the tolerance at one axis in the manipulator or station.

Move the manipulator or station to the zero position by the axis operation and check the home position alignment marks (the arrow).

Sub Code: Signifies the axis in which the alarm occurred

■ AL-4109

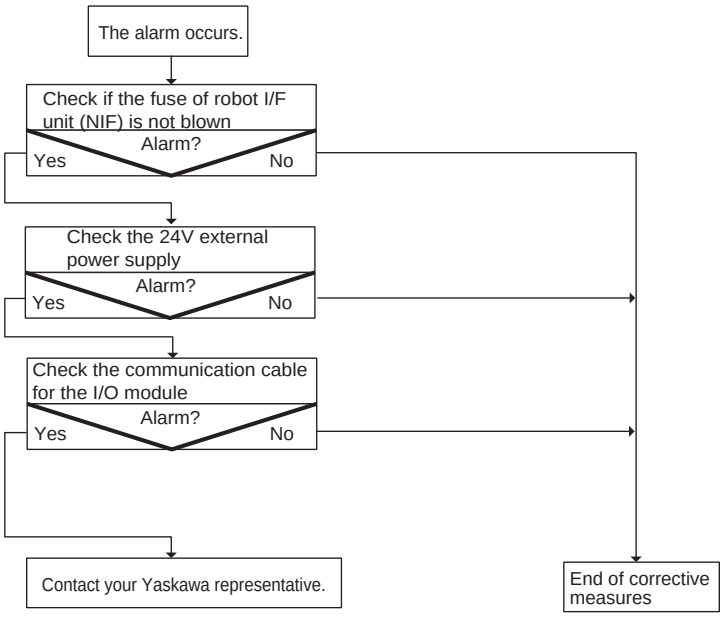
DC 24V POWER SUPPLY FAILURE (I/O)

The external 24v power for I/Os is not output.

Sub Code:

0000_0001: Fuse blown (NIF01unit)

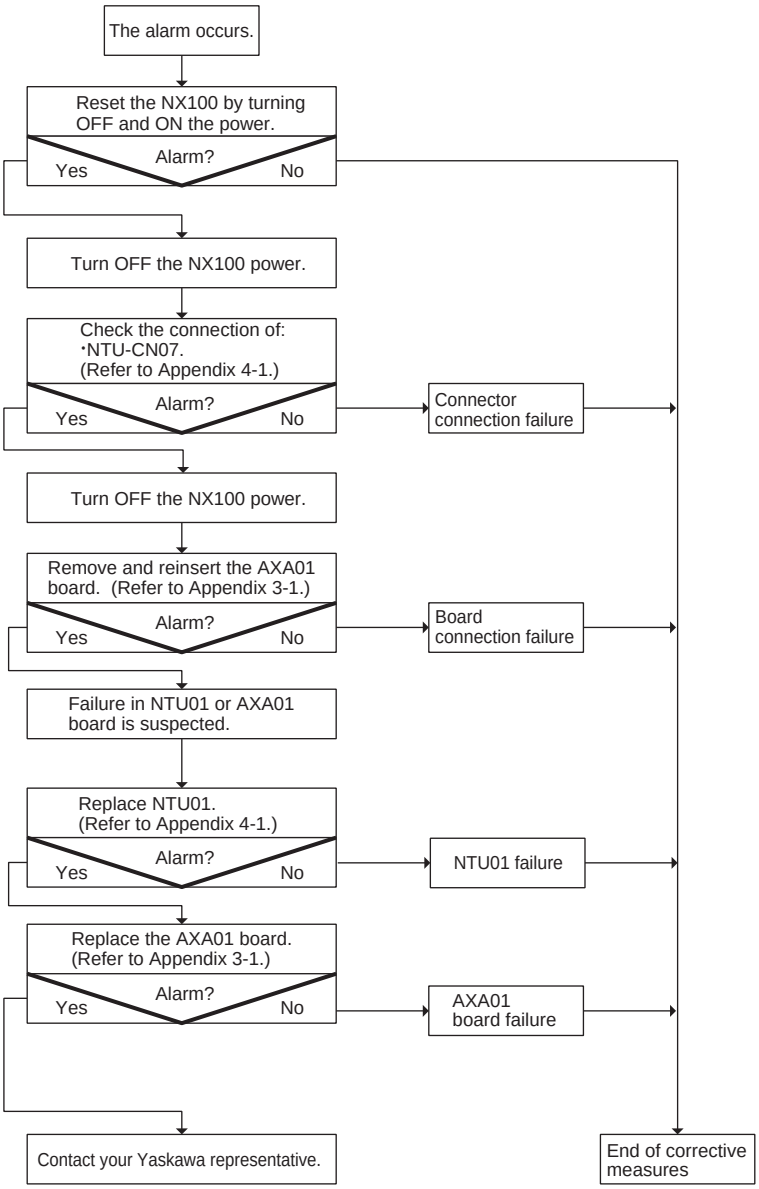
0000_0011: External 24V power supply error.



■ AL-4110

SHOCK SENSOR ACTIVATION

This alarm occurs if the shock sensor signal is detected.
Since this overrun signal is short-circuited to mask the alarm before shipment, a defective cable or board is suspected as the alarm cause.
If the shock sensor connected to the robot is active, clear it by referring to the manual.
If the shock sensor is not active, check the situation with the following procedure.
Sub Code: None



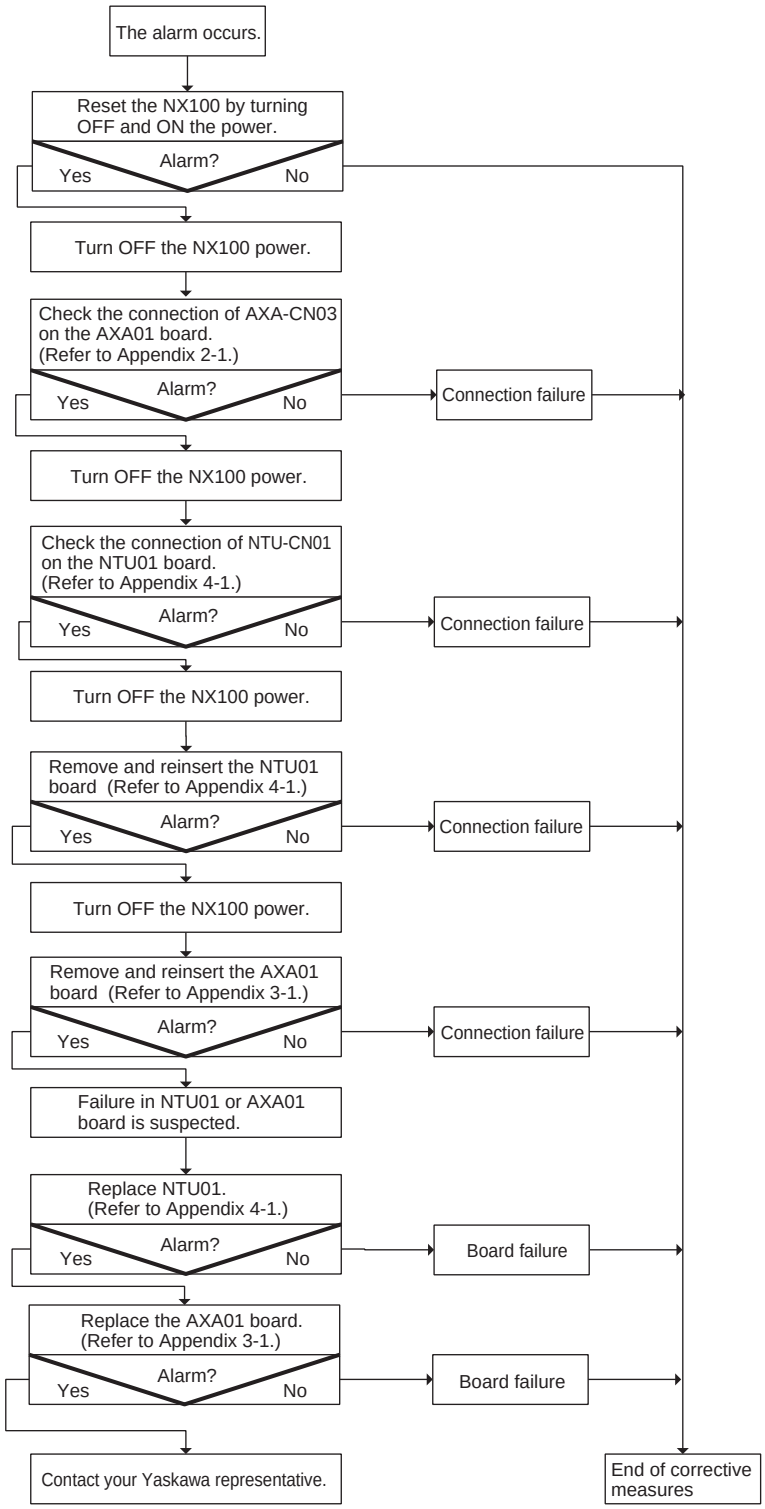
■ AL-4111

BRAKE FUSE BREAKDOWN

The NX100 system checks the brake output relay fuse of NTU01. This alarm occurs if the brake fuse blows out.

As a cause of the alarm, loose connection or noise interference is suspected as well as failure in the NTU01 board.

Sub Code: Signifies the physical No. of contactor in which the alarm occurred



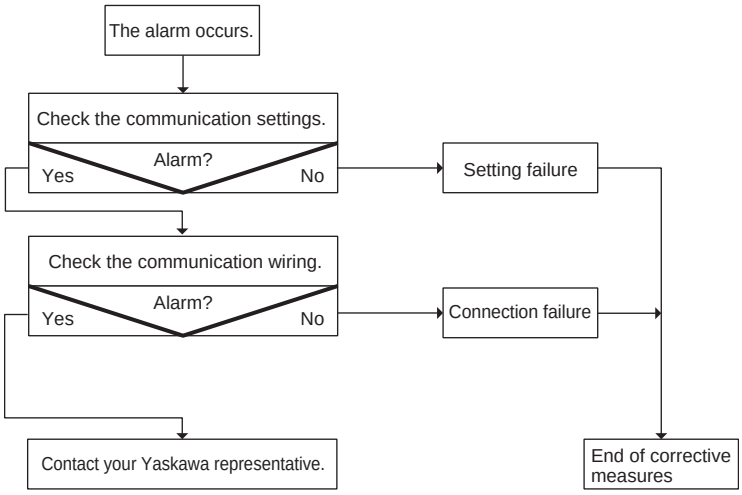
■ AL-4112

DATA SENDING ERROR

A sending error occurred during data transmission.

Sub Code:

1: Retry over of NAK
2: Retry over for timeout in timer A
3: Retry over for mutual response error



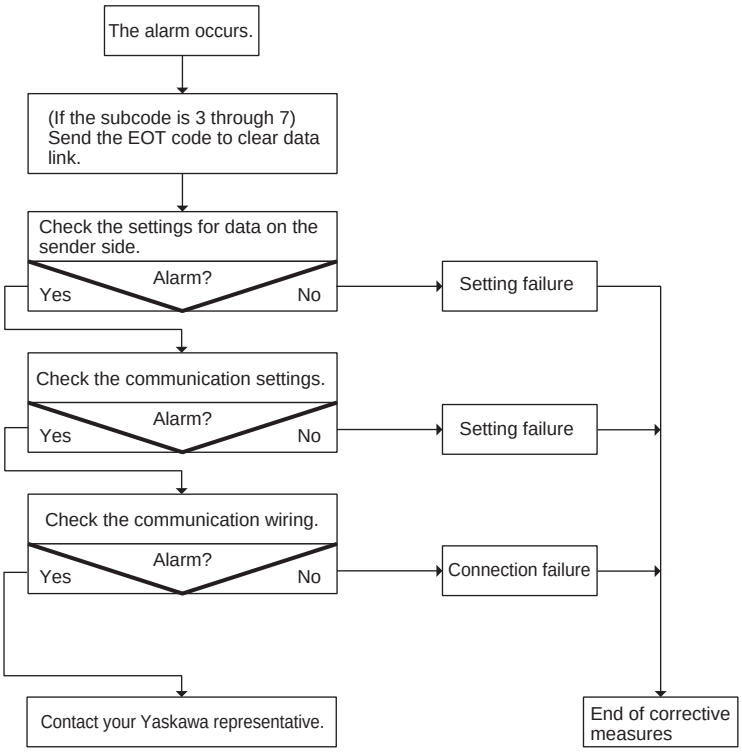
■ AL-4113

DATA RECEIVING ERROR

A receiving error occurred during data transmission.

Sub Code:

1: Reception timeout (timer A)
2: Reception timeout (timer B)
3: Heading length is too short.
4: Heading length is too long.
5: The header No. error
6: The text length exceeded 256 characters.
7: Illegal data received.



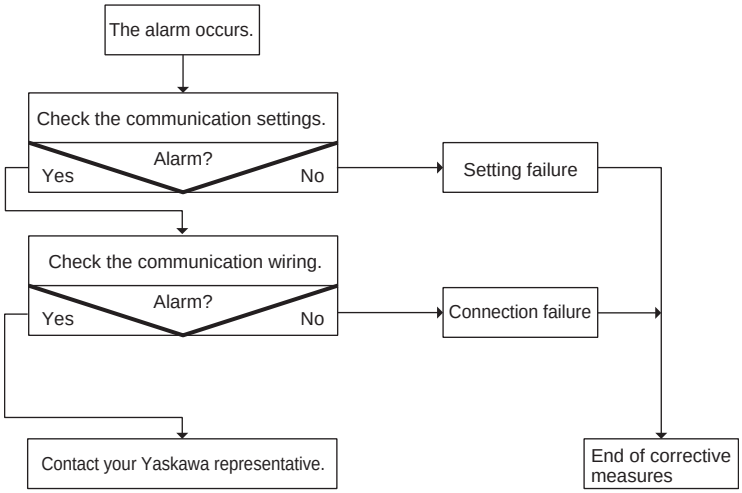
■ AL-4114

TRANSMISSION HARDWARE ERROR

A hardware-related error occurred during data transmission.

Sub Code:

1: Overrun error
2: Parity error
3: Framing error
4: Transmission timeout (timer A)
5: Transmission timeout (timer B)



■ AL-4115

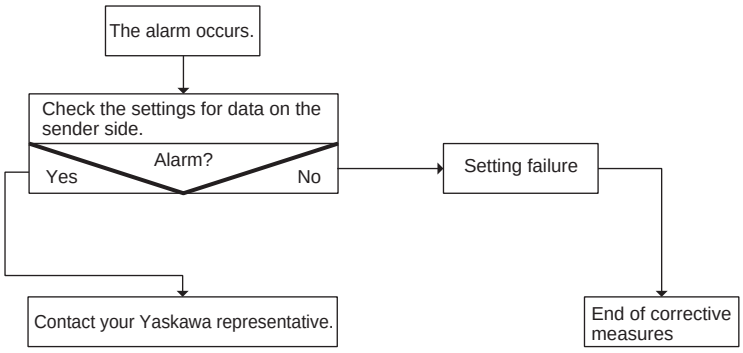
TRANSMISSION SYSTEM BLOCK

An error occurred during data transmission.

(This alarm occurs when received data cause inconsistency on the system although the transmission protocol is correct. Mainly, this alarm occurs due to an illegal transmission or erroneous report at the data sending side.)

Sub Code:

1: Received EOT while waiting ACK.
2: Received EOT while waiting ENQ.
3: Received EOT before last block reception.
4: Received codes other than EOT after last block reception.



■ AL-4116

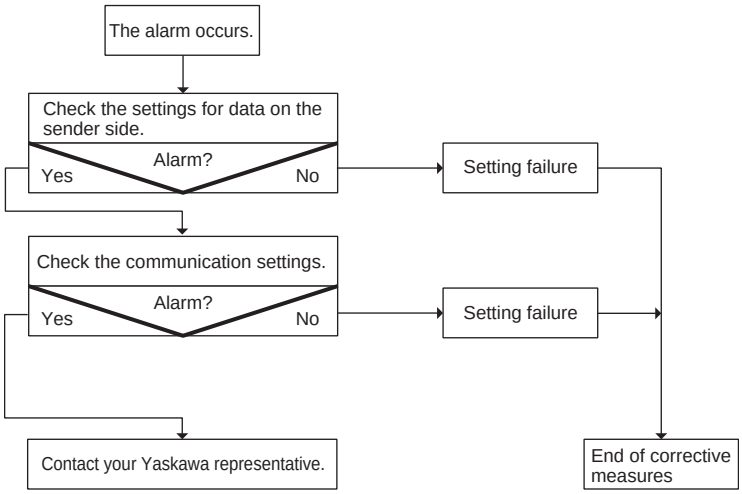
TRANSMISSION SYSTEM ERROR

An error occurred during data transmission.

Sub Code:

1: Sending data contents error

100: Trans error or protocol error



■ AL-4117

BRAKE POWER ERROR

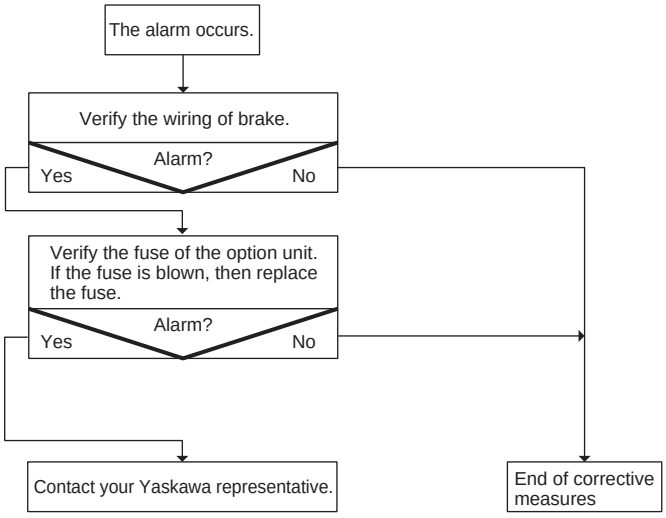
One of the following errors occurred.

- The power for brake is not output.
- An error occurred in control power supply (CPS-420F) due to ground fault or short circuit of the brake wiring.
- The fuse of optional unit (NBP□□) or brake release unit (NBU□□) is blown.

Sub Code: XY (Signifies the servo board No. and power-ON unit No.)

X : Servo board No.
1: Servo board1 (SV#1)
2: Servo board2 (SV#2)
3: Servo board3 (SV#3)
4: Servo board4 (SV#4)

Y : Power-ON unit No.
1: Power-ON unit1 (TU#1)
2: Power-ON unit2 (TU#2)
3: Power-ON unit3 (TU#3)
4: Power-ON unit4 (TU#4)
5: Power-ON unit5 (TU#5)
6: Power-ON unit6 (TU#6)



■ AL-4118

FAN CIRCUIT PROTECTOR TRIPPED

The circuit protector of the in-panel cooling fan is tripped due to ground fault or short circuit of the in-panel cooling fan, or defective fan.

Note: This alarm is displayed approximately one minute after the detection.

Verify the wiring of in-panel cooling fan.

If there is no defect in the wiring, replace the cooling fan since the fan may be defective.

Sub Code: XY (Signifies the servo board No. and power-ON unit No.)

X : Servo board No.
- 1: Servo board1 (SV#1)
- 2: Servo board2 (SV#2)
- 3: Servo board3 (SV#3)
- 4: Servo board4 (SV#4)

Y : Power-ON unit No.
- 1: Power-ON unit1 (TU#1)
- 2: Power-ON unit2 (TU#2)
- 3: Power-ON unit3 (TU#3)
- 4: Power-ON unit4 (TU#4)
- 5: Power-ON unit5 (TU#5)
- 6: Power-ON unit6 (TU#6)

■ AL-4119

FAN ERROR (IN CONTROL BOX): [XY]

As a cause of this alarm, failure in cooling fan is suspected; AL-1200 occurs if the fan for power supply malfunctions, and if the other fan malfunctions, AL-4119 may occur.
AL-4119 FAN ERROR (IN CONTROL BOX) puts out a signal #50017 to activate the alarm 10 hours after the fan malfunction.
Related Alarm: AL-1200
AL-1514
AL-4340

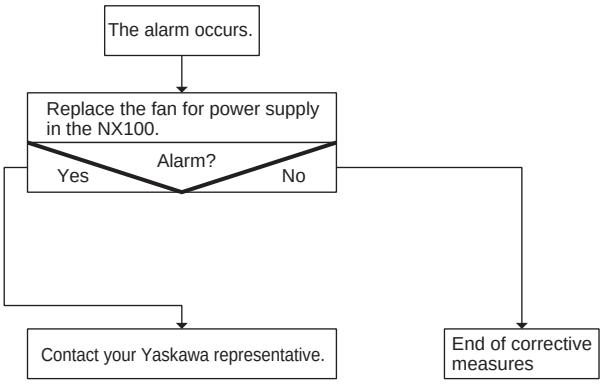
Sub Code: XY (Signifies the servo board No. and power-ON unit No.)

X: Servo board

1: Servo board 1 (SV#1)
2: Servo board 2 (SV#2)
3: Servo board 3 (SV#3)
4: Servo board 4 (SV#4)

Y: Power-ON unit

1: Power-ON unit1 (TU#1)
2: Power-ON unit2 (TU#2)
3: Power-ON unit3 (TU#3)
4: Power-ON unit4 (TU#4)
5: Power-ON unit5 (TU#5)
6: Power-ON unit6 (TU#6)



■ AL-4121

COOLING FAN1 ERROR

The rotation speed of the cooling fan 1 with the alarm sensor connected to the power-ON unit decreased.

If the rotation speed of cooling fan 1 remains slow for 45 seconds, signal# 50017 is turned ON and the message, "Cooling fan 1 stopped," is displayed or the "AL-4121 COOLING FAN1 ERROR" occurs.

As a cause of the alarm, a failure in the cooling fan 1 or wiring failure between the cooling fan 1 and the power-ON unit is suspected.

Signal #50017: cooling fan error status output

Sub Code: XY (Signifies the servo board No. and power-ON unit No.)

X : Servo board No.
- 1: Servo board1 (SV#1)
- 2: Servo board2 (SV#2)
- 3: Servo board3 (SV#3)
- 4: Servo board4 (SV#4)

Y : Power-ON unit No.
- 1: Power-ON unit1 (TU#1)
- 2: Power-ON unit2 (TU#2)
- 3: Power-ON unit3 (TU#3)
- 4: Power-ON unit4 (TU#4)
- 5: Power-ON unit5 (TU#5)
- 6: Power-ON unit6 (TU#6)

■ AL-4122

COOLING FAN2 ERROR

The rotation speed of the cooling fan 2 with the alarm sensor connected to the power-ON unit decreased.

If the rotation speed of cooling fan 2 remains slow for 45 seconds, signal# 50017 is turned ON and the message, "Cooling fan 2 stopped," is displayed or the "AL-4122 COOLING FAN2 ERROR" occurs.

As a cause of the alarm, a failure in the cooling fan 2 or wiring failure between the cooling fan 2 and the power-ON unit is suspected.

Signal #50017: cooling fan error status output

Sub Code: XY (Signifies the servo board No. and power-ON unit No.)

X : Servo board No.
- 1: Servo board1 (SV#1)
- 2: Servo board2 (SV#2)
- 3: Servo board3 (SV#3)
- 4: Servo board4 (SV#4)

Y : Power-ON unit No.
- 1: Power-ON unit1 (TU#1)
- 2: Power-ON unit2 (TU#2)
- 3: Power-ON unit3 (TU#3)
- 4: Power-ON unit4 (TU#4)
- 5: Power-ON unit5 (TU#5)
- 6: Power-ON unit6 (TU#6)

■ AL-4123

COOLING FAN3 ERROR

The rotation speed of the cooling fan 3 with the alarm sensor connected to the power-ON unit decreased.

If the rotation speed of cooling fan 3 remains slow for 45 seconds, signal# 50017 is turned ON and the message, "Cooling fan 3 stopped," is displayed or the "AL-4123 COOLING FAN3 ERROR" occurs.

As a cause of the alarm, a failure in the cooling fan 3 or wiring failure between the cooling fan 3 and the power-ON unit is suspected.

Signal #50017: cooling fan error status output

Sub Code: XY (Signifies the servo board No. and power-ON unit No.)

X : Servo board No.
- 1: Servo board1 (SV#1)
- 2: Servo board2 (SV#2)
- 3: Servo board3 (SV#3)
- 4: Servo board4 (SV#4)

Y : Power-ON unit No.
- 1: Power-ON unit1 (TU#1)
- 2: Power-ON unit2 (TU#2)
- 3: Power-ON unit3 (TU#3)
- 4: Power-ON unit4 (TU#4)
- 5: Power-ON unit5 (TU#5)
- 6: Power-ON unit6 (TU#6)

■ AL-4126

CANNOT EXECUTE AUTO PMT
An error occurred during automatic PMT.

Sub Code: Signifies the error type. Subcodes and remedys are shown below.

Sub code	Contents	Remedy
1:	System error	(1) Reset the alarm, and then re-execute the job. (2) If the alarm occurs again, save CMOS.BIN in maintenance mode, and then inform your Yaskawa representative of the occurrence status (operating procedure).
2:	Editing PBOX is currently prohibited.	Check the input status of editing prohibition signal. Such a signal cannot be input.
3:	Editing the conversion source job is currently prohibited.	Check the editing prohibition status of the conversion source job. While editing of this job is prohibited, it cannot be edited.
4:	Editing the conversion destination job is currently prohibited.	Check the editing prohibition status of the conversion destination job. While editing of this job is prohibited, it cannot be edited.
5:	Insufficient job area memory	(1) Reset the alarm. (2) If this phenomenon occurs again, delete unused jobs. (3) If this phenomenon occurs again after the above countermeasure is taken, initialize jobs in the maintenance mode and reload saved job. Delete unused jobs at this time. (4) If this phenomenon occurs again after the above countermeasure is taken, save CMOS.BIN in the maintenance mode, and then inform your Yaskawa representative of the occurrence status (operating procedure).
6:	There is no conversion source job.	Check if the specified conversion source job exists. Non-existent job cannot be specified as the conversion source job.
7:	Position data memory insufficient	(1) Reset the alarm. (2) If this phenomenon occurs again, delete unnecessary teaching points, if any. (3) If this phenomenon occurs again after the above countermeasure is taken, initialize jobs in the maintenance mode and reload saved job. Delete unused jobs at this time. (4) If this phenomenon occurs again after the above countermeasure is taken, save CMOS.BIN in the maintenance mode, and then inform your Yaskawa representative of the occurrence status (operating procedure).
8:	Conversion of currently executed job is specified.	Check the execution statuses of conversion source job and conversion destination job. A job currently executed has been specified as the conversion source job or conversion destination job. Perform the conversion after the execution of the job is completed.

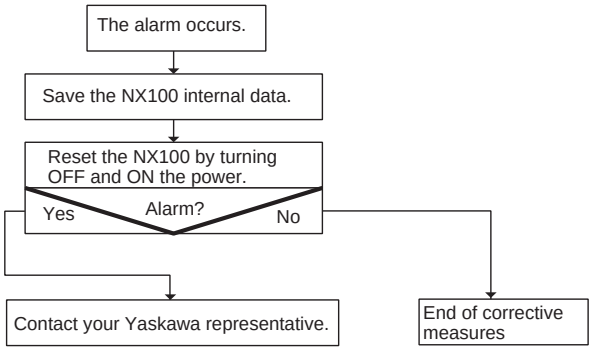
■ AL-4130

NETWORK APPLICATION PROCESS ERROR

An error occurred while the Ethernet function was used.

Sub Code: Signifies the error type

1: APP task re-initialization notification error
2: Re-initialization response receiving error
3: Abnormal termination of incomplete task of re-initialization
4: Synchronous semaphore for re-initialization receiving error
5: Re-initialization mail sending error
6: Area storage management table exclusion error
50: PCI written data size error
51: PCI data writing request reception error
52: Undefined transmission request received
53: Transmission request reception error
54: Transmission request without data received
55: Transmission request with illegal data length received



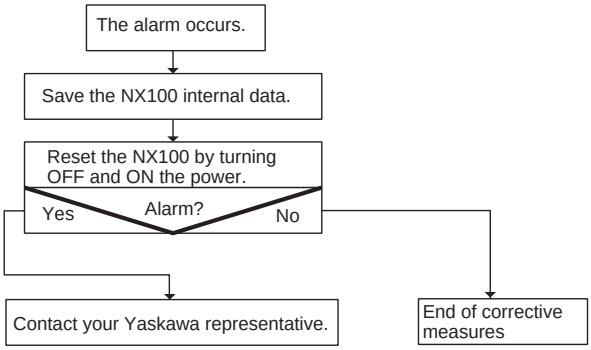
■ AL-4131

UDP PROCESS ERROR

An error occurred in the UDP process of the Ethernet function.

Sub Code: Signifies the error type

- 1: Receiving socket creation error
- 2: Transmission socket creation error
- 3: Received data error
- 4: Transmission error
- 5: Abnormal termination of SELECT operation
- 100: Re-initialization notification data length error
- 101: Re-initialization notification data reception error
- 102: Abnormal termination of PCI writing
- 103: Transmission request data length error
- 104: Transmission request data reception error



■ AL-4132

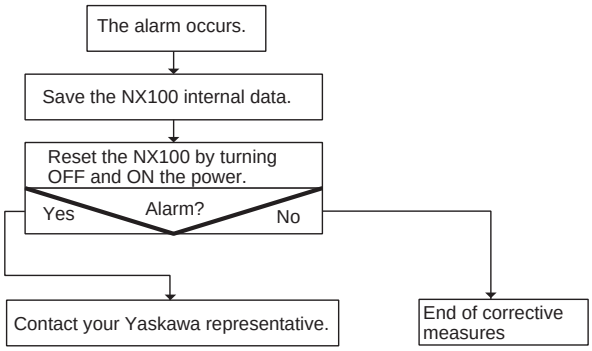
TCP PROCESS ERROR

An error occurred in the TCP process of the Ethernet function.

Sub Code: Signifies the error type

1: Socket cable creation error

2: TCP server initialization error



■ AL-4137

WRONG EXECUTION OF SETUALM INST

An error occurred at SETUALM instruction execution.

Sub Code: Signifies the error type. Subcodes and remedys are shown below.

Sub code	Contents	Remedy
1:	Alarm code specification error	Check the alarm code setting. Specify the alarm in the range 8000 to 8999.
2:	Task specification error	Check the task specification setting. Specify the task in the range 0 to 4 (7 at expansion).
3:	Motion mode specification error	Check the motion mode specification setting. Set the motion mode to 0 or 1.

■ AL-4200

SYSTEM ERROR (FILE DATA)

This alarm occurs if abnormal internal data is detected during the access to file data.
As a major cause of the alarm, system software malfunction is suspected; as another possibility, an internal RAM data error also may cause the alarm.

Sub Code

01 to 23: Signifies the internal software error

■ AL-4201

SYSTEM ERROR (JOB)

This alarm occurs if abnormal internal data is detected during the access to the job data of MOTION.
As a major cause of the alarm, system software malfunction is suspected; as another possibility, an internal RAM data error also may cause the alarm.

Sub Code

-1 to -63: Signifies the internal software error

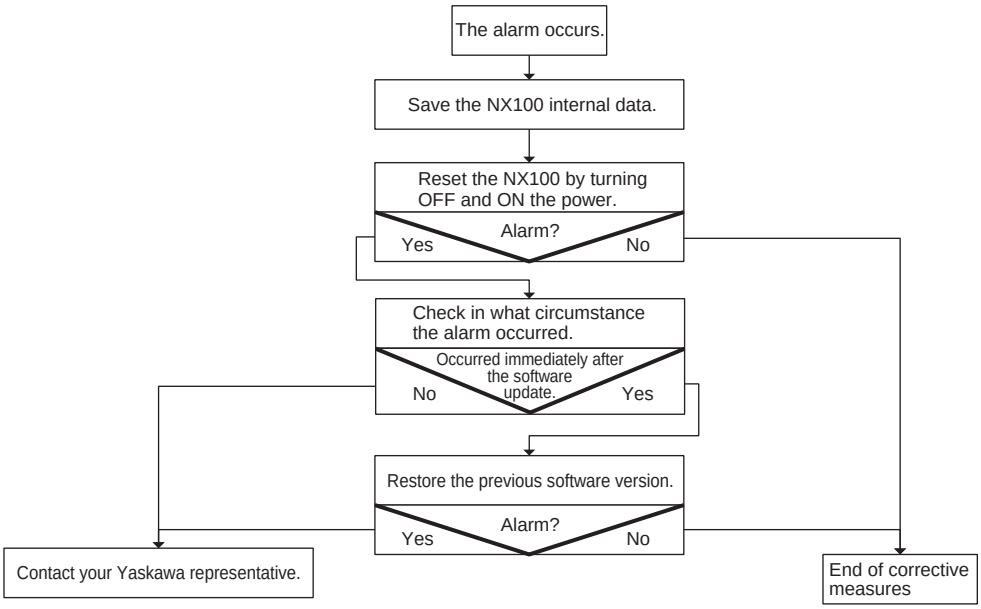
■ AL-4202

SYSTEM ERROR (JOB)

This alarm occurs if abnormal internal data is detected during the access to the job data of operating/editing software.
As a major cause of the alarm, system software malfunction is suspected; as another possibility, an internal RAM data error also may cause the alarm.

Sub Code

01 to 63: Signifies the internal software error



■ AL-4203

SYSTEM ERROR (POSITION DATA)

This alarm occurs if abnormal internal data is detected during the access to the position data of MOTION. As a major cause of the alarm, system software malfunction is suspected; as another possibility, an internal RAM data error also may cause the alarm.

Sub Code

-1 to -11: Signifies the internal software error

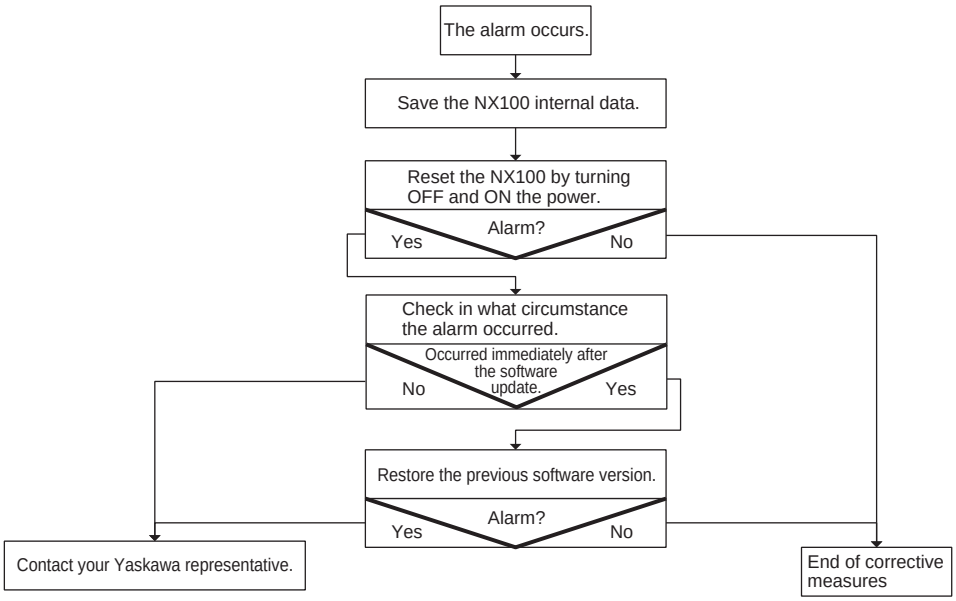
■ AL-4204

SYSTEM ERROR (POSITION DATA)

This alarm occurs if abnormal internal data is detected during the access to position data.
As a major cause of the alarm, system software malfunction is suspected; as another possibility, an internal RAM data error also may cause the alarm.

Sub Code

1 to 11: Signifies the internal software error



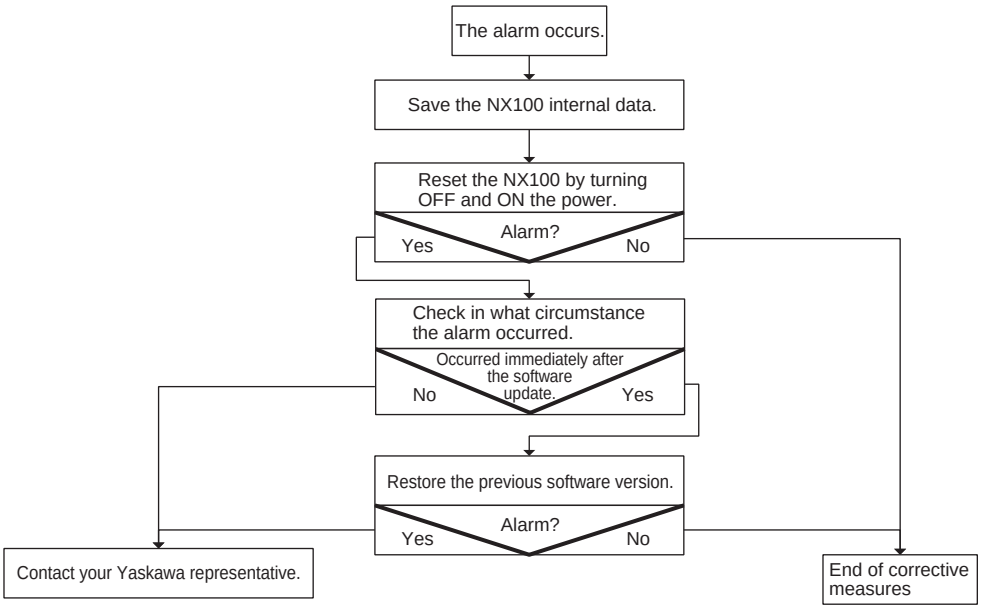
■ AL-4206

SYSTEM ERROR (TRANSMISSION)

This alarm occurs if an error is detected during data transmission.
The RAM data failure in the board may also cause this alarm.

Sub Code:

1 to 4: Signifies the internal software error



■ AL-4207

SYSTEM ERROR (MOTION)
This alarm occurs if an error is detected during MOTION processing.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 2000: Signifies the internal software error

■ AL-4208

SYSTEM ERROR (ARITH)
This alarm occurs if an error is detected during ARITH processing.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 77: Signifies the internal software error

■ AL-4209

OFFLINE SYSTEM ERROR (ARITH)
This alarm occurs if an error is detected during offline ARITH processing.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

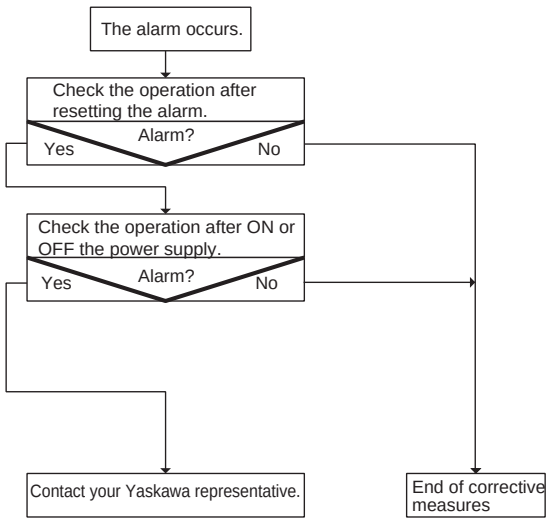
100 to 112: Signifies the internal software error

■ AL-4210

SYSTEM ERROR (LOCAL VARIABLE)
This alarm occurs if abnormal internal data is detected during the access to the local variable of MOTION.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

-1 to -12: Signifies the internal software error

4207 to 4210

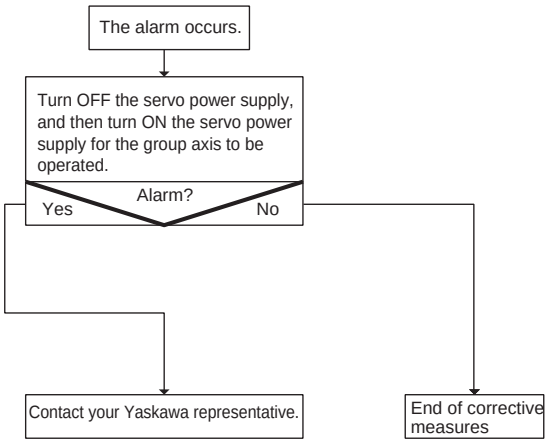


■ AL-4220/4221

SERVO POWER OFF FOR JOB

The servo power is not supplied to the job control group axis to be operated.

Sub Code: Signifies the servo power supply power-OFF control group



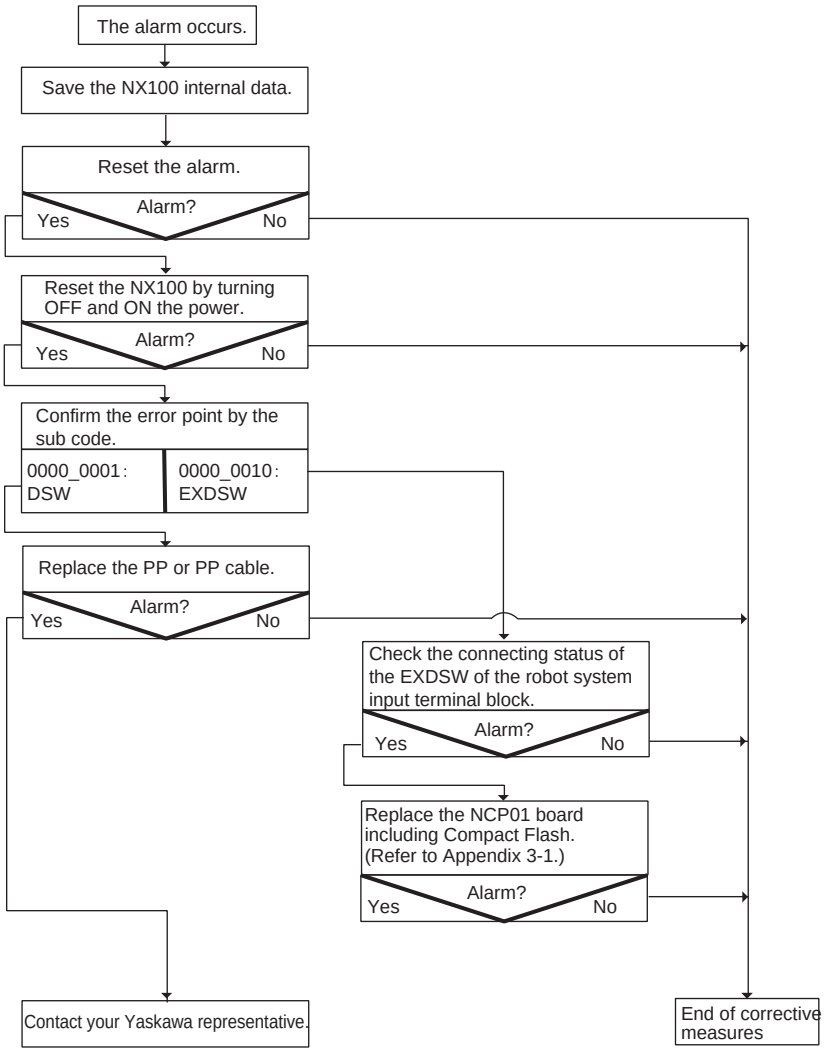
■ AL-4222

SAFETY CIRCUIT IMPUT FAULT (DSW)

This alarm occurs if a dual signal is mutually checking with the result that Enable switch (DSW and EXDSW) I/O signal does not have a match signal as a result the mutual check of a dual signal.

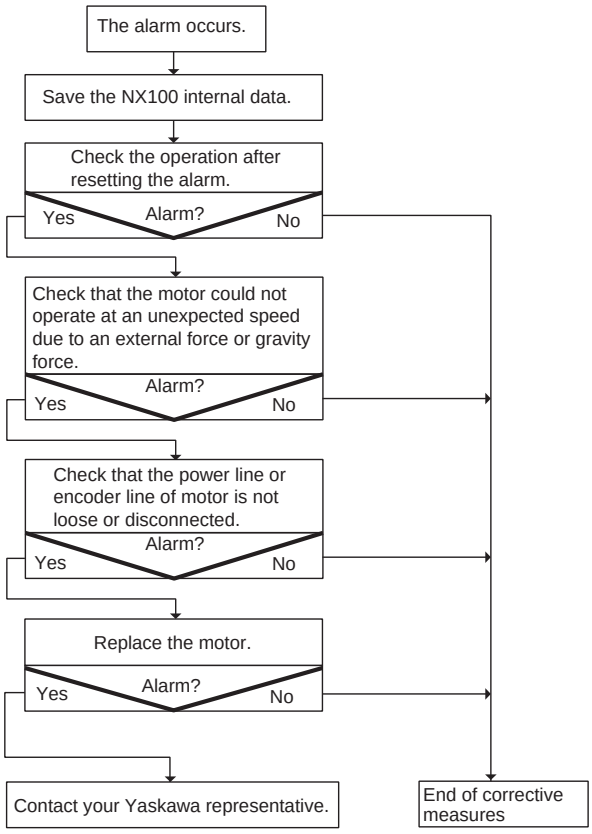
Sub Code

0000_0001: DSW
0000_0011: EXDSW



■ AL-4225

OVER SPEED (NCP01)
The NX100 monitors the actual speed of motor with feedback pulses.
The speed limit is the safe speed in the teach mode and the maximum speed in the play mode.
This alarm occurs if the speed limit is exceeded.
Sub Code: Signifies the control axis name where a speed error is detected



■ AL-4228

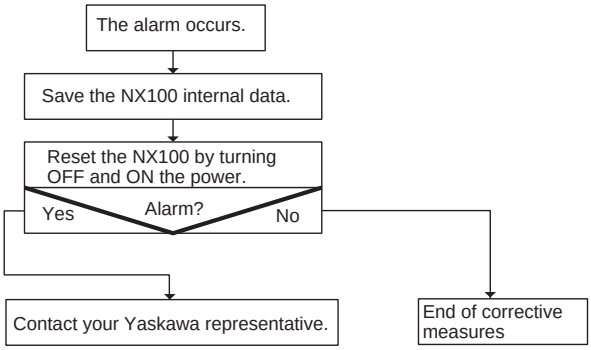
WRONG DATA

The job saves the job instruction file and position file in the different files.
This alarm occurs if it detects the inconsistency between the Job instruction and position file.
The followings are the causes of the inconsistency.
Cause 1: Single position data is chained by the plural Job instruction data. (Overlapped chain)
Cause 2: Job instruction file chaines the unregistered position data. (Unregistered position data chain)
Cause 3: Registered position data is not chained.(Unchained position data)
*For details, refer to "NX100 OPTIONS INSTRUCTIONS FOR JOB DATA SIMPLIFIED RESTORATION FUNCTION (HW 0483260)"

■ AL-4229

ETHERNET PROCESS ERROR

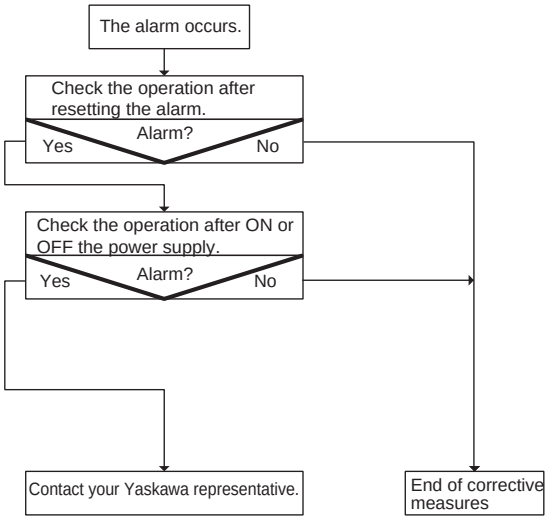
An error occurred while the Ethernet function was used.
Sub Code: None



■ AL-4230

DECELERATION MONITOR OVER

Failed to stop the operation under the hold command.
Sub Code: None



■ AL-4301

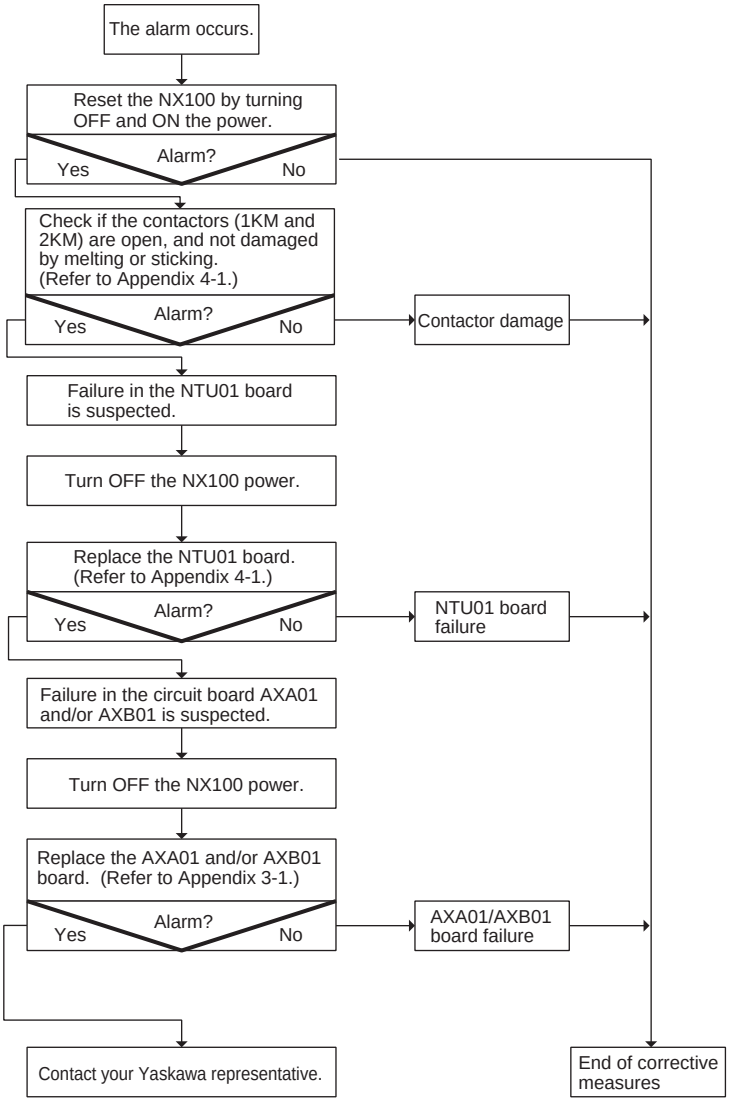
CONTACTOR ERROR

The NX100 system checks the status of the NTU01 contactors. This alarm occurs if there is an inconsistency between the control output and contactor status.
As a cause of the alarm, the NTU01 board failure or loose connection is suspected; in rare cases, the alarm may be caused by a noise interference.

Sub Code: Signifies the physical No. of contactor in which the alarm occurred

NOTE

Before performing a connection check of the wiring, turn OFF the controller power. Make sure that all the LEDs of SERVOPACK and converter are OFF, then verify that no electricity is charged using equipment such as a tester.
This process may take a few minutes after shutting off the power.



■ AL-4302

BRAKE CIRCUIT ERROR

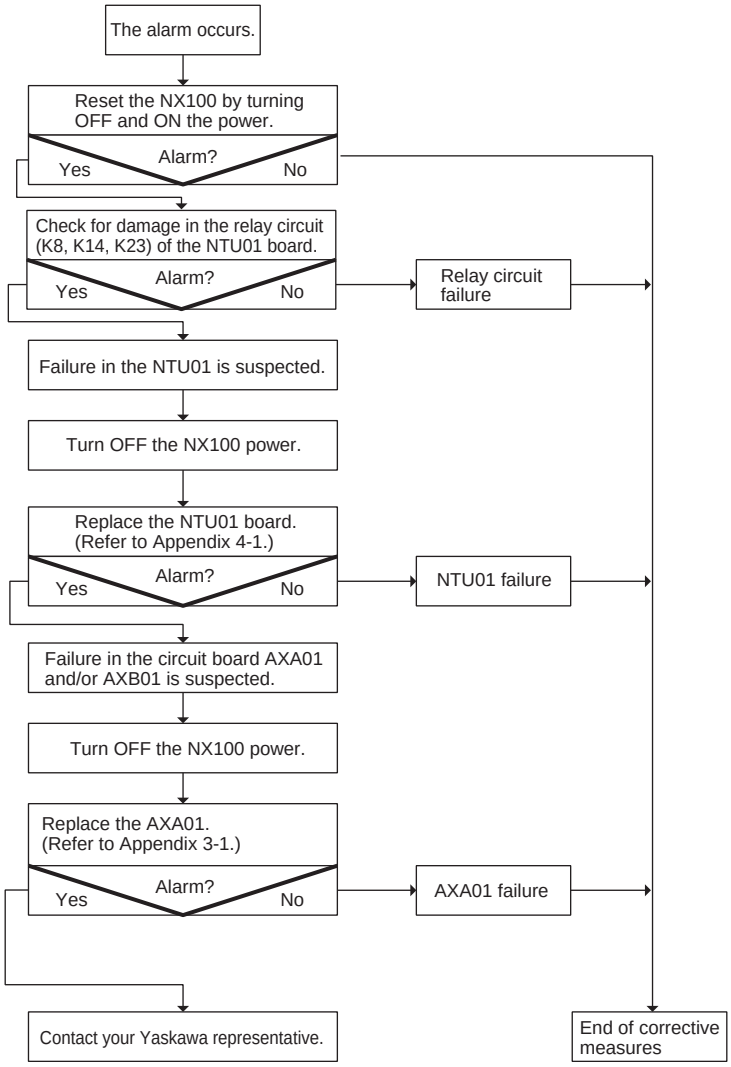
The NX100 system checks the brake status of NTU01. This alarm occurs if there is an inconsistency between the control output and brake status.

As a cause of the alarm, the NTU01 failure is suspected.

Sub Code: Signifies the physical No. of contactor in which the alarm occurred

NOTE

Before performing a connection check of the wiring, turn OFF the controller power. Make sure that all the LEDs of SERVOPACK and converter are OFF, then verify that no electricity is charged using equipment such as a tester.
This process may take a few minutes after shutting off the power.



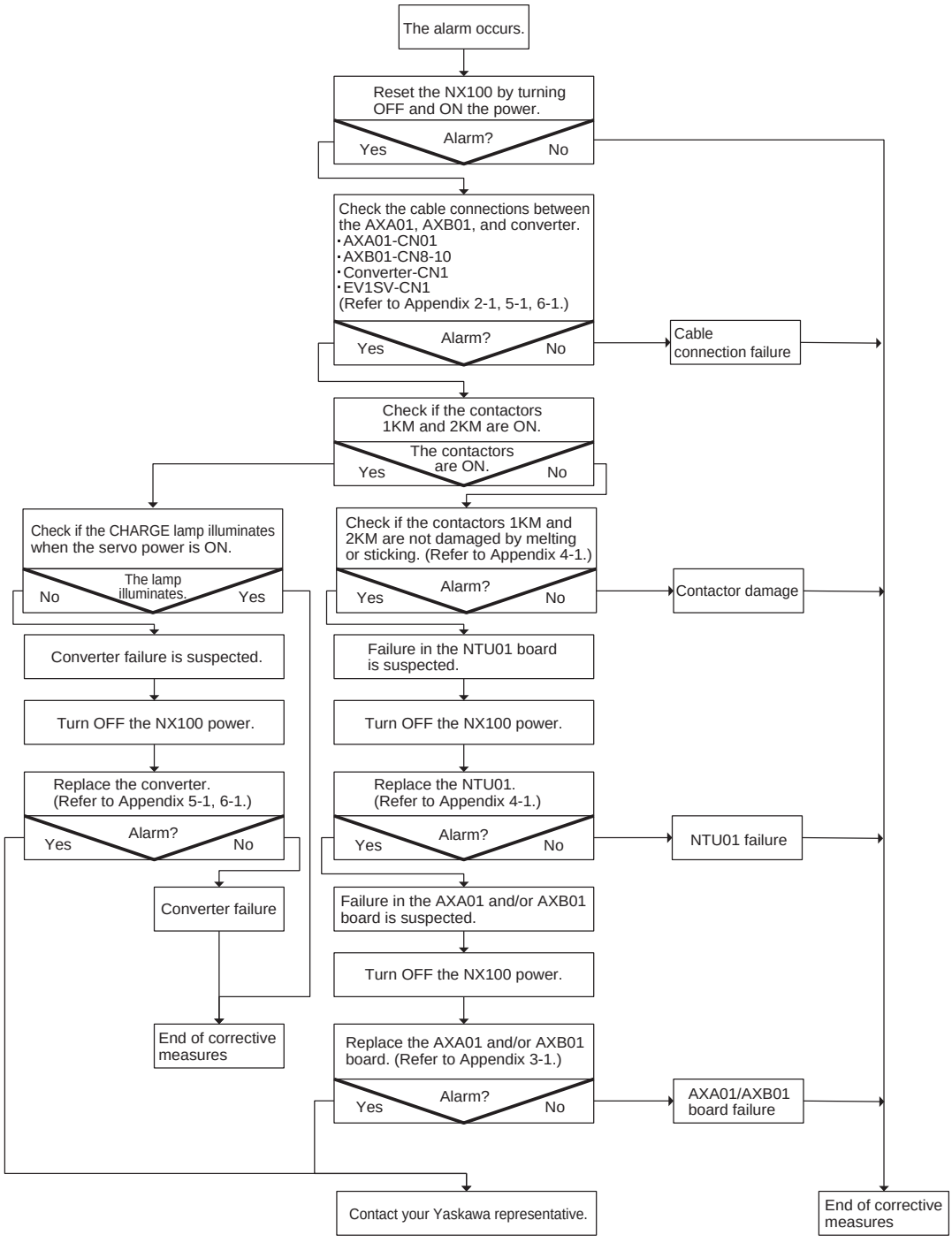
■ AL-4303

CONVERTER READY SIGNAL ERROR

The NX100 system checks the converter status. This alarm occurs if the ready state signal of converter is not ON after a certain time period from SERVO ON.

As a cause of the alarm, a failure in the NTU01, AXA01 or AXB01, or converter is suspected; as another possibility, a loose connection also may cause the alarm. In rare cases, the alarm may be caused by a noise interference.

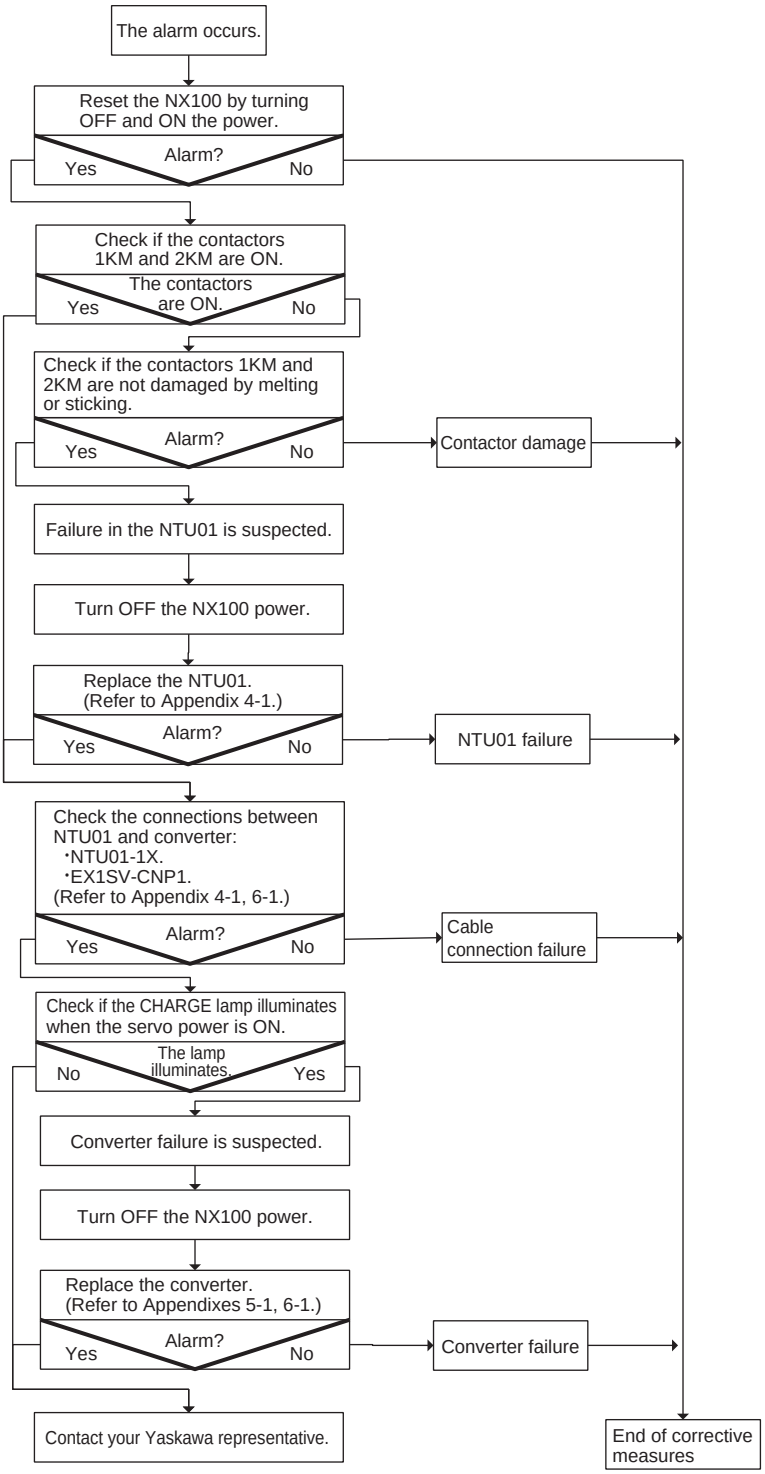
Sub Code: Signifies the physical No. of converter in which the alarm occurred



■ AL-4304

CONVERTER INPUT POWER ERROR

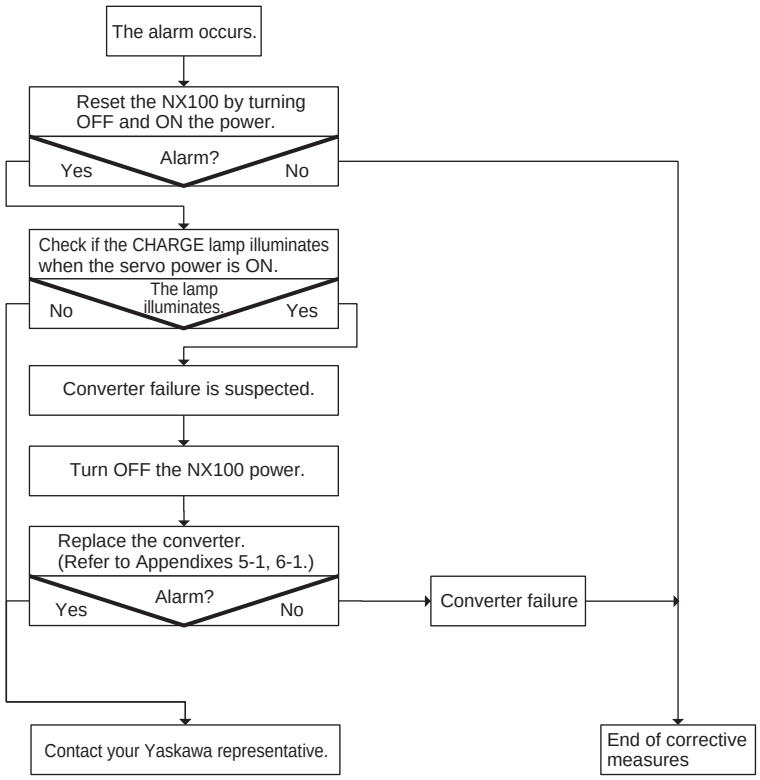
The NX100 system checks the converter status. This alarm occurs if the ready state signal of converter is not ON after a certain time period from SERVO ON.
As a cause of the alarm, a failure in the NTU01, AXA01 or AXB01, or converter is suspected; as another possibility, a loose connection also may cause the alarm. In rare cases, the alarm may be caused by a noise interference.
Sub Code: Signifies the physical No. of converter in which the alarm occurred



■ AL-4305

CONVERTER CIRCUIT CHARGE ERROR

The NX100 system checks the charge status of converter. This alarm occurs if the ready state of converter is not achieved within a certain time period.
As a cause of the alarm, a failure in the Converter1 board is suspected.
Sub Code: Signifies the physical No. of converter in which the alarm occurred

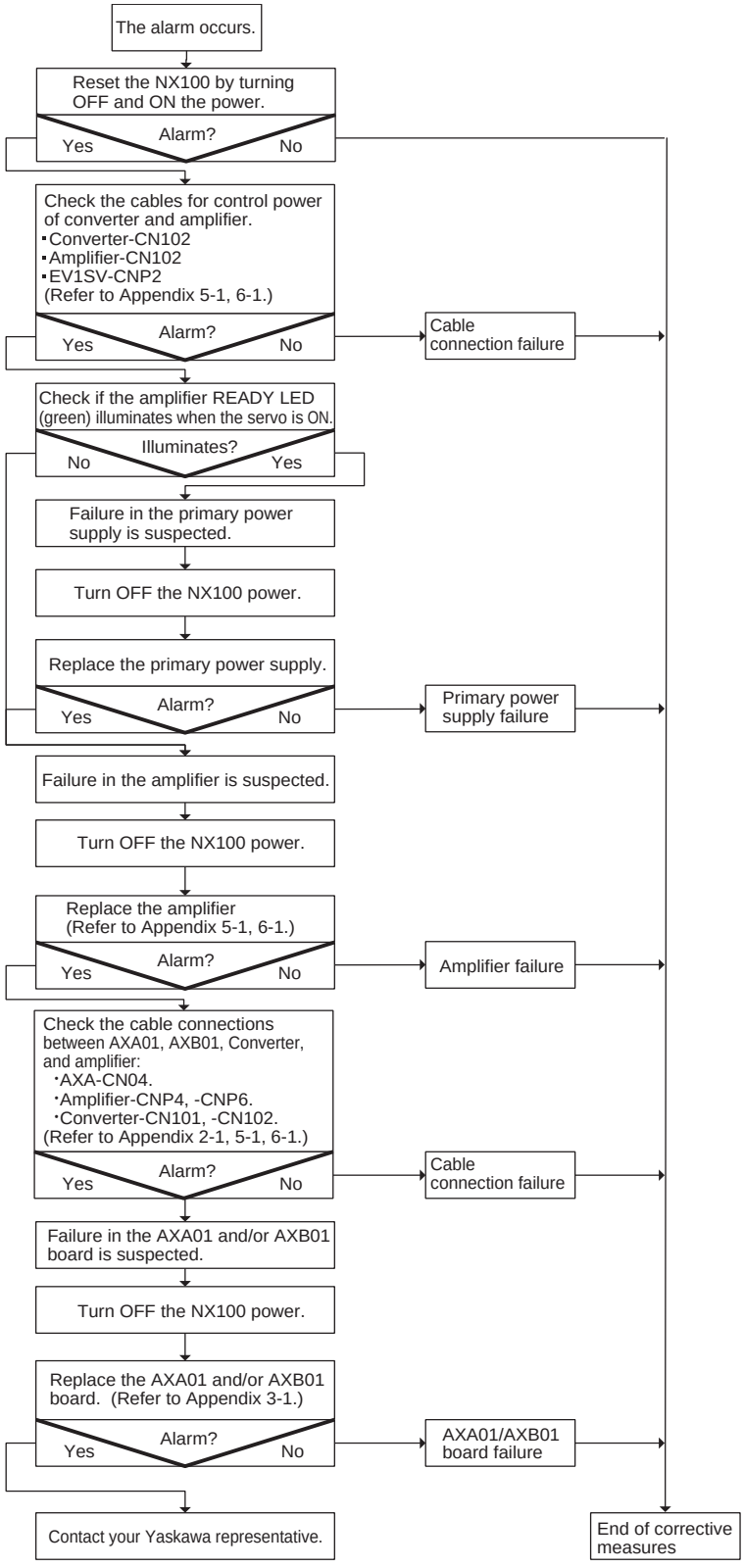


■ AL-4306

AMPLIFIER READY SIGNAL ERROR

This alarm occurs is the amplifier READY signal is OFF while the servo power is turned ON.
As a cause of the alarm, a failure in the servo control circuit board, converter, and/or amplifier is suspected.

Sub Code: Signifies the axis in which the alarm occurred

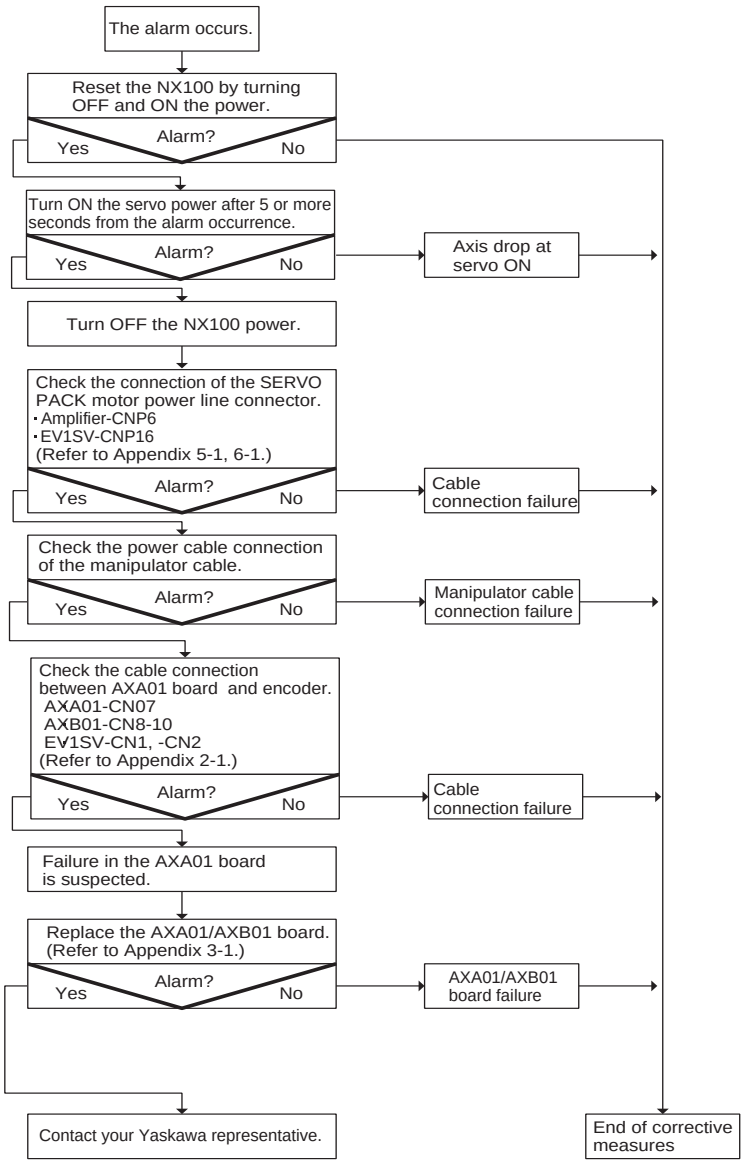


■ AL-4307

SERVO ON DEFFECTIVE SPEED

This alarm occurs if any manipulator axis moves (or any motor operates) when the servo power is ON.

Sub Code: Signifies the axis in which the alarm occurred

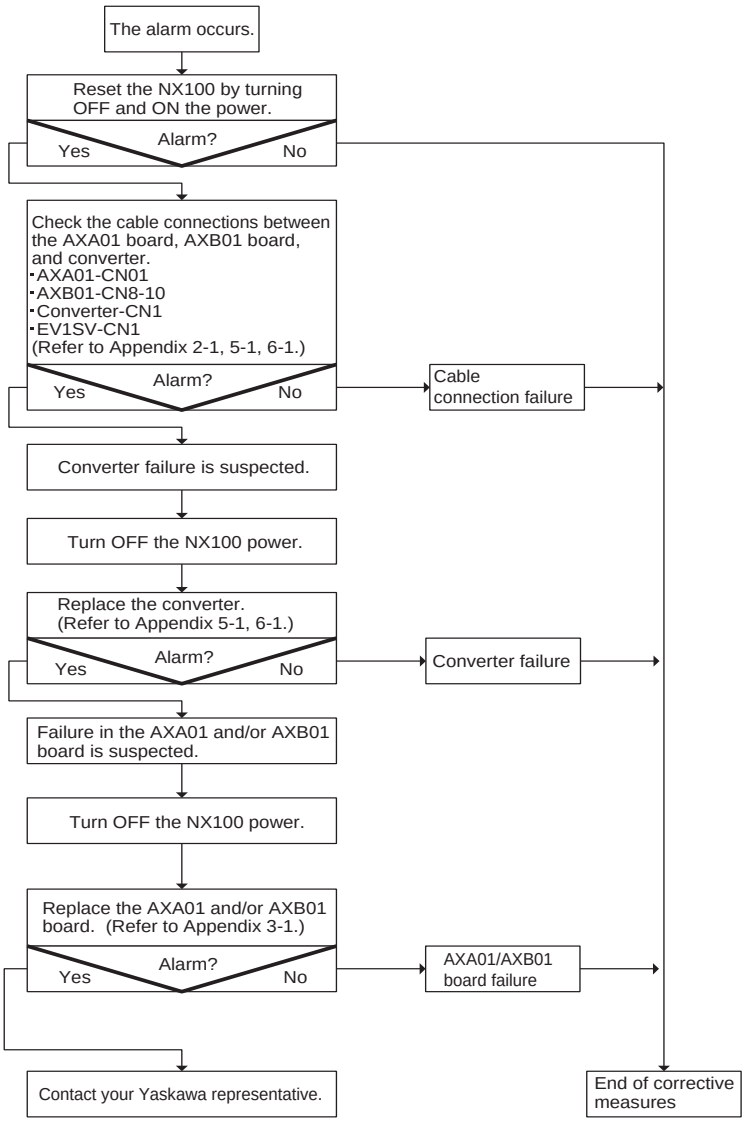


■ AL-4308

VOLTAGE DROP (CONVERTER)

This alarm occurs if there is any error in the charge status of converter when the servo power is ON. As a cause of the alarm, a failure in the converter, primary power output, or amplifier is suspected. In rare cases, the alarm may be caused by a noise interference.

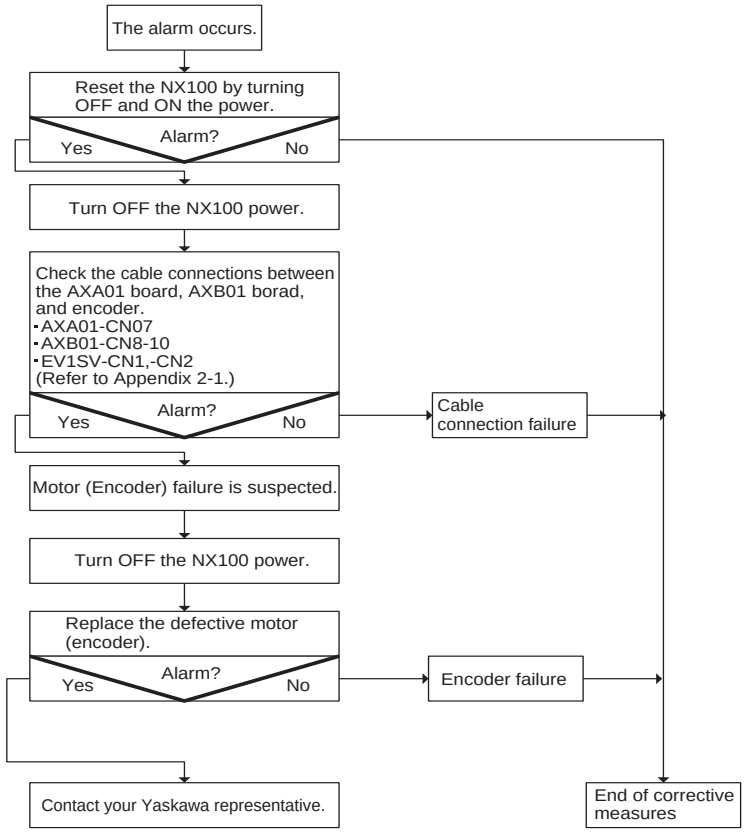
Sub Code: Signifies the physical No. of converter in which the alarm occurred



■ AL-4309

DEFECTIVE ENCODER INTERNAL DATA

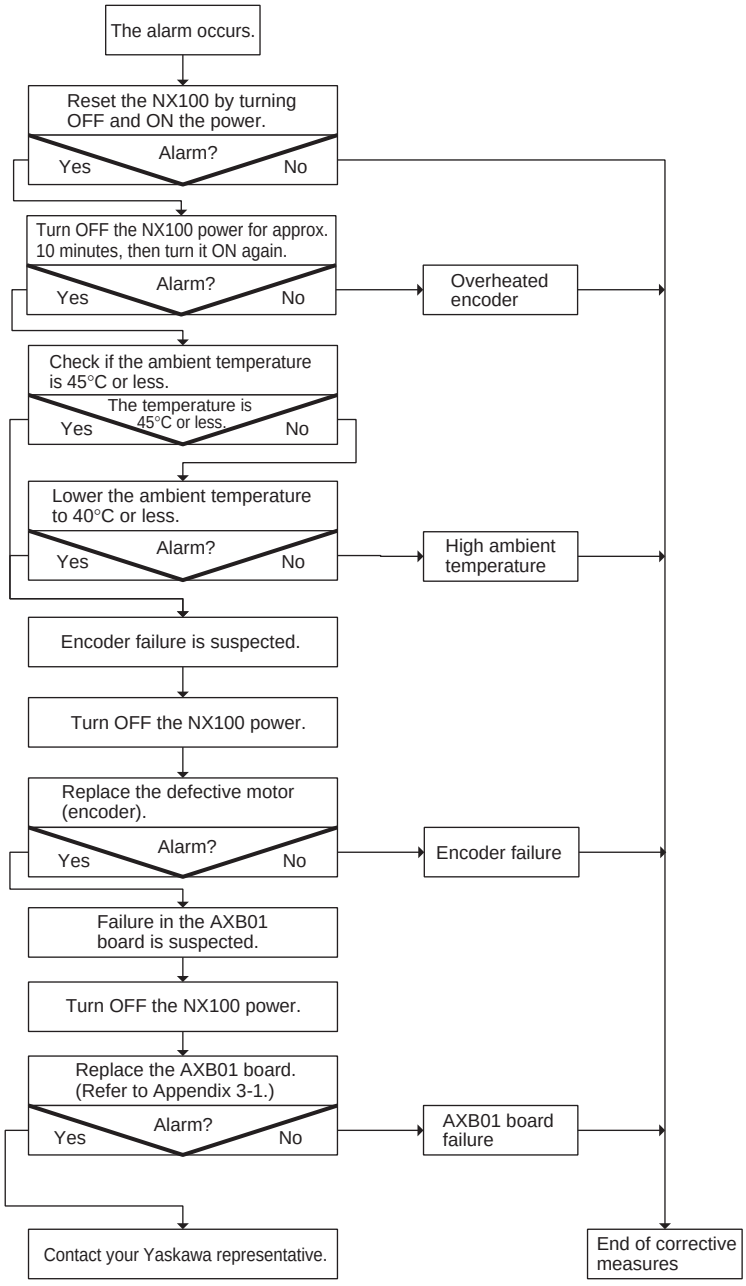
This alarm occurs if there is an internal parameter error in serial encoder.
To recover from the alarm status, reset the alarm, recalibrate the home position, then restart the NX100. If unable to restore the normal condition, replace the motor causing the alarm.
Sub Code: Signifies the axis in which the alarm occurred



■ AL-4310

ENCODER OVERHEAT

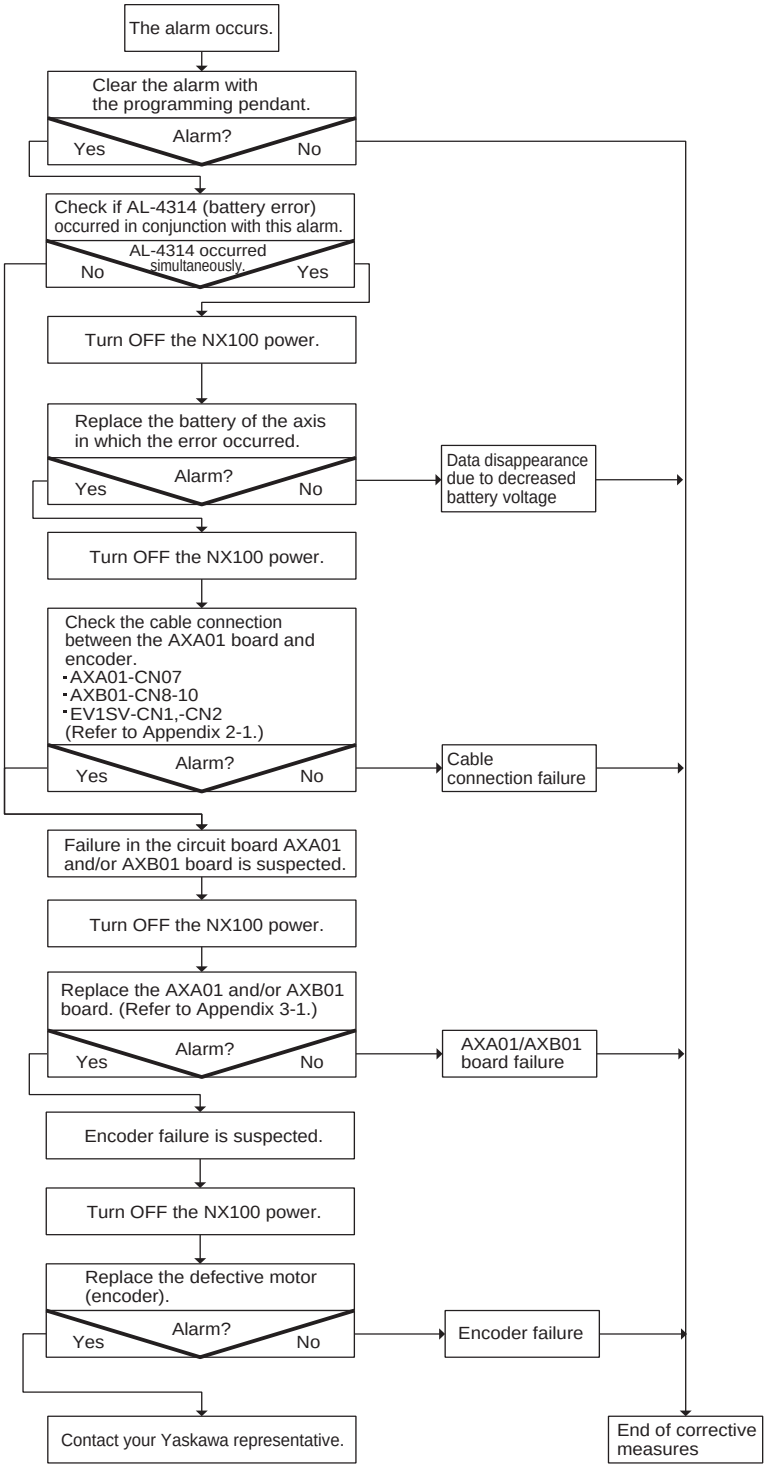
This alarm occurs if there is a failure in a motor (for motor gun) which is not equipped with a serial encoder.
The encoder monitors its own temperature with a thermometer; if the temperature exceeds the specified value, the alarm occurs.
As a cause of the alarm, overloaded encoder status or effect of the ambient environmental temperature is suspected. In rare cases, the alarm may be caused by a defective motor gun axis.
Sub Code: Signifies the axis in which the alarm occurred



■ AL-4311

ENCODER BACK-UP ERROR

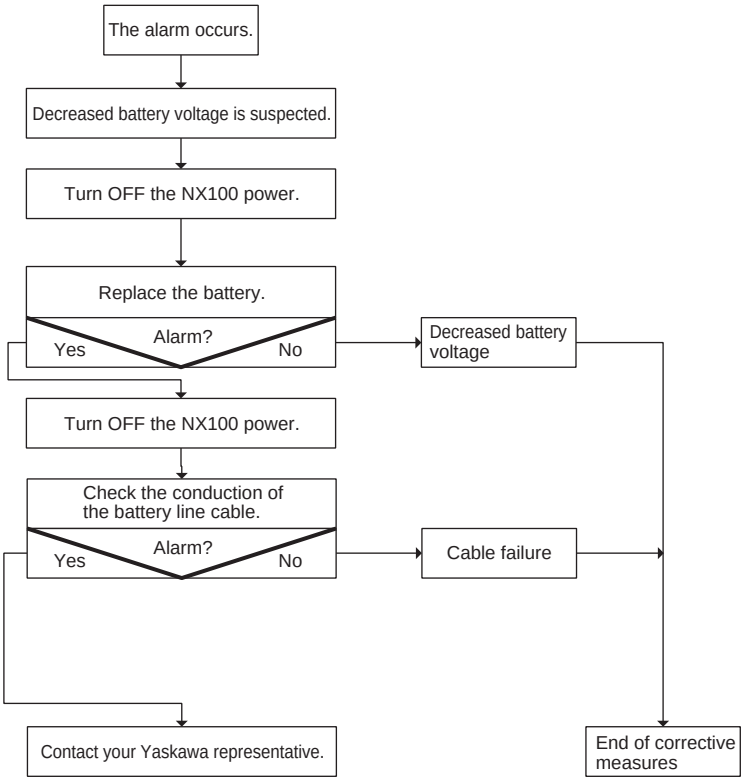
If this alarm occurs, the cumulative rotation amount data in the motor subsequently disappears.
When operating the manipulator after performing the troubleshooting procedure, calibrate the home position with due consideration to the P-point maximum envelope of manipulator.
Sub Code: Signifies the axis in which the alarm occurred



■ AL-4312

ENCODER BATTERY ERROR

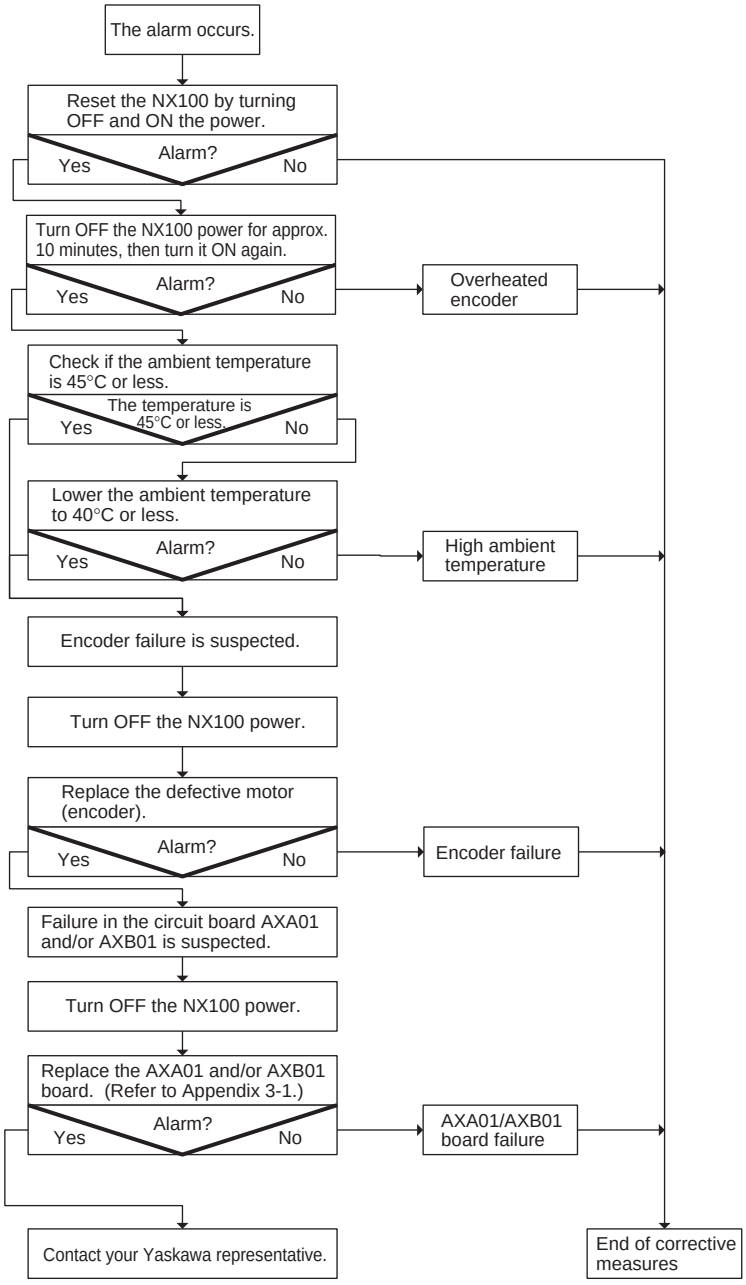
The cumulative rotation amount data in the motor disappears if its encoder backup battery voltage decreases and is left unrestored without turning on the power to the NX100; therefore, make sure to replace the battery without delay.
If the cumulative rotation amount data is lost as a result of ENCODER BACK-UP ERROR (AL-4311), calibrate the home position with due consideration to the P-point maximum envelope of manipulator when starting to operate the manipulator again.
Sub Code: None



■ AL-4313

SERIAL ENCODER OVER HEAT

The NX100 system monitors the encoder temperature with a thermometer in the encoder. This alarm occurs if the temperature exceeds the specified value.
As a cause of the alarm, overloaded encoder status or effect of the ambient environmental temperature is suspected. In rare cases, the alarm may be caused by a manipulator malfunction.
Sub Code: Signifies the axis in which the alarm occurred

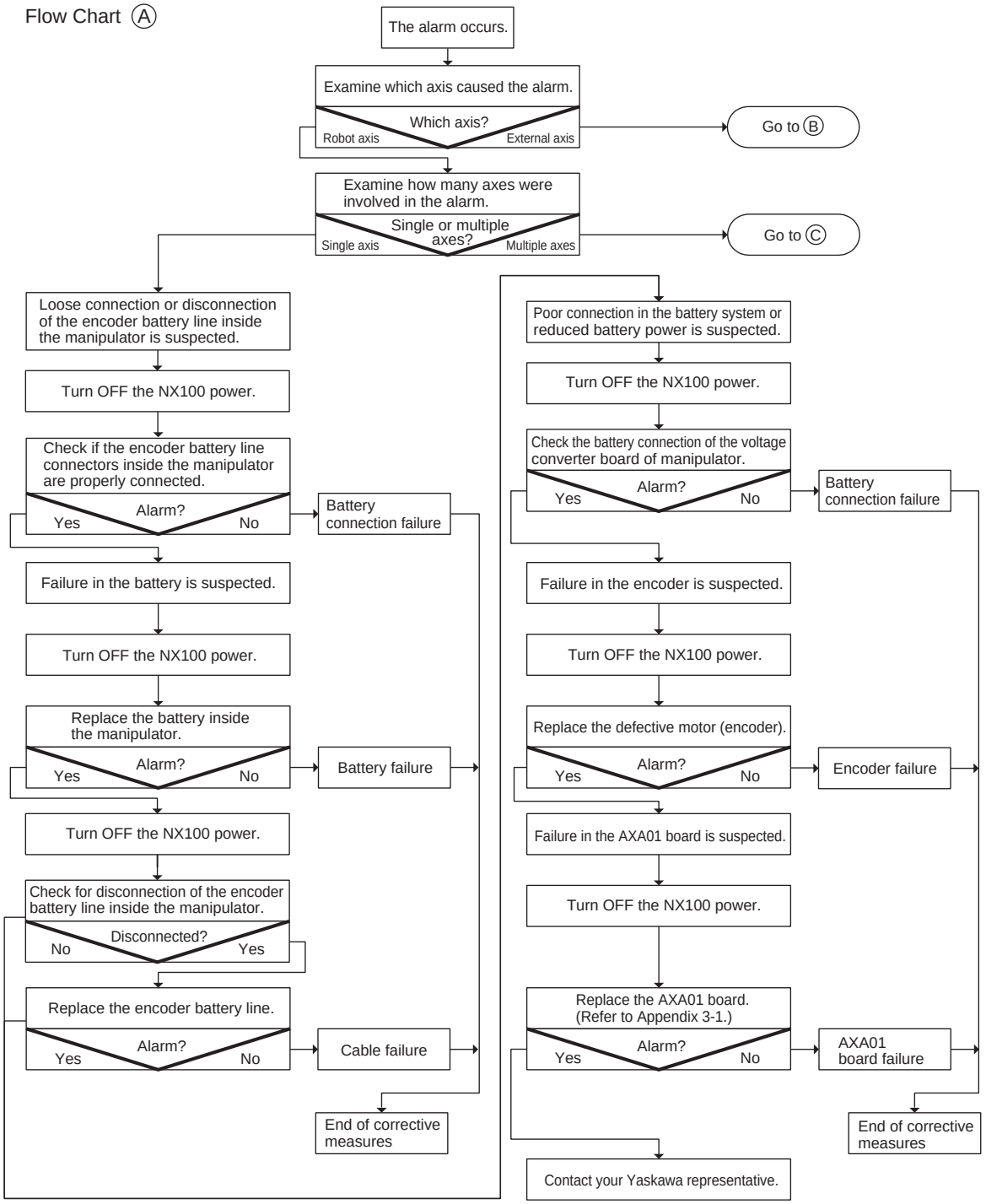


■ AL-4314

SERIAL ENCODER BATTERY ERROR

This alarm occurs when the voltage of encoder backup battery decreases to 2.8 V or less.
Since a condenser is not equipped in the motor with a serial encoder, the cumulative rotation amount data in the motor disappears when the NX100 power is turned OFF if the decreased voltage of encoder battery is left unrestored. Therefore, make sure to replace the battery using a backup battery. The same alarm occurs if the wiring between the backup battery and the motor is disconnected.
If the cumulative rotation amount data is lost as a result of ENCODER BACK-UP ERROR (AL-4311), calibrate the home position with due consideration to the P-point maximum envelope of manipulator when starting to operate the manipulator again.
Sub Code: Signifies the axis in which the alarm occurred

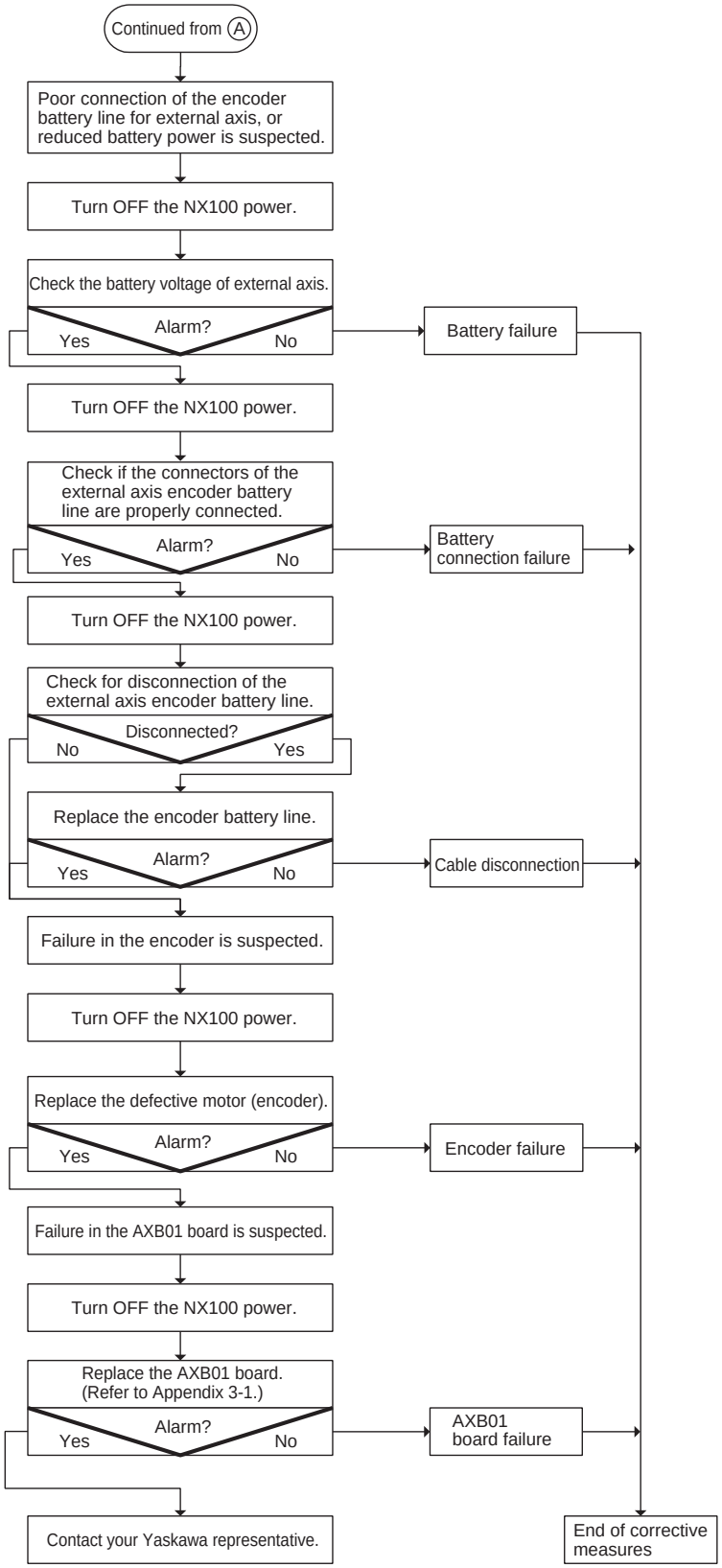
Flow Chart Ⓐ



■ AL-4314

SERIAL ENCODER BATTERY ERROR

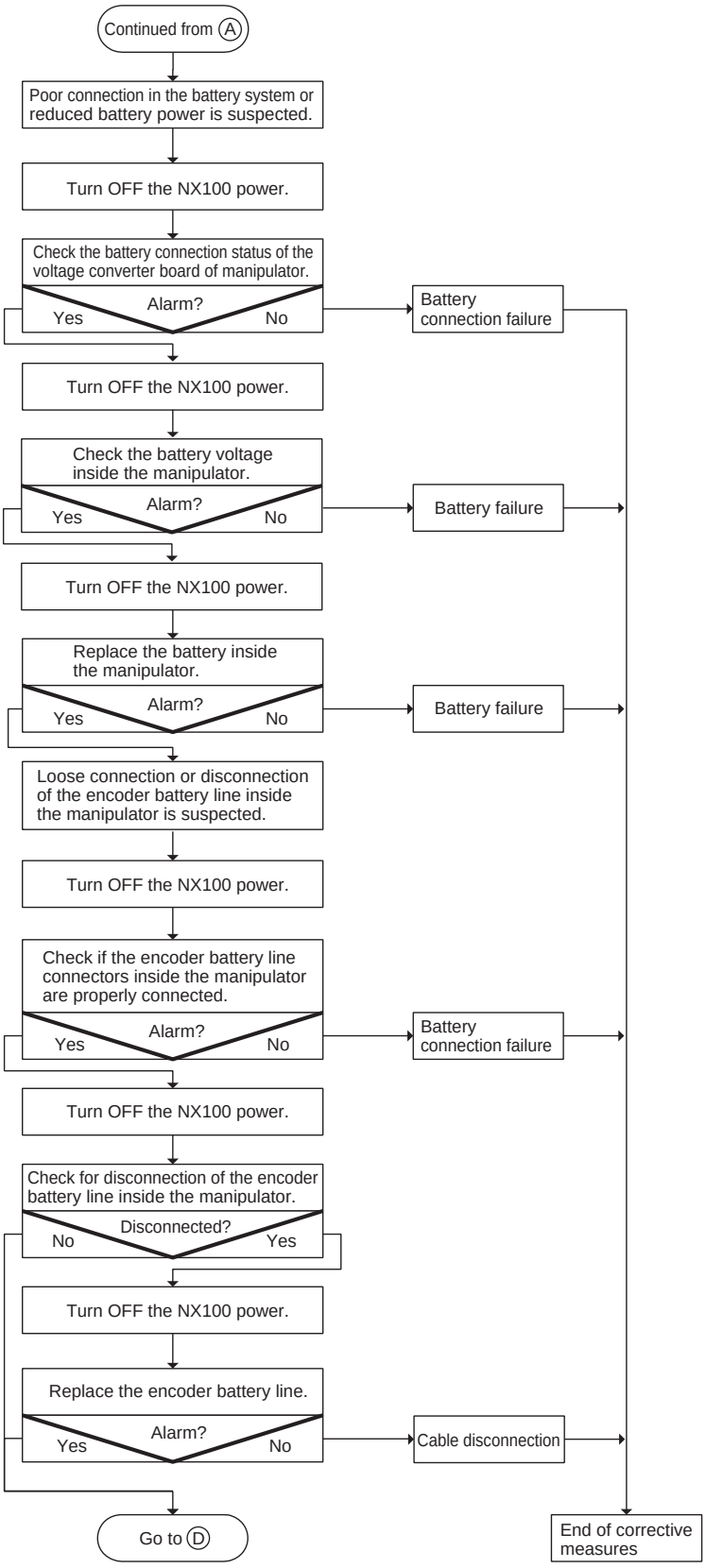
Flow Chart Ⓑ



■ AL-4314

SERIAL ENCODER BATTERY ERROR

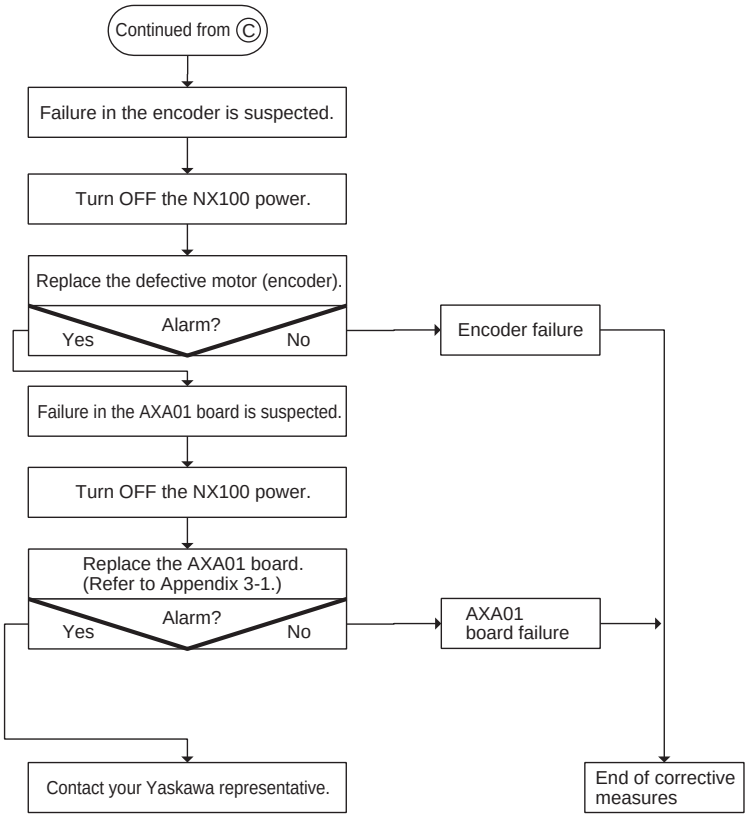
Flow Chart Ⓒ



■ AL-4314

SERIAL ENCODER BATTERY ERROR

Flow Chart Ⓓ



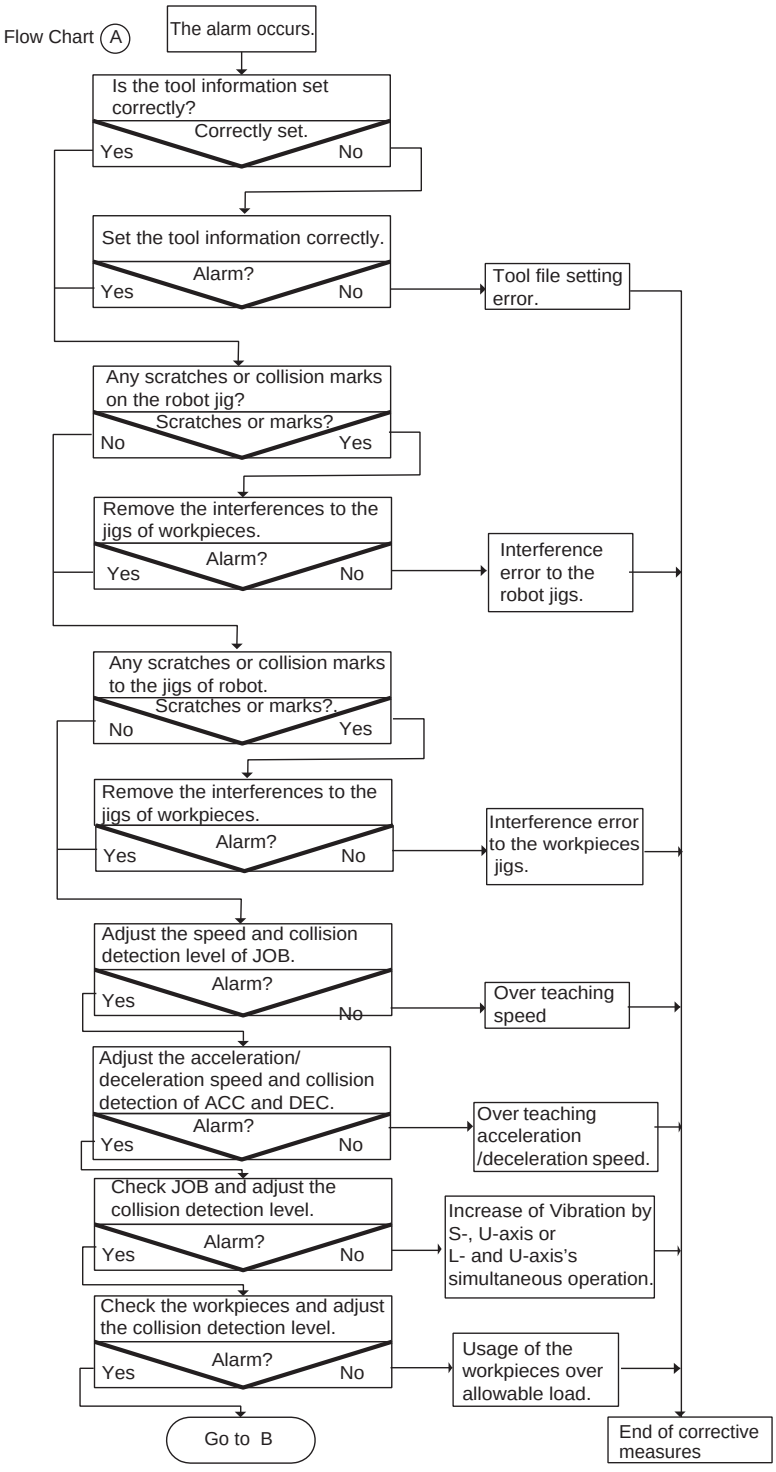
■ AL-4315

COLLISION DETECT

A collision was detected because of the interference between the manipulator and a peripheral device.

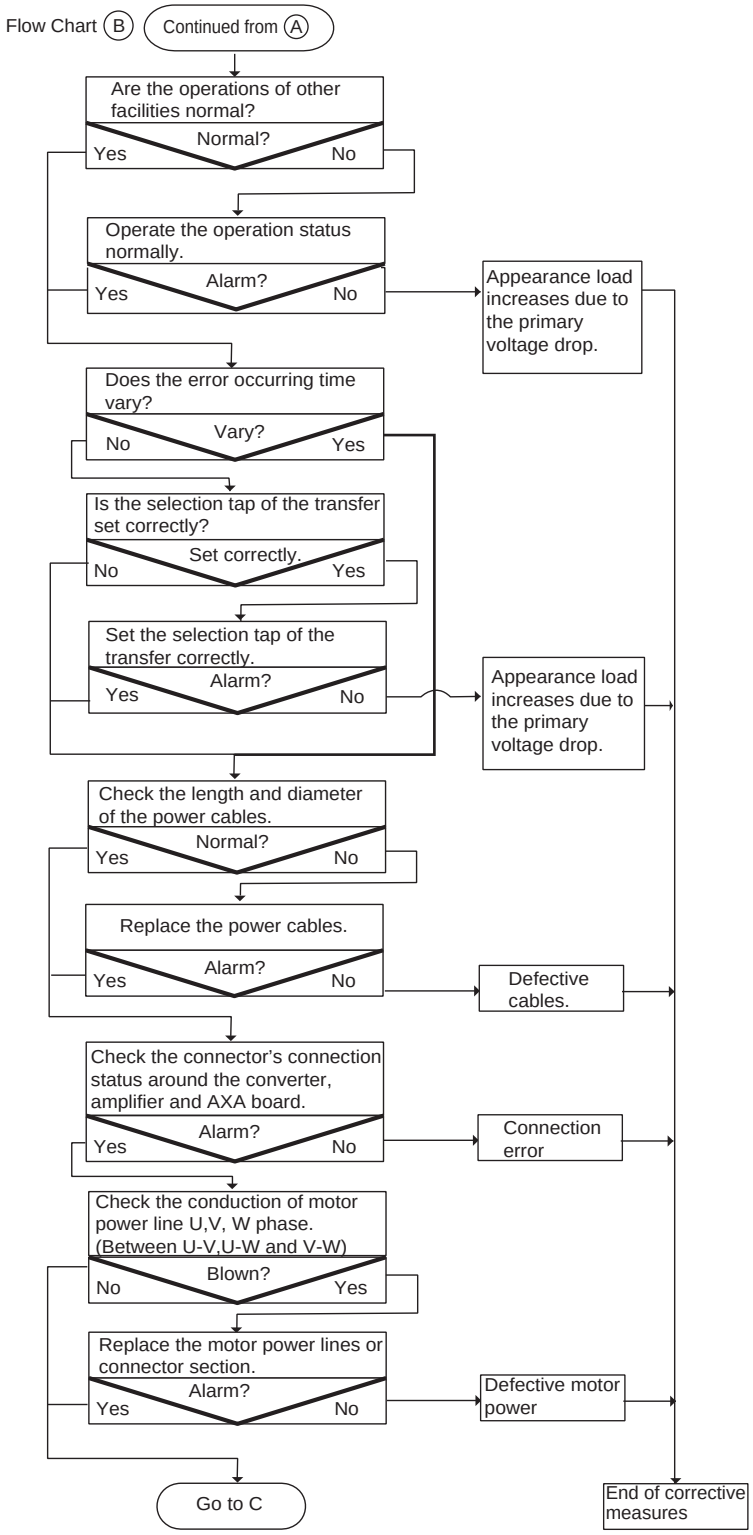
The external force applied to the robot exceeded the threshold.

Sub Code: None



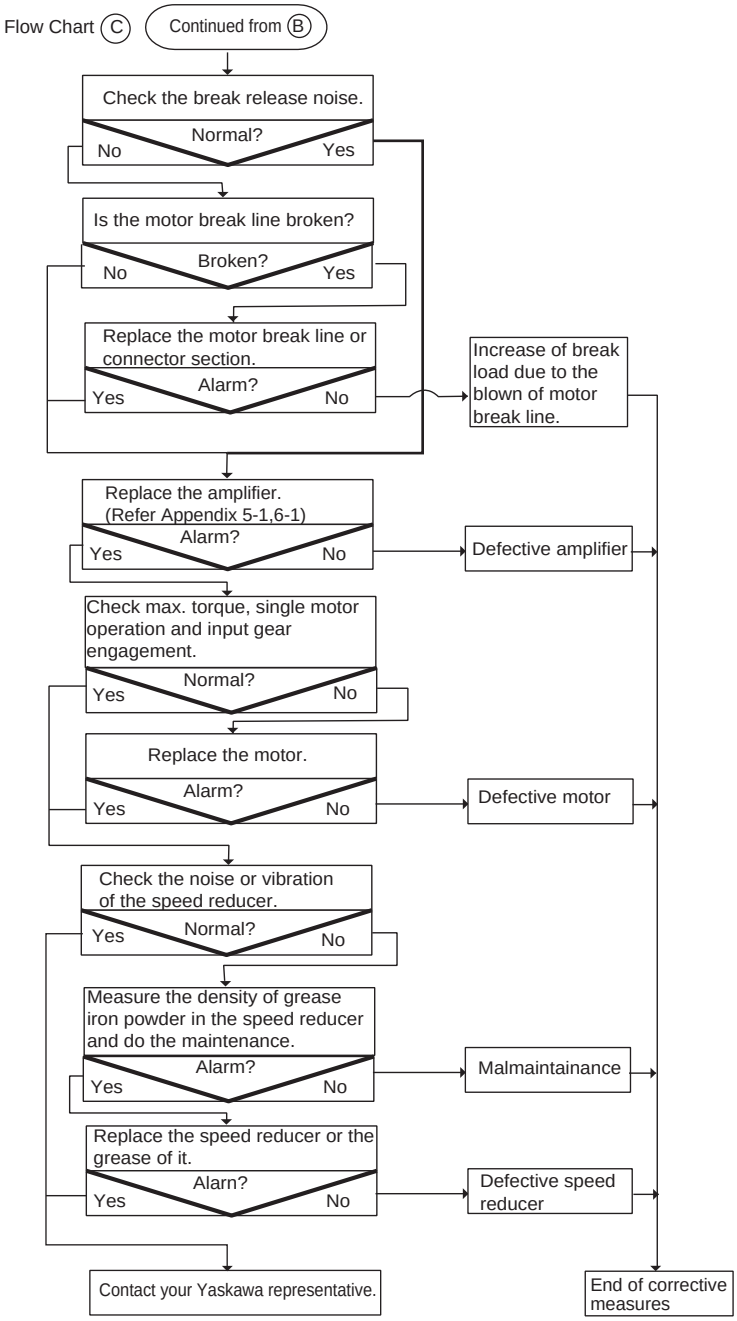
■ AL-4315

COLLISION DETECT



■ AL-4315

COLLISION DETECT

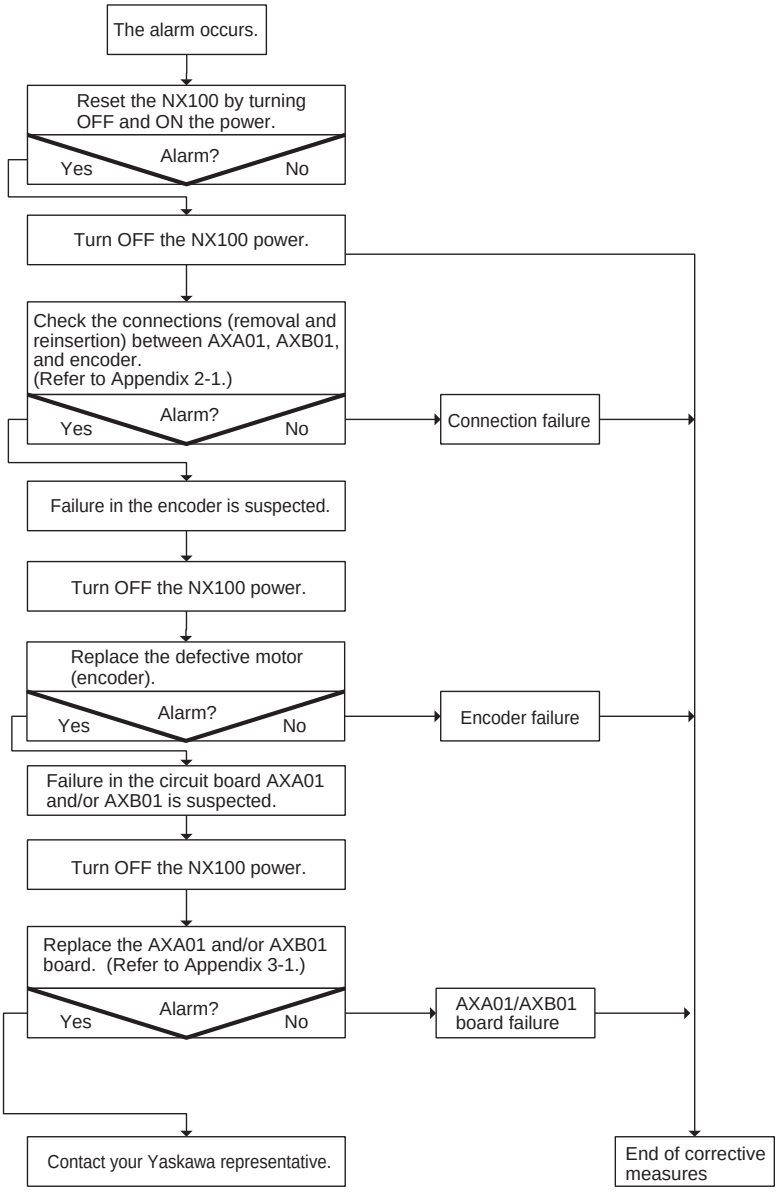


■ AL-4318

SERIAL ENCODER CORRECT LIMITATION OVER

This alarm occurs if the value generated from the position data of serial encoder exceeds the limit value.

Sub Code: Signifies the axis in which the alarm occurred

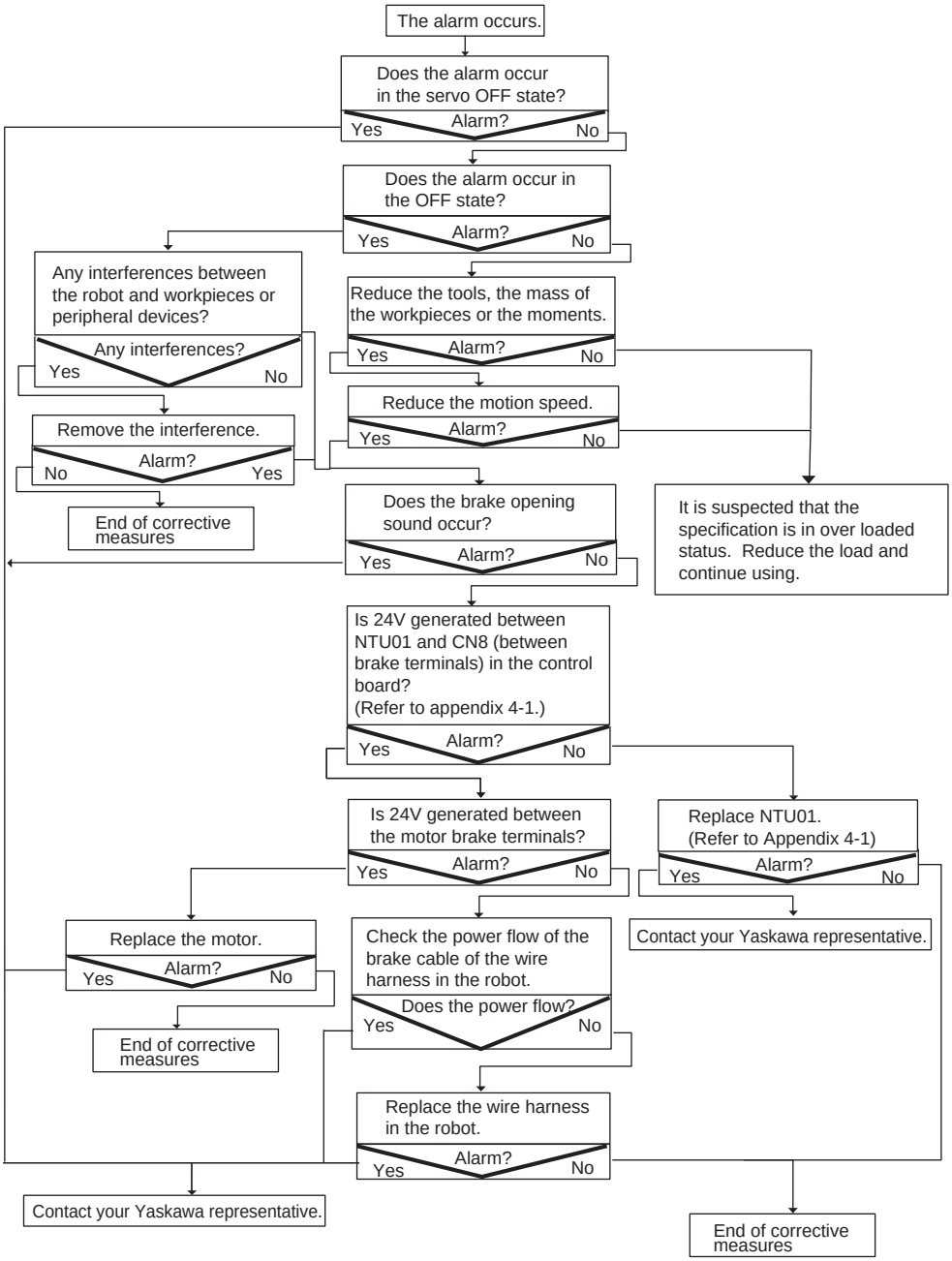


■ AL-4320

OVER LOAD (CONTINUE)

This alarm occurs to protect the the servo motor from overloading when 110%~150% of the rated torque is continuously loaded .

Sub Code: Signifies the axis in which the alarm occurred



■ AL-4321

OVER LOAD (INSTANT)

This alarm occurs instantly to protect the servo motor from overloading when 200% of the rated torque is loaded.

Sub Code: Signifies the axis in which the alarm occurred

■ AL-4322

AMPLIFIER OVER LOAD (CONTINUE)

This alarm occurs to protect the power transister of the servo pack from overloading when 110%~150% of the rated torque is continuously loaded .

Sub Code: Signifies the axis in which the alarm occurred

■ AL-4323

AMPLIFIER OVER LOAD (INSTANT)

This alarm occurs instantly to protect the power transister of the servo pack from overloading when 200% of the rated torque is continuously loaded .

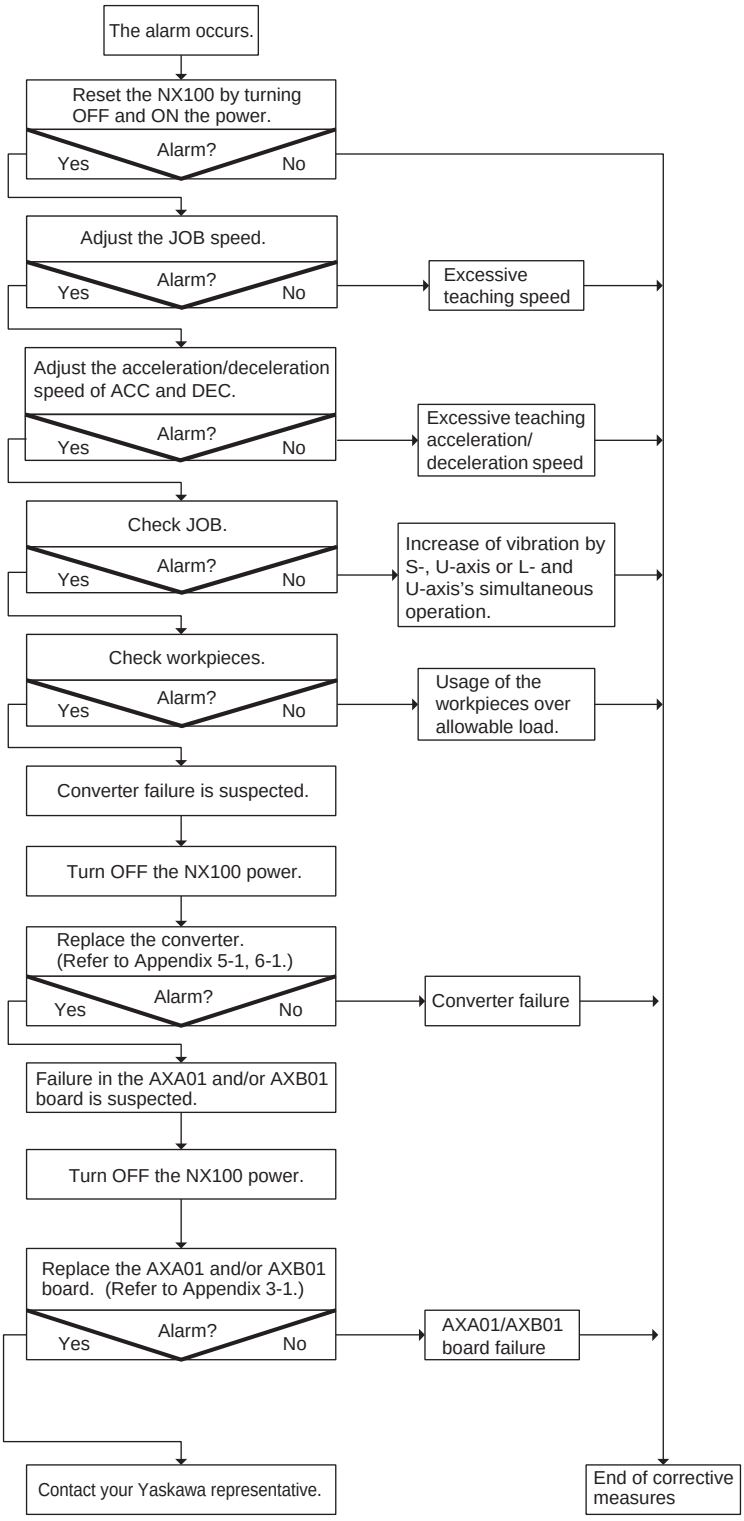
Sub Code: Signifies the axis in which the alarm occurred

■ AL-4324

CONVERTER OVERLOAD

This is a converter overload alarm detected by the converter.

Sub Code: Signifies the physical No. of converter in which the alarm occurred



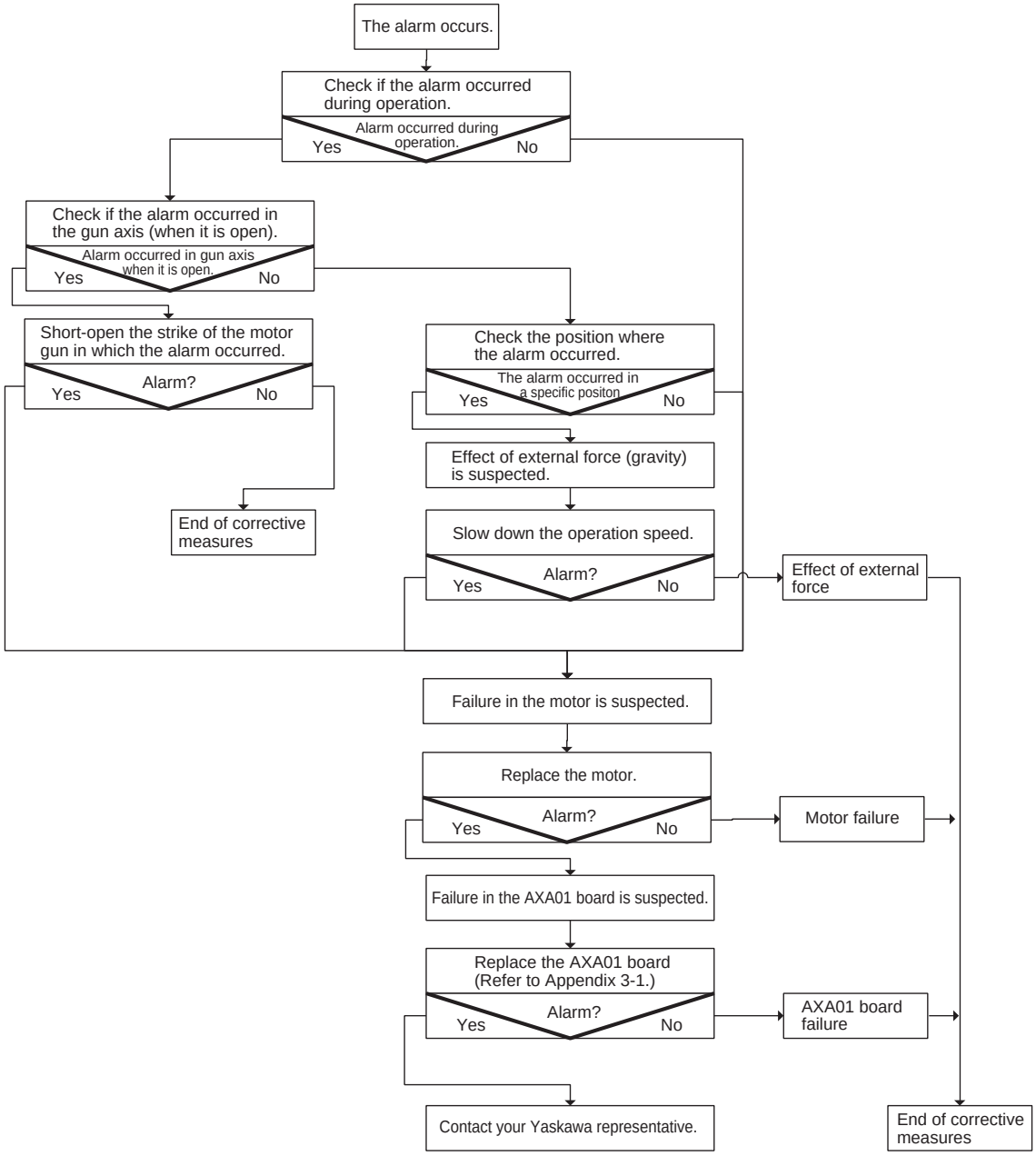
■ AL-4326

SPEED ERROR

This alarm occurs if the motor speed indicated in the axis data exceeds the allowable maximum motor speed.

As a cause of the alarm, effect of external force on the manipulator, motor power cable disconnection, defective motor, or defective board is suspected.

Sub Code: None

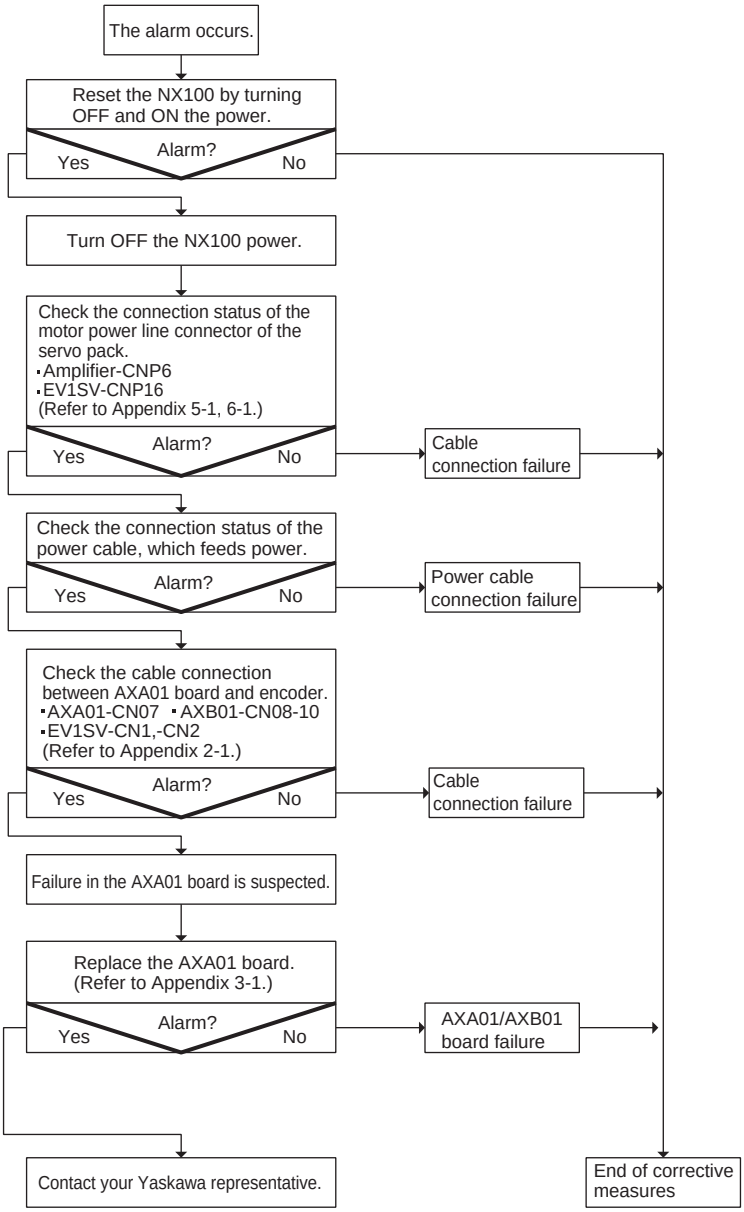


■ AL-4327

WRONG MOTOR ROTATION

This alarm occurs if the torque and speed have opposite directions during motor acceleration. As a cause of the alarm, a failure in the motor power cable or the connection cable between the encoder, AXA01 servo control board, and AXB01 servo control board is suspected.

Sub Code: None

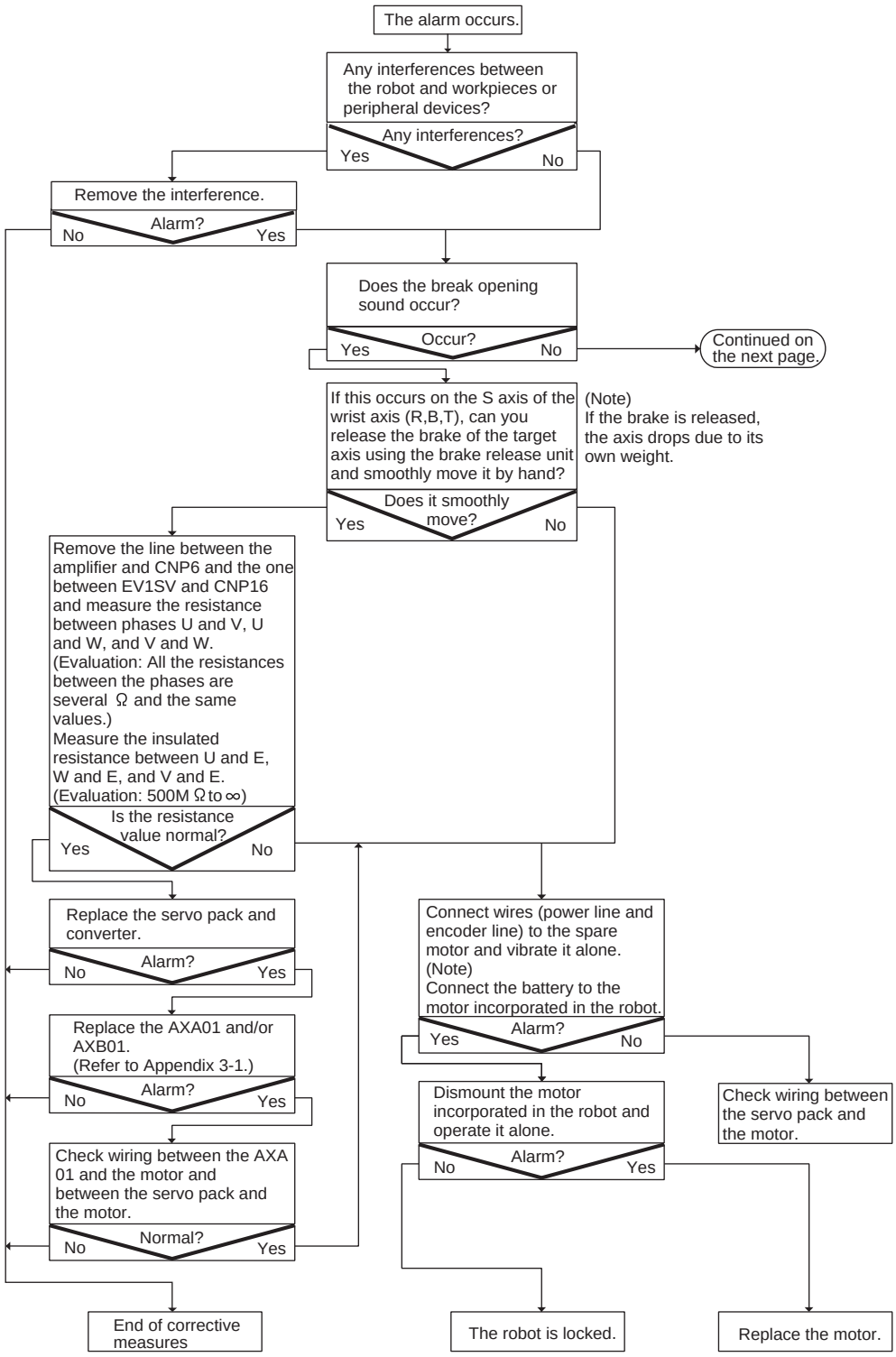


■ AL-4328

SERVO TRACKING ERROR

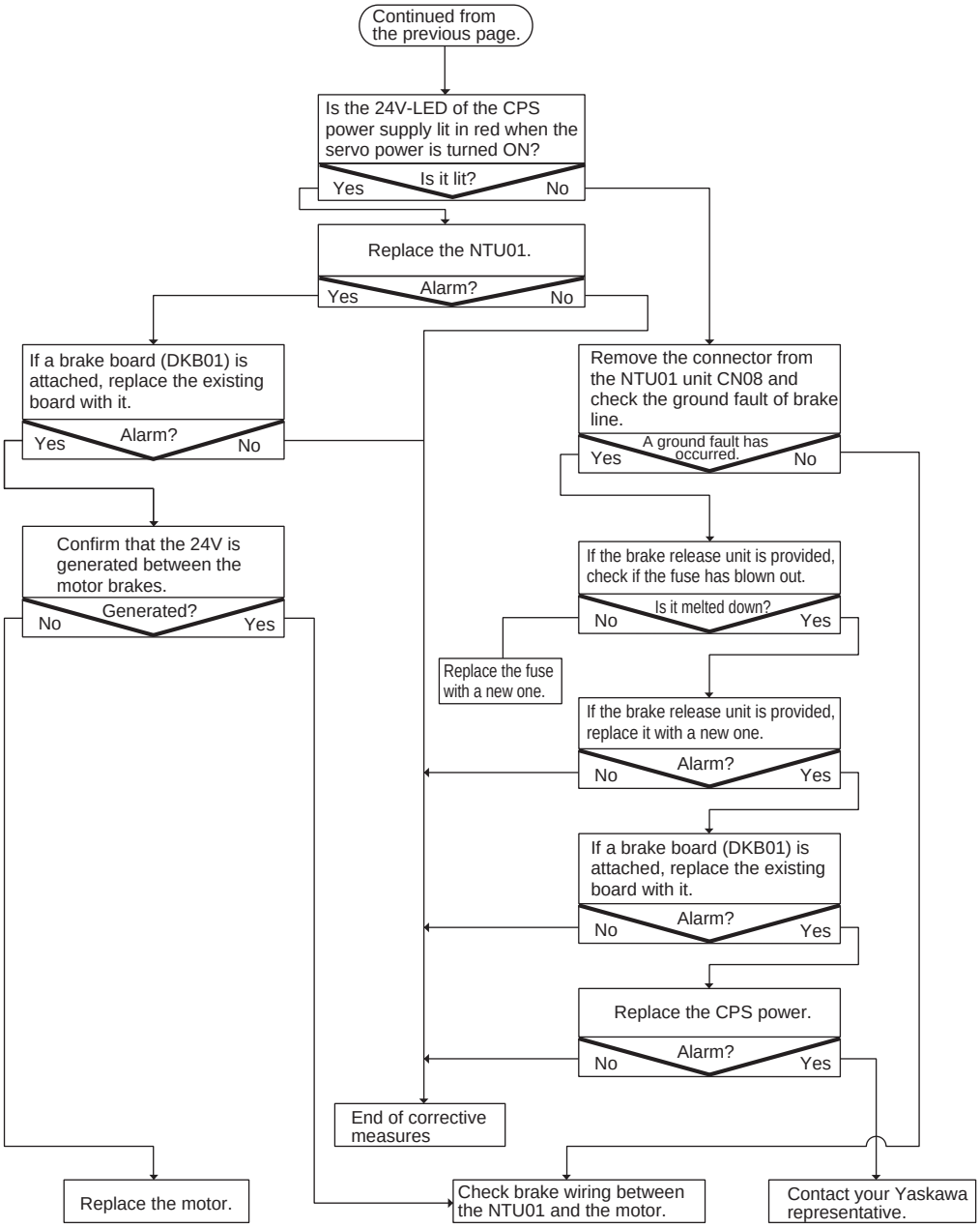
The deviation from the order pulse from the servo control board (AXA01, AXB01), or that from feedback pulse from the encoder becomes too large while the servo power is turned ON.

Sub Code: None



■ AL-4328

SERVO TRACKING ERROR

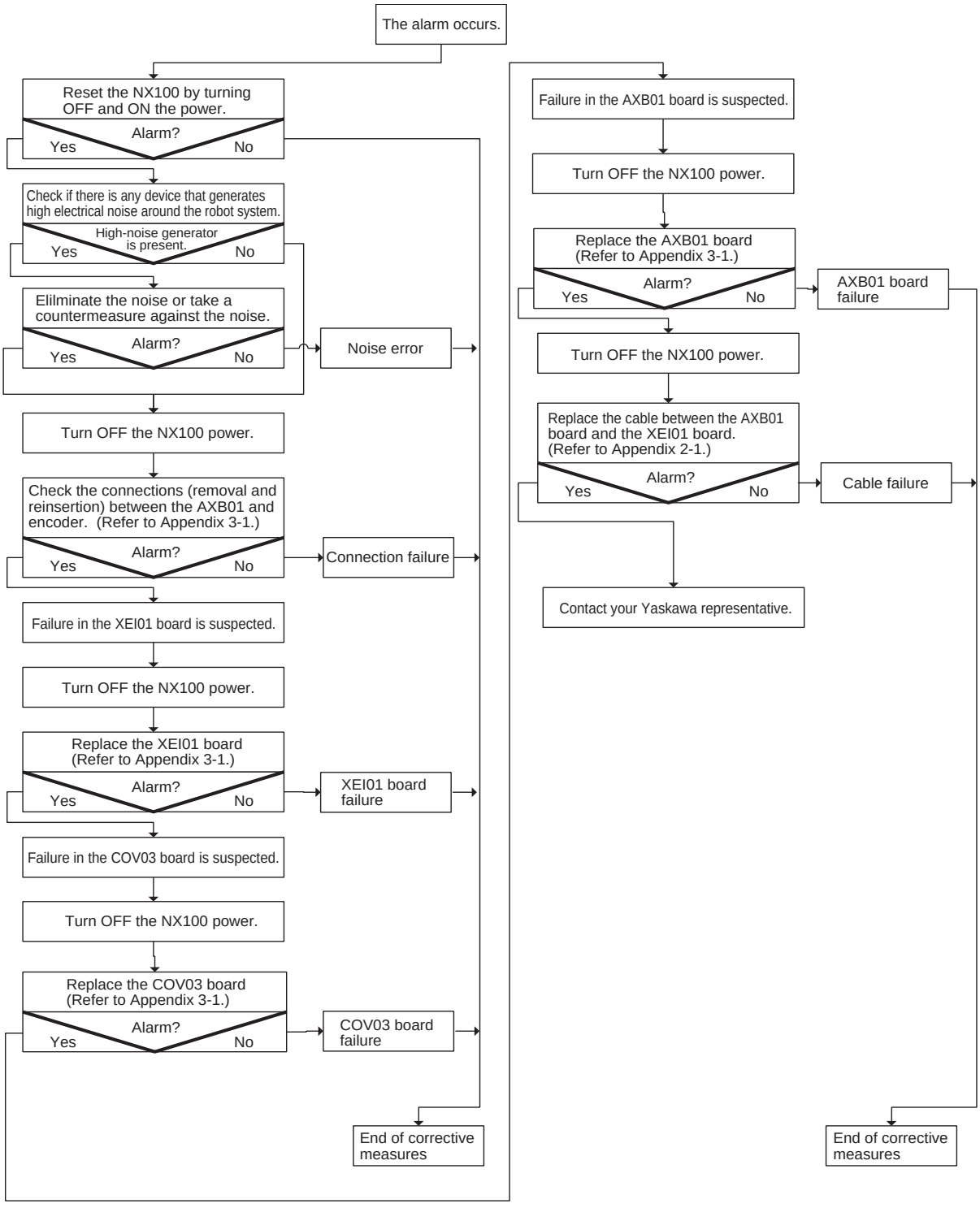


■ AL-1332

POSITION ERROR

This alarm occurs if the feedback pulse miscount is detected in the motor gun axis.
A failure in the AXB01 or XEI01 circuit board, or connection cable between those boards may cause the alarm.

Sub Code: None



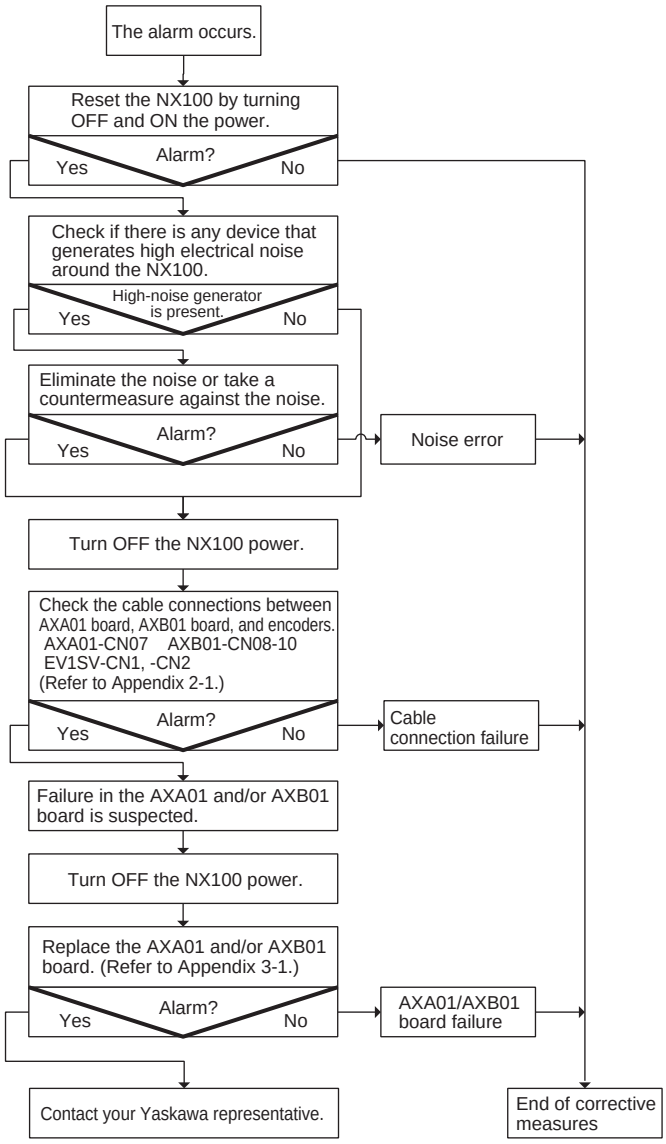
■ AL-4331

SPEED MONITOR LEVEL ERROR

The speed monitoring level value is incorrect.

(1) Reset the alarm.

(2) If the alarm occurs again, replace the servo control board (AXA01).

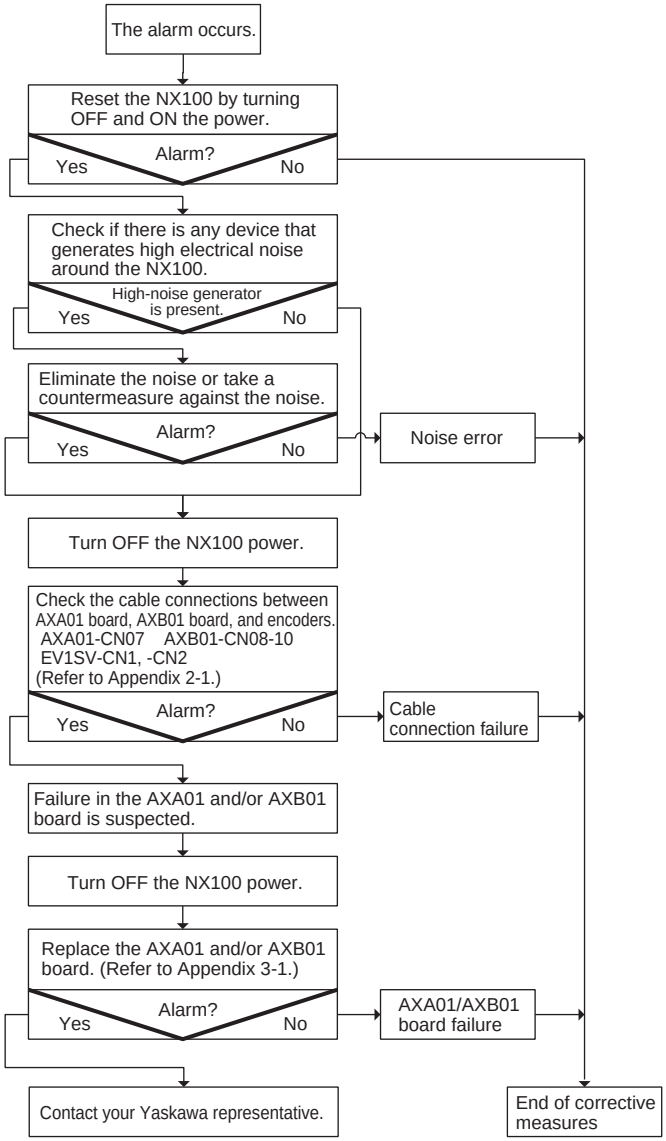


■ AL-1333

POSITION ERROR (SERIAL ENCODER)

This alarm occurs if the feedback pulse miscount is detected in a serial encoder.
As a cause of the alarm, a failure in the connection cable between the servo control board (AXA01, AXB01) and the encoder is suspected.

Sub Code: None



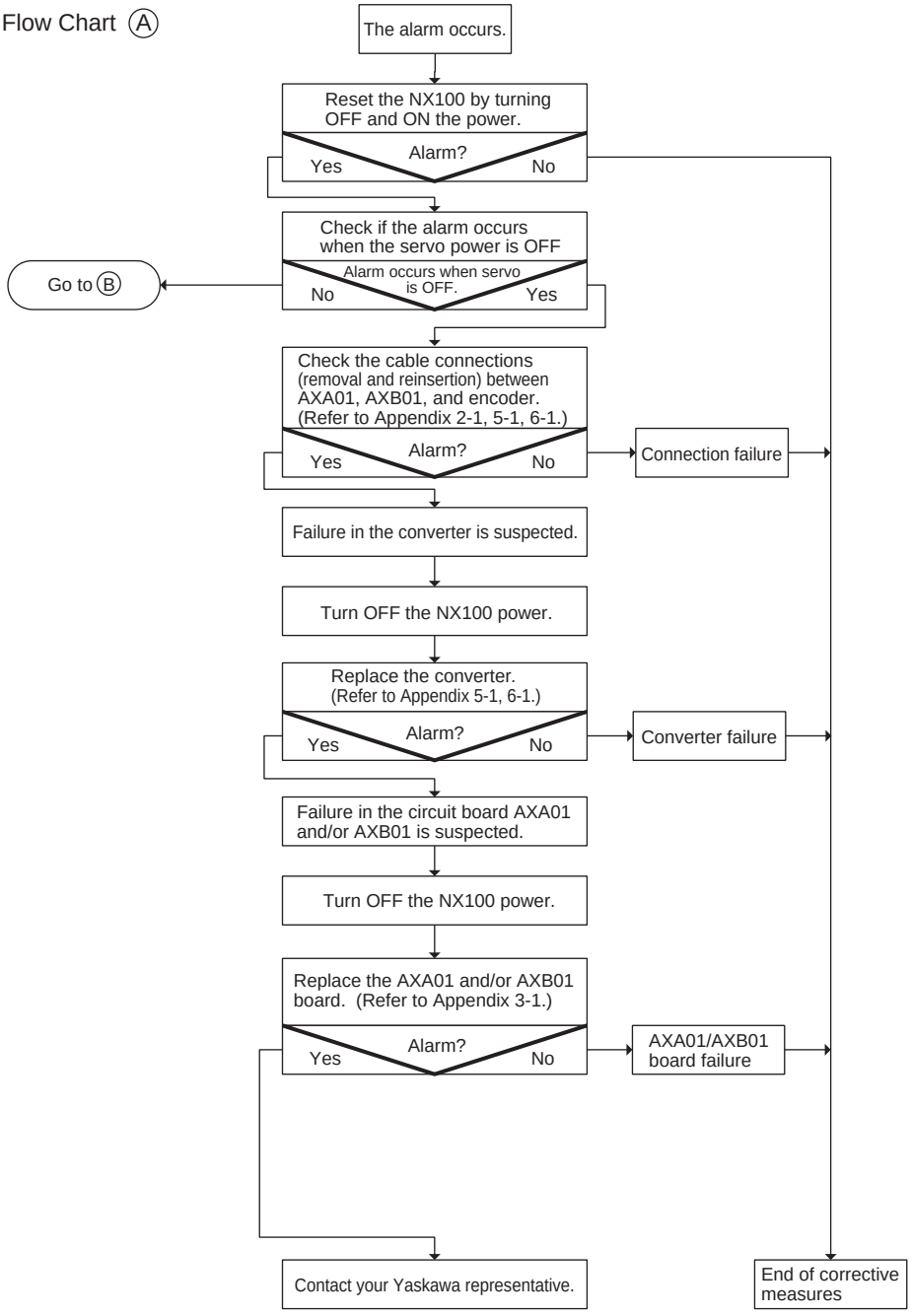
■ AL-4334

OVER VOLTAGE (CONVERTER)

This alarm occurs if the regenerative energy during the motor deceleration is too much that it cannot be processed in the regenerative circuit of SERVO PACK.

Sub Code: Signifies the physical No. of converter in which the alarm occurred

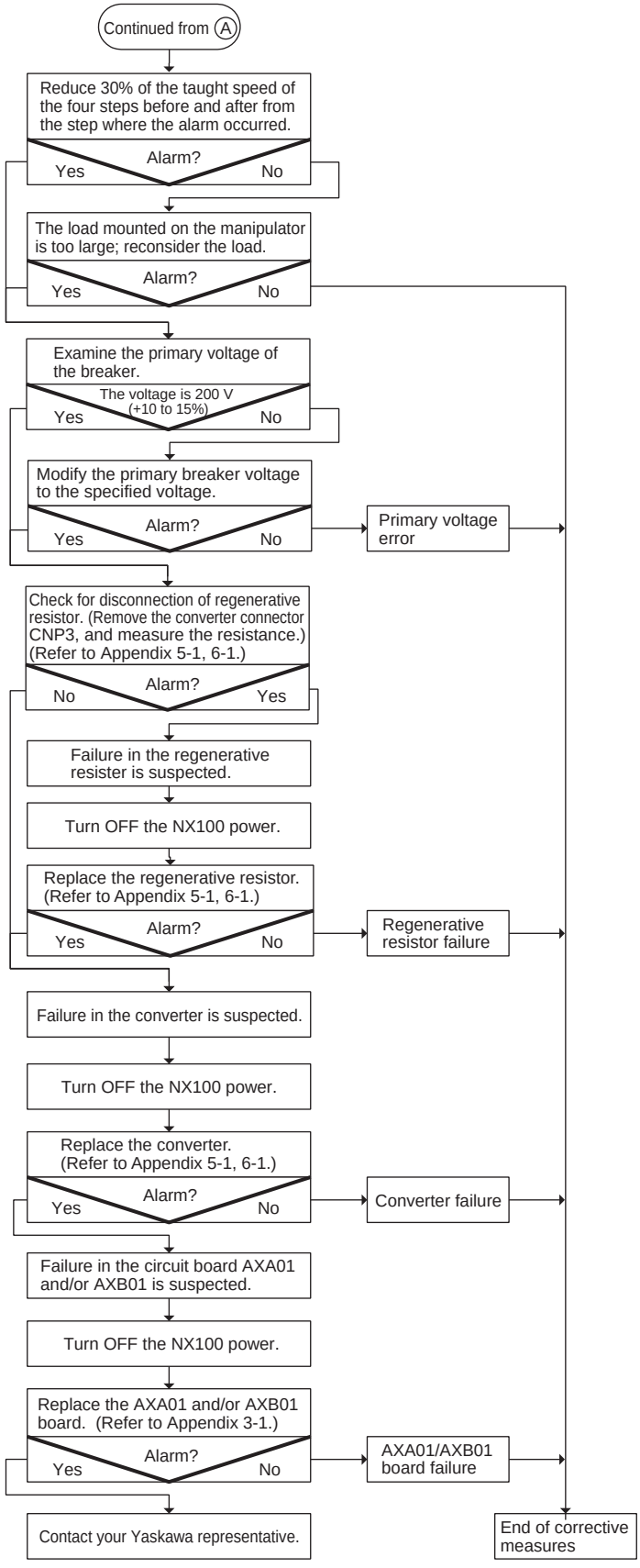
Flow Chart Ⓐ



■ AL-4334

OVER VOLTAGE (CONVERTER)

Flow Chart Ⓑ

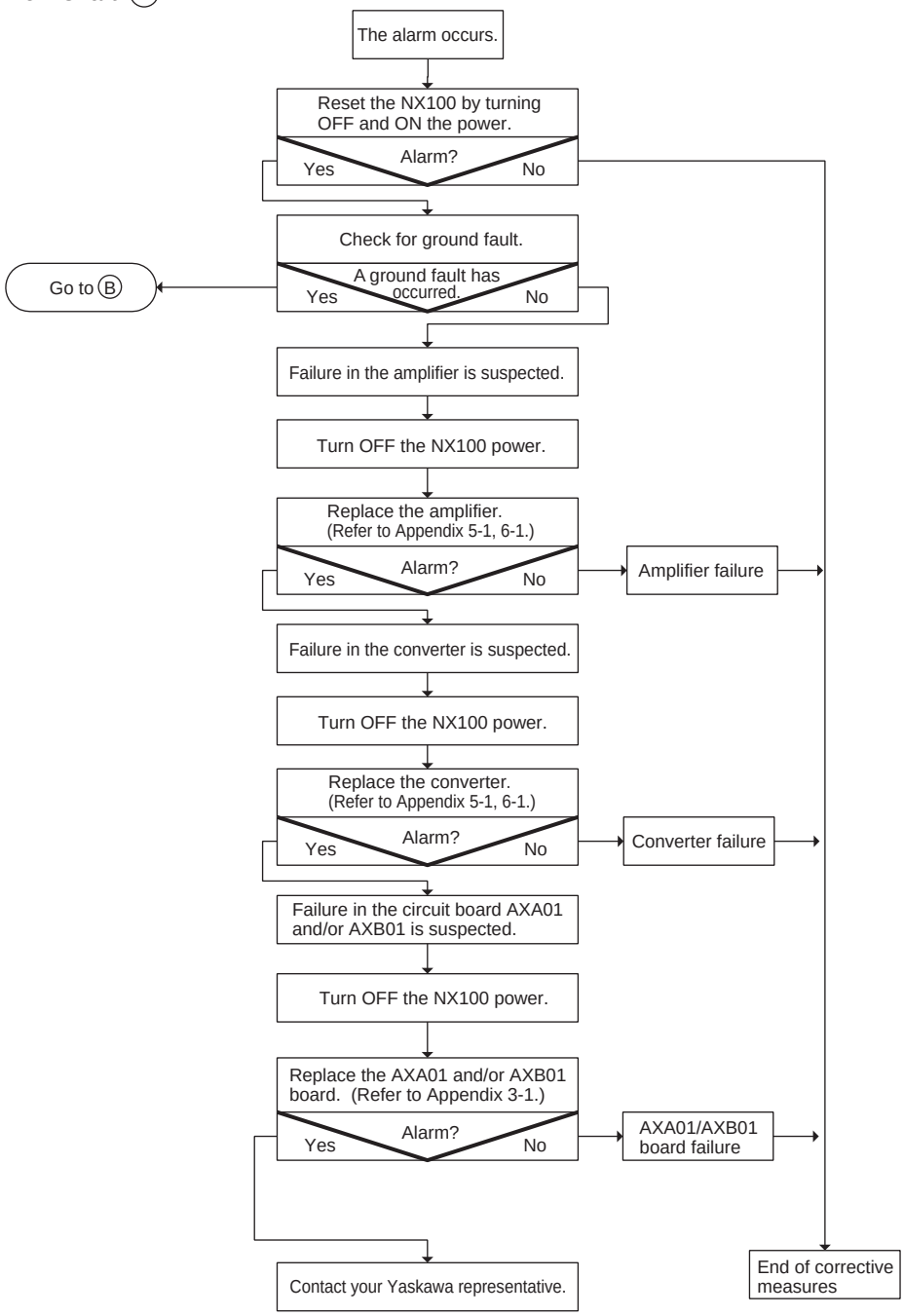


■ AL-4335

GROUND FAULT

This alarm occurs if there is a ground fault in any motor power line connected to converter.
When this alarm occurs, AL-4337 OVER CURRENT (AMPLIFIER) also occurs to all the axes driven by a common converter.
Sub Code: Signifies the axis in which the alarm occurred (If the axis is driven by a common converter, all the subject axes are signified.)

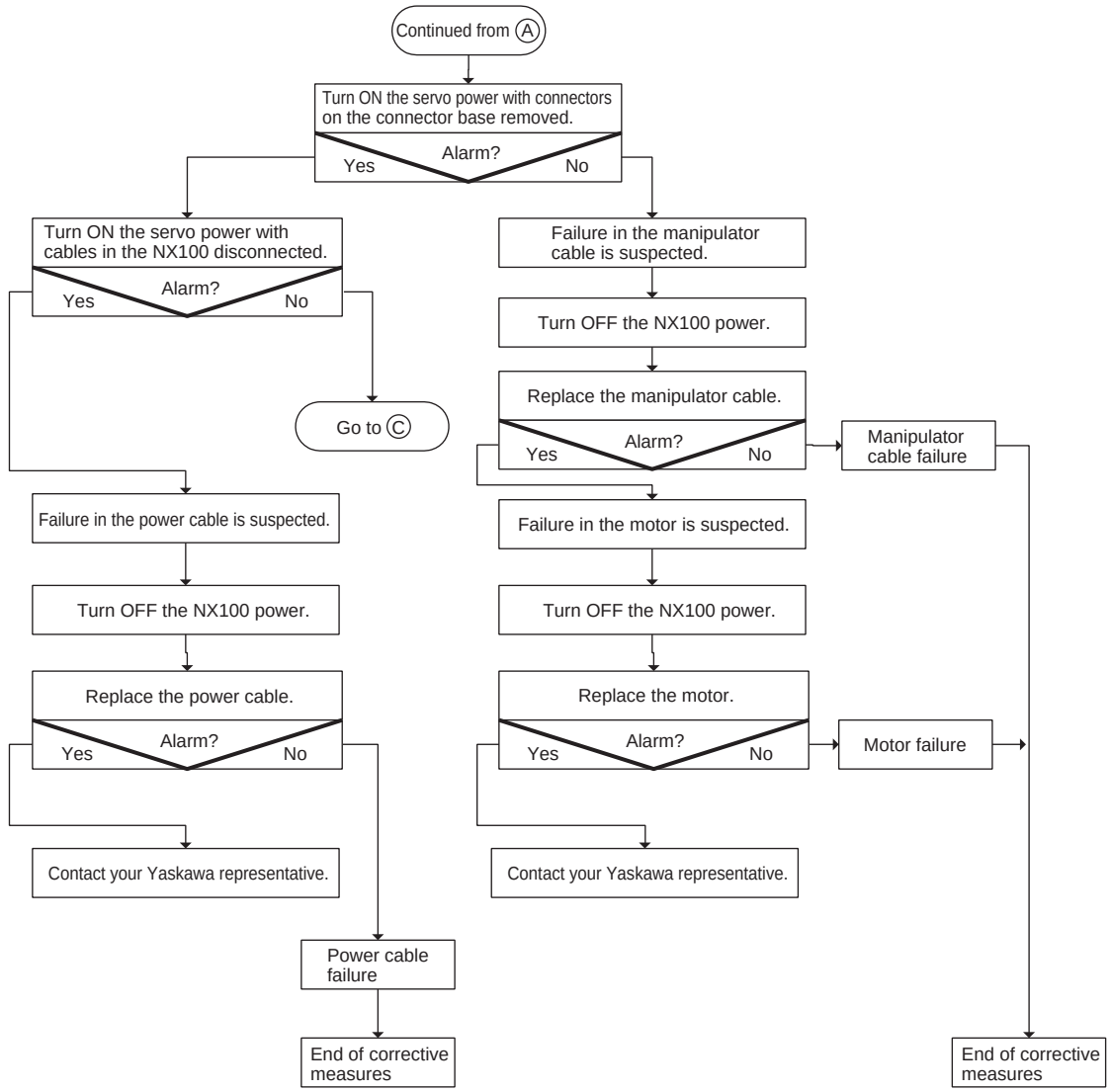
Flow Chart Ⓐ



■ AL-4335

GROUND FAULT

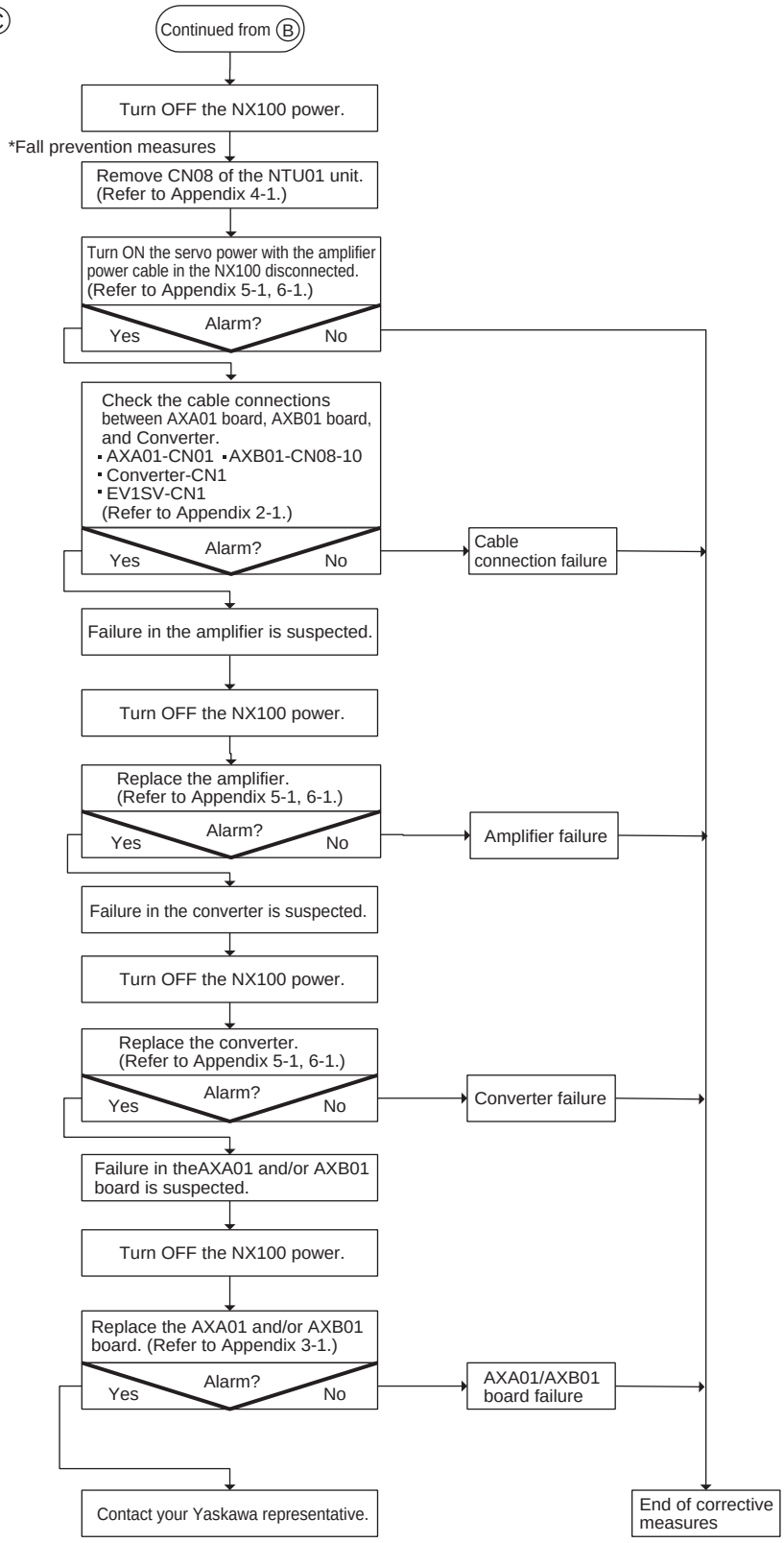
Flow Chart Ⓑ



■ AL-4335

GROUND FAULT

Flow Chart Ⓒ

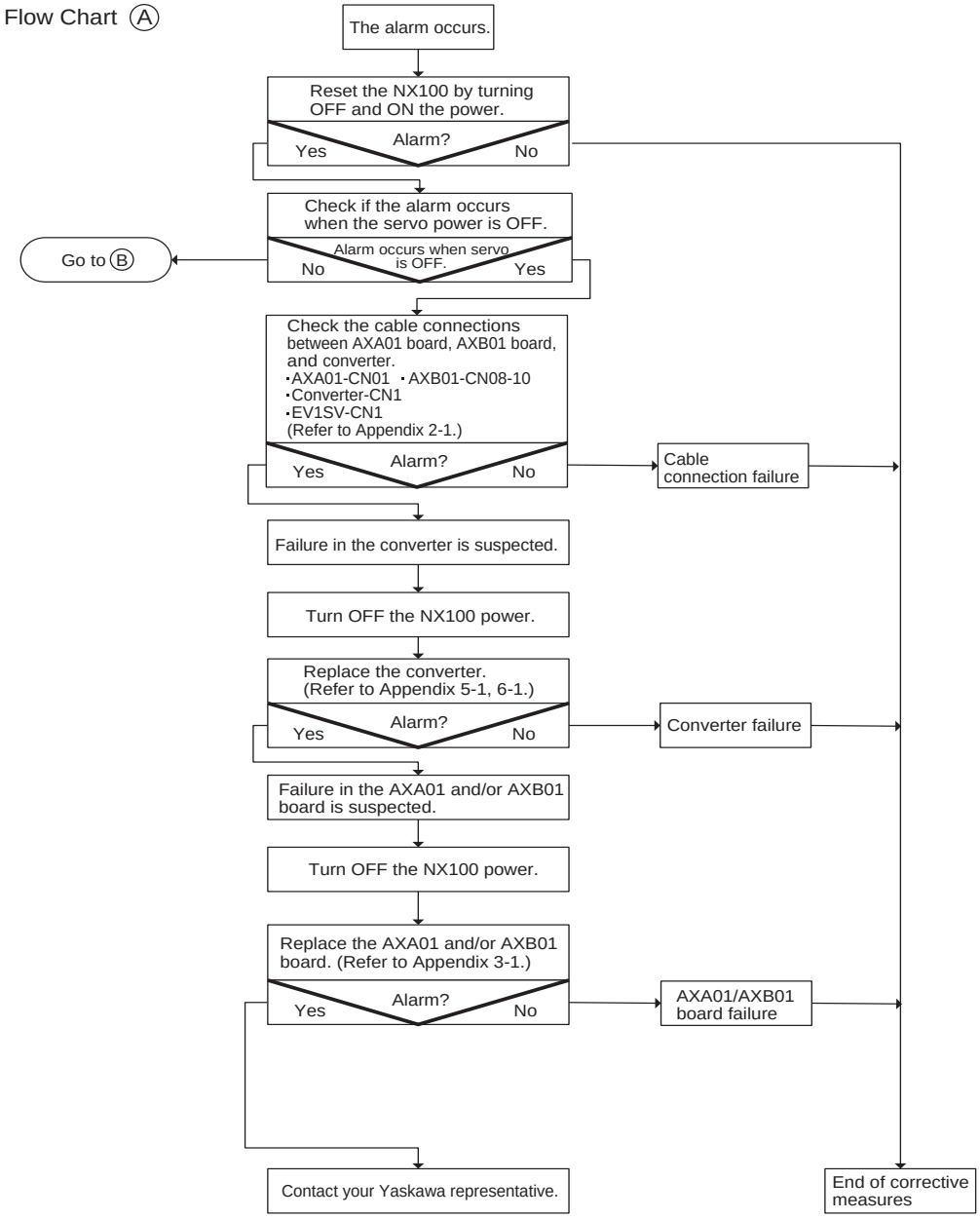


■ AL-4336

OPEN PHASE (CONVERTER)

This alarm occurs if there is an open phase in the converter input power.
As a major cause of this alarm, open phase of primary power or failure contactor is suspected.
Sub Code: Signifies the physical No. of converter in which the alarm occurred

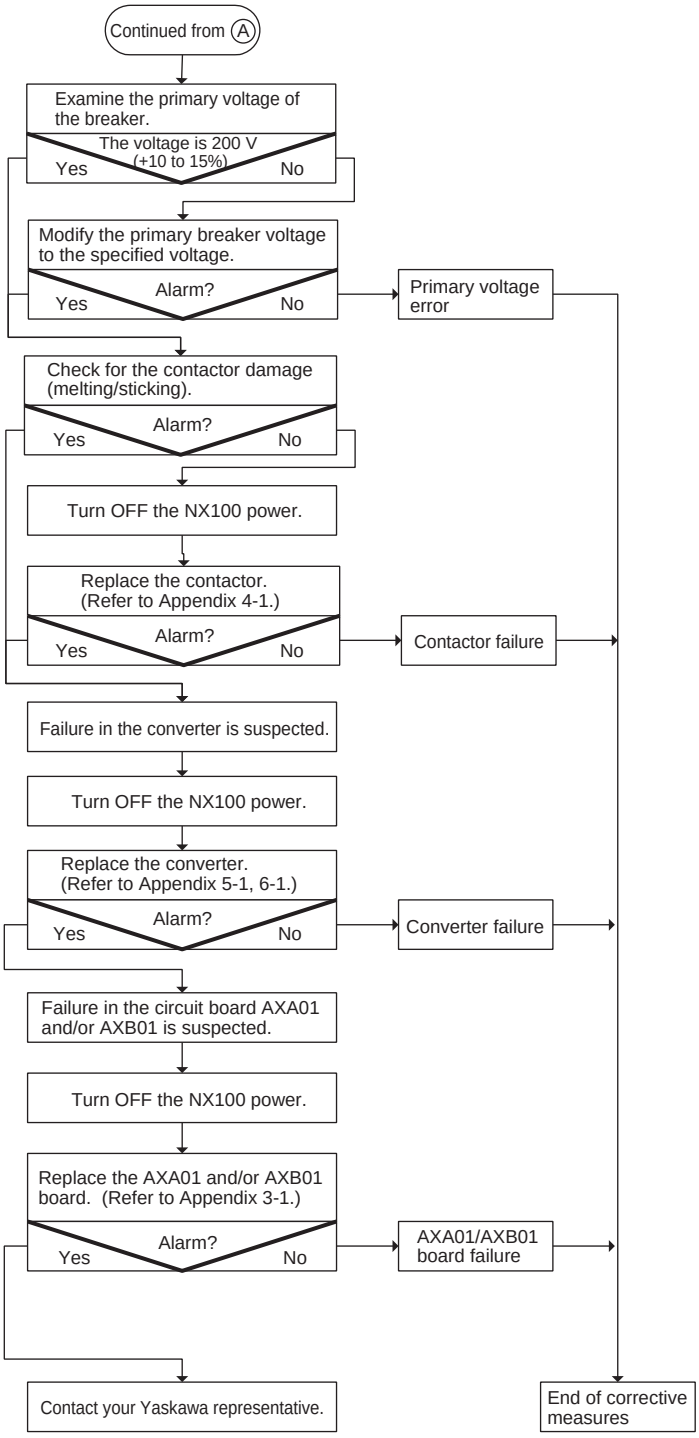
Flow Chart Ⓐ



■ AL-4336

OPEN PHASE (CONVERTER)

Flow Chart Ⓑ

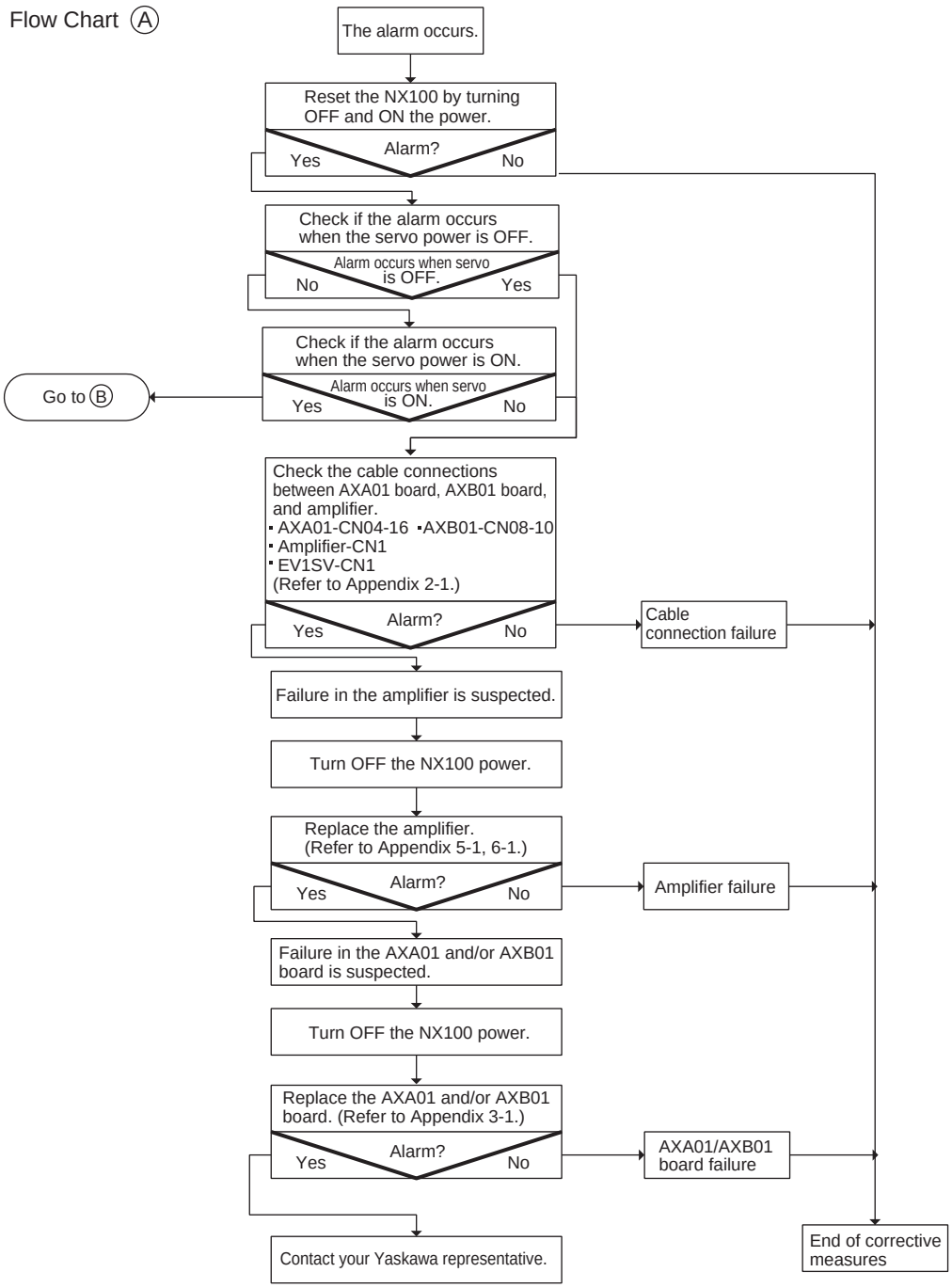


■ AL-4337

OVER CURRENT (AMPLIFIER)

This alarm occurs if a current exceeding the allowable maximum current is applied for amplifier.
As a cause of the alarm, a ground fault in the U, V, or W wire, or a short circuit between these wires is suspected.
As a cause of the alarm, the amplifier or the servo control board (AXA01, AXB01) failure is suspected.
Also, when AL-4335 GROUND FAULT alarm occurs, this alarm occurs to all the axes driven by the common converter.
Sub Code: Signifies the axis in which the alarm occurred

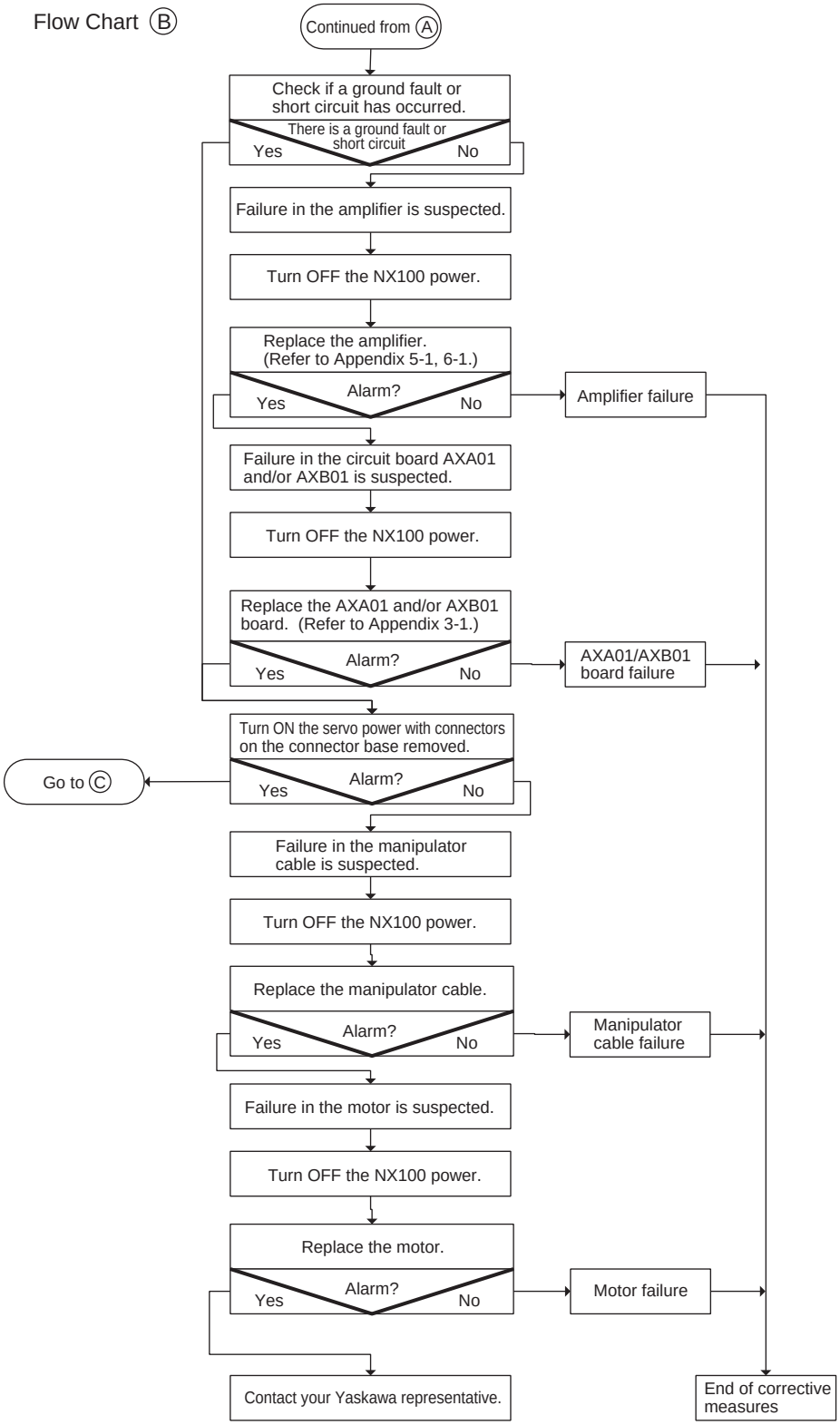
Flow Chart Ⓐ



■ AL-4337

OVER CURRENT (AMPLIFIER)

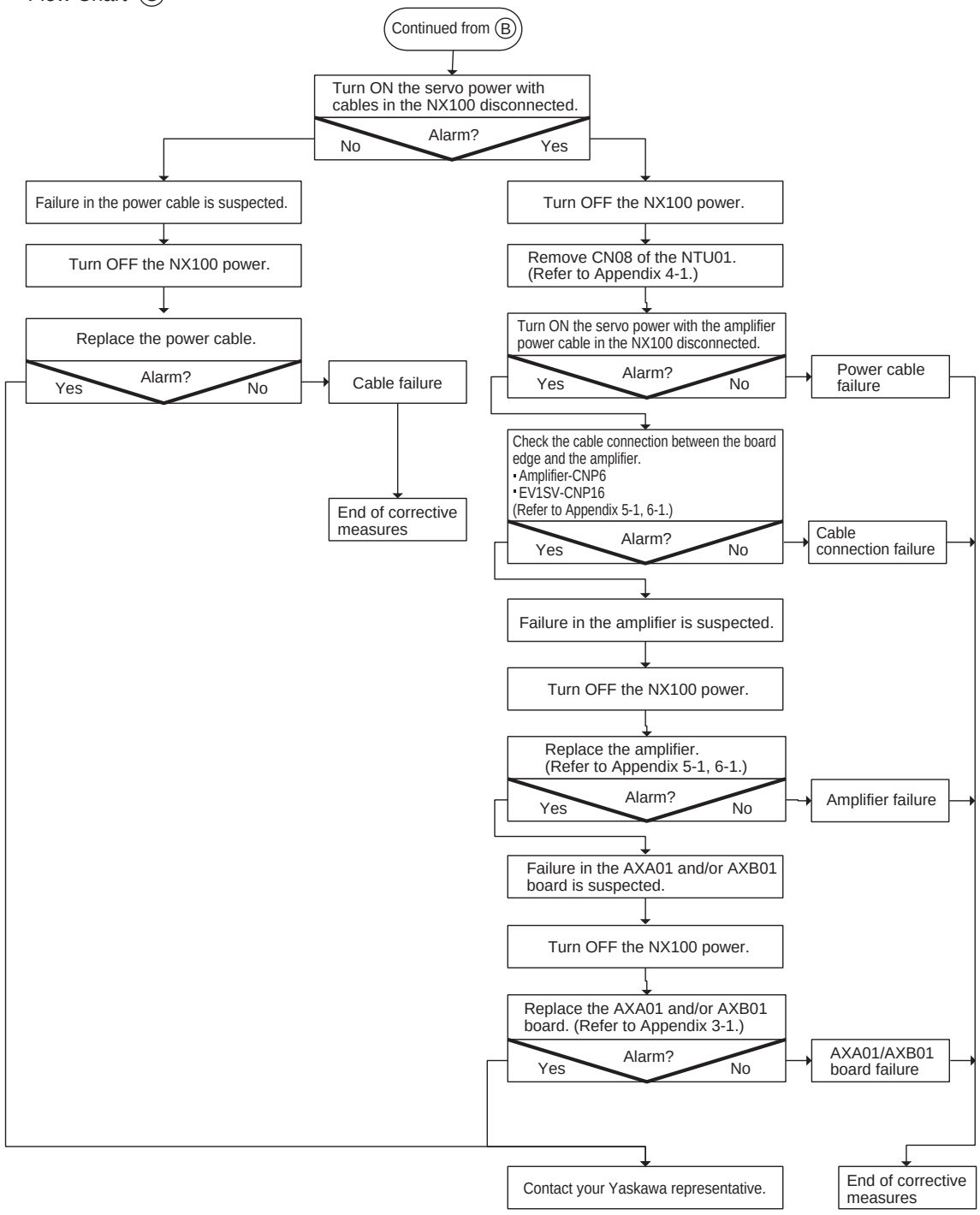
Flow Chart Ⓑ



■ AL-4337

OVER CURRENT (AMPLIFIER)

Flow Chart Ⓒ

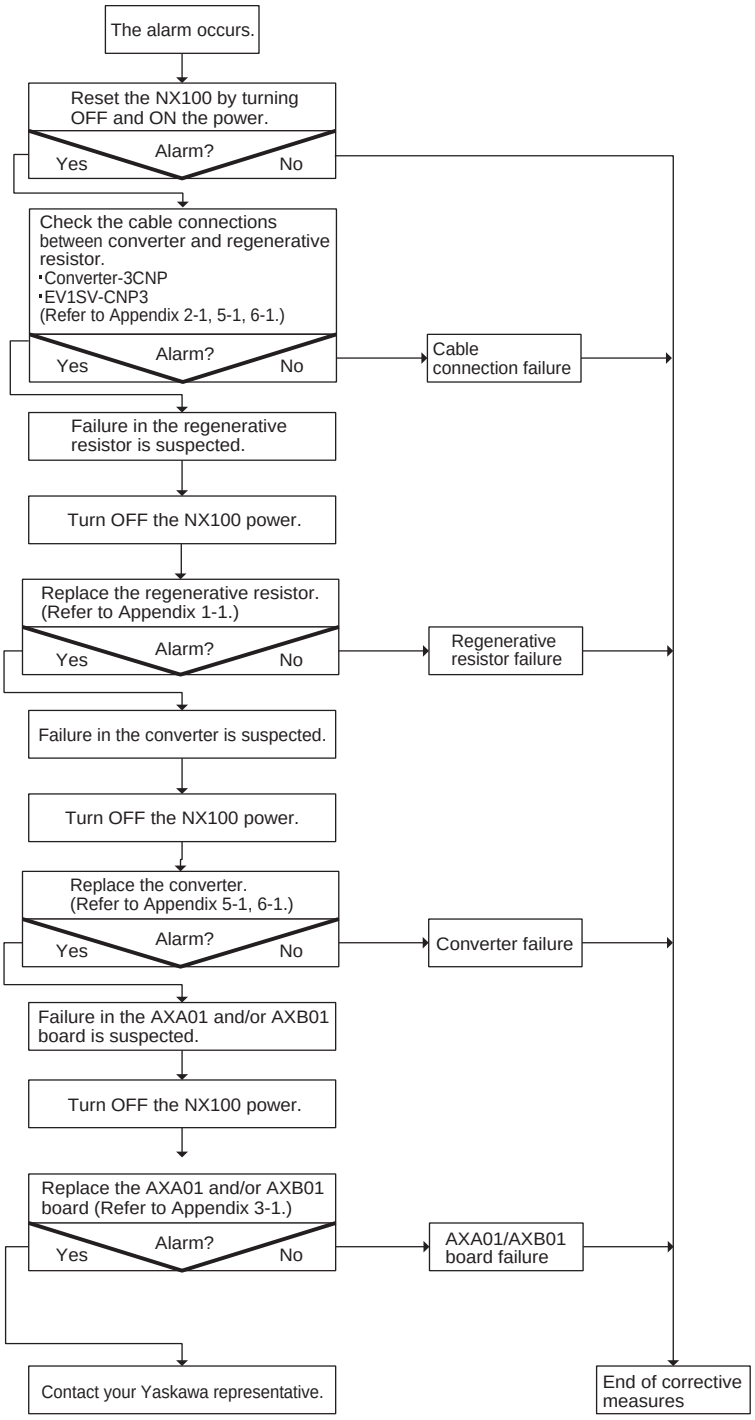


■ AL-4338

REGENERATION TROUBLE (CONVERTER)

This alarm occurs if the converter regenerative resistor cable is disconnected or short-circuited.

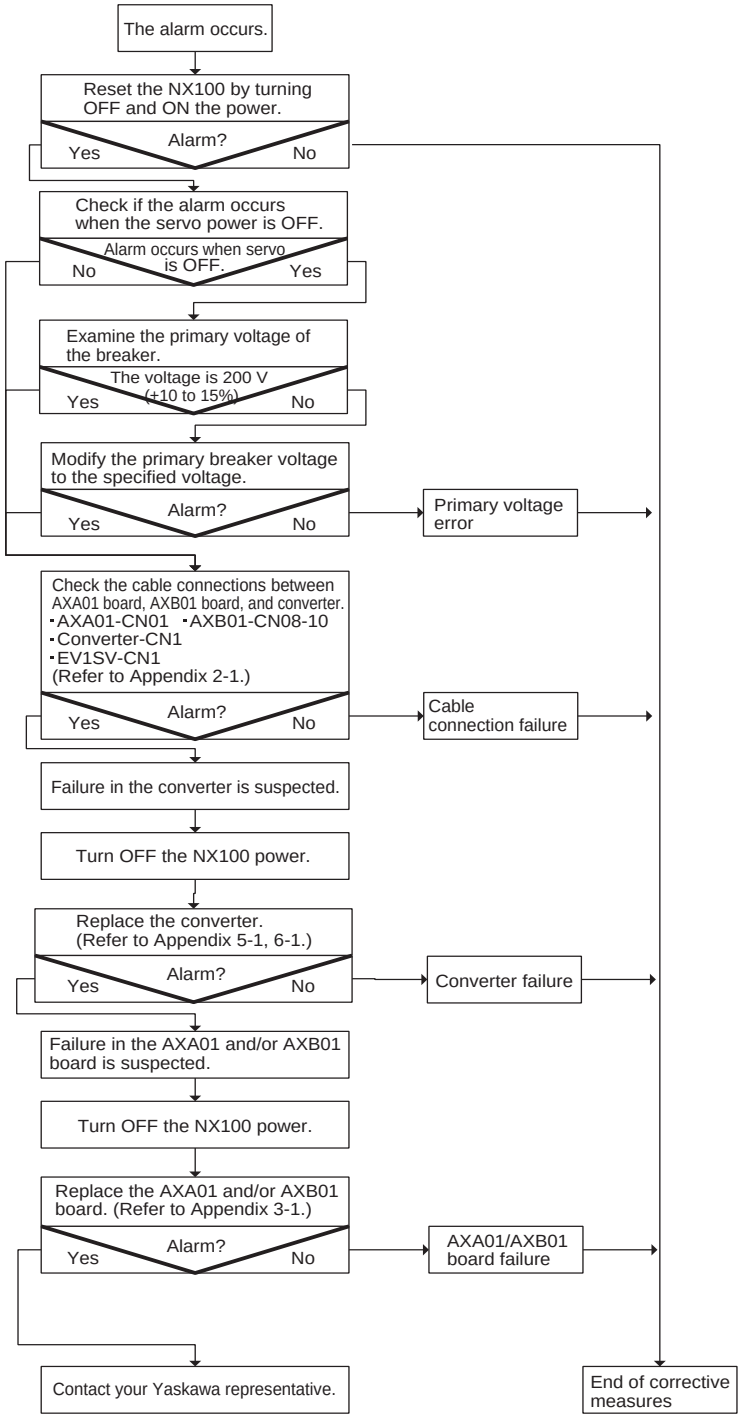
Sub Code: Signifies the axis in which the alarm occurred



■ AL-4339

INPUT POWER OVER VOLTAGE (CONVERTER)

This alarm occurs if the input voltage monitored by the NX100 exceeds 420 V.
As a cause of the alarm, a primary power supply voltage error or defective circuit board is suspected.
Sub Code: Signifies the physical No. of converter in which the alarm occurred



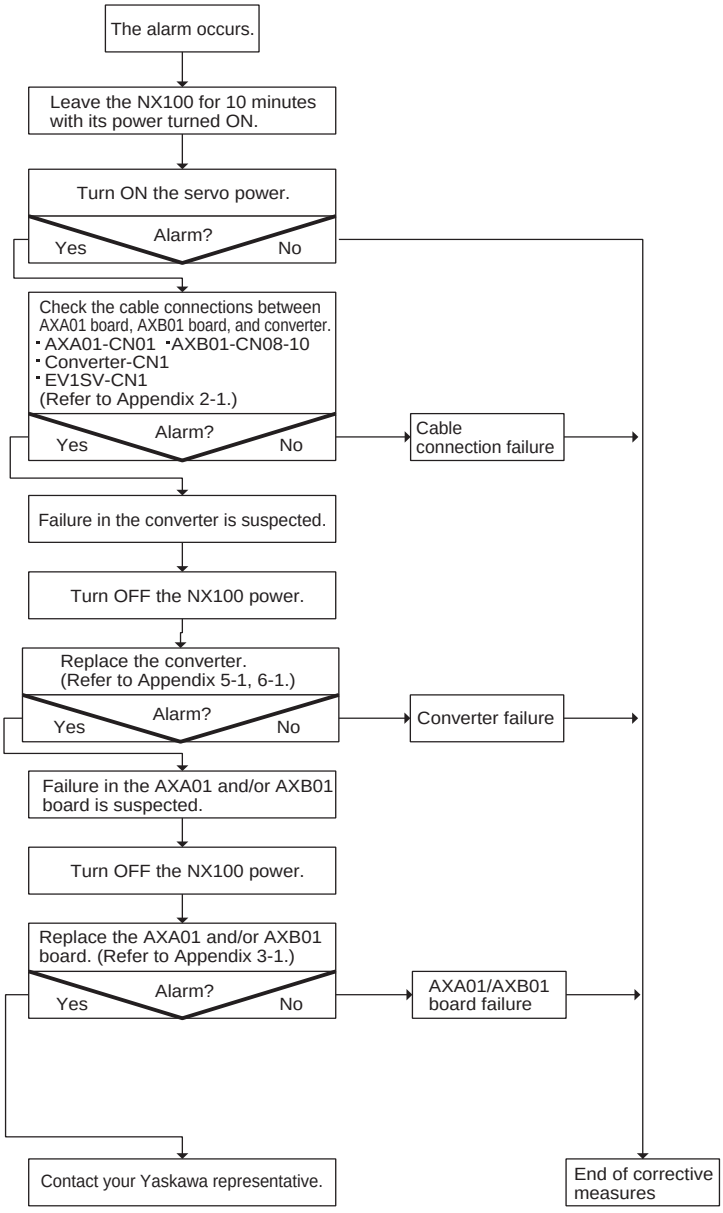
■ AL-4340

TEMPERATURE ERROR (CONVERTER)

This alarm occurs if the converter temperature monitored by the NX100 system exceeds the specified value.

As a cause of the alarm, an effect of ambient environment or temperature is suspected. In rare cases, the alarm cause is attributed to an error in the primary control power supply or noise interference.

Sub Code: Signifies the physical No. of converter in which the alarm occurred



■ AL-4344

LINEAR SERVOFLOAT TRACKING ERROR
The deviation of X, Y, and Z-axis exceeded the allowable limit while the linear servo float was in execution.
(1) Check the job.
Sub Code: Signifies the axis in which the alarm occurred

■ AL-4345

LNK SERVOFLOAT EXECUTE ERROR
The link servo float execution request was made for the axis that is already performing the linear servo float process.
(1) Check the job.
Sub Code: Signifies the axis in which the alarm occurred

■ AL-4346

LNK SERVOFLOAT TRQ LIMIT ERROR
The limit torque of the link servo float condition file is outside the specified range.
(1) Check the limit torque of the link servo float condition file.
Sub Code: Signifies the axis in which the alarm occurred

■ AL-4347

LNR SERVOFLOAT TRQ LIMIT ERROR
The limit torque of the linear servo float condition file is outside the specified range.
(1) Check the limit torque of the linear servo float condition file.
Sub Code: Signifies the axis in which the alarm occurred

■ AL-4348

LNR SERVOFLOAT COORD TYPE ERROR
The linear servo float execution request was made for the axis that is already performing the linear servo float process, specifying another coordinate system.
(1) Check the job and the linear servo float condition file.
Sub Code: Signifies the linear servo float condition file No. in which the alarm occurred

■ AL-4349

LNR SERVOFLOAT TOOL POSE ERROR
The linear servo float execution request was made for the axis that is already performing the linear servo float process, specifying different tool posture control.
(1) Check the job and the linear servo float condition file.
Sub Code: None

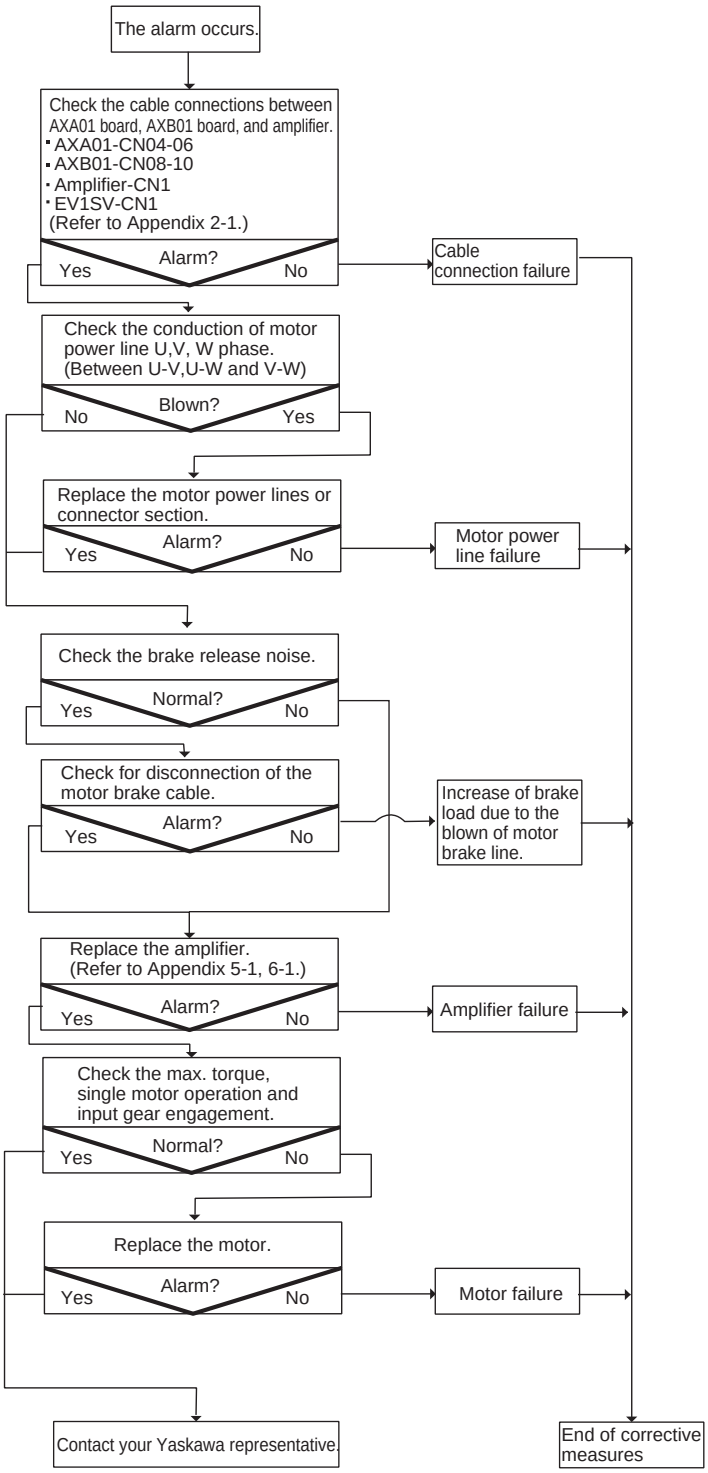
■ AL-4350

LNR SERVOFLOAT EXECUTE ERROR
The linear servo float execution request was made for the axis that is already performing the link servo float process.
(1) Check the job setting.
Sub Code: Signifies the axis in which the alarm occurred

■ AL-4352

TWIN DRIVE OVER DEVIATION

The deviation of the position error pulse from the twin drive axis exceeded the allowable limit.

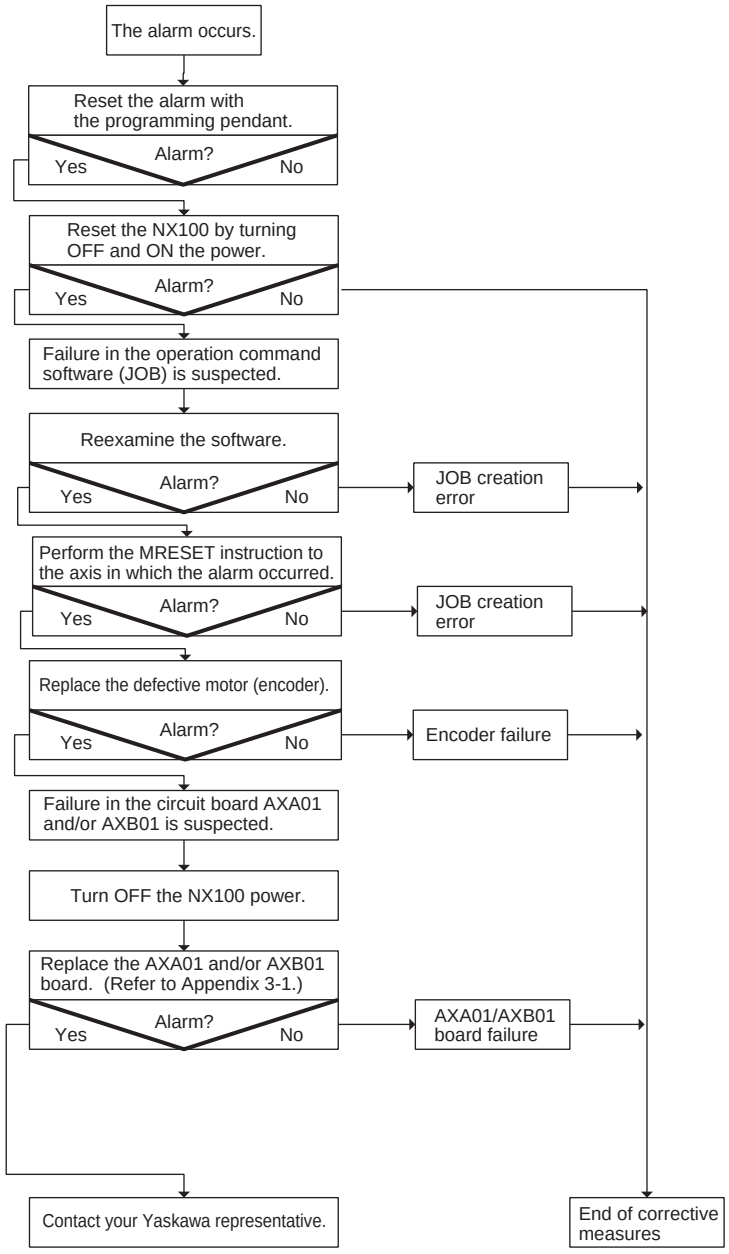


■ AL-4353

DEFECTIVE TAUGHT POINT (ENDLESS)

This alarm occurs if the feedback pulse count of endless operation axis exceeds the allowable maximum pulse count (229 = 536, 870, 912).

Sub Code: Signifies the axis in which the alarm occurred



■ AL-4355

EXTERNAL PRES DETECT (SERVOFLOAT)

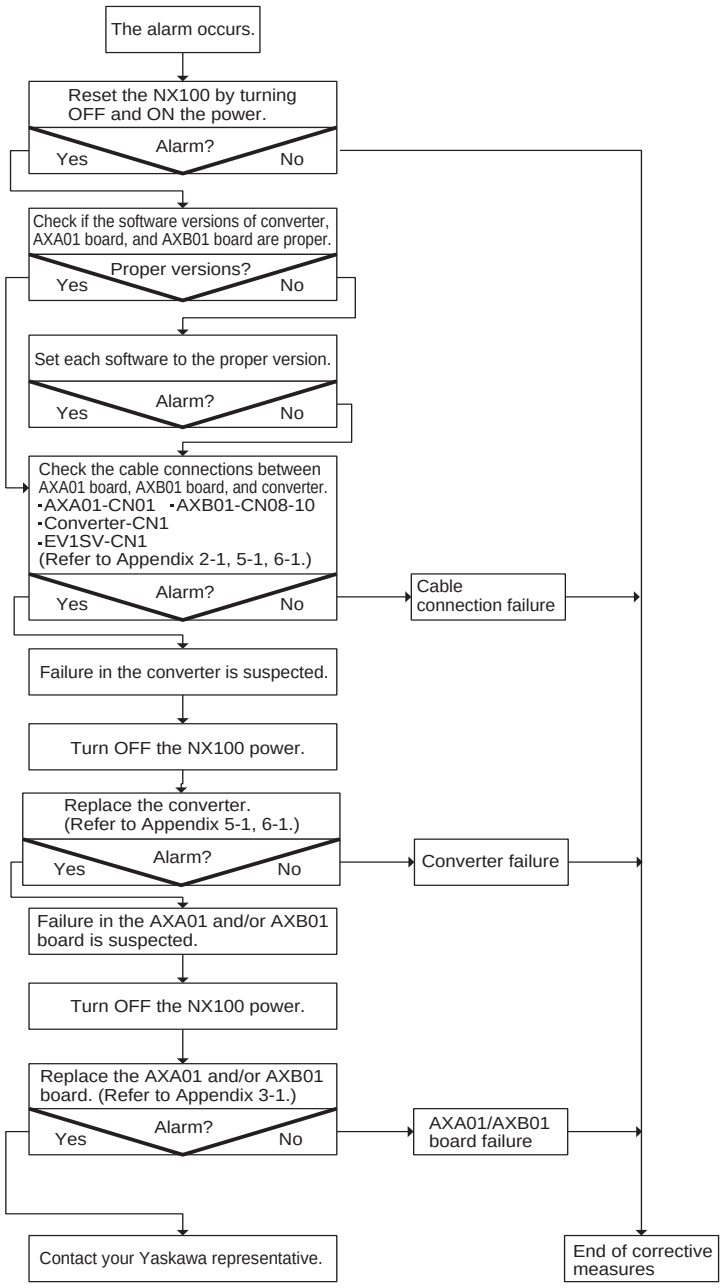
An external force above the threshold was detected on the servo-float executing axis.

(1) Check the job.

■ AL-4359

CONVERTER ERROR

Converter, AXA01, and AXB01 perform communications in the NX100 system.
This alarm occurs if the AXA01 and AXB01 have failed to identify the alarm type.
Sub Code: Signifies the physical No. of converter in which the alarm occurred

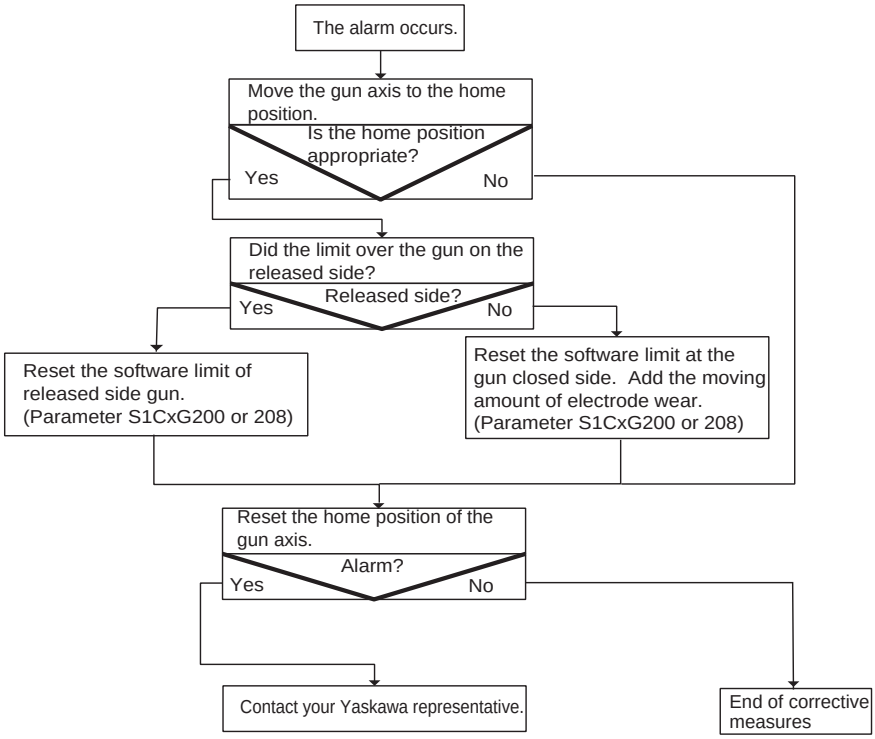


■ AL-4364

GUN AXIS SOFT LIMIT OVER

The gun axis overed the software limit while pressuring operation is excuted.

Sub Code: None

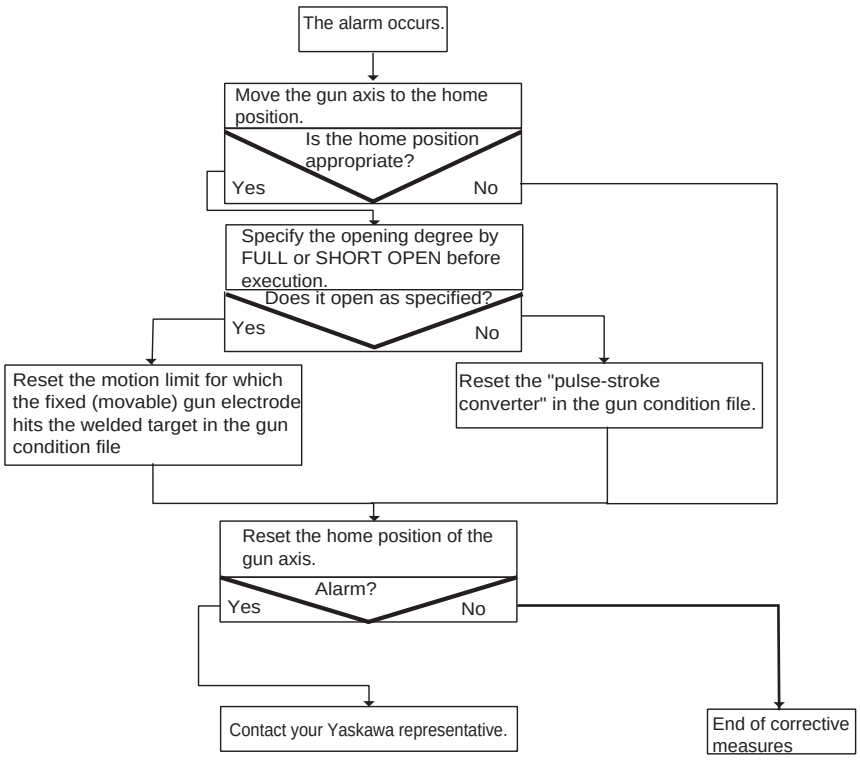


■ AL-4365

TOUCH DETECT DATA OVER

In comparison to the postion, where the gun electrode hits the welded target, at the previous wear detection, the position during pressuring exceeded the allowable limit which had been set in the motion limit for which the fixed (movable) gun electrode hits the welded target.

Sub Code: None



■ AL-4371

SYSTEM ERROR (SERVO)

Sub Code: Error Cause Data No.

2200: The independent brake control was performed in brake unit which was not supported for the control. Check the brake unit to see if it supports independent brake control.

6001: The DIN-1 signal is used by multiple functions. Check the setting.

6002: The DIN-2 signal is used by multiple functions. Check the setting.

6003: The DIN-3 signal is used by multiple functions. Check the setting.

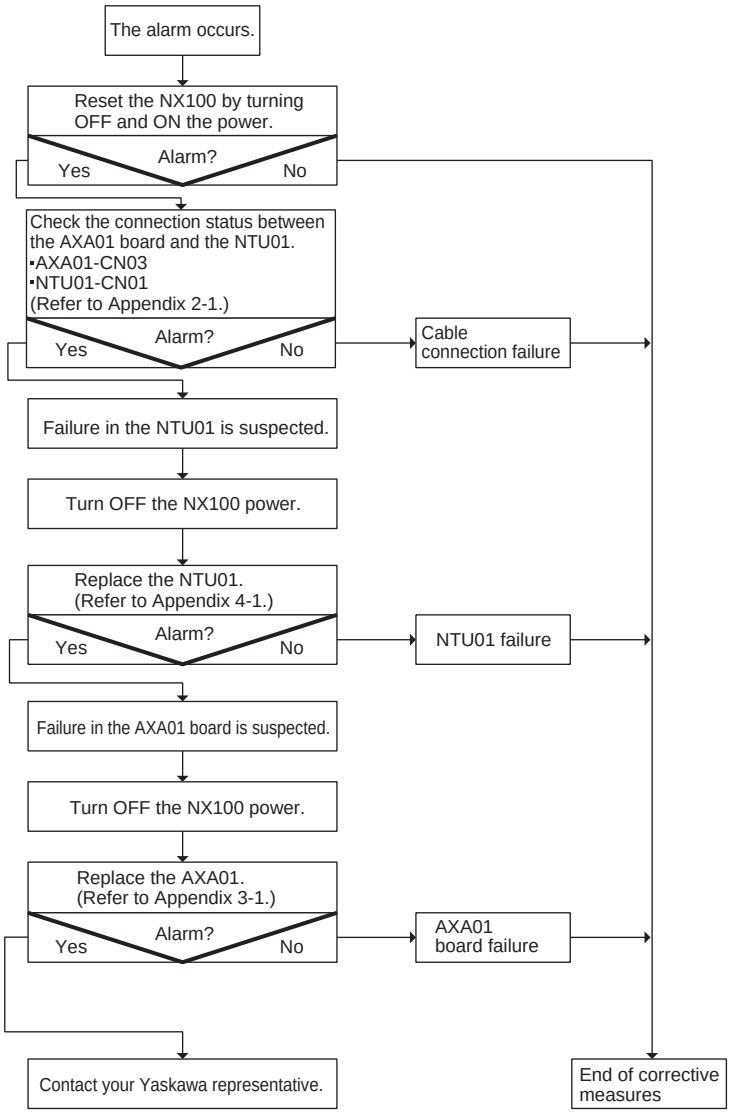
6004: The DIN-4 signal is used by multiple functions. Check the setting.

6005: The DIN-5 signal is used by multiple functions. Check the setting.

■ AL-4372

SERVO ON SIGNAL ERROR

The servo OFF status is sent from power-ON unit (NTU01).
As a cause of the alarm, a failure in the connection cable between the power-ON unit (NTU01) and the AXA01 is suspected.



■ AL-4400

NOT READY (ARITH)
This alarm occurs if the arithmetic process has not completed within regulated time.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 6: Signifies the internal software error

■ AL-4401

SEQUENCE TASK CONTR ERROR
This alarm occurs if an error is detected during job execution control.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 106: Signifies the internal software error

■ AL-4402

UNDEFINED COMMAND (ARITH)
An undefined command was issued to the arithmetic processing section.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code: None

■ AL-4404

ARITHMETIC ERROR
This alarm occurs if an error is detected in the arithmetic processing section.
Sub Code

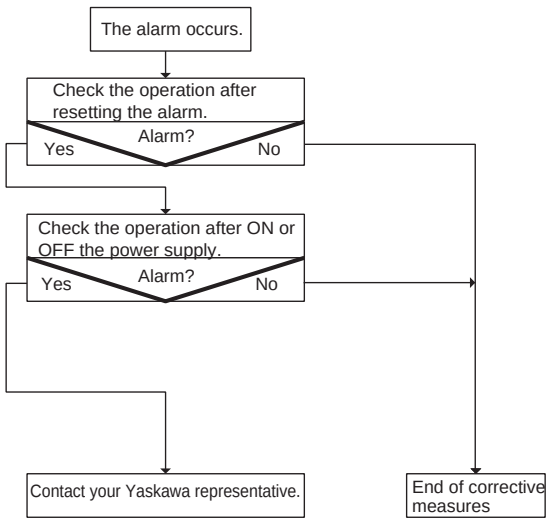
1 to 11: Signifies the internal software error

■ AL-4406

GROUP AXIS CONTROL ERROR
This alarm occurs if an error is detected in internal control group data during coordinated motion.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 12: Signifies the internal software error

4400 to 4402/4404/4406



■ AL-4407

TWO STEPS SAME POSITION (CIRC)
Among three taught points in a circular interpolation step, this alarm occurs if two or three points are on the same point.
Sub Code: None
< Remedy >
Modify the teaching points of circular interpolation step so that multiple points will not be placed at the same point and check operation.

■ AL-4408

TWO STEPS SAME POSITION (SPLINE)
Among three taught points in a spline interpolation step, this alarm occurs if two or three points are on the same point.
Sub Code: None
< Remedy >
Modify the teaching points of spline interpolation step so that multiple points will not be placed at the same point and check operation.

■ AL-4409

TWO STEPS SAME POSITION (3 STEPS)
Among three taught points to create an user coordinate system, this alarm occurs if two or three points are on the same point.
Sub Code: None
< Remedy >
Teach the different 3 points to create an user coordinate system again.

■ AL-4410

TWO STEPS SAME POSITION (WEAV)
Among three taught points to create a weaving coordinate system, this alarm occurs if two or three points are on the same point.
Sub Code: None
< Remedy >
Teach the different 3 points (start, end, and reference points) again.

■ AL-4411

TEACH ERROR (SPLINE)
This alarm occurs if the distance between the teaching points in the spline interpolation section is not equidistant.
Sub Code: None
< Remedy >
Teach the positions so that the distance between the teaching points is even.

■ AL-4412

IMPOSSIBLE LINEAR MOTION (L/U)
This alarm occurs if the form of LU axis (bending direction) differs between the start point and the end point in a move instruction other than MOVJ.
Sub Code: None
< Remedy >

- Perform the teaching again to make the form of L- and U-axes same at start point and end point.
- Perform the teaching again use a MOVJ instruction.

■ AL-4413

IMPOSSIBLE LINEAR MOTION (S/L)

This alarm occurs if the form of SL axis (bending direction) differs between the start point and the end point in a move instruction other than MOVJ.

Sub Code: None

< Remedy >

- Perform the teaching again to make the form of S- and L-axes same at start point and end point.
- Perform the teaching again use a MOVJ instruction.

■ AL-4414

EXCESSIVE SEGMANT (LOW)

This alarm occurs if the operation command which exceeds the designated max. speed is output.
It may occur when the robot operates near the singular point or when the robot is going to change its orientation widely with single control point.

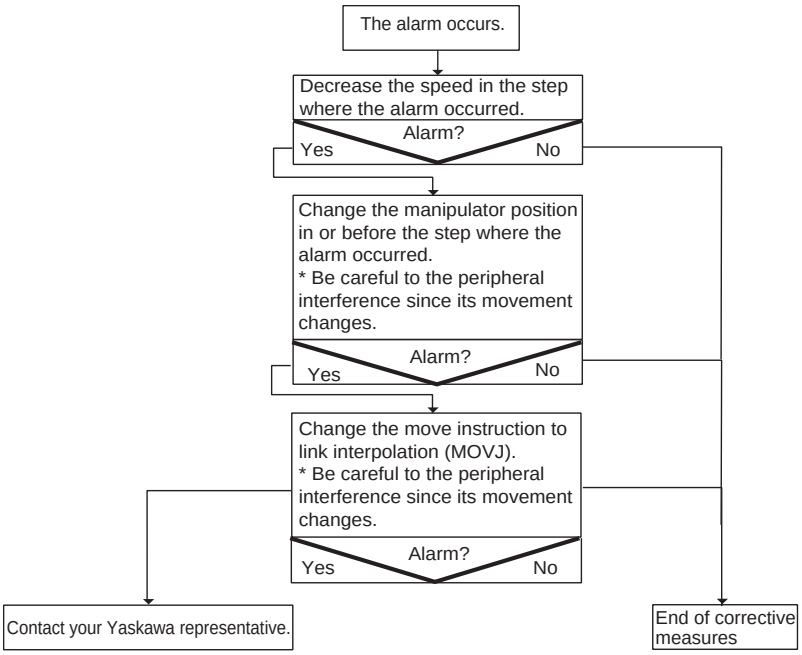
Sub Code: Signifies the control group and the axis in which the alarm occurred

■ AL-4415

EXCESSIVE SEGMENT (HIGH)

This alarm occurs if the operation command which exceeds the designated max. speed is output.
It may occur when the robot operates near the singular point or when the robot is going to change its orientation widely with single control point.

Sub Code: Signifies the control group and the axis in which the alarm occurred



■ AL-4416/AL-4417

PULSE LIMIT
This alarm occurs if each axes exceeds its motion limit (pulse limit).
Sub Code: Signifies the control group and the axis in which the alarm occurred

■ AL-4418/AL-4419

CUBE LIMIT
This alarm occurs if the manipulator TCP exceeds its motion limit (cube limit) in the negative(-) direction.
Sub Code: Signifies the control group and XYZ in which the alarm occurred

■ AL-4420/AL-4421

SPECIAL SOFT LIMIT
This alarm occurs if the manipulator axis exceeded its motion limit (special software limit).
Sub Code: Signifies the control group and the axis in which the alarm occurred

■ AL-4422/AL-4423

MECHANICAL INTERFERENCE
This alarm occurs if the intended operation direction would be interfered by the robot link.
Sub Code: Signifies the control group and the axis in which the alarm occurred

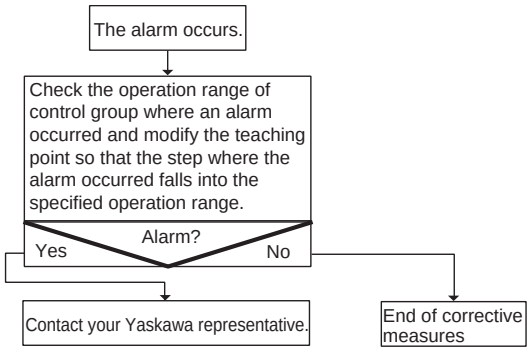
■ AL-4424/AL-4425

SPECIAL MECHANICAL INTRF
This alarm occurs if the intended operation direction would be interfered by the robot link.
Sub Code: Signifies the control group and the axis in which the alarm occurred

■ AL-4426/AL-4427

PULSE MECHANICAL LIMIT
This alarm occurs if each axes exceeds its motion limit (mechanical limit).
Sub Code: Signifies the control group and the axis in which the alarm occurred

AL-4416 TO AL-4427



■ AL-4428

SEGMENT CONTROL ERROR
This alarm occurs if an error is detected in the segment processing section.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 9: Signifies the internal software error

■ AL-4429

WRONG SPECIFIED CONTROL GROUP
This alarm occurs if an error is detected in the manipulator information at job execution.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 19: Signifies the internal software error

■ AL-4430

CPU COMMUNICATION ERROR
This alarm occurs if an error is detected in interrupt process between CPUs.
The RAM data failure in the NCP01 or YCP02 boards may also cause this alarm.
Sub Code

1 to 9: Signifies the internal software error

■ AL-4431

JHM ERROR
This alarm occurs if an error is detected in job control process.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 6: Signifies the internal software error

■ AL-4432

INSTRUCTION INTERPRETER ERROR
This alarm occurs if an error is detected in instruction interpretation/execution process.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 255: Signifies the internal software error

■ AL-4433

UNDEFINED GLOBAL VARIABLE
This alarm occurs if the memory area for global (user) variables is not reserved.
The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

1 to 7: Signifies the internal software error

■ AL-4435

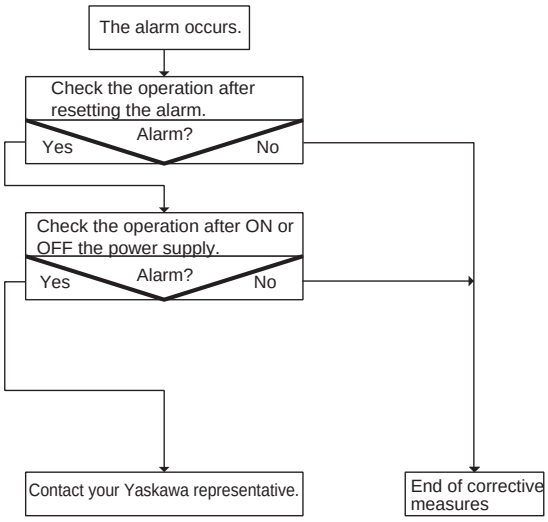
UNDEFINED LOCAL-VARIABLE

The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

1 to 7: Signifies the internal software error

4428 to 4433/4435



■ AL-4436

LESS THAN 3 STEPS (CIRCULAR)

This alarm occurs if there is no continuous three points or more for circular interpolation step.

Sub Code: None

<Remedy>

Perform teaching so that circulation interpolation steps are continuous three points or more, and then check the operation.

■ AL-4437

LESS THAN 3 STEPS (SPLINE)

This alarm occurs if there is no continuous three points or more for spline interpolation step.

Sub Code: None

<Remedy>

Perform teaching so that spline interpolation steps are continuous three points or more, and then check the operation.

■ AL-4438

UNDEFINED JOB

This alarm occurs if the job to be executed is not registered.

Sub Code: None

<Remedy>

Check that the CALL/JUMP destination job is registered. If not, delete the CALL/JUMP instruction where the alarm occurred and check operation.

■ AL-4439

UNDEFINED LABEL

This alarm occurs if the label for jump destination is not registered in the job.

Sub Code: None

<Remedy>

Check that the JUMP destination job is registered. If not, delete the JUMP instruction where the alarm occurred and check operation.

■ AL-4440

UNDEFINED RETURN JOB

This alarm occurs if the call source job does not exist in the job call stack.

Sub Code: None

<Remedy>

- If the start job incorrectly contains the RET instruction, delete it.
- Check if the RET instruction was executed when the job call stack did not contain any job. If this is the case, execute the instruction from the master (start) job.

■ AL-4441

LACK OF LOCAL-VARIABLE AREA

This alarm occurs if the memory area is lacking because too many local variables in the job are used.

Sub Code: None

<Remedy>

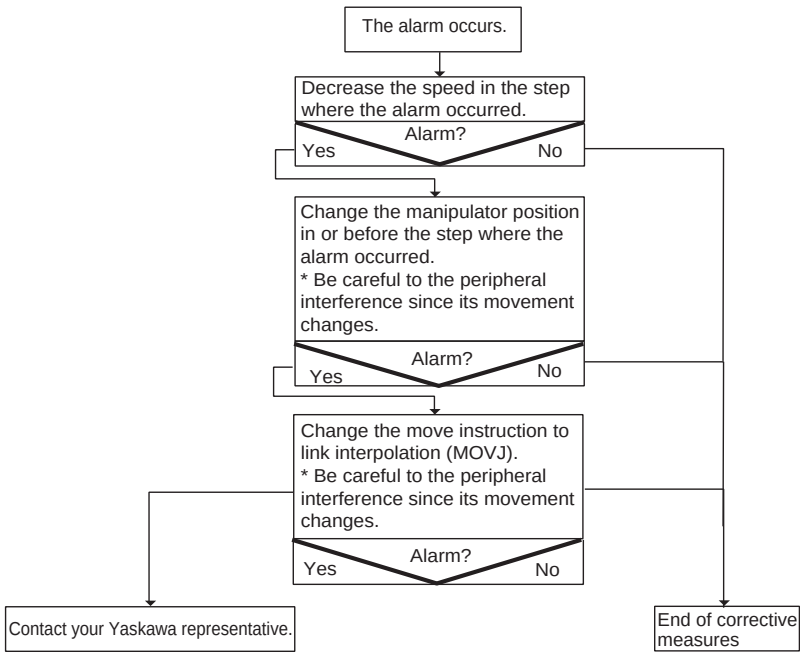
Reduce the number of local variables to be used, and then check the operation.

■ AL-4444

UMSUCCESSFUL FINE POSITIONING

When PL= 0 or an external servo turned OFF, the number of the servo error pulses did not fall in the limit range that had been set in a parameter, within the specified time.

Sub Code: None



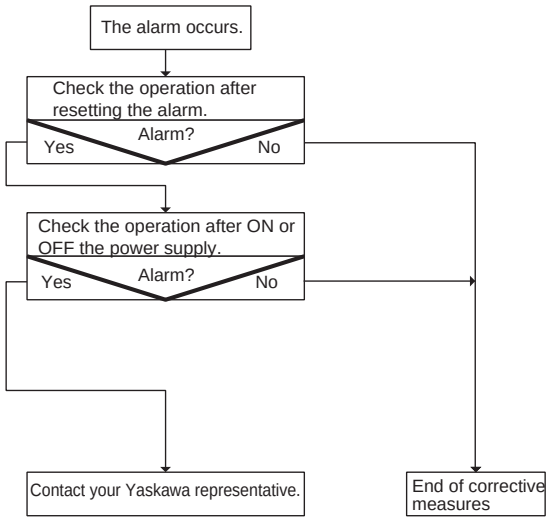
■ AL-4445

DATA PRESET ERROR

This alarm occurs if the NX100 detects data error in instruction interpretation prereading process. The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

1 to 255: Signifies the internal software error



■ AL-4446

OVER VARIABLE LIMIT

The value set in the tag for instruction exceeded the limit.

Sub Code: None

< Remedy >

Check the variable settings and modify the job so that it falls in the tag input range and check operation.

■ AL-4447

DEFECTIVE TAUGHT POINT (CIRC)

This alarm occurs if the three points taught for the circular interpolation step lie in a straight line.

Sub Code: None

< Remedy >

Teach the 3 points again so that they do not lie in a straight line, and then check the operation.

■ AL-4448

WEAVING CONTROL ERROR

This alarm occurs if an error is detected in weaving control.

Sub Code

4: When the speed is specified by weaving time in the weaving file, zero or the negative value is set for the weaving time.

< Remedy >

Reset the value 0.1 seconds or more.

5: When the speed is specified by frequency in the weaving file, zero or the negative value is set for the frequency.

< Remedy >

Reset the value 0.1 Hz or more.

6: When the timer mode is specified in the weaving file, a negative value is set for the timer value.

< Remedy >

Set a positive value for the timer value.

7: For triangle or L-type weaving, zero is set for the vertical or horizontal distance.

< Remedy >

Set a positive value for the vertical and horizontal distance.

8: The coordinate control axis designation for the reference point is different from actual control axis.

< Remedy >

Align the control group specification an the reference point with the control group specification for weaving execution.

9: The distance between the point P and the TCP could not be calculated in wrist weaving.

< Remedy >

Set the correct dimensions in the tool data.

10: The Y-direction element of circular coordinate system for circular wrist weaving could not be calculated.

< Remedy >

Set the correct dimensions in the tool data.

11: The Y-direction element of circular coordinate system for circular wrist weaving could not be calculated.

< Remedy >

Check the wall direction and horizontal direction settings.

12: The X-direction element of circular coordinate system for circular wrist weaving could not be calculated.

< Remedy >

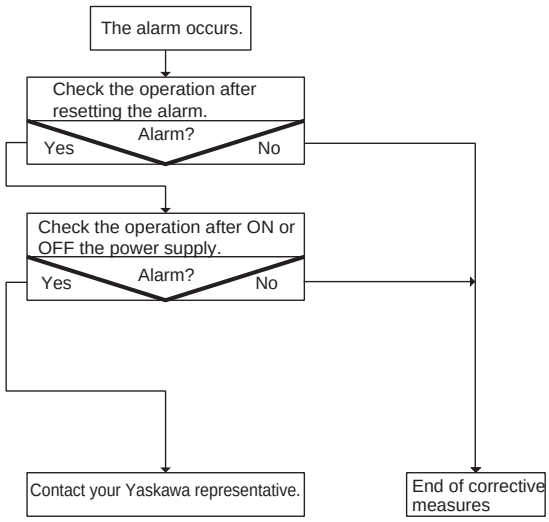
Check the wall direction and horizontal direction settings.

16: The weaving synchronizing file No. is incorrect.

< Remedy >

Correct the weaving synchronizing file No. to be used.

1,14 to 15: Signifies the internal software error



■ AL-4449

UNMATCHED POSN VAR DATA TYPE
This alarm occurs if the position type variable data type is different.
Sub Code: None
< Remedy >
Match the position type variable data type for the storage source/destination, and then check the operation.

■ AL-4450

FILE NO. ERROR
This alarm occurs if an error occurs in file number check.
Sub Code:

1: An error occurred in tool file number check.
2: An error occurred in user coordinate file number check.
3: An error occurred in calibration file number check between the manipulators.
4: An error occurred in tool calibration file number check.
5: An error occurred in reference point number check.
7: An error occurred in check for welding start condition file number.
8: An error occurred in check for welding end condition file number.
9: An error occurred in conveyor characteristic file number check.
10: An error occurred in press characteristic file number check.
12: An error occurred in conveyor calibration file number check.
13: An error occurred in argument number check.
14: An error occurred in check for motor gun condition file number.

< Remedy >
Modify the specified file number within the range, and then check the operation.

■ AL-4451

UNDEFINED REFERENCE POINT
This alarm occurs if the reference point is not registered or is insufficient.
Sub Code: Reference point No.
< Remedy >
Teach the reference point again, and then check the operation.

■ AL-4452

STACK MORE THAN 10 (JOB CALL)
This alarm occurs if an attempt is made to add more than twelve stacks in the job call stack at CALL instruction execution.
Sub Code: None
< Remedy >
Correct the job so that the number of nests for CALL instruction is twelve or less, and then check the operation.

■ AL-4453

OVER VARIABLE NO.
This alarm occurs if the variable number is out of range.
Sub Code: Variable number
< Remedy >
Correct the job using the variable number within the range, and then check the operation.

■ AL-4454

UNDEFINED POWER SOURCE COND.
This alarm occurs if the arc welding characteristic file is not set.
Sub Code: None
< Remedy >
Complete the settings for the arc welding characteristic file, and then check the operation.

■ AL-4455

UNDEFINED ARC START COND FILE
This alarm occurs if the welding start condition file is not set.
Sub Code: None
< Remedy >
Complete the settings for the welding start condition file, and then check the operation.

■ AL-4456

UNDEFINED ARC END COND FILE
This alarm occurs if the welding end condition file is not set.
Sub Code: None
< Remedy >
Complete the settings for the welding end condition file, and then check the operation.

■ AL-4457

WRONG WELDER SELECTION
This alarm occurs if the reference unit for the welding voltage and the welder type (independent/unified) do not match.
Sub Code: None
< Remedy >
Correct the reference unit for the welding voltage, and then check the operation.

■ AL-4459

EXCESSIVE INSTRUCTION EQUATION
This alarm occurs if the operation is impossible because the expression is too long.
Sub Code: None
< Remedy >
Correct the job so that the expression may shorten separating the operation expression, and then check the operation.

■ AL-4460

ZERO DIVIDED OCCURRENCE
This alarm occurs if the zero division is done.
Sub Code: None
< Remedy >
Correct the job so as not to divide by zero, and then check the operation.

■ AL-4461

UNDEFINED AUTO-WELD RELEASE COND

This alarm occurs if the number of welding release condition is zero for arc auxiliary file in automatic welding release process.

Sub Code: None

< Remedy >

Set the number of times of welding release condition other than zero, and then check the operation.

■ AL-4462

UNDEFINED POSITION FOR ARC RETRY

This alarm occurs if the arc retry has been set, but no move instruction exists following ARCON instruction.

Sub Code: None

< Remedy >

Set the move instruction following ARCON instruction, and then check the operation.

■ AL-4463

PARITY ERROR

This alarm occurs if the parity check for user I/O group detects the data error.

Sub Code: None

< Remedy >

Check the parity data setting of the user I/O group.

■ AL-4464

OVER BCD RANGE

This alarm occurs if the BCD value exceeds the limit.

Sub Code: None

< Remedy >

Correct the BCD data so as not to exceed the range.

■ AL-4465

OVER BINARY RANGE (PARITY CHECK)

The binary data exceeds the limit. This alarm occurs if an attempt is made to output a value that exceeded 127 (decimal) to the user I/O when parity check is specified.

Sub Code: None

< Remedy >

Correct the binary data so as not to exceed the range.

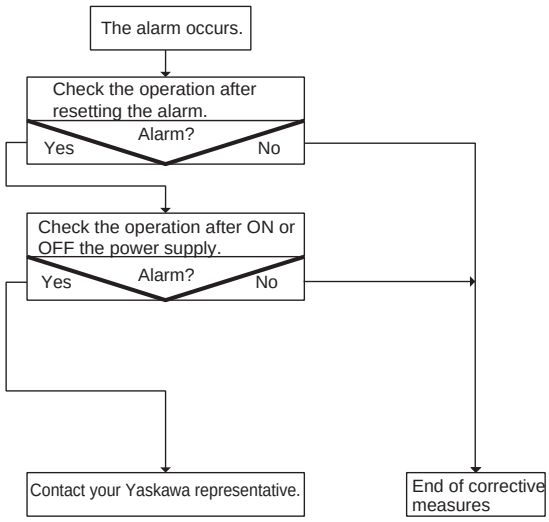
■ AL-4466

OFFLINE UNDEFINED COMMAND (ARITH)

This alarm occurs if an invalid command is issued to the offline position-data preparation section.
The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

0: Signifies the internal software error



■ AL-4467

USER COORD STEP NOT ENOUGH

This alarm occurs if the number of steps is lacking for a job for user coordinate creation.

Sub Code: None

< Remedy >

Correct the job so that the number of steps will be three or more.

■ AL-4468

ROBOT CALIBRATION DATA ERROR

This alarm occurs if an error is detected during calibration between manipulators.
The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

1 to 9: Signifies the internal software error

■ AL-4469

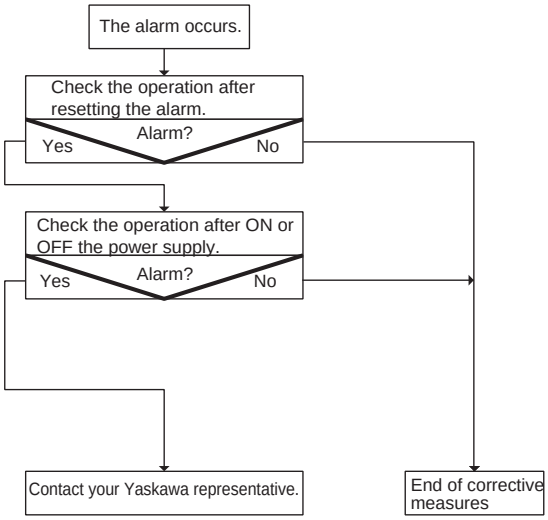
ROBOT CALIBRATION FRAME ERROR

This alarm occurs if an error is detected during calibration coordinate conversion between manipulators. The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

1 to 5: Signifies the internal software error

4468 to 4469



■ AL-4470

ROBOT CARIB STEP NOT ENOUGH

This alarm occurs if the number of steps is lacking for a job for calibration data creation between manipulators.

Sub Code: None

< Remedy >

Teach the specified number of steps again, and then check the operation.

■ AL-4471

CALIBRATION DATA ERROR

This alarm occurs if the tool calibration data cannot correctly be prepared.
The RAM data failure in the NCP01 board may also cause this alarm.

Sub Code

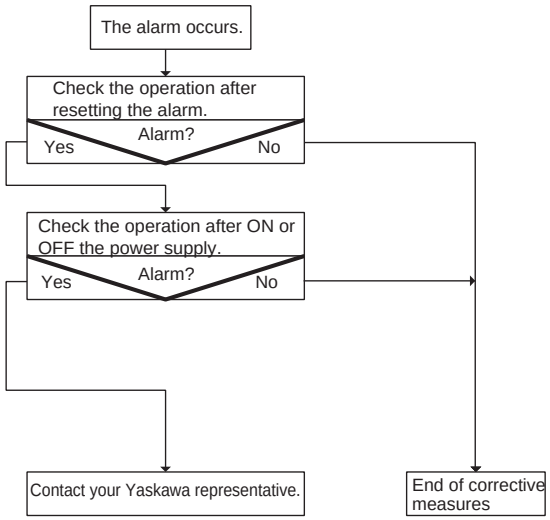
1 to 4: Signifies the internal software error

■ AL-4473

ARITHMETIC ALARM RESET ERROR

This alarm occurs if the alarm occurred in the calculation section cannot be reset.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code: None

4471/4473



■ AL-4474

WRONG CONTROL GROUP AXIS

This alarm occurs if an attempt is made to call or jump to a job whose control group cannot be controlled or to start such a job.
Sub Code: None

< Remedy >

- Make the setting in advance so that the control group of the CALL/JUMP designation job is included in that of the CALL/JUMP source job.
- Do not start the job containing a control group already operating by using the "PSTART" instruction.

■ AL-4475

CANNOT EXECUTE JOB (NO ROBOT)

This alarm occurs if the robot axis is not designated for the control-group of the job at execution of a work instruction that uses a manipulator.
Sub Code: None

< Remedy >

Add the robot axis to the control-group of the job.

■ AL-4476

CANNOT EDIT (EDIT LOCK JOB)
This alarm occurs if an attempt is made to change the data for the job prohibited from being edited.
Sub Code:

0: An attempt was made to change the tag data.
1: An attempt was made to change the speed tag data.
2: An attempt was made to change the board thickness tag data.

< Remedy >
Release the prohibition, and then check the operation.

■ AL-4477

SELECT ERROR (APPLICATION)
This alarm occurs if the application selection parameter (parameter exclusive for manufacturer) is inconsistent with the application parameter (AP parameter) in work instruction execution.
Sub Code: Application No.
< Remedy >
Specify an application appropriate to the specified robot when selecting the application in the maintenance mode.

■ AL-4480

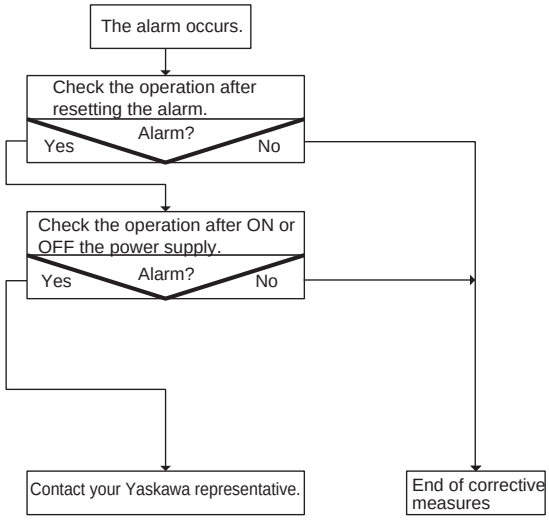
SELECT ERROR (SENSOR)
This alarm occurs if the sensor application selection parameter (parameter exclusive for manufacturer) is inconsistent with the sensor parameter (SE parameter) in work instruction execution.
Sub Code: Sensor No.
< Remedy >
Specify a sensor function appropriate to the specified robot when selecting the application in the maintenance mode.

■ AL-4487

WRONG MECH PARAMETER FILE

This alarm occurs if an error is detected in mechanical parameter for the internal control section.

Sub Code: None



■ AL-4490

DEFECTIVE TAUGHT POINT(ENDLESS)

This alarm occurs if endless operation cannot be performed due to setting failure.

Sub Code

1: After the Endless rotation completed, an attempt was made to execute an interpolation instruction such as MOVL and MOVC before executing an MRESET instruction.

< Remedy >

To perform an interpolation motion such as MOVL and MOVC after an Endless rotation, execute an MRESET instruction beforehand.

2: The base axis is set as an Endless rotation axis. The Endless function cannot be used with the base axis.

< Remedy >

Check the parameter setting that designates the Endless rotation axis.

3: An attempt was made to execute the Endless function although the endless axis was not designated.

< Remedy >

Check the parameter setting that designates the Endless rotation axis.

4: The Endless axis exceeded the maximum pulse value (Å}536870911).

< Remedy >

Correct the rotation amount so that the Endless axis does not exceed the maximum pulse value.

■ AL-4491

CORRECTIONAL DIRECTION ERROR
This alarm occurs if an error is detected while the correcting direction is created.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 5: Signifies the internal software error

■ AL-4492

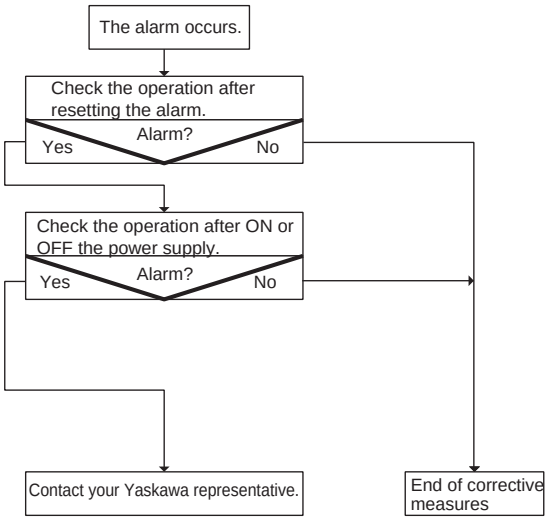
POSITION CORRECTION ERROR
This alarm occurs if an error is detected in the calculation section for the correcting direction at path correcting motion.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code

1 to 2: Signifies the internal software error

■ AL-4493

OVER TOOL FILE NO.
This alarm occurs if the tool number for internal control is 49 or more.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code: None

4491 to 4493



■ AL-4494

DEFECTIVE TAUGHT POINT (WEAV)

This alarm occurs if the weaving operation cannot be executed due to an incorrect teaching point setting.

Sub Code

1: The weaving start point and end point are on the same point.

< Remedy >

Check the position settings so that the weaving start point and end point are different.

2: Among the weaving start point, end point, and reference point, two or three points are on the same point.

< Remedy >

Check the position settings so that the weaving start point, end point, and reference point are different.

■ AL-4495

UNDEFINED ROBOT CALIBRATION

This alarm occurs if the calibration between manipulators has not executed at coordinated motion.

Sub Code: Control group for which calibration is not yet completed

< Remedy >

Execute the calibration between manipulators, and then check the operation.

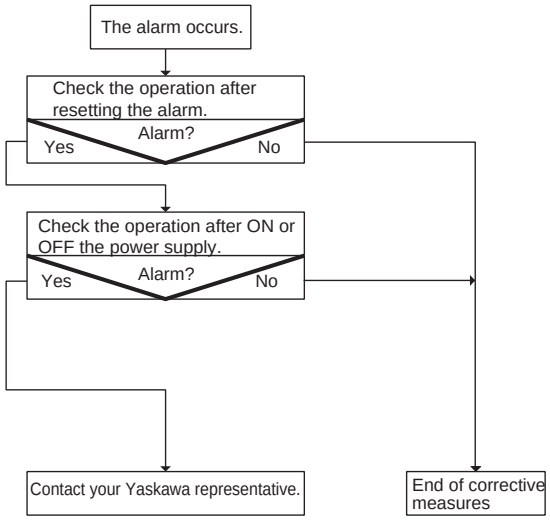
■ AL-4496

PARAMETER ERROR

This alarm occurs if a data error is detected during parameter setting.

Sub Code

1 to 140: Signifies the internal software error



■ AL-4497

DEFECTIVE TAUGHT POINT (CALIB)

This alarm occurs if teaching for calibration between manipulators is inappropriate.

Sub Code

1: Some of the teaching points for master-group are on the same point.

< Remedy >

Perform the teaching again so that the teaching points are different from one another.

2: Some of the teaching points for slave-group are on the same point.

< Remedy >

Perform the teaching again so that the teaching points are different from one another.

3: The 2nd-axis positions of C3, C4, and C5 of station axes are not the same.

< Remedy >

Perform the teaching again so that the 2nd-axis positions of C3, C4, and C5 of the station axes are the same.

4: The 1st-axis positions of C1, C2, and C3 of station axes are not the same.

< Remedy >

Perform the teaching again so that the 1st-axis positions of C1, C2, and C3 of station axes are the same.

5: The 2nd-axis positions of C1, C2, and C3 of station axes are the same.

< Remedy >

Perform the teaching again so that the teaching positions are different from one another.

6: The 1st-axis rotation direction of C3, C4, and C5 of station axes are not the same.

< Remedy >

Perform the teaching again so that the 1st-axis rotation direction of C3, C4, and C5 of station axes are the same.

7: The 1st-axis (elevation axis) positions of C1, C2, and C3 of station axes are not the same.

< Remedy >

Perform the teaching again so that the 1st-axis (elevation axis) positions of C1, C2, and C3 of station axes are the same.

8: The 1st-axis (elevation axis) positions of C3, C4, and C5 of station axes are not the same.

< Remedy >

Perform the teaching again so that the 1st-axis (elevation axis) positions of C3, C4, and C5 of station axes are the same.

■ AL-4498

CANNOT EXECUTE JOB (NO GRP AXIS)
This alarm occurs if an attempt is made to execute an instruction that cannot be executed in a job without control group.
Sub Code: None
< Remedy >
Delete the instruction where an alarm occurred, and then check the operation.

■ AL-4499

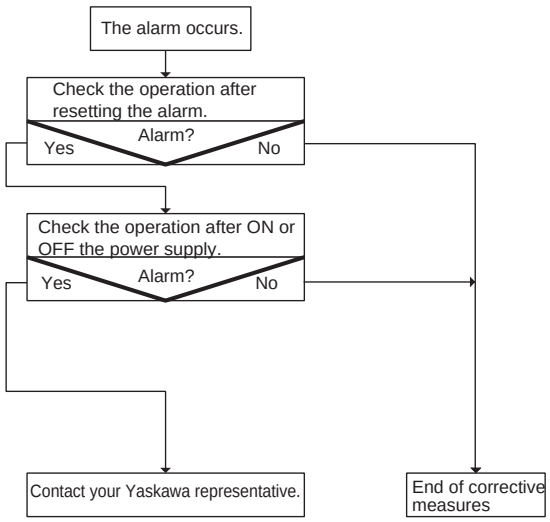
UNDEFINED POSITION VARIABLE
This alarm occurs if an attempt is made to use the position type variable that is not set.
Sub Code: Variable No.
< Remedy >
Register the position type variable used by the instruction where an alarm occurred, and then check the operation.

■ AL-4500

UNDEFINED USER FRAME
This alarm occurs if an attempt is made to use the user coordinate that is not set.
Sub Code: User coordinate No.
< Remedy >
Register the user coordinate used by the instruction where an alarm occurred, and then check the operation.

■ AL-4501

OUT OF RANGE (PARALLEL PROCESS)
This alarm occurs if the number of tasks exceeds the limit in the independent control function.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code: Task No.



■ AL-4505

UNDEFINED POSITION FOR ARC ON

This alarm occurs if the Arc retry cannot be executed because there is no step before the ARCON instruction.

Sub Code: None

< Remedy >

Register a step before the ARCON instruction.

■ AL-4506

UNDEFINED POS FOR RESTART RETURN

This alarm occurs if the Arc-restart-return cannot be executed because there is no restart-return step in the job.

Sub Code: None

< Remedy >

Change the job configuration so that the CALL job will not be used in the ARCON section.

■ AL-4507

REFP POS ERROR (SEARCH MOTION)

This alarm occurs if the search start point and the motion target point are the same, or the distance between the two points is too short.

Sub Code: None

< Remedy >

- Perform the teaching again so that the search start point and the motion target point are not the same.
- Increase the distance between the search start point and the motion target point.

■ AL-4508

SPECIFIED ERROR (COORDINATE)
This alarm occurs if an invalid coordinate system is specified.
Sub Code:

0: The specified coordinate system does not exist.
1: Designation error of the master tool coordinate system. This coordinate system cannot be used.
2: Designation error of the tool coordinate system. This coordinate system cannot be used.
3: Designation error of the traveling direction coordinate system (for shared function). This coordinate system cannot be used.
4: Designation error of the any direction coordinate system (for shared function). This coordinate system cannot be used.
5: Designation error of the approximate tool coordinate system (for shared function). This coordinate system cannot be used.
6: Designation error of the conveyor coordinate system. This coordinate system cannot be used.
8: Designation error of the COMARC coordinate system. This coordinate system cannot be used.
9: Designation error of the power sensor coordinate system. This coordinate system cannot be used.
10: Designation error of the cylindrical coordinate system. This coordinate system cannot be used.
11: Designation error of the coordinate system for the external reference point. This coordinate system cannot be used.
12: Designation error of the coordinate system for 3D shifting. This coordinate system cannot be used.
15: Designation error of the coordinate system at IMOV for 3D shifting. This coordinate system cannot be used.
16: Designation error of the H-LINK type cylindrical coordinate system. This coordinate system cannot be used.
17: Designation error of the FSER_FRAME type cylindrical coordinate system. This coordinate system cannot be used.

< Remedy >
Specify a valid coordinate system, and then check the operation.

■ AL-4509

MFRAME ERROR
This alarm occurs if an error is detected at MFRAME execution.
Sub Code

1: The master tool user coordinate is cannot be created.

< Remedy >
Execute the MFRAME instruction in the coordinate job when creating the master tool user coordinate.

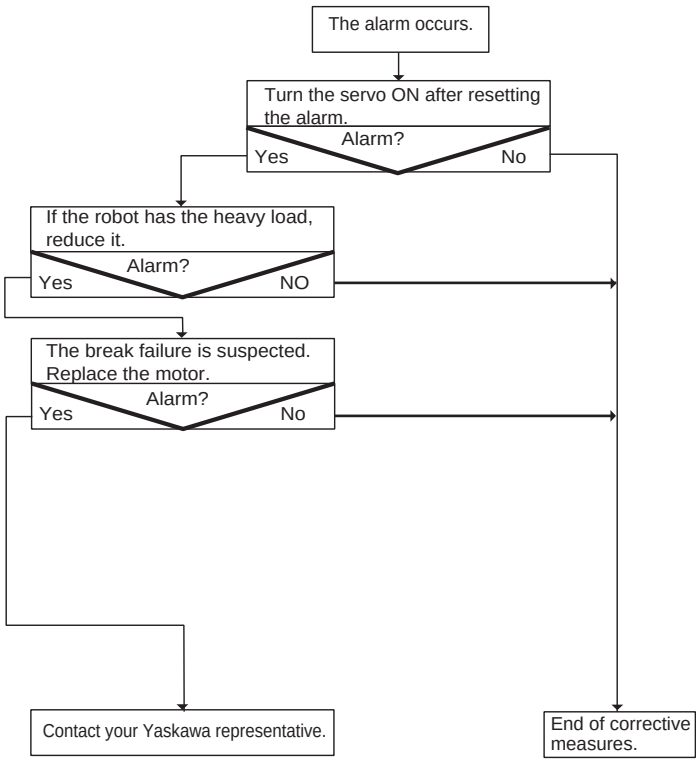
■ AL-4510

CANNOT EXECUTE INSTRUCTION (SQRT)
This alarm occurs if an attempt is made to calculate the square root of negative value at SQRT instruction execution.
Sub Code: None
< Remedy >
Change the job so that the second argument of the SQRT instruction may not be a negative value and check operation.

■ AL-4511

DROP-VALUE OUT OF RANGE

The pulse difference of the robot position exceeded the allowable value between when the servo was OFF previously and when the servo was ON this time.
The standard allowable number of pulses is 100.
Sub Code: Control group exceeding the allowable value



■ AL-4512

TWO STEPS SAME LINE (3 STEPS)

This alarm occurs if three or more teaching points are aligned in a straight line in the user coordinates for calibration between manipulators.

Sub Code: None

< Remedy >

Check the setting so that the teaching points are not aligned in a straight line.

■ AL-4513

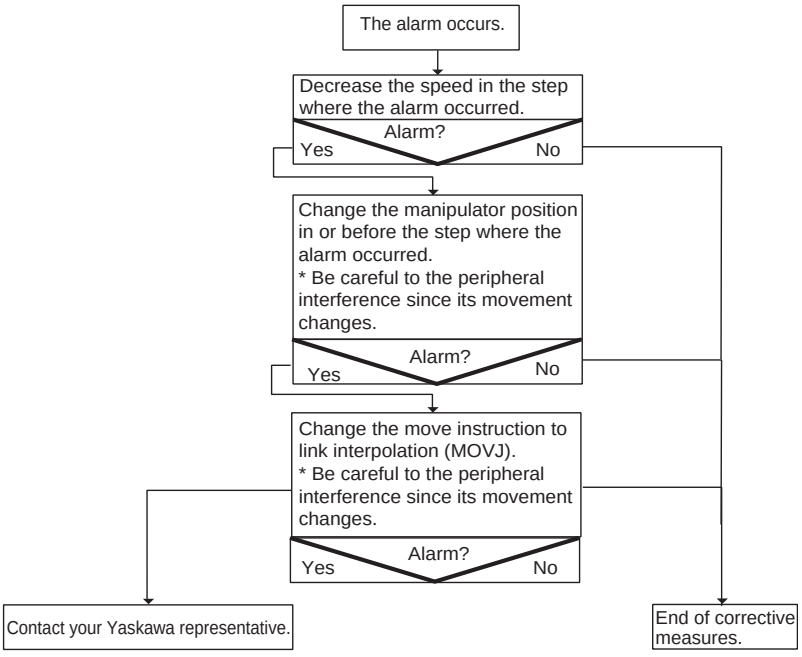
EXCESSIVE SEGMENT (SAFETY 1): LOW

This alarm occurs if the operation command which exceeds the designated max. speed is output.
It may occur when the robot operates near the singular point or when the robot is going to change its orientation widely with single control point.
Sub Code: Signifies the alarm occurring axes

■ AL-4514

EXCESSIVE SEGMENT (SAFETY 1): HIGH

This alarm occurs if the operation command which exceeds the designated max. speed is output.
It may occur when the robot operates near the singular point or when the robot is going to change its orientation widely with single control point.
Sub Code: Signifies the alarm occurring axes



■ AL-4515

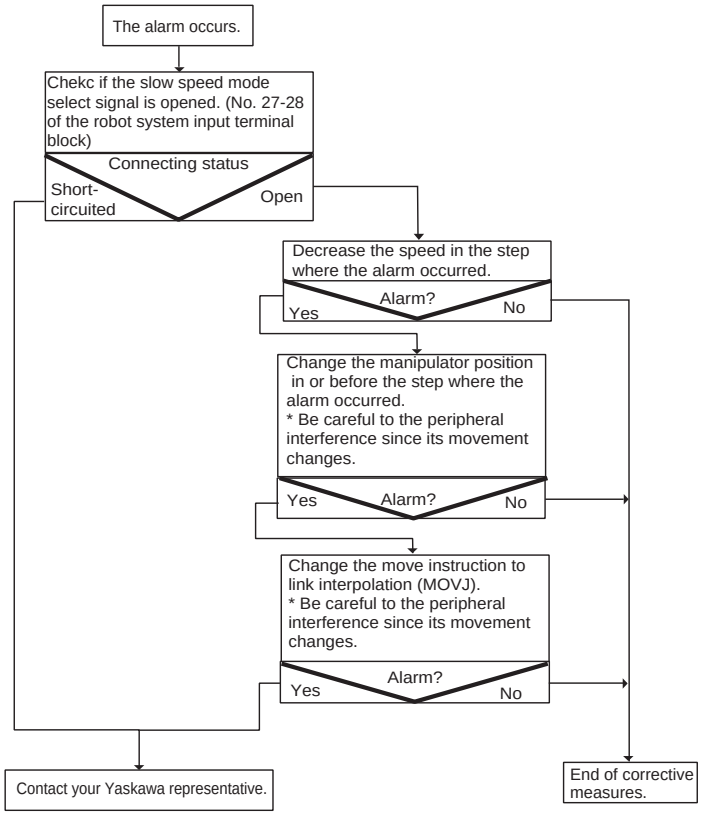
EXCESSIVE SEGMENT (SAFETY 2): LOW

This alarm occurs if the operation command which exceeds the designated max. speed is output.
It may occur when the robot operates near the singular point or when the robot is going to change its orientation widely with single control point.
Sub Code: Signifies the alarm occurring axes

■ AL-4516

EXCESSIVE SEGMENT (SAFETY 2): HIGH

This alarm occurs if the operation command which exceeds the designated max. speed is output.
It may occur when the robot operates near the singular point or when the robot is going to change its orientation widely with single control point.
Sub Code: Signifies the alarm occurring axes



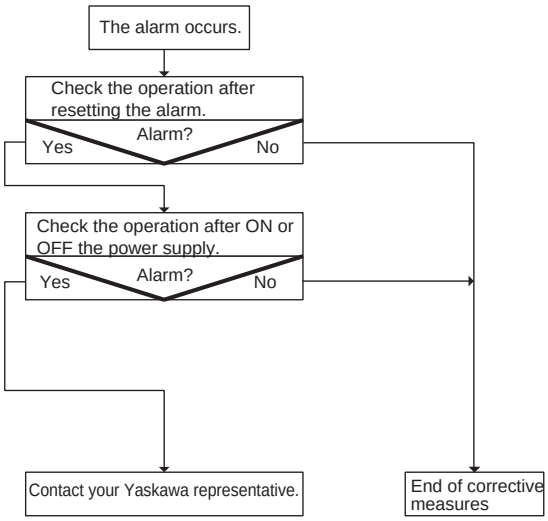
■ AL-4517

SEARCH MONITOR SET ERROR (SERVO)
This alarm occurs if an error is detected in search/monitoring mode settings.
Sub Code: Control group

■ AL-4518

SEARCH MON RELEASE ERR (SERVO)
This alarm occurs if an error is detected in search/monitoring mode releasing.
Sub Code: Control group

4517/4518



■ AL-4521

WRONG JOB TYPE
This alarm occurs if the job type is inconsistent.
Sub Code

0000_0001: A robot job was started from the concurrent job at CALL/JUMP instruction execution.
0000_1001: A concurrent job was started from the robot job at CALL/JUMP instruction execution.
1000_0001: A system job was started from the robot job at CALL/JUMP instruction execution.

< Remedy >
Change the type of job will be started before starting.

■ AL-4524

CANNOT EXECUTE INST (COUCUR JOB)
This alarm occurs if there is an unexecutable instruction such as move instruction in the concurrent job.
Sub Code: None
< Remedy >
Delete the instruction where an alarm occurred, and then check the operation.

■ AL-4527

UNDEFINED PORT NO. (AOUT)

This alarm occurs if the incorrect analog output port number is specified.

Sub Code: None

< Remedy >

Change the setting of the specified analog output port number.

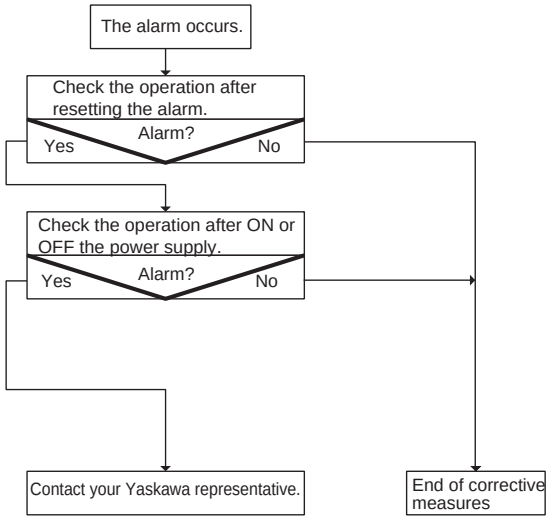
■ AL-4528

SYNTAX ERROR

This alarm occurs if an error is detected in the instruction syntax.

Sub Code

1: An error occurred in the IF statement syntax.



■ AL-4529

TWIN COORDINATED ERROR

This alarm occurs if an error is detected at twin synchronization execution.

Sub Code

1: A job without control group was started by SYNC instruction.

< Remedy >

Check the control group setting for the job started by SYNC instruction.

2: A job without master control group was started by SYNC instruction.

< Remedy >

Check the control group setting for the job started by SYNC instruction.

3: A job only with master control group axes was started by SYNC instruction.

< Remedy >

Check the control group setting for the job started by SYNC instruction.

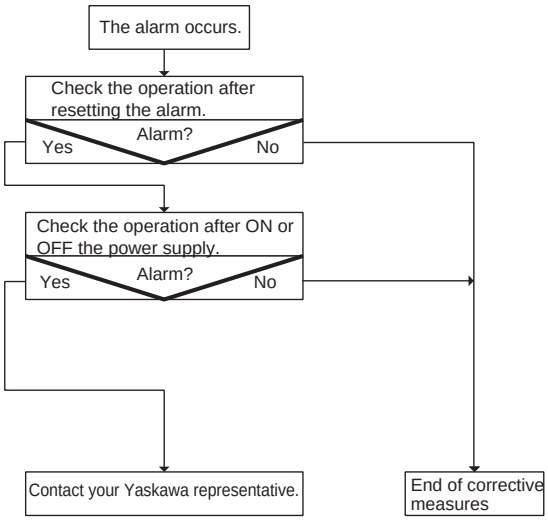
4: At full synchronization, the completion timings of move instructions for the master and the slave disagreed.

5: At full synchronization, no operation request from the master was sent.

6: At full synchronization, the execution timings of move instructions for the master and the slave disagreed.

7: The twin synchronization ID number is incorrect.

8: Triple synchronization was attempted using the sub/master specification on a different master.

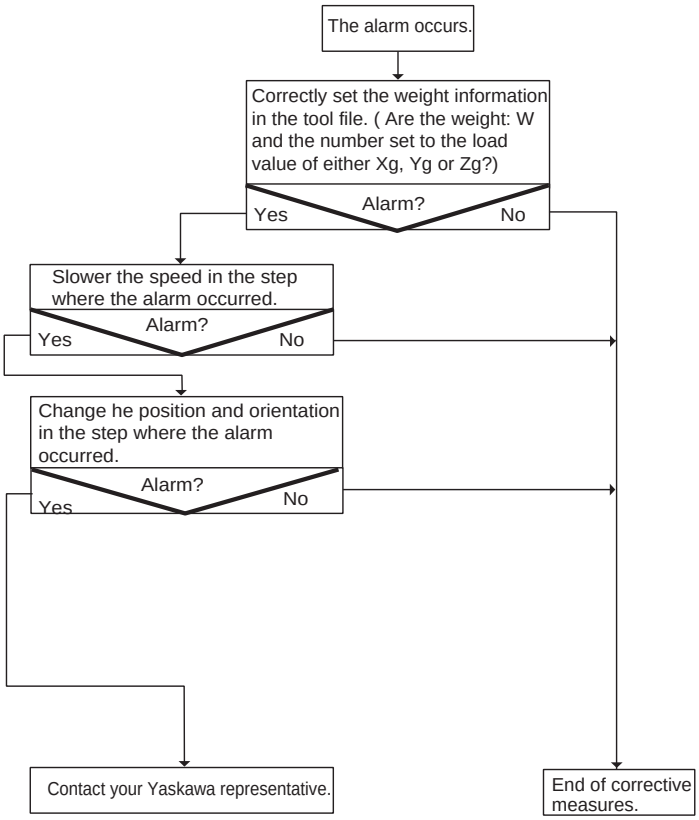


■ AL-4534

TORQUE INTERFERENCE OVER

The torque assumed value to the axis exceeded the allowable value during acceleration/deceleration operation.

Sub Code: None



■ AL-4542

MRESET ERROR

This alarm occurs if an error is detected at MRESET instruction execution.

Sub Code

1: A MRESET instruction was executed while no endless axis was designated.

< Remedy >

Designate the endless axis, and then check the operation.

■ AL-4543

STACK LESS THAN 0 (JOB CALL)

This alarm occurs if an error is detected during job call stack control.

Sub Code: None

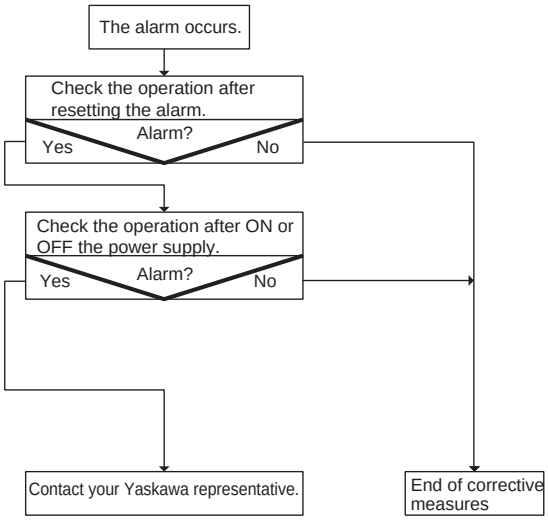
■ AL-4546

CANNOT EXCUTE SYSTEM JOB

This alarm occurs if an error is detected in the starting system number of system job.

Sub Code: System number

4543/4546



■ AL-4550

CANNOT EXECUTE INST (USER JOB)
This alarm occurs if an instruction that can be executed only in a system job is attempted in a user job.
Sub Code: System number
< Remedy >
Delete the instruction where an alarm occurred, and then check the operation.

■ AL-4551

CANNOT MEASURE TIP INSTALL COEF
This alarm occurs if the fixed-side electrode installation correction value is not measured when "SVGUNCL TWC-BE" is executed.
Sub Code: Gun number
< Remedy >
Execute "SVGUNCL TWC-AE" before executing "SVGUNCL TWC=BE."

■ AL-4566

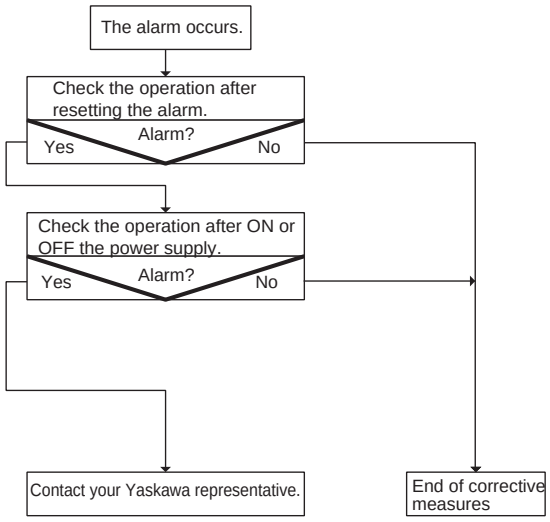
USER FRAME MAKING ERROR
This alarm occurs if an error is detected at preparation of a user coordinates.
Sub Code

1 to 6: Signifies the internal software error

■ AL-4571

SERVO FLOAT MODE RELEASE ERR
This alarm occurs if the servo float mode cannot be reset when executing a FLOATOF instruction.
Sub Code: None

4566/4571



■ AL-4572

UNDEFINED MOTOR GUN CONTROL GRP
This alarm occurs if the control group for the motor gun is not set.
Sub Code: None
< Remedy >
Specify the "motor gun axis" under control group settings in the maintenance mode.

■ AL-4573

SPOT WELDER NO. ERROR
This alarm occurs if the spot welder number is incorrect.
Sub Code: Welder number
< Remedy >
Change the "POWER SOURCE NO." in the gun condition file.

■ AL-4574

SPOT WELD COMPLETE TIME LIMIT
This alarm occurs if neither the welding completion signal nor the welding error signal is received from the timer conductor within the set time.
Sub Code: Welder number
< Remedy >
- Turn the timer contactor ON.
- If the response from the timer takes too long time due to the system layout, increase the set time.

■ AL-4575

ERROR IN WELD START TIMING SET
For motor gun, this alarm occurs if the welding timing is set to "After First Pressure" while no 2nd pressure is set.
Sub Code: None
< Remedy >
- Check the setting of the WST tag.
- Check the setting of the pressure file.

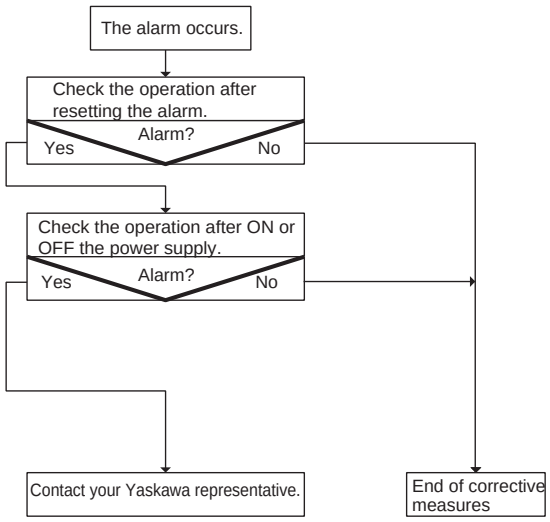
■ AL-4576

ERR IN MOTOR GUN CONT MODE
This alarm occurs if the motor gun control mode cannot be set when executing a SVSPOT/SVGUNCL instruction.
Sub Code: None

■ AL-4577

ERR IN MOTOR GUN MODE RLSE
This alarm occurs if the motor gun control mode cannot be release when executing a SVSPOT/SVGUNCL instruction.
Sub Code: None

4576/4577



■ AL-4578

SPOT WELD ERROR

This alarm occurs if an error occurs when executing welding using the specified system timer conductor.

Sub Code: Welder number

< Remedy >

Check the setting of the timer conductor where the welding error occurred.

■ AL-4579

ANTICIPATION CONTROL ERROR

This alarm occurs if an error is detected in the anticipation control processing.

Sub Code

1: No availability in anticipation control

2: The anticipation data exceeded the maximum length.

< Remedy >

Up to five anticipation control sessions can be executed at the same time. Check the job setting so that the number of control sessions may be within five.

■ AL-4581

DEFECTIVE ANTICIPATION OUT FILE

This alarm occurs if an error is detected in the anticipation output file.

Sub Code

1: Incorrect setting of OT output number for anticipation output file

2: Incorrect setting of OG output number for anticipation output file

< Remedy >

Change the OT/OG output number within the range.

■ AL-4587

MOTOR GUN CHANGE ERROR

This alarm occurs if an error is detected when changing the gun.

Sub Code

1: A GUNCHG instruction was executed in the system configuration that did not allow the gun change function.

< Remedy >

Enable the gun change function parameter.

2: A GUNCHG/PICK instruction was executed while the motor gun motor was servo ON.

< Remedy >

Execute GUNCHG/PICK instruction when the motor gun motor is servo OFF.

3: A GUNCHG/PICK instruction was executed while the ATC was in unchuck status.

< Remedy >

Execute GUNCHG/PICK instruction when the ATC is in chuck status.

4: A GUNCHG/PLACE instruction was executed while the ATC was in unchuck status.

< Remedy >

Execute GUNCHG/PLACE instruction when the ATC is in chuck status.

5: The encoder power supply could not be turned ON when executing a GUNCHG/PICK instruction.

< Remedy >

(1) Reset the alarm, and then try again.

(2) If the alarm occurs again, check the connection and insertion statuses of the following cables and connectors.

• Encoder cable of motor gun

6: The encoder power supply could not be turned OFF when executing a GUNCHG/PLACE instruction.

< Remedy >

(1) Reset the alarm, and then try again.

(2) If the alarm occurs again, check the connection and insertion statuses of the following cables and connectors.

• Encoder cable of motor gun

7: The gun number specified by the GUNCHG instruction did not agree with the gun identification signal.

< Remedy >

(1) Check the following settings.

• Change the gun condition file number of GUNCHG instruction to the target gun number.

• Change the gun identification signal to the target gun number.

■ AL-4591

SPEED CTRL MODE SET ERR (SERVO)
This alarm occurs if the speed control mode cannot be set during execution of VCON instruction.
Sub Code: None

■ AL-4592

SPEED CTRL MODE CANCEL ERR (SV)
This alarm occurs if the speed control mode cannot be release during execution of VCON instruction.
Sub Code: None

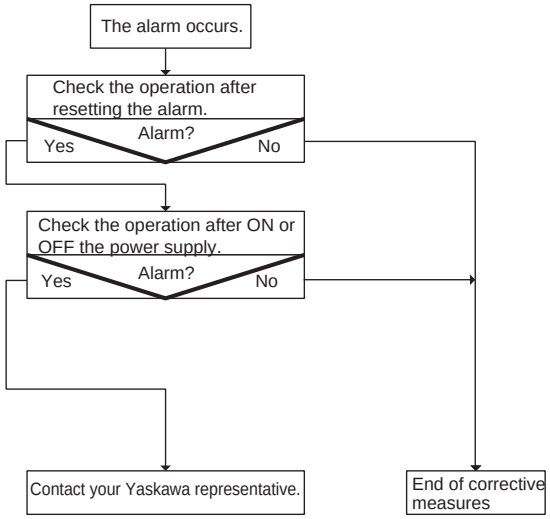
■ AL-4597

OFFLINE DATA TRANSMISSION ERROR
This alarm occurs if an internal control error occurs at offline position data conversion.
Sub Code: None

■ AL-4599

SERVO COMMAND ERROR
This alarm occurs if the operation request cannot be issued to the servo processing section.
The RAM data failure in the NCP01 board may also cause this alarm.
Sub Code: None

4591/4592/4597/4599



■ AL-4601

UNDEFINED GUN COND FILE
This alarm occurs if the motor gun condition file is not set.
Sub Code: Motor gun condition file number
< Remedy >
Complete the settings for the motor gun condition file, and then check the operation.

■ AL-4603

WIRE STICKING

This alarm occurs if wire stick was detected at the welder.

Sub Code: Welder number

< Remedy >

Remove the cause of wire stick, and then check the operation.

■ AL-4606

LACK OF GLOBAL VARIABLE AREA

This alarm occurs if an error is detected in the value of parameter that defines the number of global (user) variables.

Sub Code: None

■ AL-4607

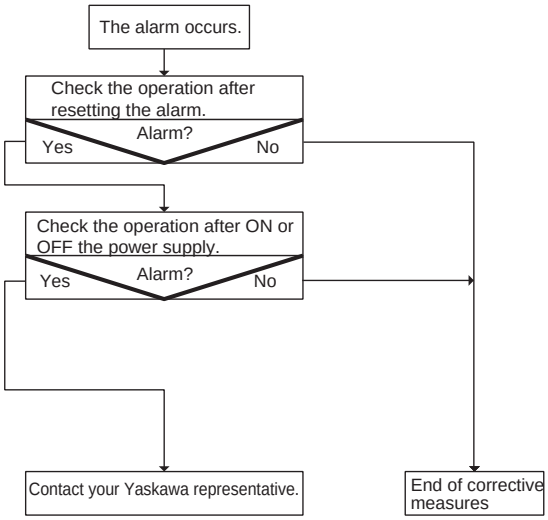
WRONG EXECUTION OF MACRO INST

This alarm occurs if an error is detected when executing a macro instruction.

Sub Code

5 to 6: Signifies the internal software error

4606/4607



■ AL-4608

WRONG EXECUTION OF GETARG INST

This alarm occurs if an error is detected when executing a GETARG instruction.

Sub Code

1: The job argument is not set.

2: No number of the specified job argument

3: The data types of job argument disagreed.

< Remedy >

Modify the job so that data passed by a macro instruction matches the tag setting of GETARG instruction.

■ AL-4612

TSYNC ERROR

This alarm occurs if an error is detected when executing a TSYNC instruction.

Sub Code: The number of synchronizations of the first executed TSYNC

< Remedy >

Modify the job so that the number of synchronizations specified by the TSYNC instruction matches the number of used TSYNCs.

■ AL-4621

WELD COMPLETE SIGNAL ERROR

This alarm occurs if the welding completion signal is ON when starting the spot welding instruction execution.

Sub Code: Welder number

Modify the job so that the spot instruction is not executed when the welding completion signal is ON.

■ AL-4622

SELF-INTERFERENCE

This alarm occurs if a robot link or tool attempts to interfere operation.

Sub Code: Robot number

Modify the teaching point to the point not interfered by the robot and check operation.

■ AL-4652

TRQ MEASURE MODE SET ERR (SV)

This alarm occurs if the constant speed torque measurement mode cannot be set during execution of MEASON instruction.

Sub Code: None

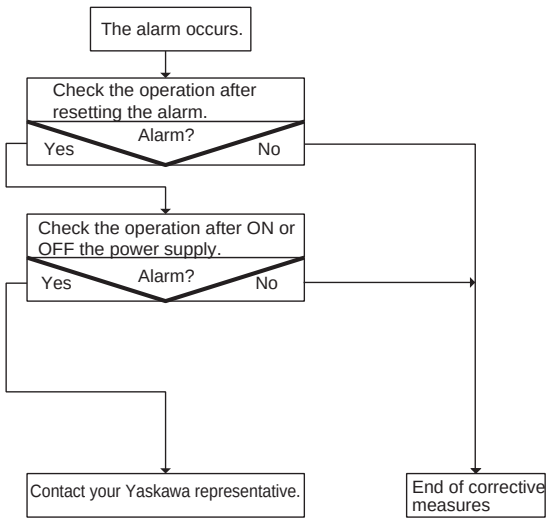
■ AL-4653

TRQ MEASURE MODE CANCEL ERR (SV)

This alarm occurs if the constant speed torque measurement mode cannot be cleared during execution of MEASOF instruction.

Sub Code: None

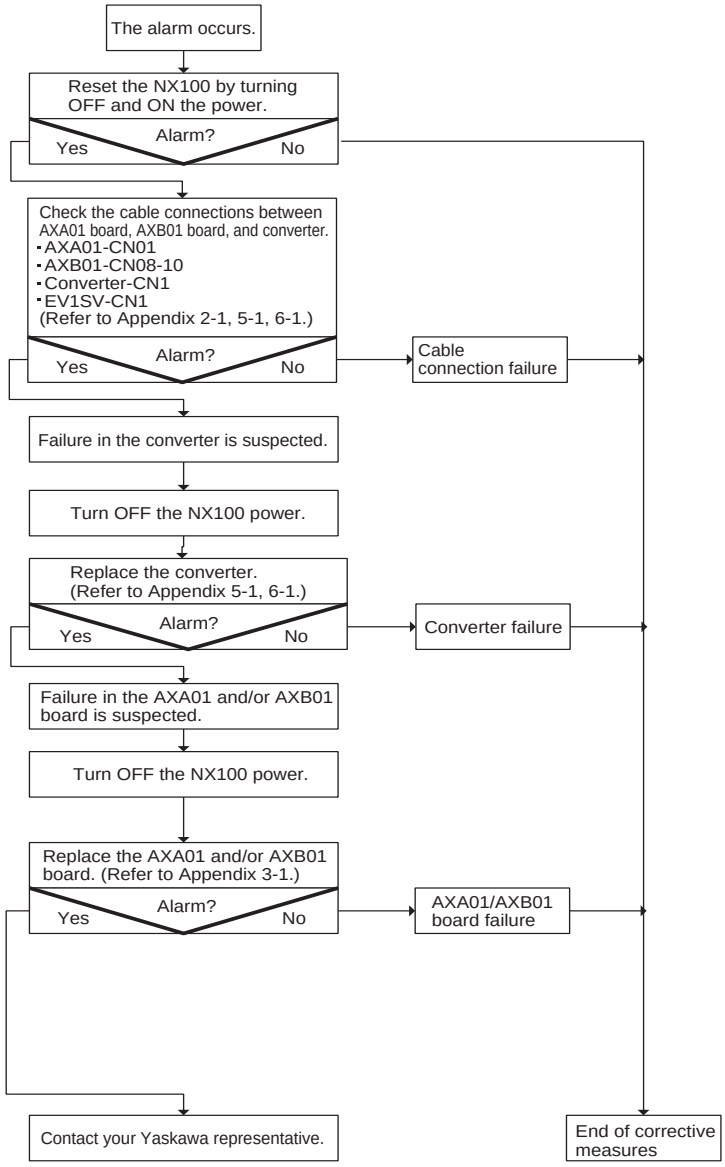
4652/4653



■ AL-4800

WDT ERROR (CONVERTER)

The NX100 system is equipped with a watchdog timer in the converter.
This alarm occurs if the watchdog time-out is detected.
As a cause of the alarm, a failure in the converter is suspected. In rare cases, the alarm may be caused by a noise interference.
Sub Code: Signifies the physical No. of converter in which the alarm occurred



■ AL-4901

CUBE/AXIS INTERFERENCE

This alarm occurs if the robot moves into the specified cube area or exceeds the maximum or minimum value of axis interference.

Sub Code: Signifies the control group, axis No., and interference area No. in which the alarm occurred

< Remedy >

- Change the teaching point so that the step where the alarm occurred (move instruction) may be out of the interference area.
- Change the interference area setting.

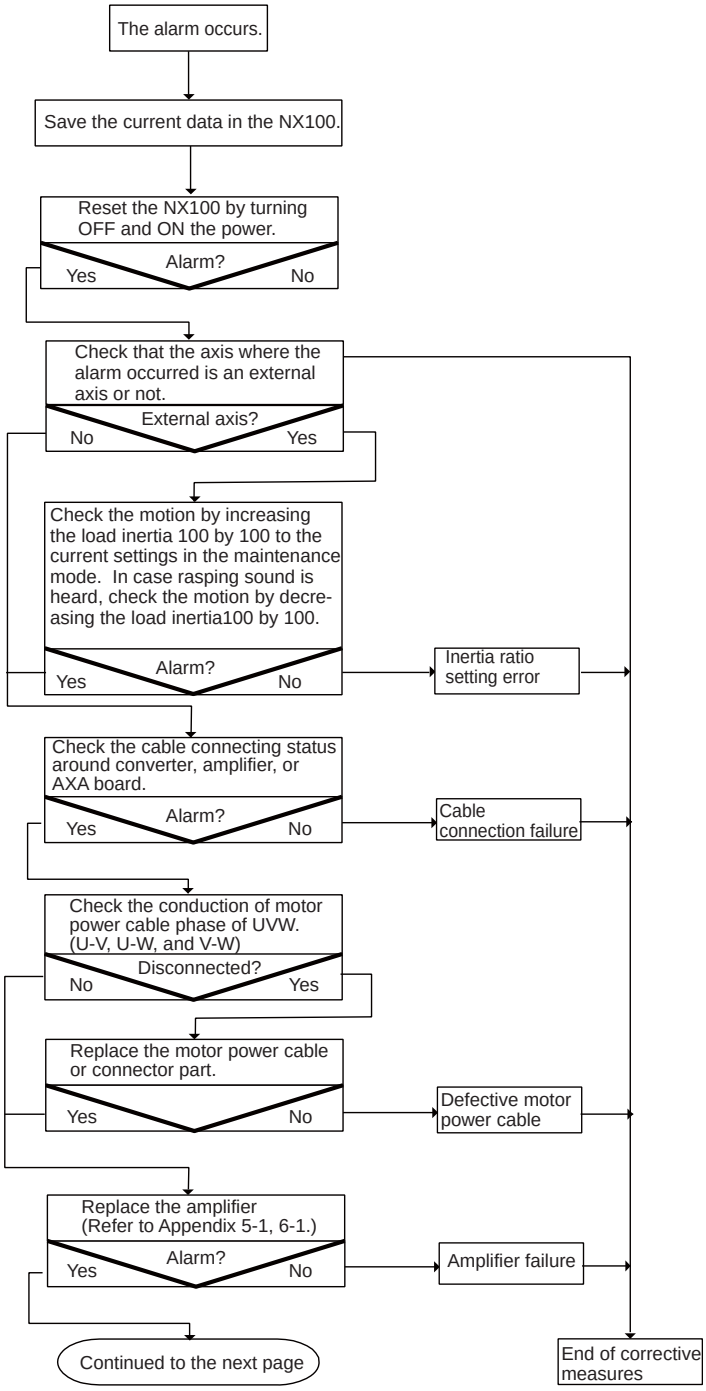
■ AL-4332

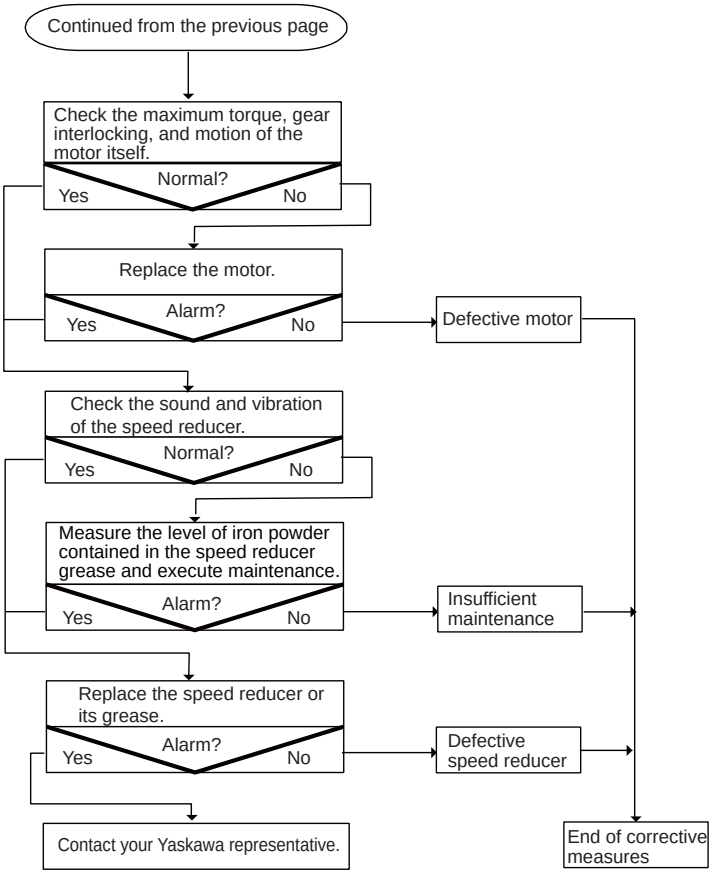
SPEED CTRL MODE CHANGE ERR (SV)

This alarm occurs if the speed control does not start after a certain time from the execution of CVON instruction.

The registered inertia ratio dose not match with the actual mechanism or defective speed reducer or motor is suspected as the alarm cause. In rare cases, however, the cause is attributed to a disconnection of motor power cables or an error in the amplifier.

Sub Code: Signifies the axis in which the alarm occurred

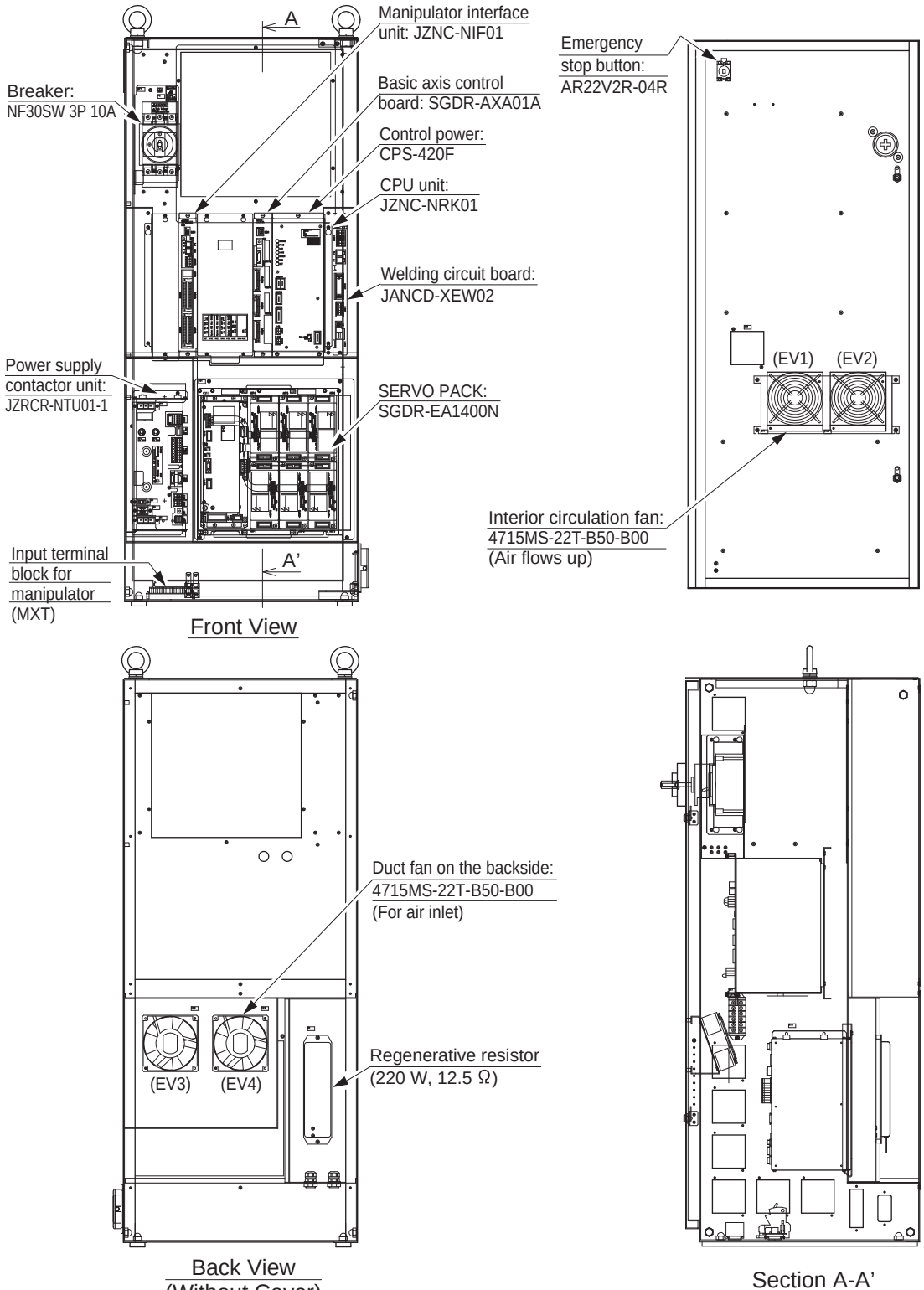



Continued from the previous page
Check the maximum torque, gear interlocking, and motion of the motor itself.
Normal?
Yes
No
Replace the motor.
Alarm?
Yes
No
Defective motor
Check the sound and vibration of the speed reducer.
Normal?
Yes
No
Measure the level of iron powder contained in the speed reducer grease and execute maintenance.
Alarm?
Yes
No
Insufficient maintenance
Replace the speed reducer or its grease.
Alarm?
Yes
No
Defective speed reducer
Contact your Yaskawa representative.
End of corrective measures

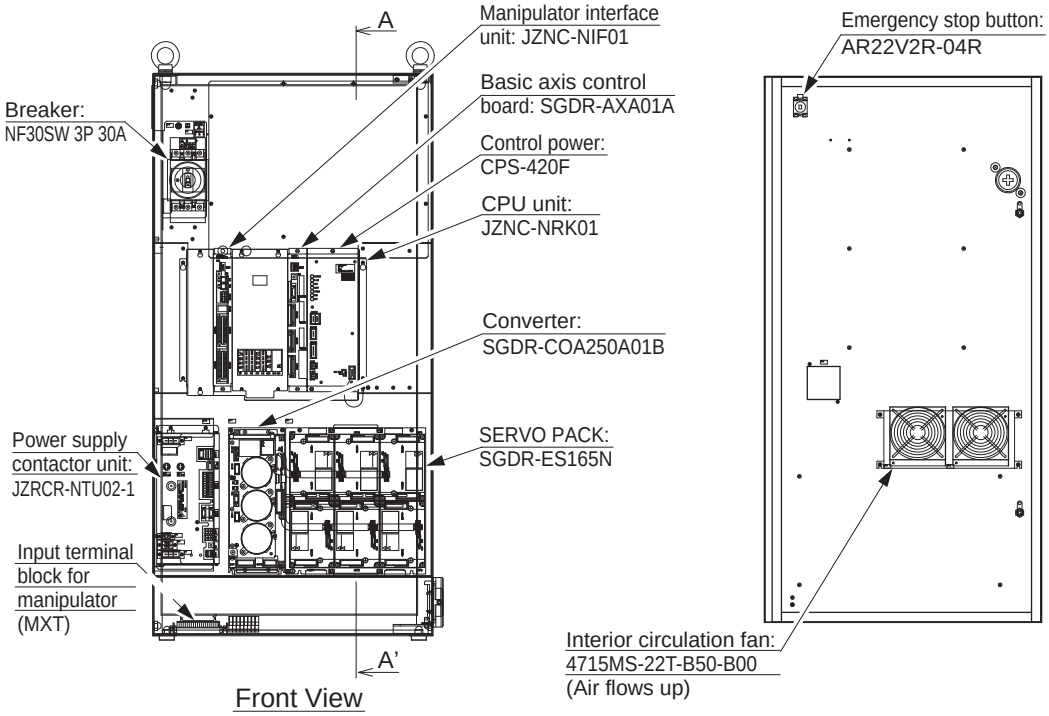
Appendixes

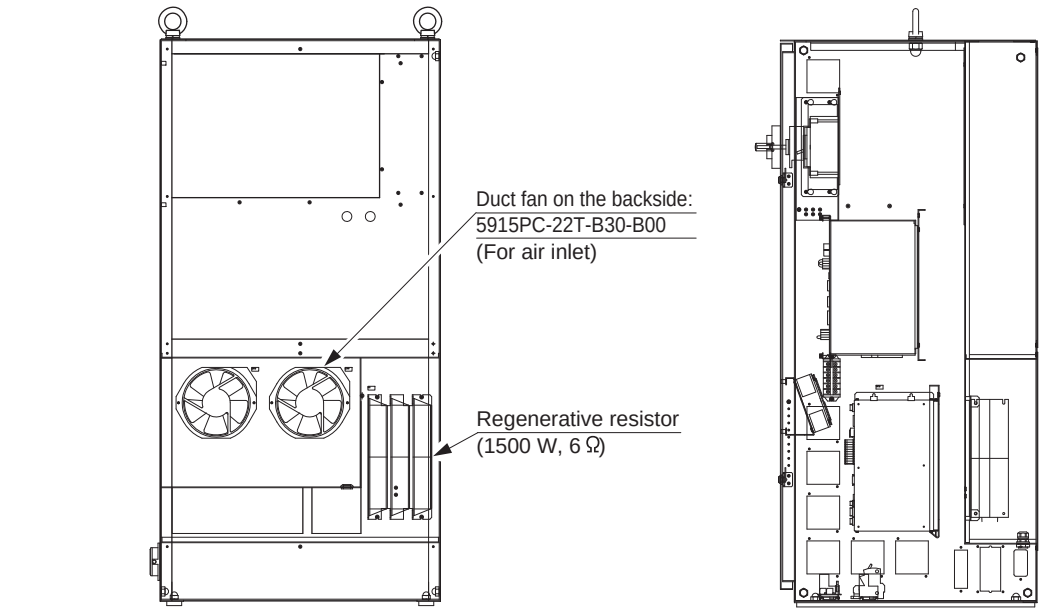
Appendix 1-1 Internal Layout Drawings

■ EA1400N



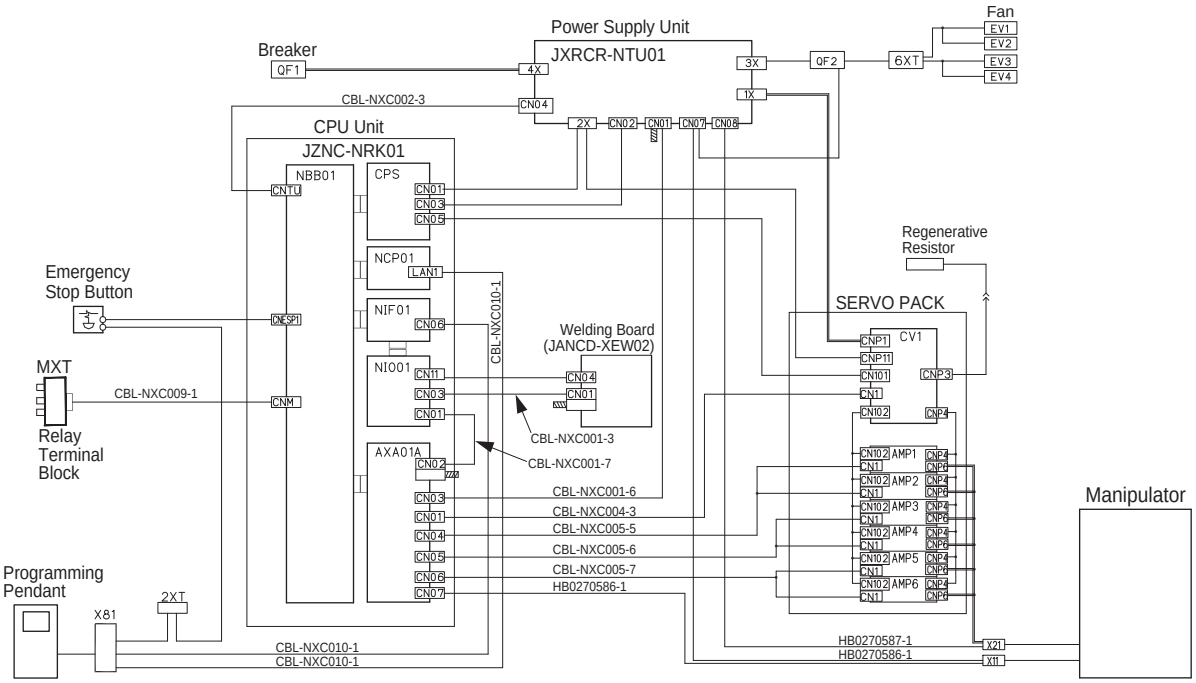
■ ES165N


A
Manipulator interface unit: JZNC-NIF01
Basic axis control board: SGDR-AXA01A
Control power: CPS-420F
CPU unit: JZNC-NRK01
Converter: SGDR-COA250A01B
SERVO PACK: SGDR-ES165N
Breaker: NF30SW 3P 30A
Power supply contactor unit: JZRCR-NTU02-1
Input terminal block for manipulator (MXT)
A'
Front View
Emergency stop button: AR22V2R-04R
Interior circulation fan: 4715MS-22T-B50-B00 (Air flows up)

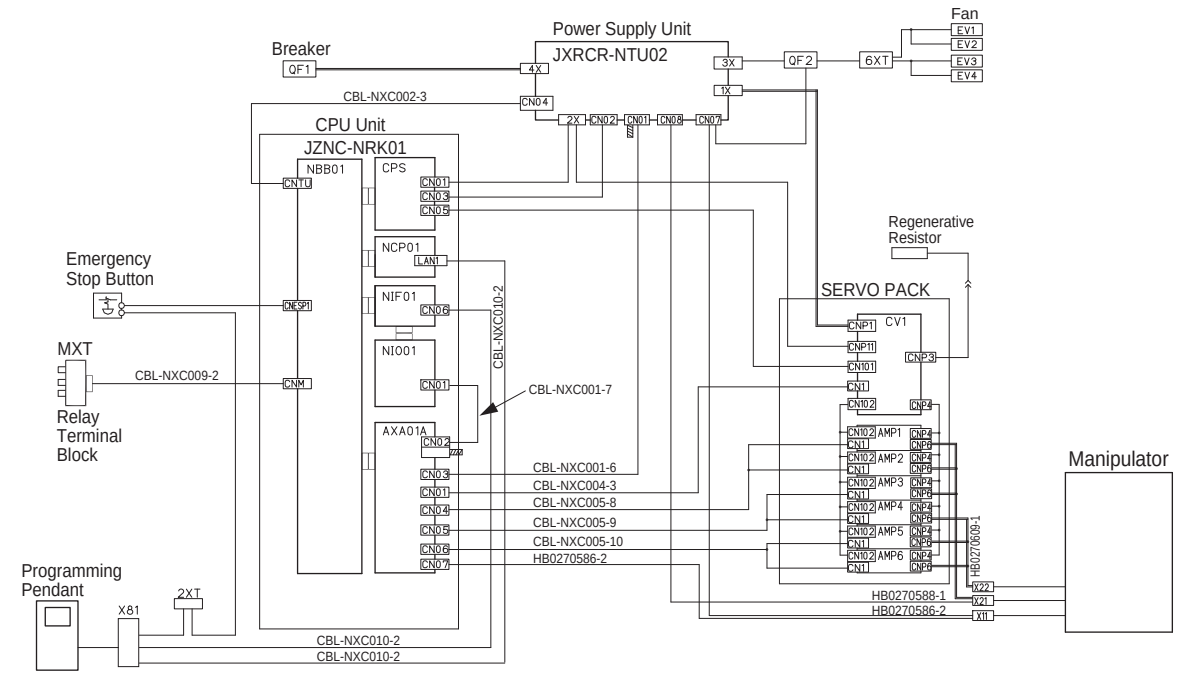

Duct fan on the backside: 5915PC-22T-B30-B00 (For air inlet)
Regenerative resistor (1500 W, 6 Ω)
Back View (Without Cover)
Section A-A'

Appendix 2-1 Block Diagrams

■ EA1400N



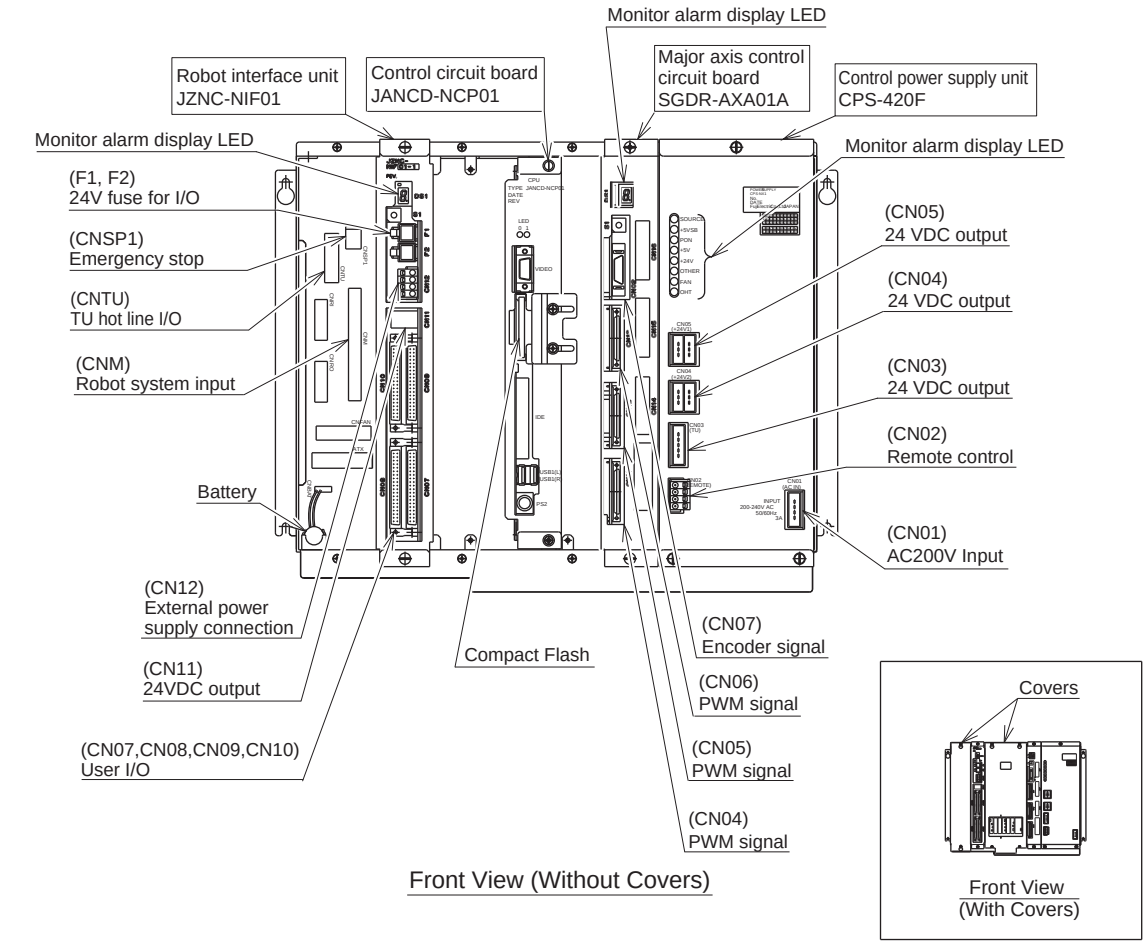
■ ES165N

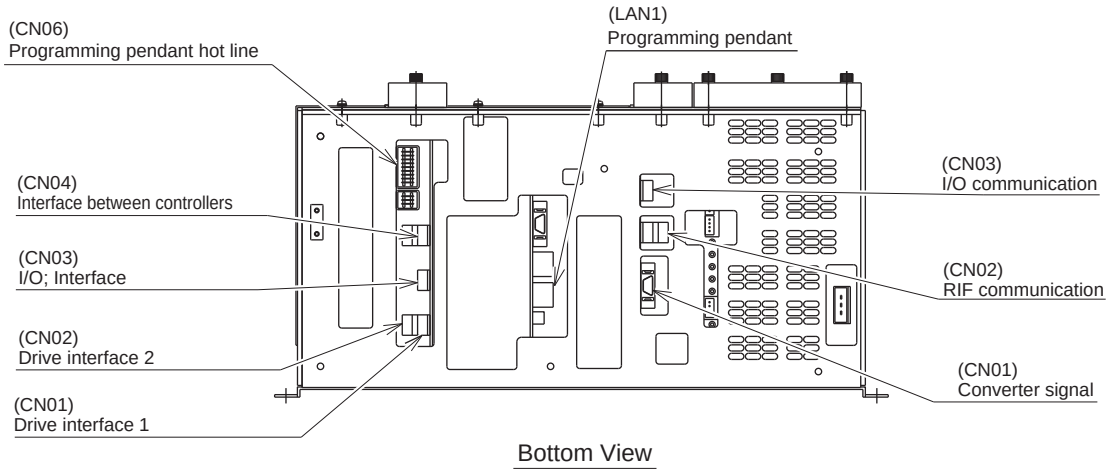


Appendix 3-1 CPU Unit

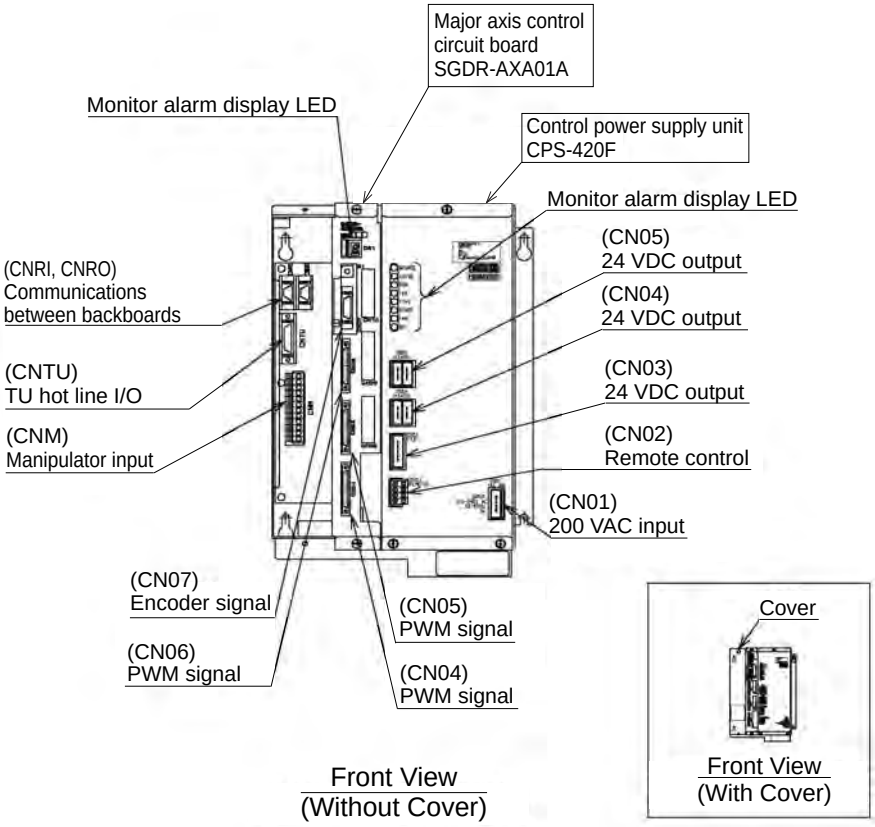
■ External View of a Typical CPU Unit

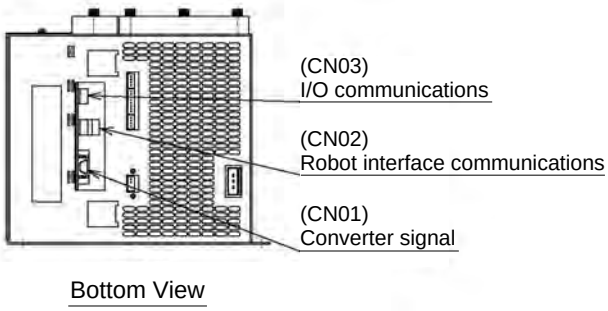
NOTE The JZNC-NRK01 CPU unit contains only circuit board racks and control circuit boards. It does not contain the control power supply, robot interface unit, and major axis control circuit board.





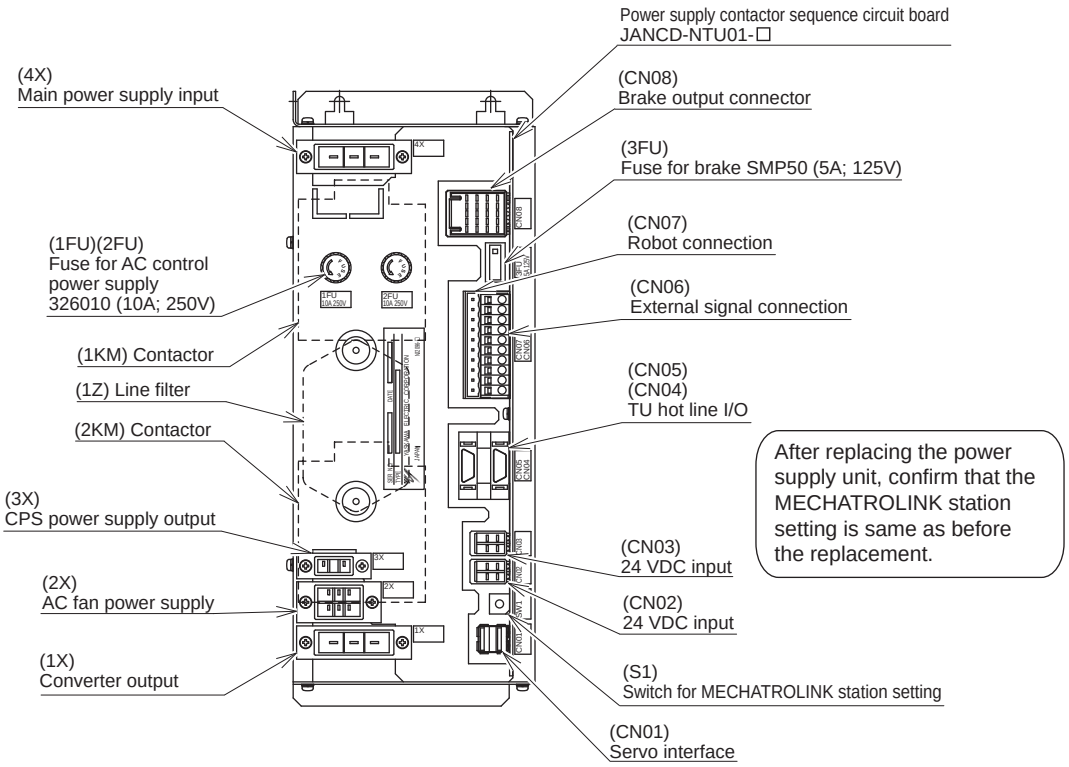
■ External View of JZNC-NRK02□-□



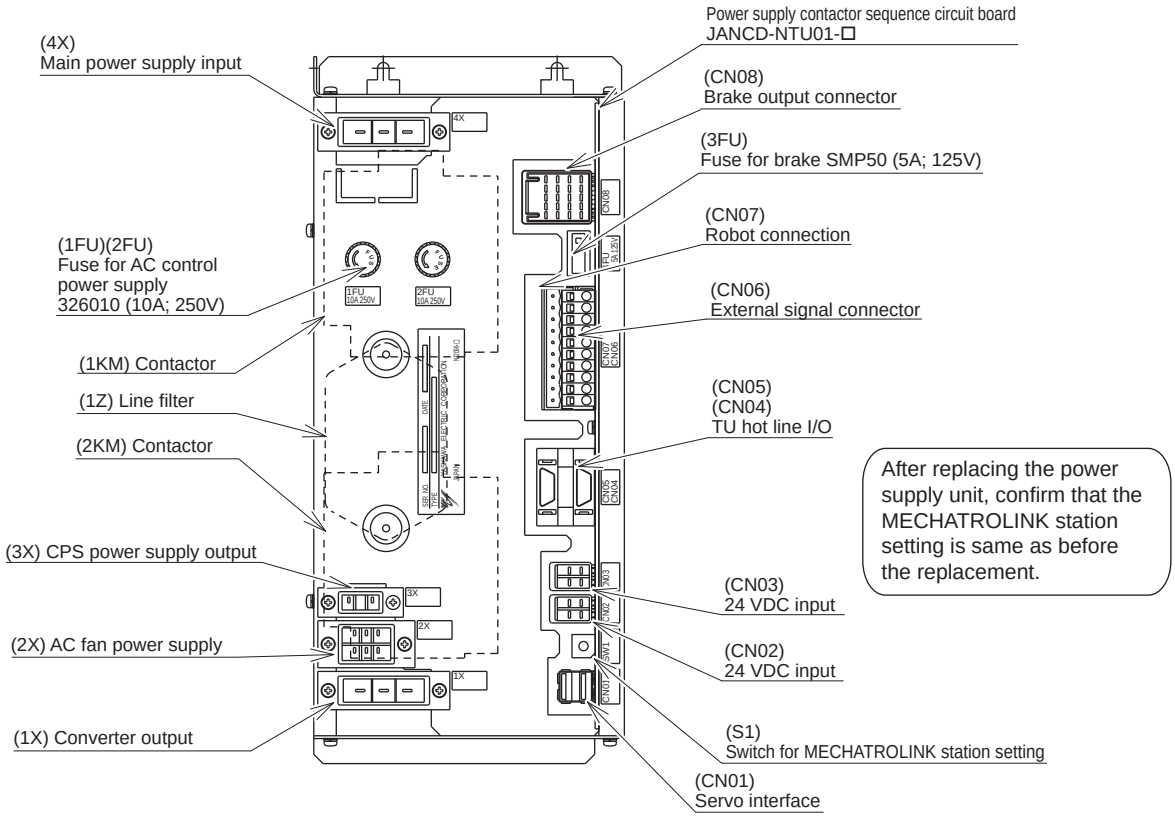


Appendix 4-1 Power Supply Contactor Unit

■ JZRCR-NTU01

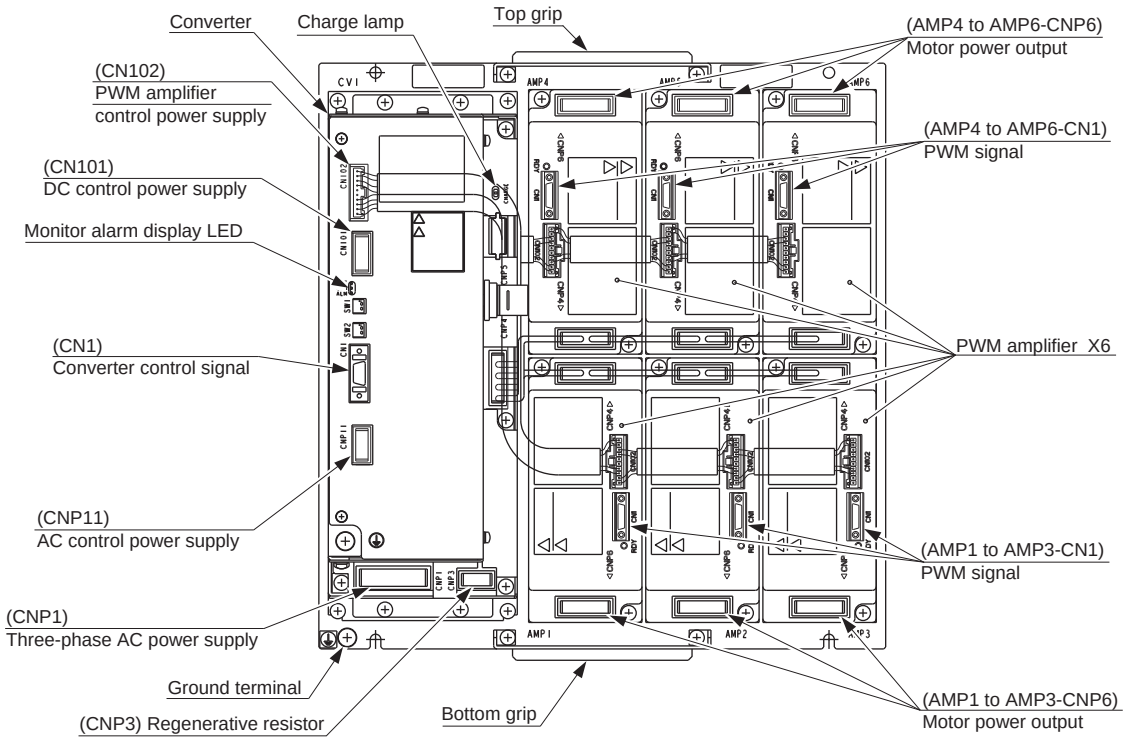


■ JZRCR-NTU02

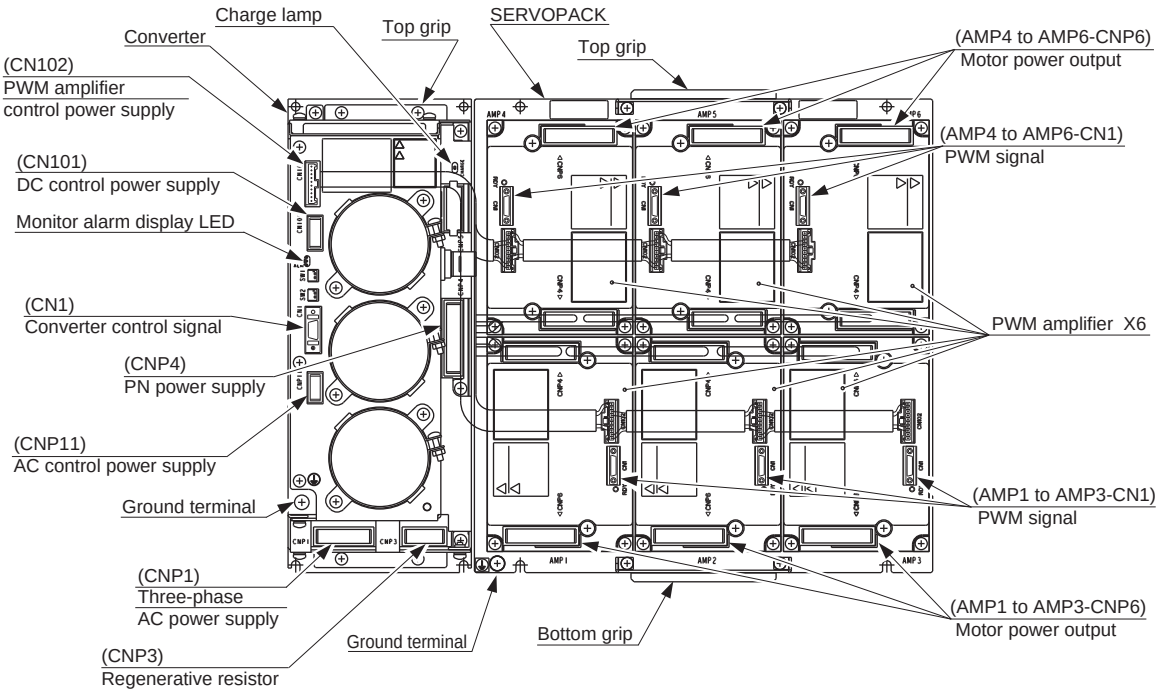


Appendix 5-1 SERVO PACK

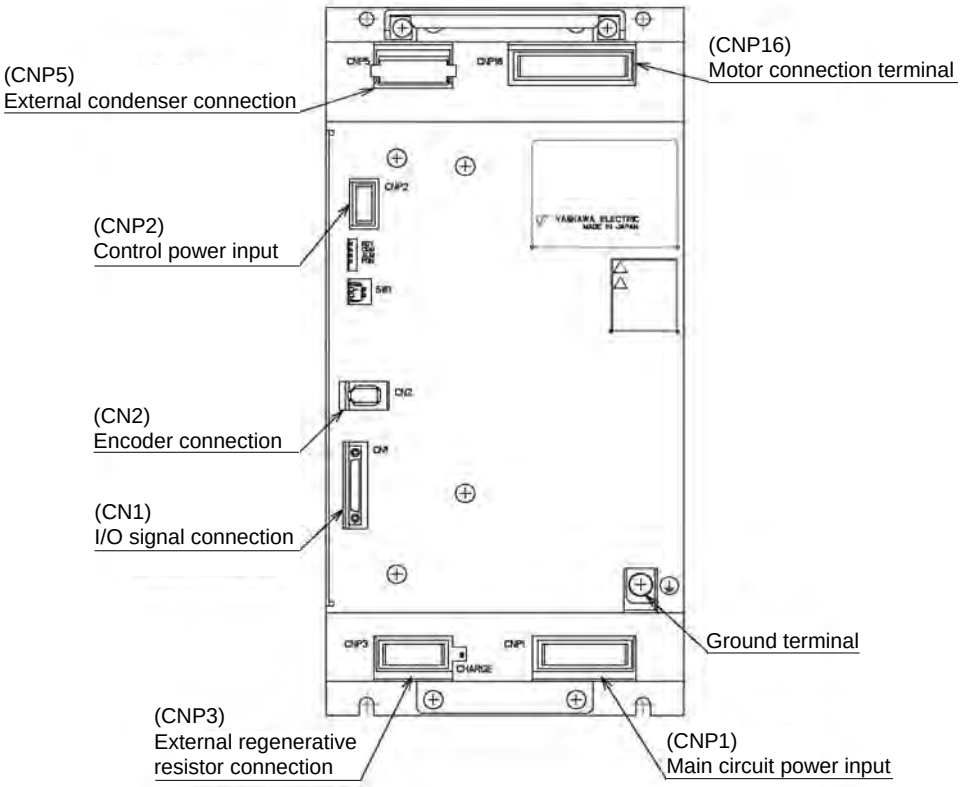
■ HP6 and EA1400N



■ ES165N and HP165



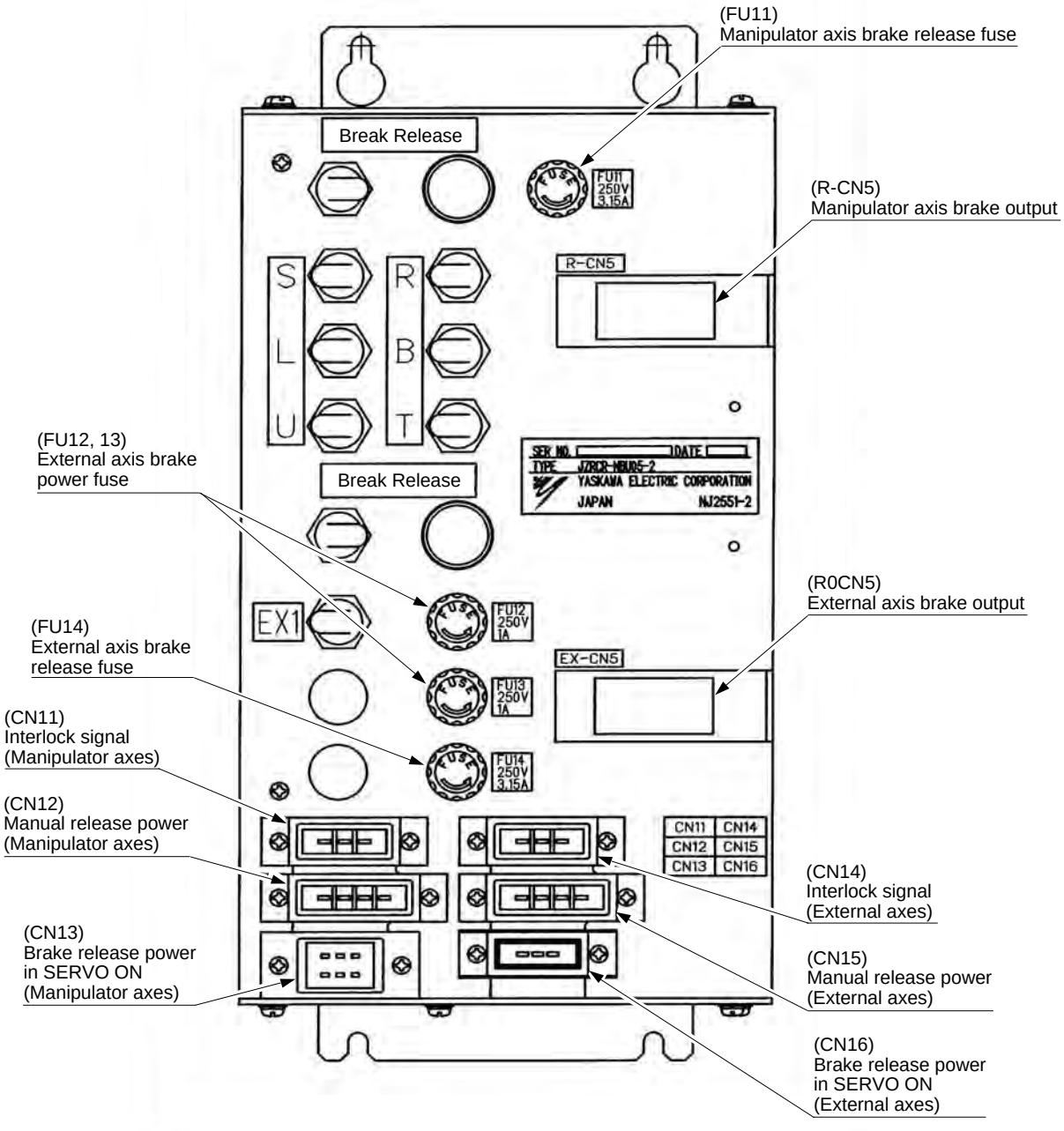
Appendix 6-1 SERVO PACK for External Axes



Appendix 7-1 Brake Release Unit

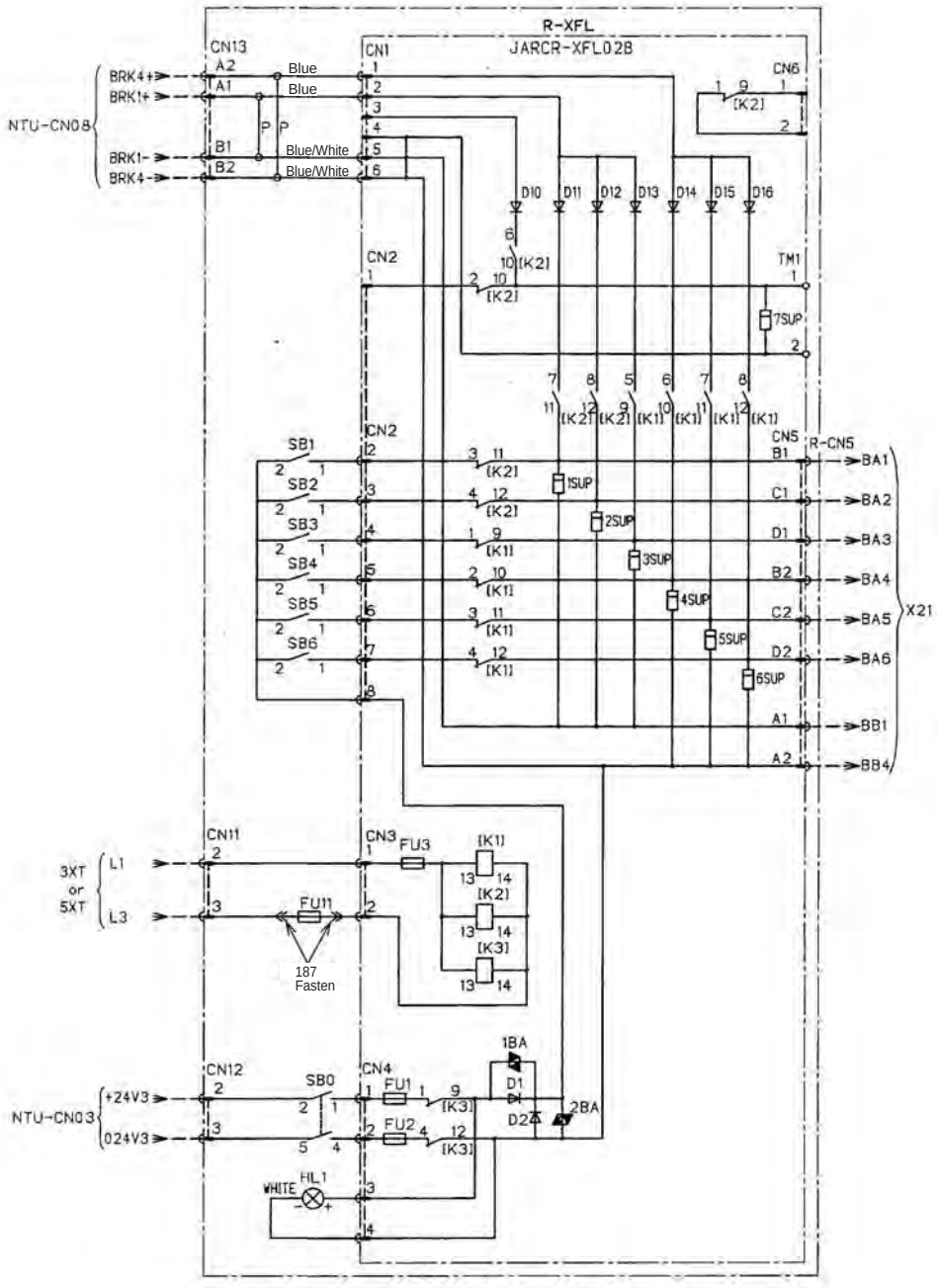
■ External View of JZRCR-NBU05-2

NOTE The following figure shows the external view of a typical brake release unit. The number of switches and fuses and their layouts may vary depending on the use of axes to be released.

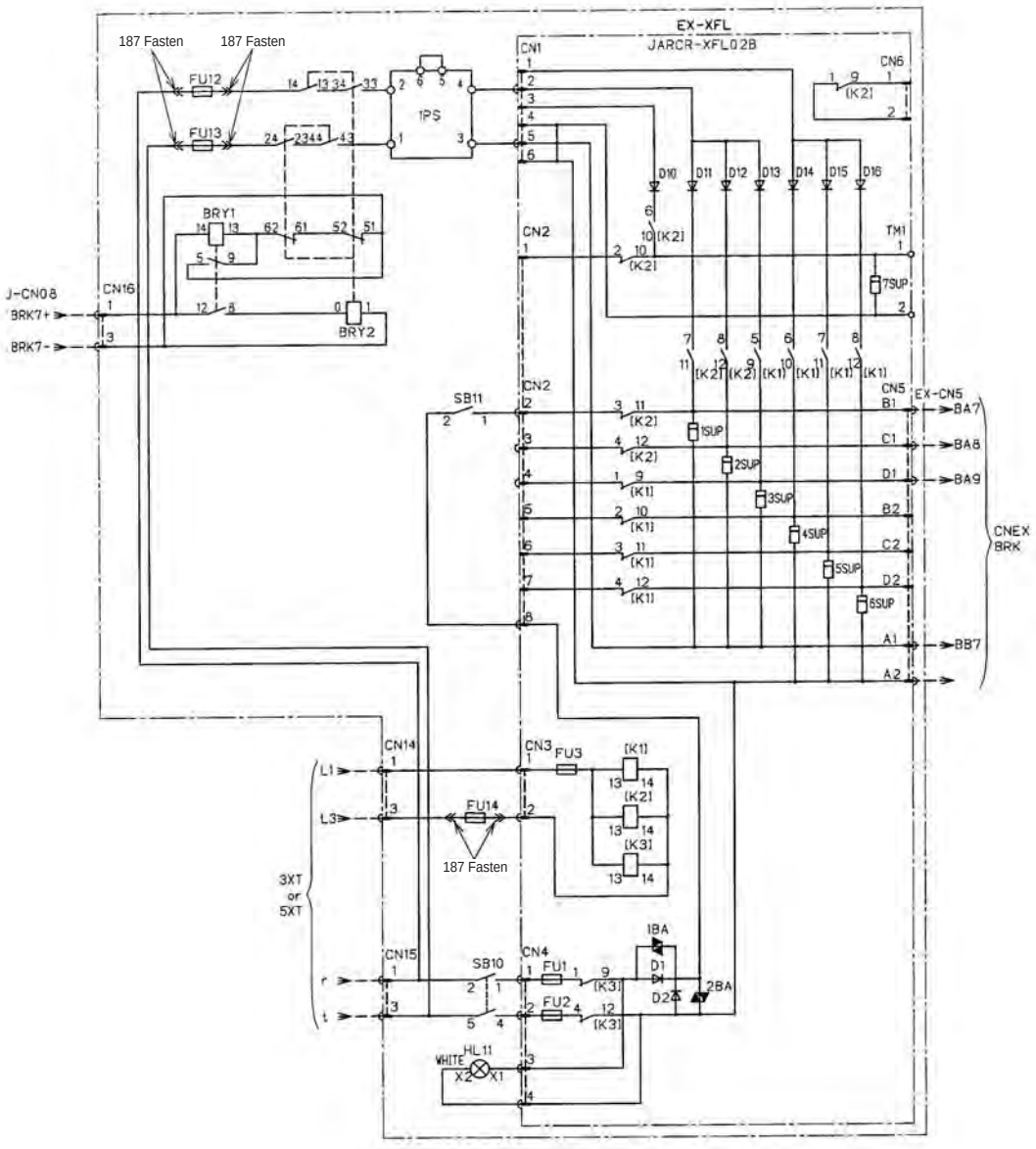


JZRCR-NBU05-2 (External View)

■ Internal Connections of JZRCR-NBU05-2 (1/2)



■ Internal Connections of JZRCR-NBU05-2 (2/2)



■ Measuring the Output Voltage of the Brake Release Unit

1. Remove the connector R-CN5 or EX-CN5 connected to the brake release unit.

2. While the servo power is ON, measure the voltage output from the R-CN5 or EX-CN5 connector on the brake release unit with a tester.

   If the brake release unit operates properly, the measured output voltage of each axis is as follows:

Output Voltage of R-CN5

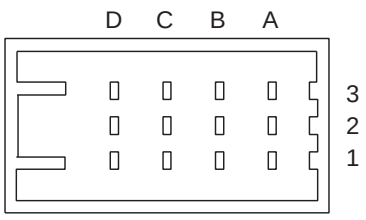
Axis Name	Pin No. (Positive Side)	Pin No. (0 V Side)	Output Voltage
S-axis	B1	A1 and A2	24 VDC
L-axis	C1		
U-axis	D1		
R-axis	B2		
B-axis	C2		
T-axis	D2		

Output Voltage of EX-CN5

Axis Name	Pin No. (Positive Side)	Pin No. (0 V Side)	Output Voltage*1
External axis 1	B1	A1 and A2	24 VDC or 90 VDC
External axis 2	C1		
External axis 3	D1		
External axis 4	B2		
External axis 5	C2		
External axis 6	D2		

1 The output voltage varies depending on the specifications of external axis motors. Since the 90 VDC is generated from the half-wave rectified 200 VAC, the output voltage may become as much as 280 VDC depending on the tester used for measurement.

Pin Assignment of R-CN5 and EX-CN5



Appendix 8-1 Internal Connections and Alarms of the NX100

Several types of connectors are used in the NX100 controller.
When removing/reinserting the connectors to replace units, ensure that each connector is properly connected to avoid the alarm caused by a loose connection.

A list on the following pages shows the alarms caused by connection failures.

- Some connectors have individual alarm signals assigned, and disconnection of such connectors may cause the same alarm as the one occurred in the connected external device. (e.g. SHOCK SENSOR ACTIVATION)
- For location of each connector, refer to Appendix 3-1 to 6-1.
- A connector code can be identified as the following example.
  e.g. NIF-CN01: CN01 of JZNC-NIF01.

Identify the rest of the connector codes in the same manner as follows:

AXA-CN☐☐: a connector on SGDR-AXA01A.

AXB-CN☐☐: a connector on SGDR-AXB01 (child circuit board of AXA).

NBB-CN☐☐: a connector on the back board.

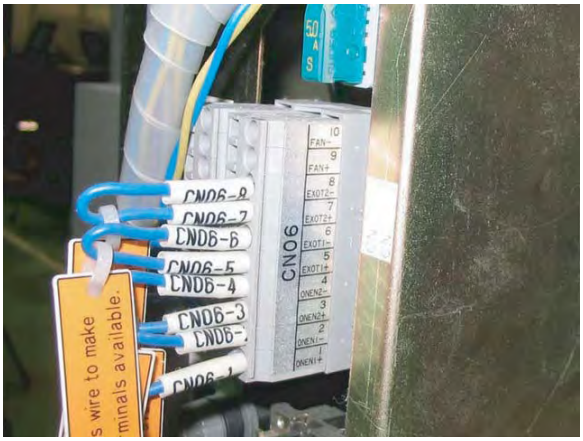
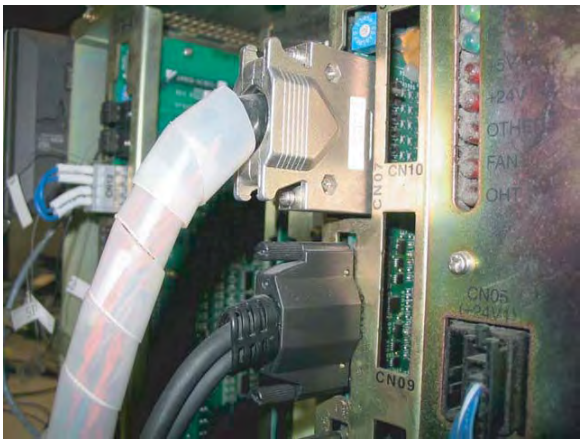
The alarms caused by connection failures

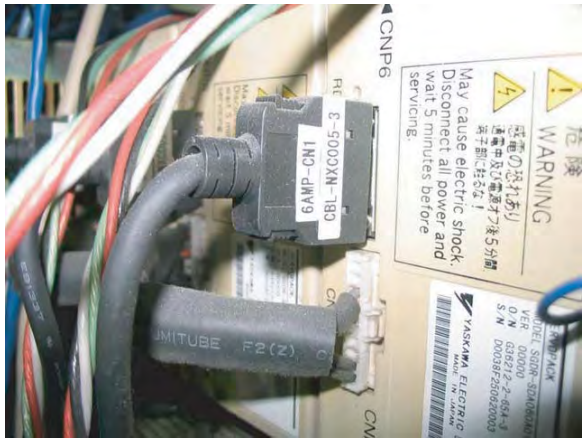
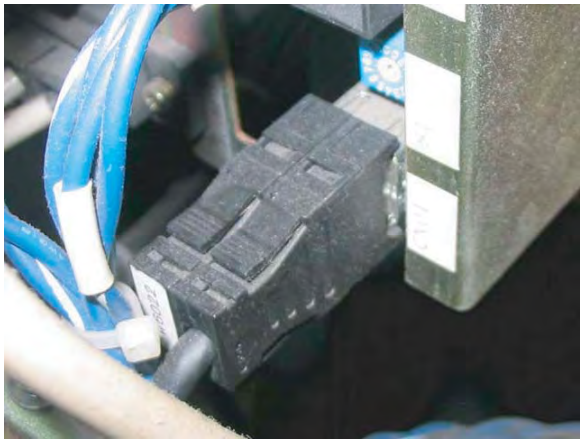
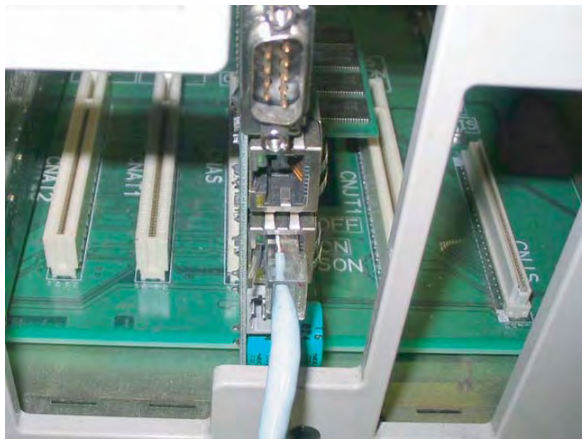
Connector Code	Alarm No. (4xxx signifies Major Failure)	Message	Sub Code	Remarks
AXA-CN01	1343	COMMUNICATION ERROR (CONVERTER)	[102]	
	4359	CONVERTER ERROR	[0000……0001]	
AXA-CN02	0021	COMMUNICATION ERROR (SERVO)	[50]	
	0100	COMMUNICATION ERROR (AXA#1)	[1]	
	0020	CPU COMMUNICATION ERROR	[50]	
AXA-CN03	1302	COMMUNICATION ERROR (SERVO I/O)	[112]	
AXA-CN04	4306	AMPLIFIER READY SIGNAL ERROR	-	
	1514	OVERHEAT (AMPLIFIER)	-	
	1306	AMPLIFIER TYPE MISMATCH	-	
AXA-CN05	4306	AMPLIFIER READY SIGNAL ERROR	-	
	1514	OVERHEAT (AMPLIFIER)	-	
	1306	AMPLIFIER TYPE MISMATCH	-	
AXA-CN06	4306	AMPLIFIER READY SIGNAL ERROR	-	
	1514	OVERHEAT (AMPLIFIER)	-	
	1306	AMPLIFIER TYPE MISMATCH	-	
AXA-CN07	1325	COMMUNICATION ERROR (ENCODER)	-	
NTU-CN01	1302	COMMUNICATION ERROR (SERVO I/O)	[112]	
NTU-CN02	1302	COMMUNICATION ERROR (SERVO I/O)	[112]	
NTU-CN04	1345	SAFETY CIRCUIT SIGNAL UNMATCH (SERVO)	[102]	
NTU-CN06	4101	OVERRUN (EXTERNAL AXIS)	[0000……0001]	
	1345	SAFETY CIRCUIT SIGNAL UNMATCH (SERVO)	[105]	
	1503	SAFETY CIRCUIT IN FAULT (SV I/O)	[0000……0001]	

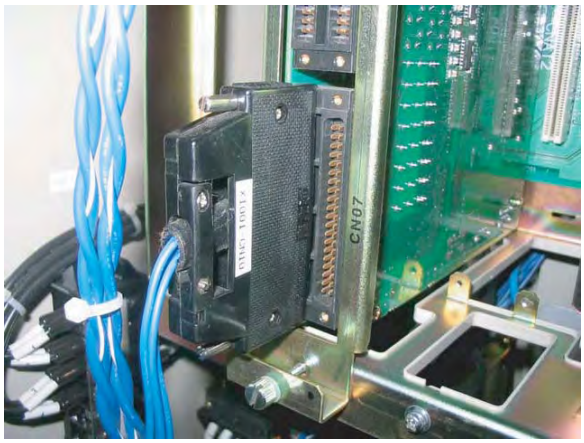
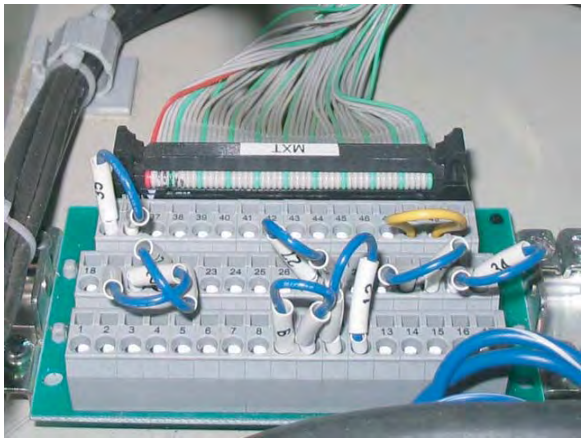
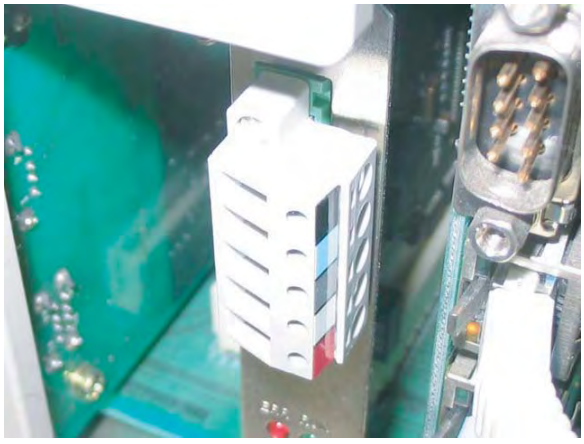
Connector Code	Alarm No. (4xxx signifies Major Failure)	Message	Sub Code	Remarks
NTU-CN07	4110	SHOCK SENSOR ACTIVATION	[0000……0001]	
	4100	OVERRUN (ROBOT AXIS)	[0000……0001]	
NTU-CN08	4328	SERVO TRACKING ERROR	-	Occurs when SERVO power is ON.
NTU-1X	4304	CONVERTER INPUT POWER ERROR	[0000……0001]	Occurs when SERVO power is ON.
NTU-2X (Upper Side)	-	-	-	Power is turned OFF when unconnected.
NTU-2X (Lower Side)	-	-	-	
NTU-3X	-	-	-	Cooling fan EV1 and EV2 stop.
NTU-4X	-	-	-	Power is turned OFF when unconnected.
NIF-CN01	0021	COMMUNICATION ERROR (SERVO)	[50]	
	0100	COMMUNICATION ERROR (AXA#1)	[1]	
NIF-CN03	1204	COMMUNICATION ERROR (IO MODULE)	[0000……0010]	
NIF-CN06	-	-	-	Power is turned OFF when unconnected.
NIF-CN11	1204	COMMUNICATION ERROR (IO MODULE)	[0000……0010]	
NIF-CN12	4109	DC 24V POWER SUPPLY FAILURE (I/O)	[0000……0011]	
NIF-F1	4109	DC 24V POWER SUPPLY FAILURE (I/O)	[0000……0011]	
NIF-F2	4109	DC 24V POWER SUPPLY FAILURE (I/O)	[0000……0011]	
CV1-CN1	1343	COMMUNICATION ERROR (CONVERTER)	[102]	
CV1-CNP11	-	-	-	
CV1-CN101	4306	AMPLIFIER READY SIGNAL ERROR	-	
	1343	COMMUNICATION ERROR (CONVERTER)	[102]	
CV1-CN102	4306	AMPLIFIER READY SIGNAL ERROR	-	
NCP-LAN1	-	-	-	A msg "Programming pendant is disconnected" appears on blank screen.

Connector Code	Alarm No. (4xxx signifies Major Failure)	Message	Sub Code	Remarks
NBB-CNSP1	-	-	-	A msg "Robot is stopped by P.PANEL emergency stop" appears.
NBB-CNPU	1345	SAFETY CIRCUIT SIGNAL UNMATCH (SERVO)	[105]	
NBB-CNM	-	-	-	
SV1-AMP*-CNP4	4306	AMPLIFIER READY SIGNAL ERROR	-	
SV1-AMP*-CNP6	4306	AMPLIFIER READY SIGNAL ERROR	-	*Signifies AMP1 to 6. (Each alarm is common for all amplifiers.)
	1514	OVERHEAT (AMPLI-FIER)	-	
	1306	AMPLIFIER TYPE MIS-MATCH	-	
	4328	SERVO TRACKING ERROR	-	Occurs when SERVO power is ON. *Signifies AMP1 to 6. (Each alarm is common for all amplifiers.)
SV1-AMP*-PN	4328	SERVO TRACKING ERROR	-	Occurs when SERVO power is ON. *Signifies AMP1 to 6. (Each alarm is common for all amplifiers.)
AXB-CN08	1325	COMMUNICATION ERROR (ENCODER)	-	
	1343	COMMUNICATION ERROR (CONVERTER)	[202]	
	1343	COMMUNICATION ERROR (CONVERTER)	[302]	
EX1SV-CNP2	1325	COMMUNICATION ERROR (ENCODER)	-	
	1343	COMMUNICATION ERROR (CONVERTER)	[202]	
EX1SV-CN1	1325	COMMUNICATION ERROR (ENCODER)	-	
	1343	COMMUNICATION ERROR (CONVERTER)	[202]	
EX1SV-CN2	1325	COMMUNICATION ERROR (ENCODER)	-	
EX1SV-CNP1	4304	CONVERTER INPUT POWER ERROR	[0000……0001]	Occurs when SERVO power is ON.
EX1SV-CNP3	4338	REGENERATION TROUBLE (CONVERTER)	[0000……0010]	Occurs when SERVO power is ON.

This section explains the typical connectors used and instructions on how to connect them.

• Tyco (AMP) Connector Insert the connector securely until it clicks.	
• WAGO Connector Insert the connector pushing all the way in. Make sure that the connector fully contacts the socket without leaving any gap.	
• SERVO Board Connector Insert the connector and fix it with two screws	

• SERVO PACK Connector Insert the connector securely until it clicks.	
• Mechatro Link Connector Insert the connector securely until it clicks.	
• Ethernet Connector for Programming Pendant Insert the connector securely until it clicks.	

• I/O Connector Insert the connector and fix it with two screws.	
• Flat Cable Connector Insert the connector pushing all the way in, and lift up the lock to fix it.	
• Connector for Communications (DeviceNet, etc.) Insert the connector and fix it with two screws.	

NX100 OPTIONS
INSTRUCTIONS

FOR TROUBLESHOOTING

HEAD OFFICE
2-1 Kurosakishiroishi, Yahatanishi-ku, Kitakyushu 806-0004 Japan
Phone +81-93-645-7745 Fax +81-93-645-7746

YASKAWA America Inc. MOTOMAN Robotics Division
100 Automation Way, Miamisburg, OH 45342, U.S.A.
Phone +1-937-847-6200 Fax +1-937-847-6277

YASKAWA Nordic AB
Verkstadsgatan 2, PO Box 504,SE-385 25 Torsås, Sweden
Phone +46-480-417-800 Fax +46-486-414-10

YASKAWA Europe GmbH Robotics Div.
Kammerfeldstr. 1, 80591 Allershausen, Germany
Phone +49-8166-90-0 Fax +49-8166-90-103

YASKAWA Electric Korea Co., Ltd
9F, KYOBO Securities Bldg., 26-4, Yeoido-Dong Yeoungeungpo-ku,Seoul, KOREA
Phone +82-2-784-7844 Fax +82-2-784-8495

YASKAWA Electric (Singapore) PTE Ltd.
151 Lorong Chuan, #04-02A, New Tech Park, Singapore 556741
Phone +65-6282-3003 Fax +65-6289-3003

YASKAWA Electric (Thailand) Co., Ltd.
252/246, 4th Floor. Muang Thai-Phatra Office Tower II Rachadaphisek Road, Huaykwang Bangkok, 10320 Thailand
Phone +66-2-693-2200 Fax +66-2-693-4200

YASKAWA Shougang Robot Co. Ltd.
No.7, Yongchang-North Road, Beijing E&T Development Area, China 100176
Phone +86-10-6788-0548 Fax +86-10-6788-0548-813

YASKAWA ELECTRIC (SHANGHAI) Co., Ltd.
No.18Xizang Zhong Road, 17F, Harbour Ring Plaza, Shanghai 200001, CHINA
Phone +86-21-5385-0655 Fax +86-21-5385-2770

YASKAWA Robotics India Ltd.
#426, Udyog Vihar, Phase- IV, Gurgaon, Haryana, India
Phone +91-124-475-8500 Fax +91-124-414-8016

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for ongoing product modifications and improvements.

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MOTOMAN ROBOTICS

MANUAL NO.
HW0483057 ⑥